УТВЕРЖДЕНО
Приказом генерального директора
ООО «Северный Ветер»
от « 28 » мая 2021 года
№ 01-165

РУКОВОДСТВО ПО ПРОИЗВОДСТВУ ПОЛЁТОВ

Часть B
Том 2 «Информация по эксплуатации воздушного судна B737 NG»

Химки, Московская область – 2021

ОГЛАВЛЕНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

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0.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

0.1. ВВЕДЕНИЕ

0.1.1. Общие положения

Часть В РПП для ВС B-737 NG является основным нормативным документом, регламентирующим летную эксплуатацию ВС B-737 NG в Авиапредприятии.

Часть В РПП для ВС B-737 NG разработана на основе РЛЭ (FCOM) и других документов (QRH, FCTM, WBM), а также Руководства по организации работы СБП для ВС B-737 NG, в соответствии с Федеральными авиационными правилами Российской Федерации, а также стандартами и рекомендуемой практикой международной организации гражданской авиации (ICAO).

В случаях, когда это не противоречит FCOM ВС B-737 NG и Федеральным авиационным правилам Российской Федерации, учтены также требования Европейской объединенной авиационной администрации (EU OPS.1) и стандарты и рекомендации IATA (IOSA Standard Manual).

РПП Часть В ВС B-737 NG включает в себя требования и процедуры, а также практические рекомендации, необходимые для летной эксплуатации ВС B-737 NG в Авиапредприятии.

Точное выполнение требований и процедур, изложенных в части В РПП, является обязательным для всех членов экипажей Авиапредприятия при эксплуатации ВС B-737 NG.

Если в части В имеются расхождения с FCOM, FCTM, QRH в плане выполнения технологических процедур и зон ответственности, то представленный в части В материал является приоритетным и обязательным к исполнению.

Стандартные слова и выражения на английском языке соответствуют принятым понятиям, используемым в AFM, FCOM, FCTM, QRH, фирмы производителя ВС B-737 NG.

0.1.2. Требования к оформлению и печати РПП

Требования к оформлению РПП подробно описаны в «Процедуре управления руководством по производству полетов», p-01-002.

Эталонный экземпляр печатается в формате А4 (двухсторонняя печать), контрольные и рабочие экземпляры печатаются на формате А5 (двухсторонняя печать) и рассылаются согласно перечню держателей РПП.

0.1.3. Индексные указатели для быстрого нахождения информации

Индексные указатели представляют собой систему деления РПП на следующие уровни: часть, глава, раздел, пункт, подпункт. При необходимости возможно применение более мелких уровней, обозначаемых буквами латинского алфавита, арабскими или римскими цифрами в круглых скобках.

Каждая часть РПП имеет оглавление, включающее перечень дополнений к данной части РПП. Главы делятся на разделы, обозначаемые двумя цифрами с номером главы и раздела. Индексные указатели по тексту в пределах одного раздела состоят из пунктов и подпунктов.

Ссылки по тексту РПП даются, как правило, полным обозначением, содержащим номер главы, раздела, пункта, подпункта и более мелкие уровни системы деления (например, 7.2.2.1 (2) (a)).

Если ссылка по тексту указывает на пункт РПП в пределах данного раздела, то обозначение раздела может не указываться (например: 2.1 (а)).

0.2. ТЕРМИНЫ, ОПРЕДЕЛЕНИЯ И СОКРАЩЕНИЯ

Основные термины, определения и сокращения, применяемые в части В соответствуют принятым понятиям и сокращениям, используемым в официальных изданиях нормативной документации гражданской авиации Российской Федерации, части А РПП Авиапредприятия, а также руководств и справочников фирмы производителя ВС Boeing (AFM, FCOM, QRH, FCTM).

0.2.1. Определение терминов Warning, Caution and Note

WARNING: AN OPERATING PROCEDURE, TECHNIQUE, ETC., THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CAREFULLY FOLLOWED.

CAUTION: An operating procedure, technique, etc., that may result in damage to equipment if not carefully followed.

NOTE: An operating procedure, technique, etc., considered essential to emphasize. Information contained in notes may also be safety related.

0.2.2. Сокращения

A	
AAL	Above Aerodrome Level
A/P	Autopilot
A/T	Autothrottle
AC	Alternating Current
ACARS	Aircraft Communications Addressing and Reporting System
ACP	Audio Control Panel
ACQ	Acquire
ADF	Automatic Direction Finder
ADIRU	Air Data Inertial Reference Unit
AFDS	Autopilot Flight Director System
AFM	Airplane Flight Manual (FAA approved)
ALT	Altitude
ALTN	Alternate
AM	Amplitude Modulation
ANP	Actual Navigation Performance
APP	Approach
APU	Auxiliary Power Unit
ARPT	Airport
ATC	Air Traffic Control

ATT	Attitude
AUTO	Automatic
AUX	Auxiliary
AVAIL	Available
	B
B/C or BCRS	Back Course
BARO	Barometric
BAT/BATT	Battery
BRT	Bright
BTL DISCH	Bottle Discharge (fire extinguishers)
BTP	Bromotrifluropropene (fire extinguishers)
	C
C	Captain, Celsius, Center
CANC/ RCL	Cancel/Recall
CB	Circuit Breaker
CDU	Control Display Unit
CG	Center of Gravity
CHKL	Checklist
CLB	Climb
COMM	Communication
CON	Continuous
CONFIG	Configuration
CRZ	Cruise
CTL	Control
	D
DDG	Dispatch Deviations Guide
DEP ARR	Departure Arrival
DES	Descent
DEU	Display Electronics Unit
DISC	Disconnect
DME	Distance Measuring Equipment
DSPL	Display
	E
E/D	End of Descent
E/E	Electrical and Electronic
EASA	European Aviation Safety Agency
EEC	Electronic Engine Control
EFIS	Electronic Flight Instrument System
EGPWS	Enhanced Ground Proximity Warning System

EGT	Exhaust Gas Temperature
ELEC	Electrical
ENG	Engine
EOSID	Engine Out Standard Instrument Departure
ESF	Estimated surface friction
EXEC	Execute
EXT	Extend
F	
F/D or FLT DIR	Flight Director
F/O	First Officer
FAF	Final Approach Fix
FAP	Final Approach Point
FCC	Flight Control Computer
FCTL	Flight Control
FCTM	Flight Crew Training Manual
FL	Flight Level
FMC	Flight Management Computer
FMS	Flight Management System
FPV	Flight Path Vector
G	
G/P or GP	Glide Path
G/S	Glide Slope
GA	Go-Around
GBAS	Ground-Based Augmentation System
GEN	Generator
GLS	GBAS Landing System
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
H	
HDG	Heading
HDG SEL	Heading Select
HYD	Hydraulic
I	
IAN	Integrated ApproachNavigation
IAS	Indicated Airspeed
IDENT	Identification
ILS	Instrument Landing System
IN	Inches
INBD	Inboard
IND LTS	Indicator Lights
INOP	Inoperative

INTC CRS	Intercept Course
ISFD	Integrated Standby Flight Display
ISLN	Isolation
K	
K	Knots
KGS	Kilograms
L	
L	Left
LAT	Latitude
LBS	Pounds
LDG ALT	Landing Altitude
LE	Leading Edge
LIM	Limit
LNAV	Lateral Navigation
LOM	Locator Outer Marker
LONG	Longitude
LVL CHG	Level Change
M	
M	Meter
MAG	Magnetic
MAN	Manual
MCP	Mode Control Panel
MDA	Minimum Descent Altitude
MEL	Minimum Equipment List
MIN	Minimum
MKR	Marker
MMO	Maximum Mach Operating Speed
N	
N1	Low Pressure Rotor Speed
NGS	Nitrogen Generation System
NM	Nautical Miles
NORM	Normal
N2	High Pressure Rotor Speed
NAV RAD	Navigation Radio
ND	Navigation Display
O	
OPT	Onboard Performance Tool
OVHD	Overhead
OVRD	Override

P	
PASS	Passenger
PCU	Power Control Unit
PERF INIT	Performance Initialization
PF	Pilot Flying
PFC	Primary Flight Computers
PM	Pilot Monitoring
PNL	Panel
POS	Position
POS INIT	Position Initialization
PRI	Primary
PWS	Predictive Windshear System
R	
R	Right
RA	Radio Altitude or Resolution Advisory
RCC	Runway Condition Code
REF	Reference
RET	Retract
RF	Refill
RH	Right Hand
RNP	Required Navigation Performance
RVSM	Reduced Vertical Separation Minimum
S	
S/C	Step Climb
SEL	Select
SPD	Speed
STA	Station
STAB	Stabilizer
STD	Standard
T	
T/D	Top of Descent
TA	Traffic Advisory
TAI	Thermal Anti-Ice
TAT	Total Air Temperature
TCAS	Traffic Alert and Collision Avoidance System
TDZE	Touch Down Zone Elevation
TE	Trailing Edge
TFC	Traffic
THR HLD	Throttle Hold
TO	Takeoff

TO/GA	Takeoff/Go–Around
U	
UTC	Universal Time Coordinated
V	
V/S	Vertical Speed
V1	Takeoff Decision Speed
V2	Takeoff Safety Speed
VHF	Very High Frequency
VMO	Maximum Operating Speed
VNAV	Vertical Navigation
VOR	VHF Omnidirectional Range
VR	Rotation Speed
VREF	Reference Speed
VSD	Vertical Situation Display
W	
WPT	Waypoint
WXR	Weather Radar

0.3. UNITS OF MEASUREMENT

Расстояние, используемое в навигации.	километры (км), морская миля (мм), метры (м).
Высота.	футы (ft), метры (м).
Горизонтальная скорость.	узлы (kt), число М.
Вертикальная скорость.	футы в минуту (ft/min).
Скорость ветра.	километры в час (км/ч), метры в секунду (м/с), узлы (kt).
Направление, используемое в навигации.	градусы от северного направления истинного или магнитного меридиана.
Направление ветра.	градусы от магнитного северного направления (для взлета и посадки).
Видимость.	километры (км), метры (м), футы (ft).
Дальность видимости на ВПП.	метры (м), километры (км).
Атмосферное давление.	гектопаскали(гПа), миллиметры ртутного столба (мм рт.ст.), дюймы ртутного столба (in m.c).
Температура.	градусы Цельсия, градусы Фарингейта.
Масса, вес.	килограммы (кг), фунты (lbs), тонны (т).
Объём.	литры (л), кубические метры (м3).
Время.	секунды (с), минуты (мин), часы (ч).

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0.4. UNITS CONVERSION TABLE

Anglo-British → Metric				Metric → Anglo-British			
Density							
lbs / US Gal	x	0,11983	= kg / Ltr	kg / Ltr	x	8,34514	= lbs/US Gal
Pressure							
lbs/in2 (psi)	x	6,89476	= kPa	kPa	x	0,145	= lbs/in2 (psi)
inchHg	x	33,8639	= hPa	hPa	x	0,02953	= inch Hg
mmHg	x	1,3332	= hPa	hPa	x	0,75	= mmHg
Weights - General							
oz (ounces)	x	28,35	= g	g	x	0,03527	= oz (ounces)
lbs (pounds)	x	0,453592	= kg	kg	x	2,20462	= lbs (pounds)
Weights of Fuel (density 0,8 kg/l)							
Ltr	x	0,8	= kg	Ltr	x	0,719	= kg
Ltr	x	1,764	= lbs	Ltr	x	1,585	= lbs
US Gal	x	3,04	= kg	US Gal	x	2,73	= kg
US Gal	x	6,7	= lbs	US Gal	x	6,0	= lbs
Imp Gal	x	3,64	= kg	Imp Gal	x	3,28	= kg
kg	x	1,25	= Ltr	kg	x	1,39	= Ltr
lbs	x	0,57	= Ltr	lbs	x	0,631	= Ltr
lbs	x	0,45359	= kg	lbs	x	0,45359	= kg
lbs	x	0,15	= US Gal	lbs	x	0,167	= US Gal
Linear Measures and Speeds							
Ft	x	0,3048	= m	m	x	3,2808	= Ft
inch	x	25,4	= mm	mm	x	0,03937	= inch
yd	x	0,9144	m	m	x	1,0936	= yd
NM	x	1,852	= km	km	x	0,53996	= NM
SM	x	1,60934	= km	km	x	0,62137	= SM
Ft / sec	x	0,3048	= m /sec	m /sec	x	3,2808	= Ft / sec
Ft / min	x	0,00508	= m /sec	m /sec	x	196,848	= Ft / min
NM / hrs	x	0,51444	= m /sec	m /sec	x	1,9438	= NM / hrs
NM / hrs	x	1,01267	= Ft / min	km / hrs	x	0,27778	= m /sec
Ft / min	x	0,98749	= NM / hrs	m /sec	x	3,6	= km / hrs
Temperature							
F° - 32 x 5 / 9			= C°	C° x 9 / 5 + 32			= F°
Volume							
US Gal	x	3,78541	= Ltr	Ltr	x	0,26418	= US Gal
US Gal	x	0,83267	= Imp Gal	Imp Gal	x	1,201	= US Gal
Imp Gal	x	4,54609	= Ltr	Ltr	x	0,21997	= Imp Gal
fluid ounce	x	29,57	= cm³	cm³	x	0,0338	= fluid ounce
Power							
PS	x	0,73549	= KW	KW	x	1,3596	= PS

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0.5. СИСТЕМА ВНЕСЕНИЯ ПОПРАВОК И ИЗМЕНЕНИЙ

РПП является одним из основных документов в системе документации по безопасности полетов Авиапредприятия и подлежит переизданию с периодичностью не реже одного раза в пять лет или при наступлении обстоятельств, требующих переиздания, а также раз в год документ должен быть проанализирован на актуальность.

Новое издание РПП согласовывается с заместителями генерального директора и руководителями структурных подразделений по направлениям деятельности, затронутым в проекте документа. РПП утверждается и вводится в действие приказом генерального директора.

В РПП вносятся изменения при:

– изменении условий эксплуатации воздушных судов;
– вступлении в силу нормативных правовых актов, регулирующих деятельность юридических лиц и индивидуальных предпринимателей, осуществляющих коммерческие воздушные перевозки;
– принятии решения руководством Авиапредприятия о совершенствовании процессов обеспечения, организации и производства полетов.

Изменения, дополнения, временные изменения в РПП проходят процедуру согласования с должностными лицами по направлениям деятельности, затронутым в проекте РПП, утверждаются и вводятся в действие приказом генерального директора Авиапредприятия.

Дата введения в действие изменения, дополнения, временного изменения отличается от даты его утверждения с учетом времени (не более 30 календарных дней), необходимого для рассылки пользователям, их ознакомления с утвержденным изменением до даты введения в действие. Дата введения в действие указывается в нижнем колонтитуле утвержденного изменения.

Бюллетени к РПП согласовываются при необходимости с руководителями структурных подразделений, затронутыми в проекте бюллетеня, утверждаются заместителем генерального директора по ОЛР – директором летного департамента, направляются в ЛМО на регистрацию, нормоконтроль, рассылку держателям, учтенных экземпляров и хранение. Бюллетени вводятся в действие с даты, указанной на титульном листе, утвержденного бюллетеня.

Лётно-методический отдел, путем направления сопроводительного письма, уведомляет уполномоченный орган о переиздании РПП или внесении изменений в РПП не позднее чем за 20 дней до вступления их в действие.

Руководители структурных подразделений своевременно организуют изучение персоналом положений руководств, в том числе при внесении в них изменений.

Изменения вносятся в действующие страницы и обозначаются с левого края вертикальной чертой.

Информация срочного характера должна предоставляться держателям РПП в виде бюллетеней и обязательна к исполнению с даты ввода в действие. Срок действия бюллетеней не должен превышать 6 месяцев. По истечении срока действия бюллетень вносится в текущее изменение или отменяется распоряжением заместителя генерального директора по ОЛР – директора летного департамента. Каждому бюллетеню присваивается порядковый номер.

Бюллетени размещаются к соответствующей странице и учитываются в перечне действующих бюллетеней.

Временные изменения действительны в течение не более одного года от даты утверждения и вносятся в РПП с заменой (добавлением) листов желтого цвета. По истечении срока действия временное изменение вносится в текущее изменение или отменяется приказом генерального директора Авиапредприятия.

Редакционные поправки вносятся датой редактирования под номером страницы с добавлением буквы R (например: 27.10.2006 0.6.2R).

Эталонный экземпляр, а также изменения, дополнения, временные изменения и бюллетеней к нему хранятся в ЛМО в соответствии со сводной номенклатурой дел Авиапредприятия. Эталонный экземпляр РПП оформляется в формате А4, контрольные и рабочие экземпляры бумажных версий РПП печатаются в формате А5 и рассылаются согласно перечню держателей РПП.

Каждый ответственный держатель РПП несет персональную ответственность за сохранность и своевременное внесение изменений в учтенный экземпляр РПП.

Подробные сведения по оформлению, согласованию и утверждению изменений, бюллетеней, временных изменений и дополнений в РПП описаны в «Процедуре управления руководством по производству полетов», p-01-002. Актуальная версия документа находится http://sirius:8080/nws_intranet; СЭД ООО «Северный Ветер».

0.6. ПЕРЕЧЕНЬ ДЕРЖАТЕЛЕЙ РПП (ЧАСТЬ В2)

Номер экземпляра	Статус экземпляра	Местоположение экземпляра	Ответственный за ведение экземпляра
001	эталонный	ЛМО	Заместитель директора летного департамента - начальник ЛМО
002	контрольный	ЦМТУ	Заместитель директора летного департамента - начальник ЛМО
003	контрольный электронная версия	ФАВТ	Заместитель директора летного департамента - начальник ЛМО
004	рабочий	ЛМО	ЛМО
005	рабочий	ИБП	Заместитель генерального директора по безопасности полетов – начальник инспекции по безопасности полетов

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0.7. ИНФОРМАЦИЯ ОБ ИЗМЕНЕНИЯХ И БЮЛЛЕТЕНЯХ

0.7.1. Запись истории документа

Номер издания	№ приказа, дата приказа об утверждении	Дата ввода в действие	Примечание
Издание 01	-	29.04.2008	Базовый документ
Издание 02	-	01.02.2013	Плановое переиздание
Издание 03	-	01.10.2015	Плановое переиздание
Издание 04	-	08.09.2018	Плановое переиздание
Издание 05	№ 01-165 от 28.05.2021	21.06.2021	Перераспределение функций в ООО «Северный Ветер», новый формат РПП на основании р-01-002

0.7.2. Изменения к пятому изданию

Номер изменения	№ и дата приказа об утверждении	Дата введения в действие
Изменение № 1	№ 01-758 от 08.12.2021	01.01.2022
Изменение № 2	№ 01-357 от 13.04.2022	16.05.2022
Изменение № 3	№ 01-055 от 08.02.2023	20.03.2023
Изменение № 4	№ 01-700 от 25.12.2024	20.01.2025

0.7.3. Действующие бюллетени

Номер бюллетеня	Глава	Дата утверждения	Примечание

0.8. ПЕРЕЧЕНЬ ДЕЙСТВУЮЩИХ СТРАНИЦ

Страница	№ Изменения	Дата ввода в действие
1-2	00	21.06.2021
3-4	04	20.01.2025
Глава 0		
0.0.1-0.0.2	04	20.01.2025
0.1.1-0.1.2	00	21.06.2021
0.2.1	01	01.01.2022
0.2.2	00	21.06.2021
0.2.3	04	20.01.2025
0.2.4	00	21.06.2021
0.2.5	04	20.01.2025
0.2.6	00	21.06.2021
0.3.1-0.3.2	00	21.06.2021
0.4.1-0.4.2	00	21.06.2021
0.5.1-0.5.2	04	20.01.2025
0.6.1-0.6.2	04	20.01.2025
0.7.1-0.7.2	04	20.01.2025
0.8.1-0.8.2	04	20.01.2025
Глава 1		
1.0.1-1.0.2	04	20.01.2025
1.1.1-1.1.2	04	20.01.2025
1.2.1-1.2.2	00	21.06.2021
1.3.1-1.3.2	01	01.01.2022
1.4.1-1.4.2	04	20.01.2025
1.5.1-1.5.5	04	20.01.2025
1.5.6	00	21.06.2021
1.5.7-1.5.8	04	20.01.2025
1.6.1	01	01.01.2022
1.6.2-1.6.3	04	20.01.2025
1.6.4	00	21.06.2021
Глава 2		
2.0.1-2.0.4	04	20.01.2025
2.1.1-2.1.4	03	20.03.2023
2.1.5-2.1.8	04	20.01.2025
2.2.1	00	21.06.2021
2.2.2-2.2.4	01	01.01.2022
2.3.1	02	16.05.2022
2.3.2-2.3.4	04	20.01.2025
2.4.1	04	20.01.2025
2.4.2	00	21.06.2021
2.4.3-2.4.4.	03	20.03.2023
2.4.5-2.4.6	00	21.06.2021

Страница	№ Изменения	Дата ввода в действие
2.5.1	04	20.01.2025
2.5.2	03	20.03.2023
2.5.3-2.5.4	04	20.01.2025
2.6.1	00	21.06.2021
2.6.2	03	20.03.2023
2.7.1	04	20.01.2025
2.7.2-2.7.4	00	21.06.2021
2.8.1	00	21.06.2021
2.8.2	04	20.01.2025
2.8.3	03	20.03.2023
2.8.4-2.8.14	04	20.01.2025
2.9.1	00	21.06.2021
2.9.2-2.9.3	04	20.01.2025
2.9.4	03	20.03.2023
2.9.5-2.9.8	04	20.01.2025
2.10.1-2.10.6	04	20.01.2025
2.11.1-2.11.28	04	20.01.2025
2.12.1-2.12.9	04	20.01.2025
2.12.10	00	21.06.2021
2.12.11	01	01.01.2022
2.12.12-2.12.29	04	20.01.2025
2.12.30	01	01.01.2022
2.12.31-2.12.41	04	20.01.2025
2.12.42	01	01.01.2022
2.12.43-2.12.54	04	20.01.2025
2.13.1	03	20.03.2023
2.13.2-2.13.3	00	21.06.2021
2.13.4	01	01.01.2022
2.13.5	00	21.06.2021
2.13.6-2.13.7	04	20.01.2025
2.13.8	01	01.01.2022
2.13.9-2.13.12	04	20.01.2025
Глава 3		
3.0.1-3.0.2	04	20.01.2025
3.1.1-.3.1.2	00	21.06.2021
Глава 4		
4.0.1-4.0.2	04	20.01.2025
4.1.1-4.1.2	04	20.01.2025
4.2.1-4.2.2	04	20.01.2025
4.3.1-4.3.4	04	20.01.2025
4.4.1-4.4.2	04	20.01.2025

Страница	№ Изменения	Дата ввода в действие	Страница	№ Изменения	Дата ввода в действие
4.5.1-4.5.5	04	20.01.2025	***Глава 6***		
4.5.6	00	21.06.2021	6.0.1-6.0.2	04	20.01.2025
4.5.7	04	20.01.2025	***Глава 7***		
4.5.8-4.5.10	00	21.06.2021	7.0.1-7.0.2	04	20.01.2025
4.6.1-4.6.3	04	20.01.2025	***Глава 8***		
4.6.4	01	01.01.2022	8.0.1-8.0.2	00	21.06.2021
4.6.5	04	20.01.2025	8.1.1-8.1.2	00	21.06.2021
4.6.6-4.6.8	04	20.01.2025	***Глава 9***		
4.6.9	02	16.05.2022	9.0.1-9.0.2	04	20.01.2025
4.6.10-4.6.16	04	20.01.2025	***Глава 10***		
4.7.1	00	21.06.2021	10.0.1-10.0.2	04	20.01.2025
4.7.2	04	20.01.2025	***Глава 11***		
4.7.3-4.7.4	00	21.06.2021	11.0.1-11.0.2	04	20.01.2025
4.8.1-4.8.2	04	20.01.2025	***Глава 12***		
4.9.1-4.9.2	04	20.01.2025	12.0.1-12.0.2	03	20.03.2023
4.10.1-4.10.2	00	21.06.2021	12.1.1-12.1.2	03	20.03.2023
4.11.1-4.11.2	00	21.06.2021	12.2.1-12.2.2	03	20.03.2023
4.12.1	03	20.03.2023	12.3.1-12.3.2	03	20.03.2023
4.12.2	00	21.06.2021	12.4.1-12.4.2	03	20.03.2023
4.13.1	00	21.06.2021	12.5.1-12.5.2	03	20.03.2023
4.13.2	03	20.03.2023	12.6.1-12.6.2	03	20.03.2023
4.14.1-4.14.2	04	20.01.2025	12.7.1	03	20.03.2023
4.15.1-4.15.2	03	20.03.20223	12.7.2	00	21.06.2021
Глава 5			12.8.1-12.8.2	03	20.03.2023
5.0.1-5.0.2	04	20.01.2025	***Глава 13***		
5.1.1-5.1.4	00	21.06.2021	13.01.-13.0.2	04	20.01.2025
5.2.1-5.2.2	00	21.06.2021	***Глава 14***		
5.2.3	04	20.01.2025	14.0.1-14.0.2	04	20.01.2025
5.2.4-5.2.6	00	21.06.2021			
5.3.1-5.3.6	00	21.06.2021			
5.4.1-5.4.10	00	21.06.2021			
5.5.1-5.5.6	00	21.06.2021			

Глава 1. General Aircraft Information and Limitations

1.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

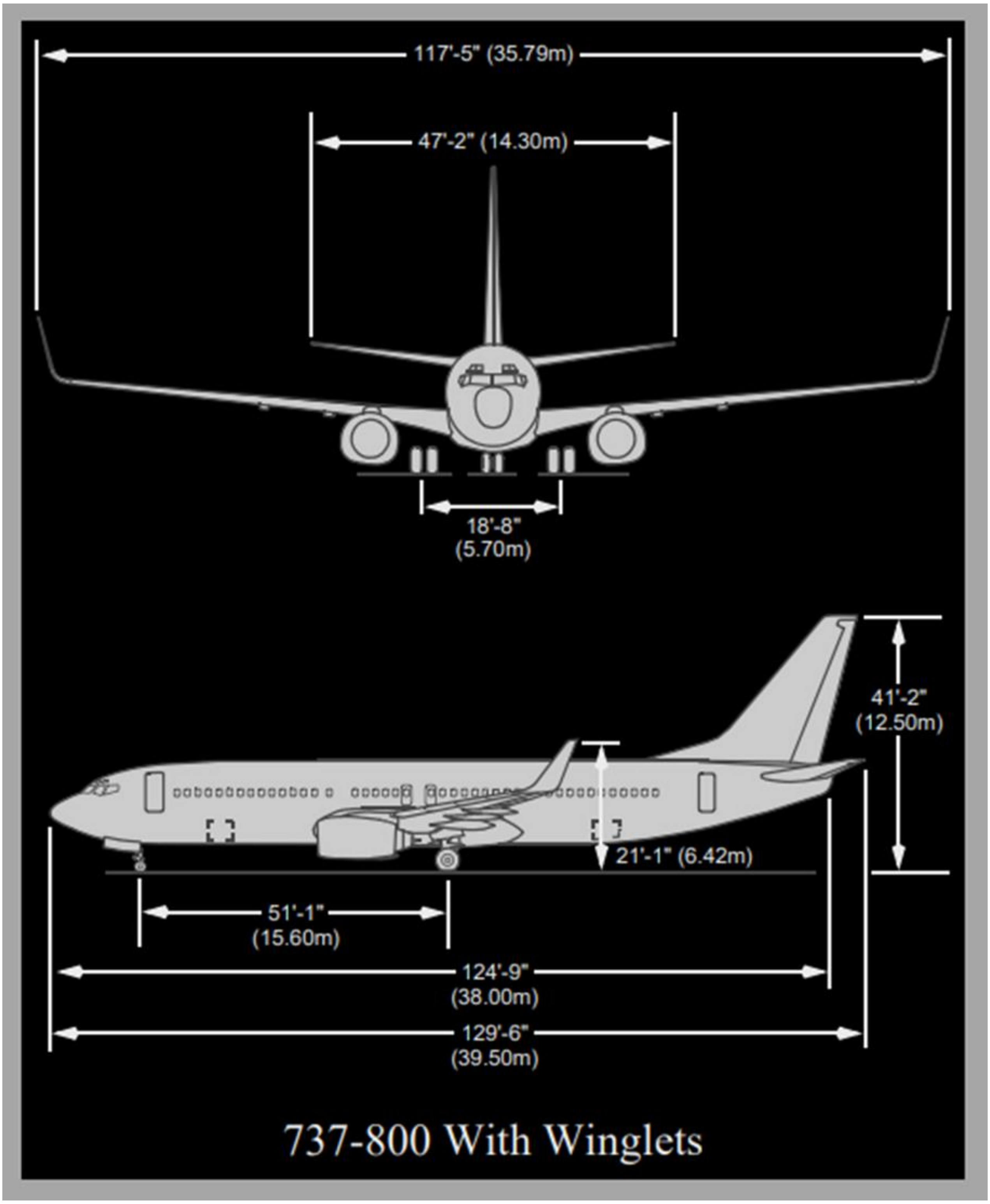
1.1. MODEL IDENTIFICATIONS

Registry Number	Serial Number	Tabulation Number	Category	Tailwind	Engines
RA-73269	30040	YK136	CAT IIIa	10	CFM56-7B24
RA-73311	40309	YF619	CAT IIIa	10	CFM56-7B24/3
RA-73312	40311	YF621	CAT IIIa	10	CFM56-7B24/3
RA-73313	35700	YL931	CAT IIIa	15	CFM56-7B26/3
RA-73314	40233	YF701	CAT IIIa	10	CFM56-7B26/3
RA-73315	40236	YF704	CAT IIIa	10	CFM56-7B26/3
RA-73316	35984	YL935	CAT IIIa	15	CFM56-7B24/3
RA-73317	40874	YL938	CAT IIIa	15	CFM56-7B26/3
RA-73318	40872	YL961	CAT IIIa	15	CFM56-7B26/3
RA-73319	42059	YV256	CAT IIIa	10	CFM56-7B26E
RA-73321	32841	YK137	CAT IIIa	10	CFM56-7B24

СТРАНИЦА НАМЕРЕНО ОСТАВЛЕНА ПУСТОЙ

1.2. AIRPLANE DIMENSIONS

1.2.1. Boeing 737-800

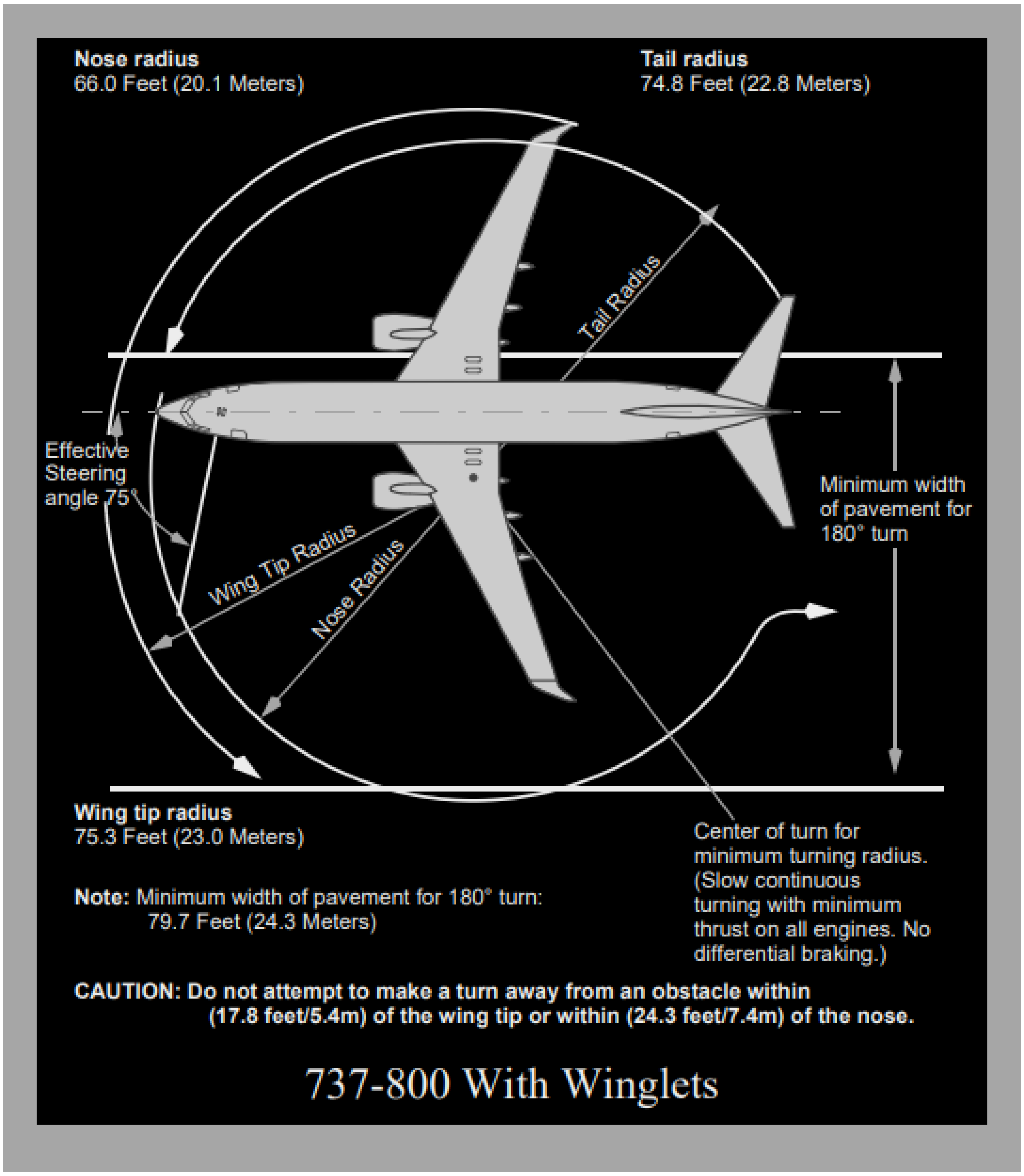


СТРАНИЦА НАМЕРЕНО ОСТАВЛЕНА ПУСТОЙ

1.3. TURNING RADIUS

The wing tip swings the largest arc while turning and determines the minimum obstruction clearance path. All other portions of the airplane structure remain within this arc.

1.3.1. Boeing 737-800



СТРАНИЦА НАМЕРЕНО ОСТАВЛЕНА ПУСТОЙ

1.4. GENERAL LIMITATIONS

1.4.1. Certification Status

The Boeing 737-800 series airplanes are certificated in the Transport Category, FAR Part 25 and Part 36 and approved by the IAC Aviation Register.

По скоростной классификации ICAO ВС B-737 800 относится к категории “D”.

По шумовым характеристикам ВС B-737 800 соответствуют:

RA-73269 - RA-73313, RA-73321

ICAO Annex 16, Volume 1, Chapter 3

RA-73314 - 73319

ICAO Annex 16, Volume 1, Chapter 4

1.4.2. Types of Approved Operations

ВС B-737 NG сертифицированы для выполнения полетов, с целью перевозки пассажиров, багажа и грузов, при условии, что установленное оборудование и системы, соответствуют требованиям авиационной безопасности и правил по производству полетов, в условиях прописанных в эксплуатационных спецификациях каждого ВС.

Для полной информации по спецификации ВС см “Эксплуатационные спецификации”.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

1.5. GENERAL AIRPLANE LIMITATIONS

This chapter contains:

- Airplane Flight Manual (AFM) limitations;
- AFM operational information;
- Non-AFM operational information.

Limitations and operational information are included if they are:

- operationally significant;
- required by FAA Airworthiness Directive;
- required by another regulatory requirement.

Limitations and operational information are not included if they are:

- incorporated into FCOM normal, supplementary, or non-normal procedures, with a few exceptions;
- shown on a placard, display, or other marking.

Limitations and operational information listed in this chapter that must be memorized (memory items) are marked with a (#) symbol. They meet the following criterion - flight crew access by reference can not assure timely compliance, e.g., severe turbulence penetration speeds. They need only be memorized to the extent that compliance is assured. Knowing the exact wording of the limitation is not required.

Assuming that the remaining items are available to the flight crew by reference, they do not need to be memorized.

Maximum Runway Slope .... ± 2 %

RA-73269 – 73312, RA-73314, RA-73315, RA-73319 – 73321

Maximum Takeoff and Landing Tailwind Component .... 10 kts

RA-73313, RA-73316 - RA-73318

Maximum Takeoff and Landing Tailwind Component .... 15 kts

Maximum Operating Altitude (pressure alt) .... 41000 ft

Maximum speeds .... observe gear and flaps placards

Maximum Takeoff and Landing Altitude	
RA-73269 - 73318, RA-73321	RA-73319
8400 ft	10000 ft

RA-73269 - 73312, RA-73314 - RA-73316, RA-73318, RA-73321

Maximum flight operating latitude is dependent on the configuration of the Magnetic Variation tables in the ADIRU as follows: 82° North and 82° South, except for the region between 80° West and 130 ° West longitude, the maximum flight operating latitude is 70° North, and the region between 120° East and 160° East longitude, the maximum flight operating latitude is 60° South.

RA-73317, RA-73319

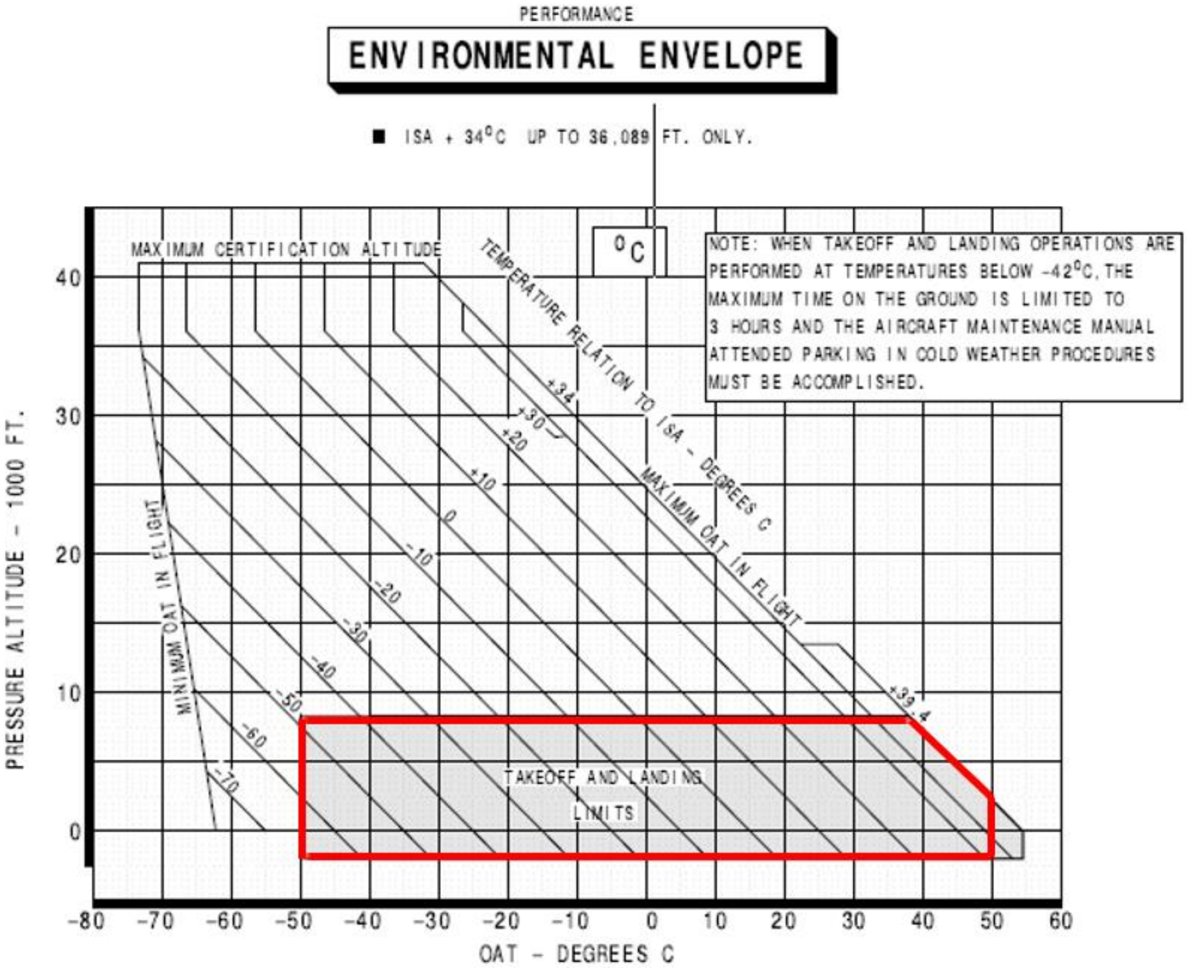
Maximum flight operating latitude is dependent on the configuration of the Magnetic Variation tables in the ADIRU as follows: 82° North and 82° South, except for the region between 80° West and 170 ° West longitude, the maximum flight operating latitude is 73° North, and the region between 120° East and 160° East longitude, the maximum flight operating latitude is 60° South.

Installation of handle covers on the overwing exits must be verified prior to departure whenever passengers are carried

RA-73269 - RA-73321

Verify that an operational check of the Flight Deck Access System has been accomplished according to approved procedures once each flight day.

TEMPERATURE FOR TAKEOFF AND LANDING



MAXIMUM WIND FOR DOOR OPERATION

Do not operate the entry or service doors with winds at the door of more than 40 knots. Do not keep the entry or service doors open when wind gusts are more than 65 knots. Strong winds can cause damage to the structure of the airplane.

SEVERE TURBULENT AIR PENETRATION SPEED

Severe Turbulent Air Penetration speed is 280 KIAS / .76M, whichever is lower. Applicable to Climb and Descent only. During Cruise, refer to FCOM SP.16, Severe Turbulence Supplementary Procedure.

ALTITUDE DISPLAY LIMITS FOR RVSM OPERATIONS

Standby altimeters do not meet altimeter accuracy requirements of RVSM airspace.

The maximum allowable in-flight difference between Captain and First Officer altitude displays for RVSM operations is 200 feet.

The maximum allowable on-the-ground altitude display differences for RVSM operations are:

Field Elevation	Max Difference Between Captain & F/O	Max Difference Between Captain or F/O & Field Elevation
Sea Level to 5,000 feet	50 feet	75 feet
5,001 to 10,000 feet	60 feet	75 feet

1.5.1. Weight Limitations

NOTE: The maximum weight limitations can be further limited as referenced in the WEIGHT LIMITATIONS section of the CERTIFICATE LIMITATIONS chapter of the AFM.

Possible conflicts between the AFM and the FCOM may occur due to separate publication release dates. In the event of a conflict between the FCOM and the AFM, the AFM shall govern.

AFM WEIGHT LIMITATIONS

MAXIMUM TAXI WEIGHT

RA-73269, RA-73316, RA-73321	72227 Kilograms
RA-73311, RA-73312	73735 Kilograms
RA-73319	79151 Kilograms
RA-73313, RA-73317, RA-73318	79227 Kilograms
RA-73314, RA-73315	79242 Kilograms

MAXIMUM TAKEOFF WEIGHT

RA-73269, RA-73316, RA-73321	72000 Kilograms
RA-73311, RA-73312	73508 Kilograms
RA-73319	78925 Kilograms
RA-73313, RA-73317, RA-73318	79000 Kilograms

MAXIMUM LANDING WEIGHT

RA-73311, RA-73312	65317 Kilograms
RA-73319	65770 Kilograms

RA-73269, RA-73313 - 73318, RA-73321	66360 Kilograms

MAXIMUM ZERO FUEL WEIGHT

RA-73311, RA-73312	61688 Kilograms
RA-73269, RA-73313 - 73321	62731 Kilograms

1.5.2. Air Systems

PRESSURIZATION

The maximum cabin differential pressure (relief valves) is 9.1 psi.

With either one or both engine bleed air switches ON, do not operate the air conditioning packs in HIGH for takeoff, approach or landing.

NOTE: The fire protection non-Normal procedures take precedence over the statement regarding no air conditioning pack in HIGH during takeoff, approach, or landing. The CARGO FIRE and SMOKE/ FUMES REMOVAL checklists require the Operating PACK switch(es) HIGH. Switch(es) need to be placed in HIGH in order to increase ventilation for smoke removal.

1.5.3. Autopilot/Flight Director System

Use of aileron trim with the autopilot engaged is prohibited.

Do not engage the autopilot for takeoff below 400 feet AAL.

RA-73314, RA-73315, RA-73319

For single channel operation during approach, the autopilot shall not remain engaged below 50 feet AAL.

RA-73319

Do not use the autopilot below 100 feet radio altitude at airport pressure altitudes above 8,400 feet.

RA-73269, RA-73311 - 73313, RA-73316 - 73318, RA-73321

Airplanes operating with EASA Certification: The Minimum Use Height (MUH) for single channel autopilot operation is defined as 158 feet AAL.

Maximum allowable wind speeds when landing weather minima are predicated on autoland operations:

- Headwind 25 knots
- Crosswind 20 knots

RA-73313, RA-73316 - 73318

- Tailwind 15 knots
- **RA-73269 – 73312, RA-73314, RA-73315, RA-73319 – 73321**
  Tailwind 10 knots.

RA-73313, RA-73316 - 73318

- Tailwind 15 knots (flaps 40).

RA-73313, RA-73316 - 73318

- Tailwind 12 knots (flaps 30).

Maximum and minimum glideslope angles for autoland are 3.25 degrees and 2.5 degrees respectively.

Autoland capability may only be used with flaps 30 or 40 and both engines operative.

Do not use LVL CHG on final approach below 1000 feet AAL.

1.5.4. Communications

Reserved

1.5.5. Electrical

Reserved

1.5.6. Engines and APU

Engine Limit Display Markings

Maximum and minimum limits are red.

Caution limits are amber.

Engine RPM

The maximum operational limits are:

N1– Low Pressure Compressor Rotor **104.0%**

N2 – High Pressure Compressor Rotor **105.0%**

Engine Ignition

Engine ignition must be on for:

- Takeoff;
- Landing;
- operation in heavy rain;
- anti-ice operation.

Thrust

Operation with assumed temperature reduced takeoff thrust is not permitted with anti-skid inoperative.

Reverse Thrust

Intentional selection of reverse thrust in flight is prohibited.

Do not use high levels of reverse thrust at low speed, unless required due to an emergency

Engine EGT

Operation Condition	Temperature Limits	Time Limit
All Altitudes	950°C	5 Minutes
All Altitudes	925°C	No Limit
Starting	725°C	No Limit

Engine Oil System

Maximum oil temperature, continuous operation, is 140 degrees C.

Maximum oil temperature, 45 minute limit, is 140-155 degrees C.

Minimum oil pressure is 13 psi. If engine oil pressure is in the yellow band with takeoff thrust set, do not takeoff.

APU

APU bleed + electrical load: max alt 10,000 ft.

APU bleed: max alt 17,000 ft.

APU electrical load: max alt 41,000 ft.

APU bleed valve must be closed when:

- ground air connected and isolation valve open;
- engine no. 1 bleed valve open;
- isolation and engine no. 2 bleed valves open.

APU bleed valve may be open during engine start, but avoid engine power above idle.

After three consecutive aborted start attempts, a 15 minute cooling period is required.

1.5.7. Fire Protection

Required airport category for rescue and firefighting – 7

1.5.8. Flight Controls

The maximum altitude with flaps extended is 20,000 ft.

Holding in icing conditions with flaps extended is prohibited.

In flight, do not extend the SPEED BRAKE lever beyond the FLIGHT DETENT.

Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw (e.g. large sideslip angles) as they may result in structural failure at any speed, including below VA.

Do not deploy the speedbrakes in flight at radio altitudes less than 1,000 feet.

Alternate flap duty cycle:

When extending or retracting flaps with the ALTERNATE FLAPS position switch, allow 15 seconds after releasing the ALTERNATE FLAPS position switch before moving the switch again to avoid damage to the alternate flap motor clutch

After a complete extend/retract cycle, i.e., 0 to 15 and back to 0, allow 5 minutes cooling before attempting another extension.

1.5.9. Flight Instruments, Displays

Reserved.

1.5.10. Flight Management, Navigation

Air Data Inertial Reference Unit (ADIRU)

ADIRU alignment must not be attempted at latitudes greater than 78 degrees 15 minutes.

All flight operations based on magnetic heading or magnetic track angle are prohibited in geographic areas where the installed IRS MagVar table errors are greater than 5 degrees.

Refer to AFM Normal Procedures/Inertial Reference System section for procedures to determine the geographic areas and magnitude of MagVar errors for the specific MagVar table installed in the IRS and if any of these limitations apply.

Look-Ahead Terrain Alerting (GPWS)

Do not use the terrain display for navigation.

Do not use the look-ahead terrain alerting and terrain display functions:

within 15 nm of takeoff, approach or landing at an airport or runway not contained in the GPWS terrain database.

NOTE: Refer to Honeywell Document 060-4267-000 for airports and runways contained in the installed GPWS terrain database.

Avoid weather radar operation in a hangar.

Avoid weather radar operation when personnel are within the area normally enclosed by the aircraft nose radome.

NOTE: The hangar recommendation does not apply to the weather radar test mode.

1.5.11. Fuel System

Maximum tank fuel temperature is 49°C.

Minimum tank fuel temperature prior to takeoff and inflight is –43°C, or 3°C above the fuel freezing point temperature, whichever is higher.

NOTE: The use of Fuel System Icing Inhibitor additives does not change the minimum fuel tank temperature limit.

Intentional dry running of a center tank fuel pump (low pressure light illuminated) is prohibited.

Fuel Balance

Lateral imbalance between main tanks 1 and 2 must be scheduled to be zero.

Random fuel imbalance must not exceed 453 kg/1000 lbs for taxi, takeoff, flight or landing.

Fuel Loading

Main tanks 1 and 2 must be full if center tank contains more than 453 kg/1000 lbs.

Fuel LOW

Fuel quantity less than 907 kg/2000 lb in related main tank.

Display remains until fuel tank quantity is increased to 1134kg/2500 lb.

Fuel configuration (CONFIG)

Center fuel tank quantity greater than 726kg/1600 lb.

Display remains until center fuel tank quantity less then 363kg/800 lb.

Fuel imbalance (IMBAL)

Main tanks differ by more then 453kg/1000 lb.

Display remains until imbalance is reduced to 91kg/200 lb.

1.5.12. Hydraulics

Hydraulic quantity indicates digital percentage (0% to 106%) of hydraulic quantity.

Hydraulic pressure indicates system pressure:

Minimum - Normal - Maximum : **2800 psi - 3000 psi - 3500 psi**.

NOTE: When both pumps for a system are OFF, the indication may read hydraulic system reservoir pressure, normally less than 100 psi.

Refill indication illuminated (white) – hydraulic quantity below 76%.

NOTE: Valid only when airplane is on ground with both engines shutdown or after landing with flaps up during taxi–in.

System A	System B
ailerons	ailerons
rudder	rudder
elevator and elevator feel	elevator and elevator feel
flight spoilers (two on each wing)	flight spoilers (two on each wing)
ground spoilers	leading edge flaps and slats, trailing edge flaps
alternate brakes	normal brakes, autobrake
No. 1 thrust reverser	No. 2 thrust reverser
autopilot A	autopilot B
normal nose wheel steering	alternate nose wheel steering
landing gear	landing gear transfer unit
power transfer unit (PTU).	autoslats
	yaw damper.

1.5.13. Landing Gear

Do not apply brakes until after touchdown.

Maximum Landing Gear Operating Speed

Maximum Landing Gear Extend Speed.. 270 kts/.82 M

Maximum Landing Gear Extended Speed.................................... 320 kts/.82 M

Maximum Landing Gear Retraction... 235 kt

1.5.14. Oxygen

Required Pressure (PSI) for 114/115 Cubic Ft. Cylinder

RA-73269 - 73321

BOTTLE TEMPERATURE		NUMBER OF CREW USING OXYGEN		
°C	°F	2	3	4
50	122	530	735	945
45	113	520	725	930
40	104	510	715	915
35	95	505	700	900
30	86	495	690	885
25	77	485	680	870
20	68	480	670	860
15	59	470	655	840
10	50	460	645	830
5	41	455	635	815
0	32	445	620	800
-5	23	440	610	785
-10	14	430	600	770

1.6. WIND LIMITATIONS

Maximum crosswind component for Boeing 737-NG depends of runway conditions. Information about runway condition is often available through three main sources:

1. Reported Runway Friction Coefficient. Measured by friction reporting vehicles designed for this purpose.
2. Reported Braking Action made by pilots (PIREPS).
3. Runway surface description - physical description of runway surface and contaminant, e.g., 6 mm of wet snow, patches of ice, compact snow.

1.6.1. Definitions of Braking Action

Good - braking deceleration is normal for the wheel braking effort applied. Directional control is normal.

Medium - Braking deceleration is noticeably reduced for the wheel braking effort applied. Directional control may be slightly reduced

Poor - Braking deceleration is significantly reduced for the wheel braking effort applied. Potential for hydroplaning exists. Directional control may be significantly reduced.

NOTE: The ICAO term Unreliable and SNOTAM code of “9” indicates contamination is outside the approved operational range for the friction measuring equipment in use and therefore mu values are not provided. This typically occurs in poor or worse conditions (greater than 1/8” of wet snow, slush or standing water) whereby a potential for hydroplaning should be expected. Use PIREPS and the depth and type of runway contaminants to assess actual braking conditions.

Takeoff and Landing crosswind limitations must be determined in depends on Runway Condition Code (RCC).

1.6.2. Maximum Crosswind limits including gusts for Runway Condition

Enter the table with given parametrs (RW Condition, ESF, Measured or Normative FC) and determine RCC in order to find Maximum crosswind component.

The basic rule of this table is to move from left to right and from top to bottom.

Example:

- If ESF is **GOOD** then you **NOT ABLE** to use **Normative FC 0.50;**
- If Wet Snow is greater then **3mm** then you **NOT ABLE** to use **GOOD ESF.**

Runway Condition Description	Braking Action (ESF)	Measured FC	Normative FC	RCC	Max. Cross Wind (kts / m/sec)	
					Takeoff	Landing
Dry	---	≥ 0,51	≥ 0.50	**6**	34/17.5	34/17.5
Frost; Wet (includes damp and 3mm depth or less of water) **3mm depth or less**: Slush; Dry Snow; Wet Snow	**Good**	0,40-0,50	0,42-0,49	**5**	25/12.9	34/17.5**
-15°C and colder: Compacted Snow	**Good to Medium**	0,36-0,39	0,40-0,41	**4**	22/11.3	34/17.5**
Slippery when wet (wet runway); Dry or Wet Snow (any depth) over Compacted Snow **Greater then 3mm depth:** Dry Snow; Wet Snow **Warmer than -15°C:** Compacted snow	**Medium**	0,30-0,35	0,37-0,39	**3**	20/10.3	25/12.9**
Greater then 3mm depth: Water; Slush	**Medium to Poor**	0,26-0,29	0,35-0,36	**2**	15/7.7	17/8.7
Ice*	**Poor**	0,17-0,25	0,30-0,34	**1**	13/6.7	15/7.7
Wet Ice; Water on top of Compacted Snow; Dry Snow or Wet Snow over Ice	**Unreliable**	<0,17	<0,30	**0**	**---**	**---**

RCC – RW Condition Code

ESF (Estimated surface friction) – рассчётное (оценочное) сцепление.

* Takeoff and landing on untreated snow or ice should only be attempted when no melting is present.

** Sideslip only (zero crab) landings are not recommended with crosswind components in excess of 17 knots at flaps 15, 20 knots at flaps 30, or 23 knots at flaps 40. This recommendation ensures adequate ground clearance and is based on maintaining adequate control margin.

NOTE: Reduce crosswind guidelines by 5 knots on wet or contaminated runways whenever asymmetric thrust is used.

1.6.3. Maximum Depth of Contaminant on Runway Surface

Contaminant	Depth	
	Takeoff / Landing	
Standing	13 mm	0.5 inches
Slush	13 mm	0.5 inches
Dry snow	100 mm	4 inches

ТАБЛИЦА СОСТАВЛЯЮЩЕЙ ВЕТРА В ЗАВИСИМОСТИ ОТ СКОРОСТИ И НАПРАВЛЕНИЯ

Скорость Ветра, уз	10^0	20^0	30^0	40^0	50^0	60^0	70^0	80^0	90^0
	ВСТРЕЧНАЯ/БОКОВАЯ СОСТАВЛЯЮЩАЯ ВЕТРА								
1	1,0/0,2	0,9/0,3	0,9/0,5	0,8/0,6	0,6/0,8	0,5/0,9	0,3/0,9	0,2/1,0	0/1
2	2,0/0,4	1,9/0,7	1,7/1,0	1,5/1,3	1,3/1,5	1,0/1,7	0,7/1,9	0,4/2,0	0/2
3	3,0/0,5	2,8/1,0	2,5/1,5	2,3/1,9	1,9/2,3	1,5/2,5	1,0/2,8	0,5/3,0	0/3
4	3,9/0,7	3,7/1,4	3,5/2,0	3,0/2,5	2,5/3,0	2,0/3,5	1,4/3,7	0,7/3,9	0/4
5	4,9/0,9	4,7/1,7	4,3/2,5	3,8/3,2	3,2/3,8	2,5/4,3	1,7/4,7	0,9/4,9	0/5
6	5,9/1,0	5,6/2,0	5,2/3,0	4,6/3,8	3,8/4,6	3,0/5,2	2,0/5,6	1,0/5,9	0/6
7	6,9/1,2	6,6/2,3	6,0/3,5	5,3/4,5	4,5/5,3	3,5/6,0	2,3/6,6	1,2/6,9	0/7
8	7,8/1,4	7,5/2,7	6,9/4,0	6,1/5,1	5,1/6,1	4,0/6,9	2,7/7,5	1,4/7,8	0/8
9	8,8/1,5	8,5/3,0	7,8/4,5	6,9/5,8	5,8/6,9	4,5/7,8	3,0/8,5	1,5/8,8	0/9
10	9,8/1,7	9,4/3,4	8,6/5,0	7,6/6,4	6,4/7,6	5,0/8,6	3,4/9,4	1,7/9,8	0/10
11	10,8/1,9	10,3/3,7	9,5/5,5	8,4/7,0	7,0/8,4	5,5/9,5	3,7/10,3	1,9/10,8	0/11
12	11,8/2,0	11,3/4,1	10,4/6,0	9,1/7,7	7,7/9,1	6,0/10,4	4,1/11,3	2,0/11,8	0/12
13	12,8/2,2	12,2/4,4	11,2/6,5	9,9/8,3	8,3/9,9	6,5/11,2	4,4/12,2	2,2/12,8	0/13
14	13,8/2,4	13,1/4,8	12,1/7,0	10,7/9,0	9,0/10,7	7,0/12,1	4,8/13,1	2,4/13,8	0/14
15	14,7/2,6	14,0/5,1	12,9/7,5	11,5/9,6	9,6/11,5	7,5/12,9	5,1/14,0	2,6/14,7	0/15
16	15,7/2,8	15,0/5,5	13,8/8,0	12,2/10,3	10,3/12,2	8,0/13,8	5,5/15,0	2,8/15,7	0/16
17	16,7/2,9	15,9/5,8	14,7/8,5	13,0/10,9	10,9/13,0	8,5/14,7	5,8/15,9	2,9/16,7	0/17
18	17,7/3,1	16,9/6,1	15,6/9,0	13,8/11,5	11,5/13,8	9,0/15,6	6,1/16,9	3,1/17,7	0/18
19	18,7/3,3	17,8/6,5	16,5/9,5	14,5/12,2	12,2/14,5	9,5/16,5	6,5/17,8	3,3/18,7	0/19
20	19,6/3,5	18,8/6,8	17,3/10,0	15,3/12,8	12,8/15,3	10,0/17,3	6,8/18,8	3,5/19,6	0/20
21	20,7/3,6	19,7/7,2	18,2/10,5	16,0/13,5	13,5/16,0	10,5/18,2	7,2/19,7	3,6/20,7	0/21
22	21,6/3,8	20,7/7,5	19,0/11,0	16,8/14,1	14,1/16,8	11,0/19,0	7,5/20,7	3,8/21,6	0/22
23	22,6/4,0	21,6/7,8	19,9/11,5	17,6/14,8	14,8/17,6	11,5/19,9	7,8/21,6	4,0/22,6	0/23
24	23,6/4,2	22,5/8,2	20,8/12,0	18,4/15,4	15,4/18,4	12,0/20,8	8,2/22,5	4,2/23,6	0/24
25	24,6/4,3	23,5/8,5	21,6/12,5	19,1/16,0	16,0/19,1	12,5/21,6	8,5/23,6	4,3/24,6	0/25
26	25,6/4,5	24,4/8,9	22,5/13,0	19,9/16,7	16,7/19,9	13,0/22,5	8,9/24,4	4,5/25,6	0/26
27	26,6/4,7	25,3/9,2	23,4/13,5	20,7/17,3	17,3/20,7	13,5/23,4	9,2/25,3	4,7/26,6	0/27
28	27,5/4,9	26,3/9,5	24,2/14,0	21,4/17,9	17,9/21,4	14,0/24,2	9,5/26,3	4,9/27,5	0/28
29	28,5/5,0	27,2/9,9	25,1/14,5	22,2/18,6	18,6/22,2	14,5/25,1	9,9/27,2	5,0/28,5	0/29
30	29,5/5,2	28,1/10,2	25,9/15,0	22,9/19,3	19,3/22,9	15,0/25,9	10,2/28,1	5,2/29,5	0/30
31	30.5/5.4	29.1/10.6	26.8/15.5	23.7/19.9	19.9/23.7	15.5/26.8	10.6/29.1	5.4/30.5	0/31
32	31.5/5.6	30.1/10.9	27.7/16.0	24.5/20.6	20.6/24.5	16.0/27.7	10.9/30.1	5.6/31.5	0/32
33	32.5/5.7	31.0/11.3	28.6/16.5	25.3/21.2	21.2/25.3	16.5/28.6	11.3/31.0	5.7/32.5	0/33
34	33.5/5.9	31.9/11.6	29.4/17.0	26.0/21.9	21.9/26.0	17.0/29.4	11.6/31.9	5.9/33.5	0/34
35	34.5/6.1	32.9/12.0	30.3/17.5	26.8/22.5	22.5/26.8	17.5/30.3	12.0/32.9	6.1/34.5	0/35
36	35.5/6.3	33.8/12.3	31.2/18.0	27.6/23.1	23.1/27.6	18.0/31.2	12.3/33.8	6.3/35.5	0/36
37	36.4/6.4	34.8/12.7	32.0/18.5	28.3/23.8	23.8/28.3	18.5/32.0	12.7/34.8	6.4/36.4	0/37
38	37.4/6.6	35.7/13.0	32.9/19.0	29.1/24.4	24.4/29.1	19.0/32.9	13.0/35.7	6.6/37.4	0/38
39	38.4/6.8	36.6/13.3	33.8/19.5	29.9/25.1	25.1/29.9	19.5/33.8	13.3/36.6	6.8/38.4	0/39
40	39.46.9	37.6/13.7	34.6/22.0	30.6/25.7	25.7/30.6	20.0/34.6	13.7/37.6	6.9/39.4	0/40

Глава 2. Normal Procedures

2.0. СОДЕРЖАНИЕ

2.1. GENERAL

2.1.1. Основные принципы организации работы экипажа

Стандартные эксплуатационные процедуры СЭП – это описание алгоритма выполнения стандартных задач и действий членов экипажа на каждом этапе полета, которые выполняются по памяти, а критически важные действия проверяются с помощью контрольного листа (Checklist). СЭП выполняются подготовленным и сертифицированным экипажем, все системы ВС исправны и возможно использование любого автоматического режима. Описанные процедуры подразумевают использование максимального уровня автоматизации. В случае понижения уровня автоматизации действия могут отличаться в соответствии с FCOM.

Настоящая глава – стандартные эксплуатационные процедуры (СЭП), разработанная на основании FCOM B-737 NG (NP) и опыта эксплуатации ВС в Авиапредприятие, является основным нормативным документом, применяемым членами летного экипажа для управления ВС, и определяет порядок их взаимодействия на ВС B-737 NG в соответствии со стандартами авиакомпании.

Некоторые стандартные процедуры, применяемые эпизодически, описаны в SUPPLEMENTARY PROCEDURES (FCOM SP). СЭП применяются членами экипажа для управления ВС в соответствующей конфигурации и определяют порядок их взаимодействия для каждого этапа полёта.

В процессе полета необходимо последовательно использовать основные принципы приоритетов:

- пилотирование;
- навигация;
- радиосвязь;
- нестандартные процедуры и NON-NORMAL CHECKLIST;
- стандартные процедуры и NORMAL CHECKLIST;
- информация бригаде бортпроводников и пассажирам в аварийной ситуации.

Процедуры выполняются по команде CPT/PF или в предписанные СЭП моменты, как правило, в установленной последовательности. В случае необходимости (особенности процедур буксировки, запуска, руления или захода на посадку) допускается изменение порядка выполнения процедур или действий внутри процедуры с обязательным согласованием всеми членами экипажа. При управлении системами ВС каждое включение или выключение должно выполняться осмысленно и раздельно, с достаточным временным интервалом для определения результата выполнения операций и взаимоконтроля. Взаимоконтроль членов экипажа является жизненно необходимым элементом обеспечения безопасности полета. Каждый член экипажа должен громко и ясно давать информацию о своих действиях по управлению ВС и работой систем, включении, выключении и изменении режимов работы FMC, изменении параметров и траектории полета, настройке навигационных средств, модификации плана полета, изменении конфигурации ВС, о своем отвлечении от управления или радиосвязи и других действиях, которые могут оказать влияние на выполнение полета, и убедиться, что его информация понята правильно.

При действующем принципе «стерильной кабины» членам кабинного экипажа доступ в кабину пилотов или вызов по внутренней связи разрешен только в аварийных ситуациях, которые по мнению бортпроводника могут оказать серьёзное влияние на безопасность выполняемого полёта (течь топлива, пожар, задымление, посторонний звук, вибрации, угроза незаконного вмешательства и т.п.) или сообщения о готовности пассажирской кабины.

Действия по управлению механизацией крыла в полете и выпуском/уборкой шасси выполняются PM по командам PF. Перед выполнением действий по управлению механизацией крыла PM должен проконтролировать соответствие фактической скорости установленным ограничениям, после чего выполнить операцию.

При выполнении руления и полета, пилоты придерживаются двух стандартных положений ног на педалях – «на тормозах» и «на полет». В положении «на тормозах» каблуки не касаются пола, а носки располагаются на верхней части педали в постоянной готовности к торможению. В положении «на полет» каблуки располагаются на полу, а носки – на нижней части педали, в положении, исключающем непреднамеренное обжатие тормозных педалей. В целях сокращения вероятности наступления неблагоприятных событий, связанных с использованием тормозов, пилотам следует избегать положения ног «на тормозах» на всех этапах выполнения полета, включая буксировку, за исключением случаев применения тормозов, руления по перрону и в непосредственной близости от препятствий.

В случае выпуска/уборки шасси на неустановленных рубежах PM должен особенно внимательно проконтролировать соответствие скорости установленным ограничениям, после чего выполнить операцию.

Перед взлетом и заходом на посадку планшет с OFP, навигационные сборники и другие посторонние вещи должны быть размещены и зафиксированы в надлежащих местах.

ЗАПРЕЩАЕТСЯ размещать какие-либо предметы на козырьке кабины (за исключением планшета с OFP на эшелоне полета) и центральном пульте.

ЗАПРЕЩАЕТСЯ членам экипажа в течение всего полета при нахождении на рабочем месте чтение любой литературы, не относящейся к выполнению полета.

Если после нажатия кнопки-табло RECALL присутствует сигнализация, которая гаснет после процедуры MASTER CAUTION RESET:

- определить с помощью MEL возможность вылета;
- выполнение соответствующего NON-NORMAL CHECKLIST не требуется.

Порядок действий экипажа при обнаружении выхода из строя или появления информационного сообщения о неисправности какого-либо прибора, оборудования или системы после закрытия дверей воздушного судна приведен в РПП Авиапредприятия часть А глава 8.

Порядок использования радиостанций и гарнитур

В процессе буксировки, руления и в полете на высотах ниже 10,000 ft все члены летного экипажа обязаны использовать для ведения радиосвязи головные гарнитуры.

Снятие наушников рекомендовано производить в момент прослушивания радиообмена для проверки факта хорошей слышимости через динамики. Радиообмен должен внимательно прослушиваться в течение всего полета.

Не рекомендуется выключать LOUDSPEAKER или чрезмерно уменьшать громкость.

В случае кратковременного покидания своего рабочего места оставшийся член летного экипажа для ведения радиосвязи должен использовать головную гарнитуру.

При возникновении сомнений у кого-либо из членов экипажа в правильности принятой команды от диспетчера ее необходимо уточнить.

В отдаленных районах, когда охват VHF не доступен, экипаж необходимо прослушивать частоту 123.45 или предписанную частоту.

Радиостанции используются в следующем порядке

- VHF-1 – для ведения радиосвязи с органами ОВД;
- VHF-2 – для прослушивания аварийной частоты 121.5, получения информации ATIS, информации коммерческого характера и т.д.;
- VHF-3 – для связи по каналам ACARS (режим DATA) или на случай отказа VHF-1 и VHF-2.

Стандартная процедура настройки VHF-1 подразумевает установку частоты на RTP-1, VHF-2 – на RTP-2, VHF-3 – на RTP-3.

На предполетной подготовке следует контролировать нейтральное положение переключателя Push–to–Talk Switch на Audio Control Panel (ACP). Иметь в виду, что на некоторых ВС возможна фиксация переключателя в положении I/C (Intercom). В этом случае громкость LOUDSPEAKER существенно уменьшается, что может привести к потере радиосвязи.

2.1.2. Распределение ответственности и обязанностей

Обязанности на земле до начала руления перед взлетом и после окончания руления после посадки распределены между Captain (C) и First officer (F/O). Обязанности в полете, с начала руления перед взлетом и до окончания руления после посадки, распределены между Pilot Flying (PF) и Pilot Monitoring (PM). Обязанности PF и PM могут меняться во время полета. Например, КВС может быть PF во время руления, но быть PM от взлета до посадки.

Действия выполняются в зоне ответственности каждого члена экипажа, по принципу, указанному в схеме SCAN FLOW в AREAS OF RESPONSIBILITY.

Каждый член экипажа действует самостоятельно и отвечает за выполняемые операции в своей зоне ответственности.

КВС может давать указания на действия за пределами зоны ответственности члена экипажа.

Член летного экипажа ВС, занимающий левое пилотское сиденье, всегда является PF при:

- выполнении руления ВС;
- метеорологических условиях, не позволяющих второму пилоту пилотировать согласно ППЧЛЭ;
- выполнении прерванного взлета;
- отказе двух двигателей;
- отказе двух генераторов;
- разгерметизации ВС;
- при выполнении автоматической посадки;
- всех случаях по решению КВС.

Примечание: КВС может передать управление на рулении второму пилоту на прямолинейных участках руления.

При потере индикации пространственного положения ВС одним из пилотов пилотирующим ВС остается пилот, имеющий индикацию. Если потеряна индикация пространственного положения у обоих пилотов, пилотирующим становится КВС. При ручном пилотировании ВС PF не рекомендуется выполнять какие-либо другие действия кроме включения/выключения автопилота, автомата тяги и действий на EFIS CTL PANEL с предварительной информацией о своих действиях. Настройку радиостанций, установку кода ответчика, действия на MCP и CDU, оповещение пассажиров и другие операции, связанные с отвлечением от пилотирования, как правило, выполняет PM.

При выполнении полета в ручном режиме PM производит все изменения режимов на MCP по команде PF. Установка курса и высоты на MCP по команде диспетчера может выполняться PM без команды PF с обязательным докладом о своих действиях. PF подтверждает полученную информацию установленной фразой.

Экипаж ВС должен постоянно придерживаться принципа ONE HEAD UP и в случае необходимости отвлечения PF (подготовка к снижению, проведение брифинга и т.п.) он должен передать свои обязанности другому члену экипажа в установленном порядке.

Распределение обязанностей на земле	
Captain	**First officer**
- анализирует метеообстановку и изучает NOTAM с учетом запасных; - проводит брифинг с кабинным экипажем; - определяет техническое состояние ВС; - готовит пилотскую кабину; - контролирует правильность ввода данных в FMC; - отвечает за определение количества заправляемого топлива; - проверяет и подписывает загрузочную ведомость и центровочный график; - выполняет расчет взлетно-посадочных характеристик; - отвечает за оповещение пассажиров; - взаимодействует с кабинным экипажем и наземным персоналом; - выполняет руление ВС.	- анализирует метеообстановку и изучает NOTAM с учетом запасных; - готовит пилотскую кабину; - вводит данные в FMC; - проверяет наличие необходимых документов и аварийного оборудования; - заполняет/проверяет загрузочную ведомость и центровочный график; - отвечает за ведение радиосвязи; - выполняет расчет взлетно-посадочных характеристик; - заполняет автоматизированный навигационный расчет полета (OFP); - осуществляет контроль параметров руления (скорости, режима работы двигателей и т.д.); - своевременно информирует об отклонениях.

Распределение обязанностей пилотирующего и не пилотирующего пилотов начинаются с момента начала взлета и заканчиваются после окончания пробега на посадке.	
ПИЛОТИРУЮЩИЙ ПИЛОТ (PF):	**НЕПИЛОТИРУЮЩИЙ ПИЛОТ (PM):**
- устанавливает необходимые параметры на MCP или подает команды на их установку при ручном пилотировании; - выдерживает траекторию и осуществляет контроль параметров полета (высоты, скорости, курса, крена, режима работы двигателей и т.д.); - принимает решение об изменении конфигурации ВС (PF дает команды, PM выполняет); - контролирует точность навигации; - подает команды и контролирует выполнение контрольных листов (NC)	- проверяет соответствие озвученных изменений и значений фактическим; - осуществляет чтение, выполнение и контроль правильности выполнения Normal Checklist в полном объеме; - ведет радиосвязь; - выполняет команды PF; - озвучивает изменения на FMA; - осуществляет программирование FMC; - собирает метеоинформацию по основному и запасным аэродромам; - заполняет автоматизированный навигационный расчет полета (OFP); - осуществляет контроль параметров полета (высоты, скорости, курса, крена, режима работы двигателей и т.д.); - своевременно информирует об отклонениях от расчетных параметров полёта

2.1.3. Golden Rules

The following golden rules are designed to assist pilots (trainees and experienced) in maintaining basic flying skills.

Automated aircraft can be flown like any other aircraft:

In line operations each pilot should keep in mind that the PF always retains the authority and capability to revert:

- to a lower level of automation; or,
- to hand flying — directly controlling the aircraft trajectory and energy condition by using “yoke, rudder and throttles”.

Aviate (Fly), Navigate, Communicate and Manage – in that order during an abnormal condition or an emergency condition, PF and PM tasks sharing should be adapted to the situation (in accordance with the Company Policy and QRH), and tasks should be accomplished with this four-step strategy:

Aviate – The PF must fly the aircraft to stabilize the aircraft’s pitch attitude, bank angle, vertical flight path and horizontal flight path. The PM must back up the PF (by monitoring and by making call-outs) until the aircraft is stabilized.

Navigate – Upon the PF’s command, the PM should select or should restore the desired mode for lateral navigation and/or vertical, being aware of terrain and minimum safe altitude. Navigate can be summarized by the following:

- know where you are;

- know where you should be; and,
- know where the terrain and obstacles are.

Communicate – After the aircraft is stabilized and the abnormal condition or emergency condition has been identified, flight crew should inform ATC of the situation and of intentions. If the flight is in a condition of distress or urgency, flight crew should use standard phraseology: “Pan-Pan, Pan-Pan, Pan-Pan” or, “Mayday, Mayday, Mayday”. Discuss situation among flight crew members. Advice cabin crew as needed.

Manage – The next priority is management of the aircraft systems and performance of the applicable abnormal procedures or emergency procedures

Practice task sharing and back-up each other

All procedures should be performed as set in the SOP or QRH. These should be performed in accordance with the published task-sharing, CRM and standard phraseology. Critical actions or irreversible actions should be accomplished by the PM after verbal confirmation by the PF.

Know your FMA guidance at all times

The A/P, F/D and A/T control panel(s) and the FMS CDU are the primary interfaces for the crew to communicate with the aircraft systems (to arm modes or select modes and to enter targets - e.g., airspeed, heading, and altitude). The PFD, the ND and particularly the FMA are the primary interfaces for the aircraft to communicate with the crew to confirm that the aircraft system has accepted correctly the crew’s mode selections and target entries. Any action on the A/P, F/D and A/T control panel(s) or on the FMS CDU should be confirmed by crosschecking the corresponding FMA annunciation or data on the FMS display unit and on the PFD/ND. At all times, the PF and the PM should be aware of the guidance modes that are armed or selected and of any mode changes and should be confirmed by CALLOUT “CHECKED”.

Crosscheck the accuracy of the FMS with raw data

When in area with low navigation aid coverage areas or if course deviation is suspected or position update is insufficient or before commencing any type of approaches other than RNAV approaches the FMS navigation accuracy should be cross-checked with raw data (data received directly, not via the AFDS or FMC, from basic navigation aids - e.g., ADF, VOR, DME, barometric altimeter) in accordance with published procedures (see FCOM, SP Chapter; Flight Management, Navigation Section, FMC Navigation Check procedure).

One head up at all times

Significant changes to the FMS should be performed by the PM. The changes then should be cross-checked by the other pilot after transfer of aircraft control to maintain one head up at all times.

When things do not go as expected – take control

If the aircraft does not follow the desired horizontal flight path or vertical flight path and time does not permit analyzing and solving the anomaly, revert without delay from FMS guidance to selected guidance or to handflying.

Use the optimum level of automation for the task

On highly automated and integrated aircraft (like Boeing-737), several levels of automation are available.

2.1.4. Normal Procedures Philosophy and Assumptions

Normal procedures verify for each phase of flight that:

- the airplane condition is satisfactory;
- the flight deck configuration is correct.

Normal procedures are done on each flight. Refer to the Supplementary Procedures (SP) chapter for procedures that are done as needed, for example the adverse weather procedures.

Normal procedures are written for a trained flight crew and assume:

- all systems operate normally;
- the full use of all automated features (LNAV, VNAV, autoland, autopilot, and autothrottle). This does not preclude the possibility of manual flight for pilot proficiency, where allowed.

Normal procedures also assume coordination with the ground crew before:

- hydraulic system pressurization; or
- flight control surface movement; or
- airplane movement.

Normal procedures do not include steps for flight deck lighting and crew comfort items.

Normal procedures are done by memory and scan flow. The panel illustration in this section shows the scan flow. The scan flow sequence may be changed as needed.

2.1.5. Duty Transfer

Transfer of control is initiated by a command and followed by an acknowledgment.

"I HAVE CONTROL" is the command that the other pilot is to transfer control and begin performing PM duties; or the acknowledgment by the other pilot that he has taken PF duties.

"YOU HAVE CONTROL" is the command that the other pilot is to take control and begin performing PF duties; or the acknowledgment by the other pilot that he has taken PM duties.

In case of flight control transfer from PF to PM (takeoff and landing roll) pilots must do the following:

- PF calls: **"YOU HAVE CONTROL"** and continues performing PF duties;
- PM takes control, calls: **"I HAVE CONTROL"** and begins performing PF duties;
- PF makes sure, that an aircraft is controlled by another pilot and begins performing PM duties.

If PM needs to be distracted from his responsibilities (for example, for listening to weather information, filling in flight documentation, informing passengers about the flight, etc.), he transfers radio communication to PF with the phrase: "**YOU HAVE COMMUNICATIONS**". PF begins to perform the duties of PM and calls: "**I HAVE COMMUNICATIONS**".

If the PF/PM is ready to resume his duties, the crewmember shall report it.

2.1.6. Transition to the Manual Control

When PF transits from automatic flight (A/P engaged and A/T engaged) to the manual control of the aircraft, those callouts has to be used:

"**MANUAL THRUST**" – when disengaging A/T;

"**MANUAL CONTROL**" – when disengaging A/P.

2.1.7. Use of Triggers

A «trigger», is an event that initiates a procedure or task. Procedures start with a trigger statement. For example, the trigger used to commence the Steps during Takeoff procedure (Retractable Lights, Engine bleeds, Air conditioning packs….) after flaps retraction and before AFTER TAKEOFF CHECKLIST is:

"**FLAPS UP, NO LIGHT**".

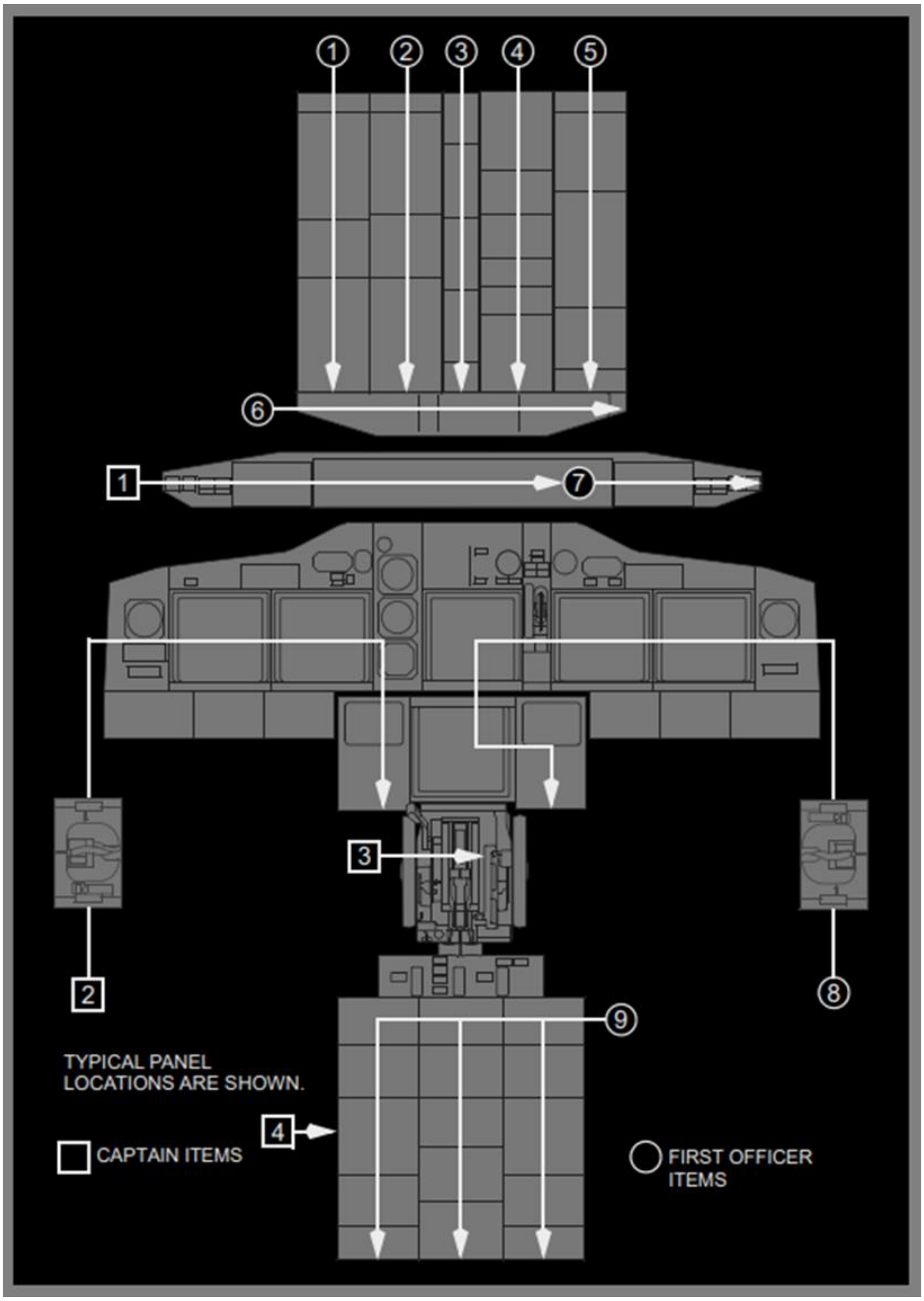
Triggers are also used within a procedure to define when a task or a series of tasks are normally accomplished. Operational flexibility may dictate adjusting a particular trigger.

2.2. AREAS OF RESPONSIBILITY

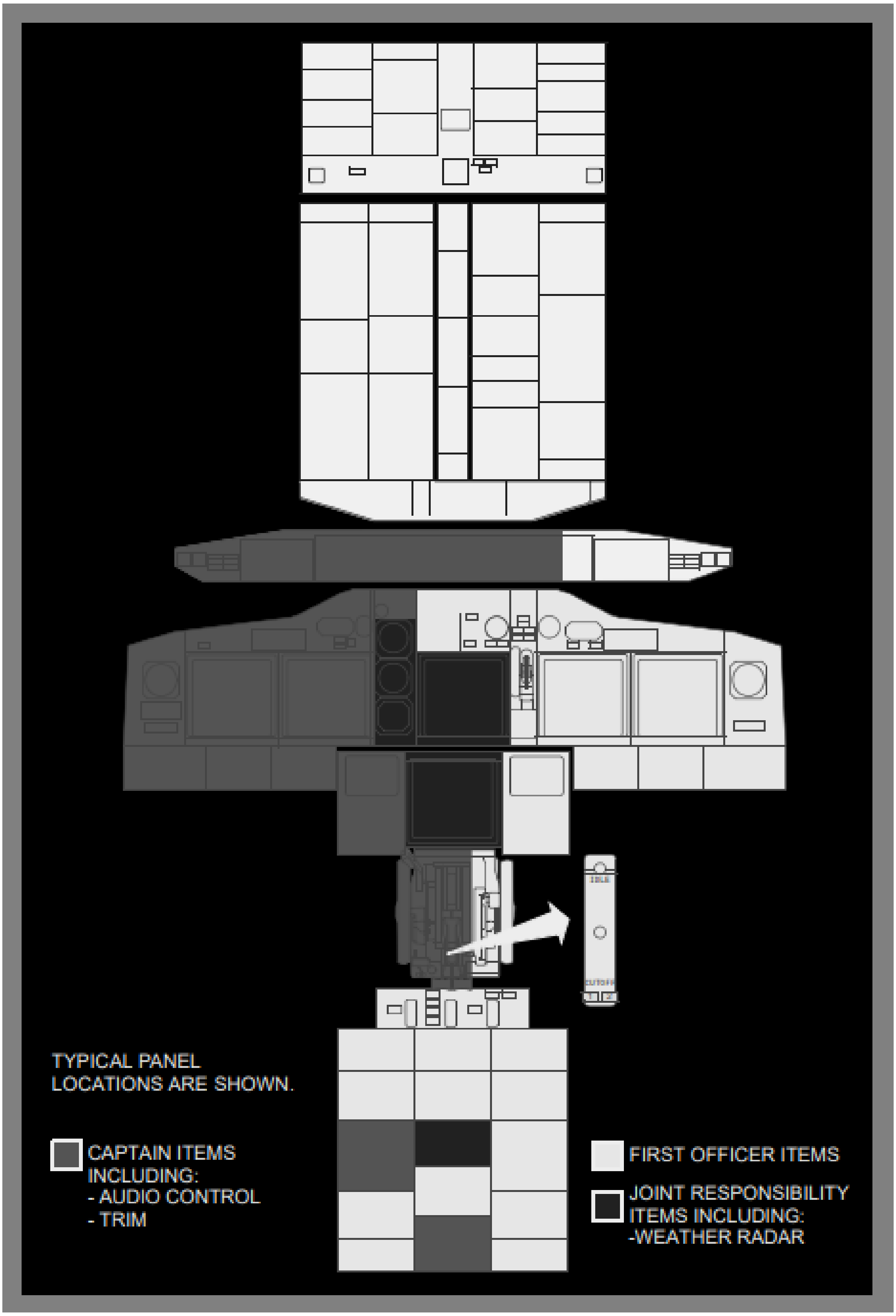
The scan flow and areas of responsibility diagrams shown below are representative and may not match the configuration(s) of your airplanes.

The scan flow diagram provides general guidance on the order each flight crew member should follow when doing the preflight and postflight procedures. Specific guidance on the items to be checked are detailed in the amplified Normal Procedures. For example, preflight procedure details are in the Preflight Procedure - Captain and Preflight Procedure - First Officer.

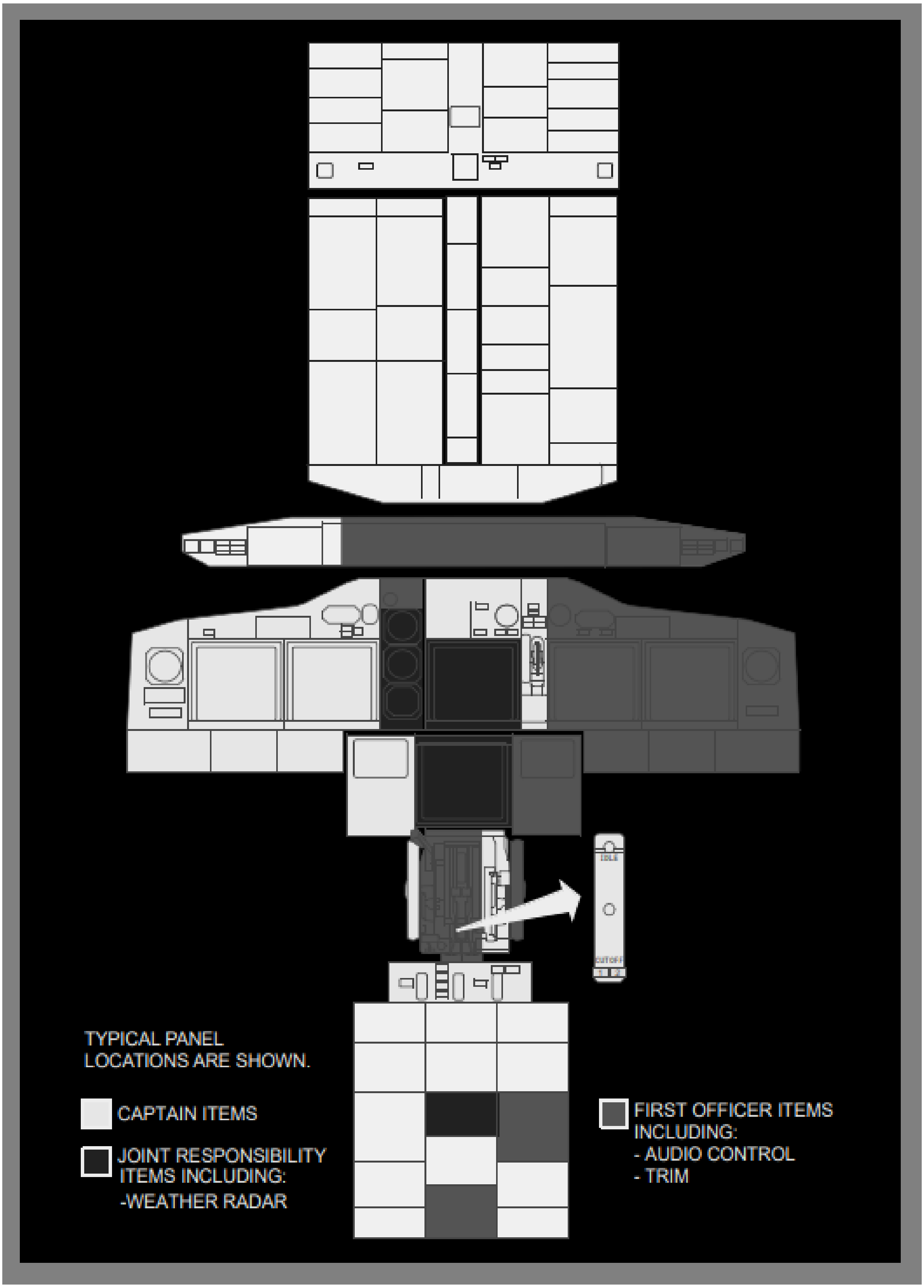
2.2.1. Preflight and Postflight Scan Flow



2.2.2. Areas of Responsibility - Captain as Pilot Flying or Taxiing



2.2.3. Areas of Responsibility - First Officer as Pilot Flying or Taxing



СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

2.3. CONTROL DISPLAY UNIT (CDU) PROCEDURES

2.3.1. General

The Flight Management System provides the crew with navigation and performance information that can result in a significant crew workload reduction. This workload reduction is fully realized when the system is operated as intended, including proper preflight and timely changes in flight. FMC guidance must always be monitored after any in flight changes.

The Captain or First Officer may make CDU entries. The other pilot must verify the entries.

Make CDU entries before taxi or when stopped, when possible. If CDU entries must be made during taxi, the PM makes the entries. The PF must verify the entries before they are executed. To attract PF attention PM calls: **"CONFIRM?"** PF verifies entries and confirms calling: **"EXECUTE"**.

In flight, the PM usually makes the CDU entries. The PF may also make simple, CDU entries when the workload allows. The pilot making the entries executes the change only after the other pilot verifies the entries.

During high workload times, for example departure or arrival, try to reduce the need for CDU entries. Do this by using the MCP heading, altitude, and speed control modes. The MCP can be easier to use than entering complex route modifications into the CDU. If flight plan changes occur during periods of high workload or in areas of high traffic density, the crew should not hesitate to revert to modes other than LNAV/VNAV.

2.3.2. Preflight Page Sequence

Preflight flow continues in this sequence:

- identification (IDENT) page;
- position initialization (POS INIT) page;
- route (RTE) page;
- DEPARTURES page (no automatic prompt);
- performance initialization (PERF INIT) page;
- N1 limit page;
- TO REF pages.

Some of these pages are also used in flight.

During preflight, a prompt in the lower right of the CDU page automatically directs the crew through the minimum requirements for preflight completion. Pushing the prompt key for the next page in the flow presents new entry requirements. Additional entries are made on pages to refine the performance and route calculations. If a required entry is missed, a prompt on the TAKEOFF page leads the crew to the preflight page that is missing data.

The airplane inertial position is required for FMC preflight and flight instrument operation.

A route must be entered and activated. The minimum route information is origin and destination airports and a route leg.

Performance information requires the airplane weight and cruising altitude.

Supplementary Pages

Supplementary pages are sometimes required. These pages must be manually selected. Manual selection interrupts the normal automatic sequence. Discussions of each normal page include methods to display the page when the automatic sequence is interrupted.

When the route includes SIDs and STARs, they can be entered using the DEPARTURES or ARRIVALS pages.

Entering Created Waypoints on the Nav Data Pages

NOTE: Created waypoints entered on the SUPP NAV DATA pages (permitted on the ground only) are stored in the supplemental navigation data base for an indefinite time period; those entered on REF NAV DATA pages are stored in the temporary navigation data base for one flight only.

INIT/REF key .... Push

Observe INDEX prompt displayed.

INIT/REF INDEX page .... Select

Observe the NAV DATA prompt displayed. To access the SUPP NAV DATA page, enter SUPP into the scratch pad.

NAV DATA page .... Select

(If the SUPP NAV DATA page is selected, observe the EFF FRM date line displayed. If an effective date had not been previously entered, box prompts are displayed. The effective date must be entered before proceeding. If required, enter the current or appropriate date on EFF FRM line and execute.)

Data ....Enter

Enter a crew-assigned identifier on either the WPT IDENT, NAVAID IDENT, or AIRPORT IDENT line, as appropriate. Use the navaid category only for stations with DME.

For a WPT IDENT entry, define the waypoint with entries for either latitude and longitude, or with entries for REF IDENT and RADIAL/DIST (REF IDENT identifier must already be stored in one of the FMC data bases).

For a NAVAID IDENT or AIRPORT IDENT entry, enter appropriate data.

EXEC key illuminates when data has been entered into all box prompts.

EXEC key .... Push

Repeat above steps to define additional created waypoints as desired. To enter a new identifier in the same category, simply overwrite the previous identifier.

2.3.3. FMC Route Verification Technique

After entering the route and SID into the FMC, the crew should verify that the entered route and SID are correct. The crew should always compare:

- the filed flight plan with the airways and waypoints entered on the ROUTE pages;
- the published charts with the entered SID (waypoint sequences, tracks, distances, and altitude or speed constraints);
- the computer flight plan total distance and estimated fuel remaining with the FMC-calculated distance to destination and the calculated fuel remaining at destination on the PROGRESS page.

If there is a discrepancy noted in any of the above, correct the LEGS page to match the filed flight plan legs. A crosscheck of the map display using the PLAN mode may also assist in verification of the flight plan.

When pilots are evaluating the charted procedure against the navigation database, the areas of primary concern are: waypoint sequence, speed and altitude constraints, and no unexpected discontinuities. Minor differences between the magnetic heading or track on a navigation chart and the heading or track in the FMC may exist. Primarily, this is because the FMC has a lookup table for magnetic variation, but chart designers apply a local magnetic variation. Minor differences may also result from equipment manufacturer's application of magnetic variation. These minor differences are operationally acceptable.

During CDU preflight procedure route verification Captain reads CDU and FO cross check CFP data.

During Takeoff Briefings PF reads chart data and PM cross check CDU (use ND PLAN mode and STEP option on CDU if needed).

Approach route verification performed by PM during descend preparation.

2.3.4. FMC Performance Predictions

FMC performance predictions are based on the airplane being in a normal configuration and at normal thrust settings.

If operating in a non-normal configuration, such as gear down, flaps extended, spoilers extended, gear doors open, etc., or if operating at reduced thrust due to a non-normal condition, the FMC performance predictions are inaccurate.

FMC fuel predictions are based on a clean configuration at normal thrust settings.

Fuel consumption may be significantly higher than predicted in other configurations. Fuel consumption can be significantly different than predicted when operating at a reduced thrust setting. Estimates of fuel remaining at waypoints, the destination or an alternate can be computed by the crew based on current fuel flow indications and should be updated frequently.

An accurate ETA is available if the current speed or Mach is entered into the VNAV cruise page.

Performance predictions for gear down altitude capability and gear down cruise performance are available in the Performance Inflight (PI) Chapter of the Quick Reference Handbook (QRH).

NOTE: VNAV PTH operation for approaches is usable for non-normal configurations.

Holding time available is accurate only in the clean configuration provided the FMC holding speed is maintained.

2.3.5. Takeoff Speeds

Proper takeoff speeds (V_1, V_R, and V_2) are based on takeoff weight, flaps setting, thrust rating and assumed temperature, ambient temperature, QNH, wind, runway surface condition, and performance options. Uplinked or OPT computed takeoff speeds (if available) should be used. The FCOM and FMC computed takeoff speeds (if enabled) are only valid for dispatch performance based on balanced field length, no improved climb, the most forward CG limit, and dry runway. Wet runway takeoff speeds may also be available for airplanes with wet runway takeoff performance in the AFM.

The FCOM and FMC computed takeoff speeds do not consider runway length available, minimum engine-out climb gradient capability, or obstacle clearance requirement. The FCOM and FMC computed takeoff speeds can only be used when compliance of these requirements has been verified separately with a takeoff analysis (runway/airport analysis), another approved source, or by dispatch. The FCOM and FMC computed takeoff speeds are not valid for dispatch performance based on optimized V_1 (unbalanced field length), Improved Climb, alternate forward CG limit, or contaminated or slippery runway.

2.3.6. CDU Pages Usage

In accordance with flight crew duties (PF or PM) the following CDU pages are recommended to be used in different phases of the flight:

TAKEOFF: PF – TAKEOFF, PM – LEGS;

CLIMB: PF – CLIMB, PM – LEGS;

CRUISE: PF – CRZ, PM – LEGS;

DESCENT: PF – DESCENT, PM – LEGS;

LANDING (after G/S Capture, during final descent): PF – APPROACH REF, PM – LEGS. PROGRESS page may be used by PF and PM as needed.

2.4. AFDS GUIDELINES

2.4.1. General

Crewmembers must coordinate their actions so that the airplane is operated safely and efficiently.

The Normal Procedures states that normal procedures are written for the trained flight crew and assume full use of all automated features. This statement is not intended to prevent pilots from flying the airplane manually. Manual flight is encouraged to maintain pilot proficiency, but only when conditions and workload for both the pilot flying and pilot monitoring are such that safe operations are maintained taking into account OM Part A restrictions.

Autopilot engagement should only be attempted when the airplane is in trim, F/D commands (if the F/D is on) are essentially satisfied and the airplane flight path is under control. The autopilot is not certified nor designed to correct a significant out of trim condition or to recover the airplane from an abnormal flight condition and/or unusual attitude.

2.4.2. Autopilot Flight Director System (AFDS) Procedures

The crew must always monitor:

- airplane course;
- vertical path;
- speed.

When selecting a value on the MCP, verify that the respective value changes on the flight instruments, as applicable.

The crew must verify manually selected or automatic AFDS changes. Use the FMA to verify mode changes for the:

- autopilot;
- flight director;
- autothrottle.

During LNAV and VNAV operations, verify all changes to the airplane's:

- course;
- vertical path;
- thrust
- speed.

2.4.3. AFDS Callouts

PM should call all FMA changes in flight. PF should cross check FMA and call: **"CHECK"**.

In all cases if an FMA indicates fails to engage or disengages expected Mode then the PM (or PF) shall call the mode that failed preceded by the word **"NO ___".** E.g. **"NO FLARE".**

AFDS ROLL MODES (ARMED)	
VOR/LOC	"VOR/LOC ARMED"
LNAV	"LNAV ARMED"
FAC	"FAC ARMED"
AFDS ROLL MODES (ENGAGED)	
HDG SEL	"HEADING SELECT"
LNAV	"LNAV"
VOR/LOC	"VOR/LOC CAPTURE"
FAC	"FAC"
B/CRS	"BACK COURSE"
CWS R	"CWS ROLL"
AFDS PITCH MODES (ARMED)	
VNAV	"VNAV ARMED"
G/S	"GLIDESLOPE ARMED"
V/S	"VERTICAL SPEED ARMED "
G/P	"GLIDEPATH ARMED"
FLARE	"FLARE ARMED"
AFDS PITCH MODES (ENGAGED)	
TOGA	"TOGA"
V/S	"VERTICAL SPEED"
MCP SPD	"MCP SPEED"
ALT HOLD	"ALT HOLD"
ALT/ACQ	"ALT ACQUIRE"
VNAV PTH	"VNAV PATH"
VNAV SPD	"VNAV SPEED"
VNAV ALT	"VNAV ALT"
G/S	"GLIDESLOPE CAPTURE"
G/P	"GLIDEPATH CAPTURE"
CWS P	"CWS PITCH"
FLARE	"FLARE"
AFDS STATUS	
FD	"FD"
CMD	"COMMAND"
SINGL CH	"SINGLE CHANNEL"
AUTO-THROTTLE MODES (ENGAGED)	
N1	"N ONE"
GA	"GO AROUND"
THR HOLD	"THROTTLE HOLD"
RETARD	"RETARD"
MCP SPD	"MCP SPEED"
FMC SPD	"FMC SPEED"
ARM	"ARM"

2.4.4. Autothrottle Use

Autothrottle use is recommended during takeoff and climb in either automatic or manual flight. During all other phases of flight, autothrottle use is recommended only when the autopilot is engaged in CMD.

2.4.5. Manual Flight

The PM should make AFDS mode selections at the request of the PF. Heading and altitude changes from air traffic clearances and speed selections associated with flap position changes may be made without specific directions. However, these selections should be announced, such as, **"HEADING 170 SET"**. The PF must be aware such changes are being made. This enhances overall safety by requiring that both pilots are aware of all selections, while still allowing one pilot to concentrate on flight path control.

Ensure the proper flight director modes are selected for the desired maneuver. If the flight director commands are not to be followed, the flight director should be turned off.

2.4.6. Automatic Flight

Autoflight systems can enhance operational capability, improve safety, and reduce workload. Automatic approach and landing, Category III operations, and fuel-efficient flight profiles are examples of some of the enhanced operational capabilities provided by autoflight systems. Maximum and minimum speed protections are among the features that can improve safety. LNAV, VNAV, and instrument approaches using VNAV are some of the reduced workload features. Varied levels of automation are available. The pilot decides what level of automation to use to achieve these goals by selecting the level that provides the best increase in safety and reduced workload.

Automatic systems give excellent results in the vast majority of situations. Deviations from expected performance are normally due to an incomplete understanding of their operations by the flight crew. When the automatic systems do not perform as expected, the pilot should reduce the level of automation until proper control of path and performance is achieved. For example, if the pilot failed to select the exit holding feature when cleared for the approach, the airplane will turn outbound in the holding pattern instead of initiating the approach. At this point, the pilot may select HDG SEL and continue the approach while using other automated features. A second example, if the airplane levels off unexpectedly during climb or descent with VNAV engaged, LVL CHG may be selected to continue the climb or descent until the FMC can be programmed.

Early intervention prevents unsatisfactory airplane performance or a degraded flight path. Reducing the level of automation as far as manual flight may be necessary to ensure proper control of the airplane is maintained. The pilot should attempt to restore higher levels of automation only after airplane control is assured. For example, if an immediate level-off in climb or descent is required, it may not be possible to comply quickly enough using the AFDS. The PF should disengage the autopilot and level off the airplane manually at the desired altitude. After level off, set the desired altitude in the MCP, select an appropriate pitch mode and re-engage the autopilot.

During LNAV and VNAV operations, verify all changes to the airplane's:

- course;
- vertical path;
- thrust;
- speed.

2.4.7. Recommended Pitch and Roll Modes

If the LEGS page and map display reflect the proper sequence and altitudes, LNAV and VNAV are recommended. If LNAV is not used, use an appropriate roll mode. When VNAV is not used, the following modes are recommended:

- LVL CHG is the preferred mode for altitude changes of more than 1,000 feet;
- V/S is preferred if the altitude change is 1,000 feet or less.

If unplanned speed or altitude restrictions are imposed during the arrival, the continued use of VNAV may induce an excessive workload. If this occurs, use LVL CHG or V/S as appropriate.

When using VNAV for published instrument departures, arrivals, and approaches, the following recommendations should avoid unnecessary level-offs while ensuring minimum altitudes are met. If waypoints with altitude constraints are not closely spaced, the normal MCP altitude setting technique is recommended. If waypoints with altitude constraints are closely spaced to the extent that crew workload is adversely affected and unwanted level-offs are a concern, an alternate MCP altitude setting technique can be used.

When the alternate MCP altitude setting technique is used, the selection of a pitch mode other than VNAV PTH or VNAV SPD will result in risk of violating altitude constraints.

Normal MCP Altitude Setting. The following MCP altitude setting technique is normally used during published instrument departures, arrivals, and approaches when waypoints with altitude constraints are not closely spaced:

- during climbs, maximum or hard altitude constraints should be set in the MCP. Minimum crossing altitudes need not be set in the MCP. The FMC alerts the crew if minimum altitude constraints will not be satisfied
- during descent, set the MCP altitude to the next constraint or clearance altitude, whichever will be reached first
- just prior to reaching the constraint, when compliance with the constraint is assured, and cleared to the next constraint, reset the MCP to the next constraint.

Alternate MCP Altitude Setting. MCP altitude setting technique may be used during published instrument departures, arrivals, and approaches where altitude constraints are closely spaced (5 n.m. or less) to the extent that crew workload is adversely affected and unwanted level-offs are a concern:

- for departures, set the highest of the closely-spaced constraints
- for arrivals, initially set the lowest of the closely spaced altitude constraints or the FAF altitude, whichever is higher

2.4.8. AFDS Mode Control Panel Faults

In-flight events have occurred where various AFDS pitch or roll modes, such as LNAV, VNAV or HDG SEL became un-selectable or ceased to function normally.

Typically, these types of faults do not generate a failure annunciation. These faults may be caused by an MCP hardware (switch) problem.

If an AFDS anomaly is observed where individual pilot-selected AFDS modes are not responding normally to MCP switch selections, attempt to correct the problem by disengaging the autopilot and selecting both flight director switches to OFF. This clears all engaged AFDS modes. When an autopilot is re-engaged or a flight director switch is selected ON, the AFDS default pitch and roll modes should engage. The desired AFDS pitch and roll modes may then be selectable.

If this action does not correct the fault condition, the desired flight path can be maintained by selecting an alternate pitch or roll mode. Examples are included in the following table:

Inoperative or Faulty Autopilot Mode	Suggested Alternate Autopilot Mode or Crew Technique
HDG SEL	Set desired heading, disengage AFDS and manually roll to the desired heading, and re-engage the AFDS. The AFDS will hold the established heading.
LNAV	Use HDG SEL to maintain the airplane track on the active route.
VNAV SPD or VNAV PTH (climb or descent)	Use LVL CHG or V/S.
VNAV PTH (cruise, descent)	Use altitude hold. If altitude hold is not directly selectable, use LVL CHG to automatically transition to altitude hold
VNAV PTH (approach)	Use V/S. Monitor and fly the approach referencing glidepath.
VOR/LOC	Use HDG SEL. Monitor and fly the approach referencing localizer raw data.
G/S	Use V/S to descend on an ILS or GLS approach. Monitor and fly the approach referencing glide slope raw data.

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2.5. AIRCRAFT CONFIGURATION

2.5.1. General

Regardless of autopilot use PM is responsible for flaps and gear extension and retraction. PF commands for flaps and gear operation. To prevent undesired operation, PM shall always acknowledge all PF commands for flaps and gear operations. If PM is busy with communication PF shall not command for configuration change and shall maintain airspeed for current configuration.

All aircraft configurations are checked by respective checklists.

2.5.2. Flap Operation

Before setting flap lever in desired position PM shall verify actual airspeed for next flaps position, and call: **“SPEED CHECK, FLAPS ___”**.

PM shall monitor normal flaps extension and retraction on flaps position indicator. After flaps movement complete PM shall announce **“FLAPS .... SET”.**

FLAP RETRACTION SCHEDULE

The minimum altitude for flap retraction is 400 feet.

During training flights, 1,000 feet AAL is normally used as the acceleration height to initiate thrust reduction and flap retraction. For noise abatement considerations during line operations, accelerate approximately at 1,000 feet AAL and reduce thrust during flaps retraction, or as specified by individual airport procedures.

During flap retraction, selection of the next flap position is initiated when reaching the maneuver speed for the existing flap position.

Therefore, when the new flap position is selected, the airspeed is below the maneuver speed for that flap position. For this reason, the airspeed should be increasing when selecting the next flap position. During flap retraction, at least adequate maneuver capability or 30° of bank (15° of bank and 15° overshoot) to stick shaker is provided at the flap retraction speed. Full maneuver capability or at least 40° of bank (25° of bank and 15° overshoot) is provided when the airplane has accelerated to the recommended maneuver speed for the selected flap position.

Begin flap retraction at V2 + 15 knots, except for a flaps 1 takeoff. For a flaps 1 takeoff, begin flap retraction when reaching the flaps 1 maneuver speed.

With airspeed increasing, flap retractions should be initiated when airspeed reaches the maneuver speed for the existing flap position. The maneuver speed for the existing flap position is indicated by the numbered flap maneuver speed bugs on the airspeed display.

Takeoff Flap Retraction Speed Schedule

T/O Flaps	At Speed "Display"	Flap Retraction Speed	Select Flaps
25	V2+15	V2 + 15	15
	"15"	Vref 40 + 20	5
	"5"	Vref 40 + 30	1
	"1"	Vref 40 + 50	UP
15 or 10	V2+15	V2 + 15	5
	"5"	Vref 40 + 30	1
	"1"	Vref 40 + 50	UP
5	V2+15	V2 + 15	1
	"1"	Vref 40 + 50	UP
1	"1"	Vref 40 + 50	UP
Flaps UP maneuvering speed Vref 40 + 70 Limit bank angle to 15° until reaching V2 + 15 knots			

For flaps up maneuvering, maintain at least "UP".

Flap Extension

During flap extension, selection of the flaps to the next flap position should be made when approaching, and before decelerating below, the maneuver speed for the existing flap position. The flap extension speed schedule varies with airplane weight and provides full maneuver capability or at least 40° of bank (25° of bank and 15° overshoot) to stick shaker at all weights.

Flap Extension Schedule

Current Flap Position	At Speedtape "Display"	Select Flaps	Command Speed for Selected Flaps
UP	"UP"	1	"1"
1	"1"	5	"5"
5	"5"	15	"15"
15	"15"	30 or 40	(Vref 30 or Vref 40) + wind additives

2.5.3. Landing Gear Operation

After takeoff or go-around gear retraction shall be started after ensuring positive rate of climb on baro altimeters. PM calls: **“POSITIVE RATE”**, PF verifies positive rate and calls: **“GEAR UP”**. PM confirms command calling: **“GEAR UP”** and sets the landing gear lever to UP. When all landing gear lights extinguished PM announce landing gear status: **“GEAR UP”.**

During approach gear must be extended prior to selecting flaps more than 15.

PF calls: **“GEAR DOWN”**. PM confirms command calling: **“GEAR DOWN”** and sets the landing gear lever to DOWN. When all green landing gear lights ON and all red landing gear lights extinguished PM announce landing gear status: **“GEAR DOWN”.**

2.5.4. Speedbrake Operation

The PF should keep a hand on the speedbrake lever when the speedbrakes are used in-flight. This helps prevent leaving the speedbrake extended when no longer required.

Use of speedbrakes between the down detent and flight detent can result in rapid roll rates and normally should be avoided. While using the speedbrakes in descent, allow sufficient altitude and airspeed margin to level off smoothly. Lower the speedbrakes before adding thrust.

NOTE: In flight, do not extend the speedbrake lever beyond the FLIGHT detent.

The use of speedbrakes with flaps extended should be avoided, if possible. With flaps 15 or greater, the speedbrakes should be retracted. If circumstances dictate the use of speedbrakes with flaps extended, high sink rates during the approach should be avoided. Speedbrakes should be retracted before reaching 1,000 feet AAL.

After landing PM verifies automatic speedbrake extension and calls: **“SPEEDBRAKES UP”** or **“SPEEDBRAKES NOT UP”**. If on landing speedbrakes fail to extend automatically captain shall manually extend speedbrakes and call: **“SPEEDBRAKES UP”**.

The flaps are normally not used for increasing the descent rate. Normal descents are made in the clean configuration to pattern or instrument approach altitude.

When descending with the autopilot engaged and the speedbrakes extended at speeds near VMO/MMO, the airspeed may momentarily increase to above VMO/MMO if the speedbrakes are retracted quickly. To avoid this condition, smoothly and slowly retract the speedbrakes to allow the autopilot sufficient time to adjust the pitch attitude to maintain the airspeed within limits.

When the speedbrakes are retracted during altitude capture near VMO/MMO, a momentary overspeed condition may occur. This is because the autopilot captures the selected altitude smoothly by maintaining a fixed path while the thrust is at or near idle. To avoid this condition, it may be necessary to reduce the selected speed and/or descent rate prior to altitude capture or reduce the selected speed and delay speedbrake retraction until thrust is increased to maintain level off airspeed.

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2.6. WHEEL BRAKES OPERATION

2.6.1. Automatic Brakes

The autobrake system provides automatic braking at preselected deceleration rates for landing and full pressure for rejected takeoff.

Takeoff Autobrake Usage

RTO shall be used for every takeoff.

The RTO mode can be selected only when on the ground. Upon selection, the AUTO BRAKE DISARM light illuminates for one to two seconds and then extinguishes, indicating that an automatic self–test has been successfully accomplished.

To arm the RTO mode prior to takeoff the following conditions must exist:

- airplane on the ground;
- antiskid and autobrake systems operational;
- AUTO BRAKE select switch positioned to RTO;
- wheel speed less than 60 knots;
- forward thrust levers positioned to IDLE.

With RTO selected, if the takeoff is rejected prior to wheel speed reaching 90 knots autobraking is not initiated, the AUTO BRAKE DISARM light does not illuminate and the RTO autobrake function remains armed. If the takeoff is rejected after reaching a wheel speed of 90 knots, maximum braking is applied automatically when the forward thrust levers are retarded to IDLE.

The RTO mode is automatically disarmed when both air/ground systems indicate the air mode. The AUTO BRAKE DISARM light does not illuminate and the AUTO BRAKE select switch remains in the RTO position. To reset or manually disarm the autobrake system, position the selector to OFF. If a landing is made with RTO selected (AUTO BRAKE select switch not cycled through OFF), no automatic braking action occurs and the AUTO BRAKE DISARM light illuminates two seconds after touchdown.

Landing Autobrake Usage

Use of the autobrake system is recommended whenever the runway is limited, when using higher than normal approach speeds, landing on slippery runways, or landing in a strong crosswind.

For normal operation of the autobrake system select a deceleration setting.

Settings include:

- MAX AUTO: Used when minimum stopping distance is required. Deceleration rate is less than that produced by full manual braking;
- 3: Should be used for wet or slippery runways or when landing rollout distance is limited. If adequate rollout distance is available, autobrake setting 2 may be appropriate;
- 1 or 2: These settings provide a moderate deceleration suitable for all routine operations.

Experience with various runway conditions and the related airplane handling characteristics provide initial guidance for the level of deceleration to be selected.

Immediate initiation of reverse thrust at main gear touchdown and full reverse thrust allow the autobrake system to reduce brake pressure to the minimum level. Since the autobrake system senses deceleration and modulates brake pressure accordingly, the proper application of reverse thrust results in reduced braking for a large portion of the landing roll.

The importance of establishing the desired reverse thrust level as soon as possible after touchdown cannot be overemphasized. This minimizes brake temperatures and tire and brake wear and reduces stopping distance on very slippery runways.

After touchdown, crewmembers should be alert for autobrake disengagement annunciations. The PM should notify the PF by calling: **"AUTOBRAKE DISARM"**, anytime the autobrakes disengage.

If stopping distance is not assured with autobrakes engaged, the PF should immediately apply manual braking sufficient to assure deceleration to a safe taxi speed within the remaining runway.

2.6.2. Transition to Manual Braking

The speed at which the transition from autobrakes to manual braking is made depends on airplane deceleration rate, runway conditions and stopping requirements. Normally the speedbrakes remain deployed until taxi speed, but may be stowed earlier if stopping distance within the remaining runway is assured. When transitioning to manual braking, use reverse thrust as required until taxi speed.

When transitioning from the autobrake system to manual braking, the PF should notify the PM: **"AUTOBRAKE DISARM"**. Techniques for release of autobrakes can affect passenger comfort and stopping distance. These techniques are:

- stow the speedbrake handle. When stopping distance within the remaining runway is assured, this method provides a smooth transition to manual braking, is effective before or after thrust reversers are stowed, and is less dependent on manual braking technique;
- smoothly apply brake pedal force as in a normal stop, until the autobrake system disarms. Following disarming of the autobrakes, smoothly release brake pedal pressure. Disarming the autobrakes before coming out of reverse thrust provides a smooth transition to manual braking.

2.6.3. Using of Parking Brakes.

Both pilots should verify:

- Hydraulic Brake Pressure Indicator indicates normal pressure (green band 2800-3500 psi).
- Parking Brake Warning Light illuminated (red).
- Aircraft not moving.

Captain calls**:" PARKING BRAKE SET"**. Then F/O: **"CHECKED".**

Captain calls:" PARKING **BRAKE RELEASED"**. Then F/O: **"CHECKED"**.

NOTE: Fully charged (3000-psi) brake accumulator keeps the brakes pressurized at least eight hours. Use chocks around both outboard main landing gear wheels when you park the airplane more than eight hours.

2.7. REVERSE THRUST OPERATION

2.7.1. Reverse Thrust Callouts

When both REV indications are green, call: **“REVERSERS NORMAL”.**

If there is no REV indication(s) or the indication(s) stays amber, call: **“NO REVERSER ENGINE №1”** or **“NO REVERSER ENGINE №2”**, or **“NO REVERSERS”.**

2.7.2. Reverse Thrust Operations

The position of the hand should be comfortable, permit easy access to the autothrottle disconnect switch, and allow control of all thrust levers, forward and reverse, through full range of motion.

Put the hand at reverse thrust levers during RTO or landing only after Speedbrakes deployment. After touchdown, with the thrust levers at idle, raise the reverse thrust levers up and aft to the interlock position, then to the number 2 reverse thrust detent. Conditions permitting, limit reverse thrust to the number 2 detent. The PM should monitor engine operating limits and call out any engine operational limits being approached or exceeded, any thrust reverser failure, or any other abnormalities.

Maintain reverse thrust as required, up to maximum, until stopping on the remaining runway is assured.

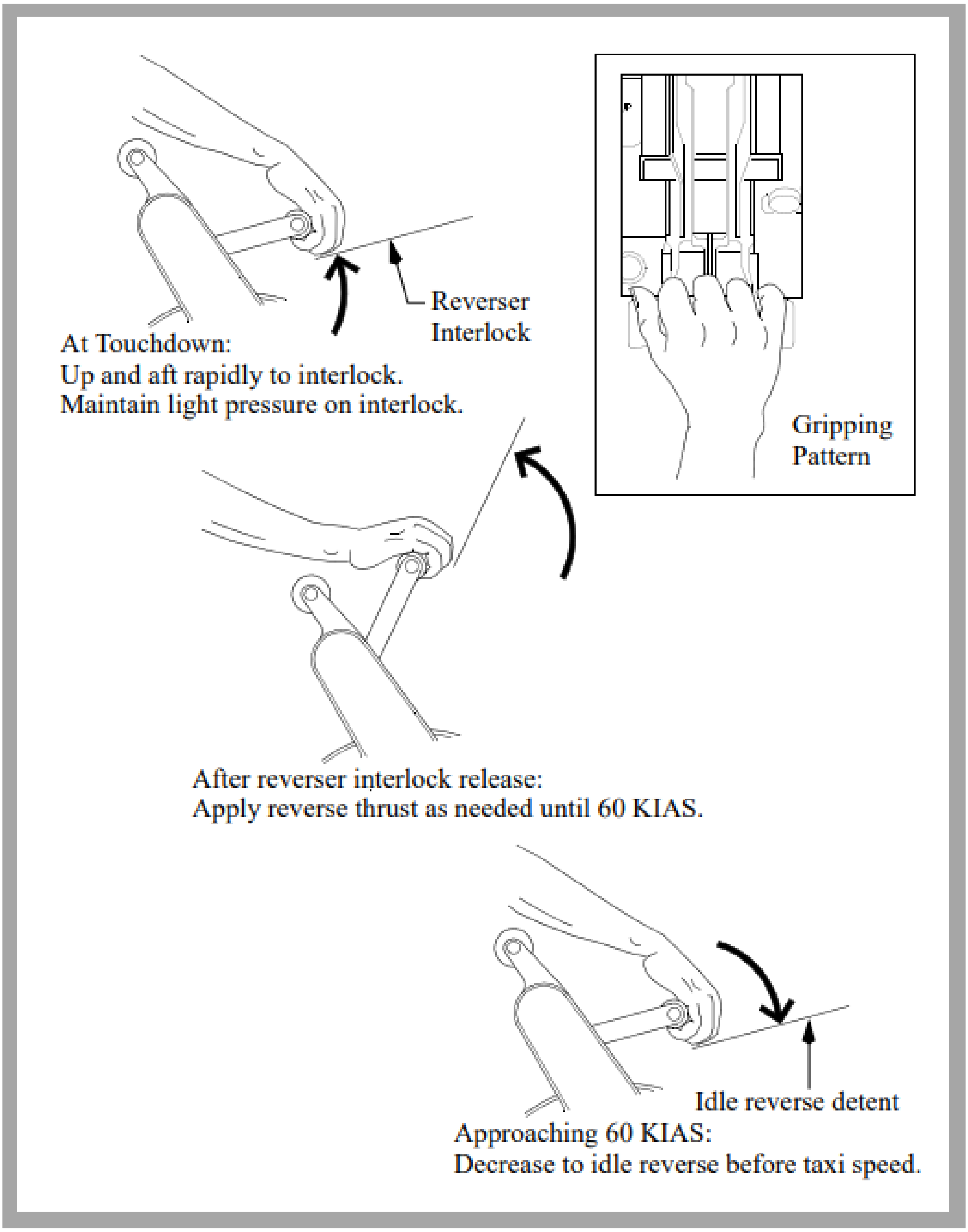
When stopping is assured and the airspeed approaches 60 KIAS start reducing the reverse thrust so that the reverse thrust levers are moving down at a rate commensurate with the deceleration rate of the airplane. The reverse thrust levers should be positioned to reverse idle by taxi speed, then to full down after the engines have decelerated to idle. Reverse thrust is reduced to idle between 60 KIAS and taxi speed to prevent engine exhaust re-ingestion and to reduce the risk of FOD. It also helps the pilot maintain directional control in the event a reverser becomes inoperative.

NOTE: Reverse thrust is most effective at high speeds.

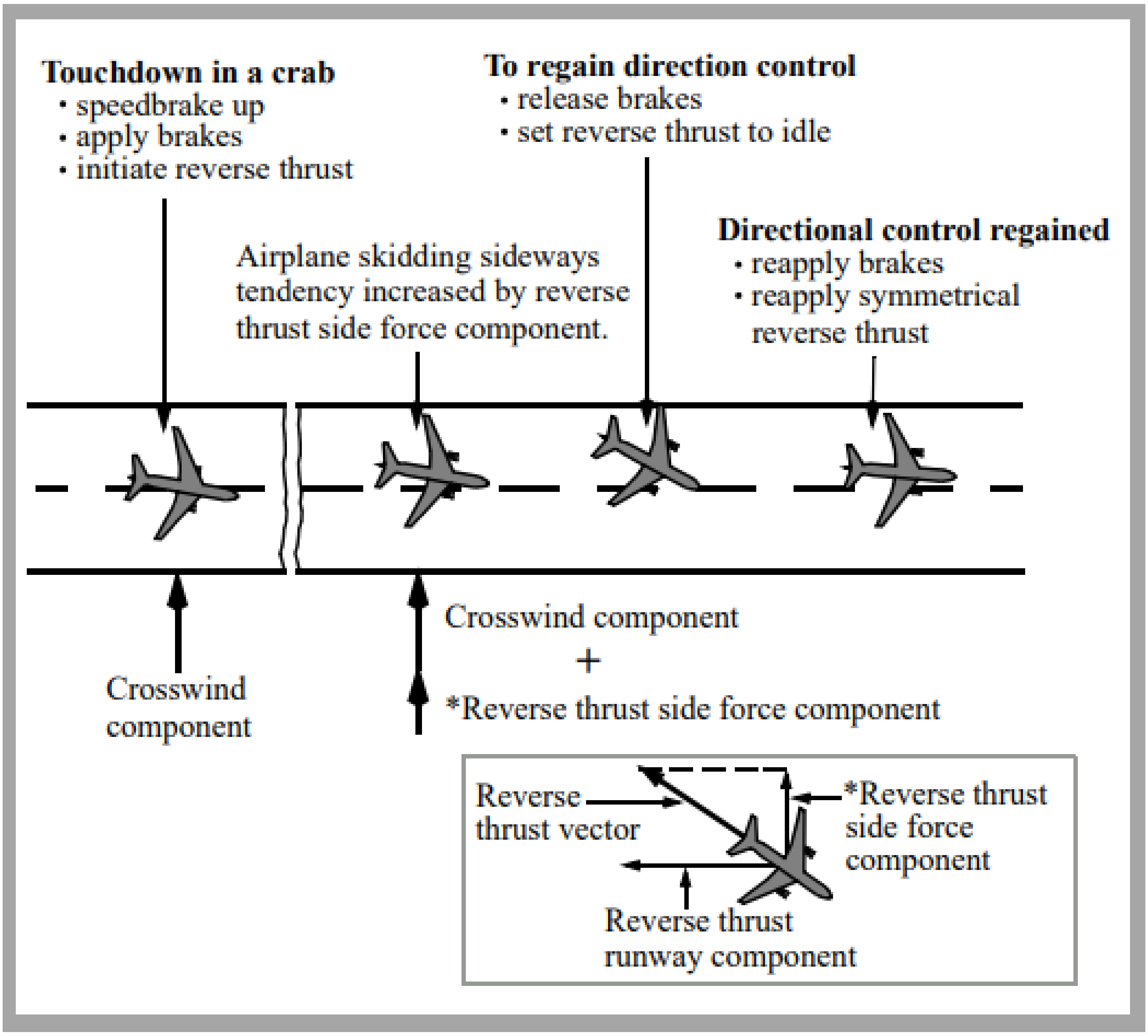
NOTE: If high levels of reverse thrust were used at low speed (below 60 kts) it should be noted in ATL.

NOTE: If an engine surges during reverse thrust operation, quickly select reverse idle on both engines.

The PM should call out 60 knots to assist the PF in scheduling the reverse thrust. The PM should also call out any inadvertent selection of forward thrust as reverse thrust is canceled.


Reverser
Interlock
At Touchdown:
Up and aft rapidly to interlock.
Maintain light pressure on interlock.
Gripping
Pattern
After reverser interlock release:
Apply reverse thrust as needed until 60 KIAS.
Idle reverse detent
Approaching 60 KIAS:
Decrease to idle reverse before taxi speed.

2.7.3. Reverse Thrust and Crosswind (All Engines)



To correct back to the centerline, release the brakes and reduce reverse thrust to reverse idle. Releasing the brakes increases the tire-cornering capability and contributes to maintaining or regaining directional control. Setting reverse idle reduces the reverse thrust side force component without the requirement to go through a full reverser actuation cycle. Use rudder pedal steering and differential braking as required, to prevent over correcting past the runway centerline. When directional control is regained and the airplane is correcting toward the runway centerline, apply maximum braking and symmetrical reverse thrust to stop the airplane.

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2.8. COMMUNICATION

2.8.1. General

Radio communication is carried out in accordance with:

- Federal aviation rules "Осуществление радиосвязи в воздушном пространстве Российской Федерации";
- Annex 10 to the Convention on International Civil Aviation "Aeronautical Telecommunications Volume II Communication Procedures including those with PANS status";
- Manual of radiotelephony (Doc. 9432).

On the ground communication with ATC is carried out by First Officer. Communication with ground handling personnel is carried out by Captain.

In flight communication with ATC is carried out by PM. PF shall monitor all communications with ATC. When F/O has to consult to the Captain before responding ATC the CPT may take communications and respond.

Dialing frequencies is the responsibility of the PM.

2.8.2. Communication with Ground Personnel

Cockpit	Ground
Initial ground contact	
"Ground from cockpit" (Земля – Кабине)	**"Go ahead"** (Передавайте)
"Request clearance to pressurize hydraulics" (Прошу разрешения на подключение гидравлических систем)	**"You are cleared to pressurize hydraulics"** (Разрешено подключать давление в гидросистемах)

Cockpit	Ground
“Are you ready for pushback (to start up)” (Вы готовы к выталкиванию/буксировке (запуску двигателей))	**“Ready for push (to start up)”** (Готов к выталкиванию / буксировке (запуску двигателей)). Эта фраза означает, что: • все наземное оборудование, наземный источник электропитания, наземное кондиционирование, оборудование по загрузке и разгрузке багажа, специальные технические приспособления из-под самолета убраны; • наземные пины безопасности удалены (GEAR PINS); • колодки из-под шасси убраны; • (в случае буксировки) водило и буксир подсоединены, буксировочный штырь в механизме поворота носовой стойки шасси установлен; • двери люки закрыты.
Pushback	
“Ready for pushback to ___” (К выталкиванию готов.) При необходимости указывается маршрут буксировки, место запуска, направление установки.	
“Brakes released” (Тормоза выключены.)	**“Confirm brakes released.”** Дается команда на снятие с тормозов.
“Brakes set” (Тормоза включены.)	**“Pushback completed, confirm brakes set.”** (Буксировка закончена, подтвердите включение тормозов.)
	“Tow bar removed, steering normal” (Водило убрано, буксировочный штырь снят)
Engine Start	
“Ready to start up” (Готов к запуску двигателей)	**“All engines are clear”** (Запуск разрешаю)
Сообщает последовательность запуска двигателей. **“Start sequence number two, then one (one, two)”** Дает информацию: **“Starting number two” (number one)**	

Before Taxi Out	
“Prepare airplane for taxi and give hand signal on left/right side” (Приготовьте самолет к выруливанию, дайте сигнал рукой о готовности с левой/правой стороны)	**“Airplane ready for taxi. Wait for hand signal on left/right side”** (ВС к рулению готово, ожидайте сигнал рукой слева/справа). Эта фраза означает, что ВС полностью готово к вылету, зона за самолетом свободна, красные проблесковые огни наблюдаются, буксировочный штырь в механизме поворота носовой стойки шасси убран. Выпускающий должен показать буксировочный штырь вместе с подаваемым рукой сигналом.
Use of external electrical power	
“Connect external electric” (Подключите наземный источник питания)	**“External electric connected”** (Наземный источник подключен)
(when aircraft AC power established) **“Remove external electric”** (Отсоедините наземный источник питания)	**“External electric removed”** (Наземный источник питания отсоединен)
Use of external ground air	
“Ready for ground air” (Подключите УВЗ)	**“Ground air available”** (УВЗ подключено, давление подается)
“Remove ground air” (Отсоедините УВЗ)	**“Ground air removed”** (УВЗ отсоединено)

2.8.3. Communication with Cabin Crew and Passengers

Communication with cabin crew and passengers should be made in accordance with OM Part A.

Before making an announcement, plan it thoroughly. Using good communication skills with your passengers can vastly improve satisfaction, and may even put anxious fliers at ease. If you deliver a professional announcement with a reassuring voice, people who are a little bit uncomfortable while flying will get the sense that the aircraft is in the control of a very capable individual.

The information must be close to the forms developed and approved by the Airline, but may contain other necessary and relevant information.

2.8.4. Standard Callouts

General concept of callouts

Both crewmembers should be continuously aware of current airplane altitude, position, energy state, configuration, and maintain situational awareness appropriate for the phase of flight.

Avoid nonessential conversation during critical phases of flight, particularly during taxi, takeoff, approach and landing. Unnecessary conversation reduces crew efficiency and alertness and is not recommended when below 10,000 feet MSL / FL100. At high altitude airports, adjust this altitude upward, as required.

Both pilots should check the flight instruments and Flight Mode Annunciations (FMAs) at regular intervals to verify the selections made are correct for the phase of flight. Both pilots should crosscheck their MCP selections with the FMAs to ensure the airplane is responding as expected. Unexpected FMAs should be announced, evaluated and addressed appropriately.

Procedural callouts are required.

The Pilot Monitoring (PM) makes callouts based on instrument indications, FMAs or observations for the appropriate condition. The PF should verify the condition/location from the flight instruments and acknowledge. If the PM does not make the callout within 10 seconds after condition/location or indication has changed the PF should make it.

The PM calls out significant deviations from command airspeed or flight path. Either pilot should call out any abnormal indications of the flight instruments (flags, loss of deviation pointers, etc.).

One of the basic fundamentals of Crew Resource Management is that each crewmember must be able to supplement or act as a back-up for the other crewmember. Proper adherence to recommended callouts is an essential element of a well-managed flight deck. These callouts provide both crewmembers required information about airplane systems and about the participation of the other crewmember. The absence of a callout at the appropriate time may indicate a malfunction of an airplane system or indication, or indicate the possibility of incapacitation of the other pilot.

The PF should acknowledge all GPWS voice callouts except altitude callouts during approach while below 500 feet AAL. No callout is necessary from the PM if the GPWS voice callout has been acknowledged by the PF. The callout of **“CONTINUE”** or **“GO-AROUND”** at minimums is not considered an altitude callout and should always be made. If the automatic electronic voice callout is not heard by the flight crew, the PM should make the callout radio altitude at 100 feet, 50 feet, 30 feet 20 feet and 10 feet (or other values as required) to aid in developing an awareness of eye height at touchdown.

Captain or PF may also request additional callouts by pre-briefing.

Callout “SET”

The call **“SET”** is used to set a value by selector (e.g. on MCP), but not modes changes. **“SET”** means rotation of appropriate selector.

- **“SET MISSED APPROACH ALTITUDE”;**
- **“SET FL ___”;**
- **“SET HDG ___”;**
- **“SET SPEED ___”;**

- **“SET FLAPS ___ SPEED”.**

After setting required value on MCP the pilot should check instrument indication and announce it.

- **“MISSED APPROACH ALTITUDE ___ SET”;**
- **“FL ___ SET”;**
- **“HDG ___ SET”;**
- **“SPEED ___ SET”;**
- **“FLAPS ___ SPEED SET”.**

Callout “ARM”

The command “ARM” means to arm a system, such as by pushing the specified MCP button:

- **“ARM VOR/LOC”;**
- **“ARM GS”.**

Callout **“ON” / “OFF”**

The commands “ON” and “OFF” are used for engagement or disengagement of autopilot and for F/D command bars switches etc.

- **“FLIGHT DIRECTOR OFF”.**

Callout “CHECK”

- **“CHECK ___”:** a command for the other pilot to check an item**.**
- **“___ CHECK”:** a response that an item has been checked**.**
- **“CROSSCHECKED”:** a call verifying information from both pilot stations**.**

2.8.5. Normal Operation Callouts

Phase	Callout	Crew member
To do Preflight Procedure	PREFLIGHT PROCEDURE	C
After completion of Preflight Procedure	PREFLIGHT PROCEDURE COMPLITED	FO
To do Preflight Checklist	PREFLIGHT CHECKLIST	C
After completion of Preflight Checklist	PREFLIGHT CHECKLIST COMPLETE	F/O
Callout for Takeoff Briefing	TAKEOFF BRIEFING	PF
After completion of Takeoff Briefing	TAKEOFF BRIEFING COMPLETE	PF
To implement Before Start Procedure	BEFORE START PROCEDURE	C
After completion of Before Start Procedure	BEFORE START PROCEDURE COMPLETE	F/O
To do Before Start Checklist	BEFORE START CHECKLIST	C
After completion of Before Start Checklist	BEFORE START CHECKLIST COMPLETE	F/O

Phase	Callout	Crew member
To start respective engine	START ENGINE №1 or №2	C
Confirmation that the Start valve has opened	START VALVE OPEN	F/O
Confirmation that there is N1 rotation	N1 ROTATION	F/O
Confirmation that the oil pressure increases.	OIL PRESSURE	F/O
At 25% or max. motoring, set Eng. Start Lever to IDLE	IDLE	C
If there is no callout “IDLE” from Captain	25 (MAX MOTORING)	F/O
Confirmation that there is Fuel flow and EGT indication	FUEL FLOW; EGT	F/O
Confirmation that the Starter cutout	STARTER CUTOUT	F/O
After engine stabilized at idle	STABILIZED	F/O
To implement Before Taxi Procedure and extend flaps	FLAPS ___ BEFORE TAXI PROCEDURE	C
After completion of Before Taxi Procedure	BEFORE TAXI PROCEDURE COMPLETE	F/O
To do Before Taxi Checklist	BEFORE TAXI CHECKLIST	C
After completion of Before Taxi Checklist	BEFORE TAXI CHECKLIST COMPLETE	F/O
Before taxi out or before intersection	LEFT SIDE CLEAR	C
Before taxi out or before intersection	RIGHT SIDE CLEAR	FO
To switch taxi and runway turnoff light switches ON	TAXI LIGHTS ON	C
To implement Before Takeoff Procedure	BEFORE TAKEOFF PROCEDURE	C
To check baro altimeters data before takeoff	ALTIMETERS	FO
Check altimeter setting, current altitude	QNH__, ALTITUDE	Both
After completion of Before Takeoff Procedure	BEFORE TAKEOFF PROCEDURE COMPLETE	F/O
To do Before Takeoff Checklist	BEFORE TAKEOFF CHECKLIST	C

Phase	Callout	Crew member
After completion of Before Takeoff Checklist	BEFORE TAKEOFF CHECKLIST COMPLETE	F/O
After RW is identified	RW ___ IDENTIFIED	Both
To start flight timing and to switch landing lights on	TAKEOFF, LIGHTS ON	PF
Set Thrust Levers to N1 40-60%	CHECK ENGINE STABILISED	PF
To confirm that engines are stabilized before takeoff	STABILIZED	PM
The correct takeoff thrust is set	TAKEOFF THRUST SET	PM
Verify 80 knots and FMA changed to HOLD	80 KNOTS, THROTTLE HOLD	PM
If thrust controlled manually during takeoff	MANUAL THRUST	PF
If RTO is initiated	REJECT (I HAVE CONTROL)	C
At V1	V1 (if no automatic call)	PM
At V_R	ROTATE	PM
When barometric altitude is increasing	POSITIVE RATE	PM
Call to retract the landing gear	GEAR UP	PF
Confirmation of command to retract the landing gear	GEAR UP	PM
After gear UP (no lights)	GEAR UP	PM
Passing 400 feet AAL	400	PM
When autopilot engaged	COMMAND	PM
Passing thrust reduction altitude or flap position	THRUST REDUCTION	PM
Call for thrust reduction If A/P and VNAV not engaged	N1	PF
Passing acceleration altitude	ACCELERATION	PM
At acceleration height flaps up maneuvering speed set by PF	FLAPS UP SPEED SET	PF

Phase	Callout	Crew member
Call to set flaps up maneuvering speed at acceleration height	SET FLAPS UP SPEED	PF
Call to retract flaps	FLAPS ___	PF
Confirmation of command to retract flaps	SPEED CHECK, FLAPS ___	PM
After flaps set to desired position	FLAPS___SET	PM
After flaps retraction is complited	FLAPS UP, NO LIGHTS	PM
Call to do After Takeoff Checklist	AFTER TAKEOFF CHECKLIST	PF
After completion of After Takeoff Checklist	AFTER TAKEOFF CHECKLIST COMPLETE	PM
Passing transition altitude	TRANSITION ALTITUDE	PM
Call to set the altimeters to standard	SET STANDARD	PF
After setting standard	STANDARD SET, CROSSCHECKED	PM
Passing FL100 / 10,000 ft MSL	FL100 or 10,000 FEET	PM
Passing FL100 / 10,000 ft MSL during clime	LIGHTS OFF	PF
1,000 ft. above / below assigned altitude / flight level	ONE THOUSAND TO LEVEL OFF	PM
To make route modification or CDU entry	LOAD CDU ____	PF
When CDU modification complete	CONFIRM?	PM
Confirmation of route modification or CDU entry	EXECUTE	PF
Callout for Approach Briefing	APPROACH BRIEFING	PF
After completion of Approach Briefing	APPROACH BRIEFING COMPLETE	PF
Call to do Descent Checklist	DESCENT CHECKLIST	PF
After completion of Descent Checklist	DESCENT CHECKLIST COMPLETE	PM

Phase	Callout	Crew member
Passing FL100 / 10,000 ft MSL	FL100 or 10,000 FEET	PM
Passing FL100 / 10,000 ft MSL during decsent	LIGHTS ON	PF
Passing transition level	TRANSITION LEVEL	PM
Call to set altimeters	SET QNH ___	PF
After setting QNH on all altimeters	QNH ___ SET, CROSSCHECKED	PM
To do Approach Checklist	APPROACH CHECKLIST	PF
After completion of Approach Checklist	APPROACH CHECKLIST COMPLETE	PM
After ILS/GLS is identified	ILS (GLS) IDENTIFIED	PM
After arming localizer	VOR/LOC ARMED	PF
After arming approach	APPROACH ARMED	PF
First positive inward motion of localizer pointer	LOCALIZER ALIVE	PM
When RW heading or GO-AROUND heading set during approach	RW (GO-AROUND) HDG SET	PF
First positive motion of glideslope pointer	GLIDESLOPE ALIVE	PM
3 NM before the FAF/FAP	APPROACHING GLIDE PATH	PM
Call to extend gear	GEAR DOWN	PF
Call to extend flaps	FLAPS ___	PF
Confirmation of command to extend flaps	SPEED CHECK, FLAPS ___	PM
After flaps set to desired position	FLAPS___SET	PM
Call to do Landing Checklist	LANDING CHECKLIST	PF
After completion of Landing Checklist	LANDING CHECKLIST COMPLETE	PM & PF
At FAF/FAP, verify the crossing altitude	PASSING FAF/FAP, ___ FEET	PM
Passing 1,000 / 500 ft AFE	1,000 / 500	PM
500 feet AAL (during Autoland)	500, FLARE ARMED	PM

Phase	Callout	Crew member
Confirmation of Passing 500 ft (during Autoland)	CHECK, AUTOLAND	PF
To recycle Flight Directors during approach	RECYCLE FD's	PF
If suitable visual reference is established at circling MDA(H)	CONTINUE CIRCLE	PF
If approach meets stabilized approach requirements (1,000' for instrument approach; 500' for visual approach and circle-to-land)	STABILIZED	PM
If approach does not meet stabilized approach recommendations	UNSTABILIZED, GO-AROUND	PM
100 ft. above DA(H) / MDA(H)	APPROACHING MINIMUMS	PM
At DA(H) / MDA(H)	MINIMUMS	PM
At DA(H) / MDA(H) – Suitable visual reference established	CONTINUE	PF
At DA(H) / MDA(H) – Suitable visual reference NOT established	GO-AROUND, FLAPS 15, CHECK THRUST	PF
Verify that the thrust is sufficient for the go-around or adjust as needed	THRUST SET	PF
At GA when barometric altitude is increasing	POSITIVE RATE	PM
At GA call to retract the landing gear	GEAR UP	PF
At GA confirmation of command to retract the landing gear	GEAR UP	PM
After gear UP (no lights)	GEAR UP	PM
At 50 feet RA (during Autoland)	FLARE or NO FLARE	PM
At 27 feet RA (during Autoland)	RETARD or NO RETARD	PM

Phase	Callout	Crew member
After Touchdown when the Speedbrake Lever is UP / Not UP	SPEEDBRAKES UP / SPEEDBRAKES NOT UP	PM
After landing PF use manual braking	MANUAL BRAKING	PF
After Touchdown announce Autobrakes System Status	AUTOBRAKES DISARM	PM
When both REV indications are green	REVERSERS NORMAL	PM
If there is no REV indication(s) or the indication(s) stays amber	NO REVERSER ENGINE №1 or NO REVERSER ENGINE№2 or NO REVERSERS	PM
At IAS of 60 knots	60 KNOTS	PM
To implement After Landing Procedure	AFTER LANDING PROCEDURE	C
After completion of After Landing Procedure	AFTER LANDING PROCEDURE COMPLETE	F/O
When APU GEN OFF BUS (blue) illuminated	APU AVAILABLE	FO
Before entering the parking gate (stand) or verify VGSD	Stand ___ / Boeing 737 identified	FO
Turning into parking stand	TAXI LIGHTS OFF	C
After engines shutdown and pushback completed	SHUTDOWN PROCEDURE	C
After completion of Shutdown Procedure	SHUTDOWN PROCEDURE COMPLETE	C
Call to do Shutdown Checklist	SHUTDOWN CHECKLIST	C
After completion of Shutdown Checklist	SHUTDOWN CHECKLIST COMPLETE	F/O
After passenger disembarkation completed and whenever the flight crew has finished flight before transferring airplane to another crew or maintenance personnel	SECURE PROCEDURE	C
Call to do Secure Checklist	SECURE CHECKLIST	C
After completion of Secure Checklist	SECURE CHECKLIST COMPLETE	FO

2.8.6. Non-Normal Operation Callouts

Condition	Callout	Crew member
Speed below V2 after liftoff Flaps retract command below maneuver speed Flaps extension command when the speed is above the flap limit speed Speed too high or too low. +10 / -5 with increasing trend	CHECK SPEED	PM
Rate of descent more than 1000 ft/min below 1,000 ft, or briefed value plus 200 ft/min	SINK RATE	PM
Altitude deviation 100 ft or more	CHECK ALTITUDE	PM
Pitch too high or too low Excessive pitch up moment during liftoff More than 8° attitudes before touch down Unusual pitch up moment after touch down	CHECK PITCH	PM
Bank angle more than 5° during approach <400’ Bank angle more than 30° for other phases of flight	CHECK BANK	PM
Heading not correct or not within 5° Heading deviation	CHECK HEADING	PM
If difference more 1/2 dot for VOR Radial or 5° for ADF bearing	CHECK COURSE	PM
Glideslope deviation 1/2 dot or more	CHECK GLIDE SLOPE	PM
Localizer deviation 1/2 dot or more	CHECK LOCALIZER	PM
Localizer failure (ground station or aircraft receiver)	NO LOCALIZER	PM
Localizer does not capture	NO LOCALIZER CAPTURE	PM
Glideslope failure (ground station or aircraft receiver)	NO GLIDESLOPE	PM
Glideslope does not capture	NO GLIDESLOPE CAPTURE	PM
FLARE not indicated on FMA at 40 ft RA	NO FLARE	PM
RETARD not indicated on FMA at 25 ft RA	NO RETARD	PM
No speedbrake extension during rejected takeoff No speedbrake extension after landing	SPEEDBRAKES NOT UP	PM
Any time Autobrake disengage	AUTOBRAKE DISARM	PM

Condition	Callout	Crew member
If there is no REV indication(s) or the indication(s) stays amber	NO REVERSER ENGINE №1, or NO REVERSER ENGINE №2, or NO REVERSERS	PM
At minimums callout - If no response from PF	“I HAVE CONTROL _____” (state intentions)	PM
Approach to Stall or Stall Recovery	STALL	PF
GPWS Warning	PULL UP, MAXIMUM THRUST	PF
Traffic Avoidance	TCAS	PF
Traffic Avoidance for a climb RA in landing configuration	TCAS GO AROUND	PF
Upset Recovery	UPSET RECOVERY	PF
Windshear Warning	WINDSHEAR, MAXIMUM THRUST	PF
Malfunction occurs before V1, for which the Captain does not intend to reject	GO	C
If the decision is to reject the takeoff	REJECT (I HAVE CONTROL)	C
When NNC memory items is needed	(NNC title) MEMEORY ITEMS	PF
To select engine out cruze page for driftdown procedure	SELECT ENGINE OUT CRUISE PAGE	PF
Anounce Max. Alt and Speed for driftdown procedure	MAX altitude ___, speed___	PM
First who notices alert or decompression	OXYGEN MASK ON	ANY
If cabin altitude controllable	Cabin Altitude Controllable	PM
If cabin altitude is **uncontrollable**	Cabin Altitude Uncontrollable	PM
2000 ft above level off altitude during emergency descend	2000 TO LEVEL OFF	PM
1000 ft above level off altitude during emergency descend	1000 TO LEVEL OFF	PM

СТРАНИЦА НАМЕРЕНО ОСТАВЛЕНА ПУСТОЙ

2.9. CHECKLIST OPERATION

2.9.1. General

Normal checklists are used after doing all respective procedural items.

The following table shows which pilot calls for the checklist and which pilot reads the checklist. **Both pilots** visually **verify** that each item is in the needed configuration or that the step is done. The far-right column shows which pilot gives the response. This is different than the normal procedures where the far-right column can show which pilot does the step.

For the checklist items identified as "Both", PF in flight and Captain on ground replies first.

Checklist	Call	Read	Verify	Respond
PREFLIGHT	Captain	First Officer	Both	Area of responsibility
BEFORE START	Captain	First Officer	Both	Area of responsibility
BEFORE TAXI	Captain	First Officer	Both	Area of responsibility
BEFORE TAKEOFF	Captain	First Officer	Both	Pilot Flying
AFTER TAKEOFF	Pilot Flying	Pilot Monitoring	Both	Pilot monitoring
DESCENT	Pilot Flying	Pilot Monitoring	Both	Area of responsibility
APPROACH	Pilot Flying	Pilot Monitoring	Both	Area of responsibility
LANDING	Pilot Flying	Pilot Monitoring	Both	Pilot Flying
SHUTDOWN	Captain	First Officer	Both	Area of responsibility
SECURE	Captain	First Officer	Both	Area of responsibility

If the airplane configuration does not agree with the needed configuration:

- stop the checklist
- complete the respective procedure steps
- continue the checklist

If it becomes apparent that an entire procedure was not done:

- stop the checklist;
- complete the entire procedure;
- do the checklist from the start.

Try to do checklists before or after high work load times. The crew may need to stop a checklist for a short time to do other tasks. If the interruption is short, continue the checklist with the next step. If a pilot is not sure where the checklist was stopped, do the checklist from the start. If the checklist is stopped for a long time, also do the checklist from the start.

After completion of each checklist, the pilot reading the checklist calls: **"____ CHECKLIST COMPLETE".**

2.9.2. Normal Checklist

PREFLIGHT	
Gear pins & Cover.Removed, Onboard	C
Oxygen .... Tested, 100%	All
Navigation transfer and display switches.. NORMAL, AUTO	FO
Window heat .... ON	FO
Pressurization mode selector .....AUTO	FO
Flight instrument … HDG__Altimeter__	B
Speed brake ....DOWN	C
Parking brake ....Set	C
Engine start levers ....CUTOFF	C

BEFORE START	
Flight deck door .... Closed & locked	FO
Fuel .... ___ LBS or KGS, Pumps ON	FO
Passenger signs .... ON	FO
Hydraulic panel .... SET	FO
Doors .... Closed	FO
Windows .... Locked	B
MCP .... V2__ HDG__ ALT__	C
Takeoff speeds .... V1__ VR__ V2__	B
CDU preflight .... Completed	C
Rudder & aileron trim .... Free & 0	C
Transponder ....____	FO
Taxi & takeoff briefing .... Completed	C
Anti-collision light .... ON	FO

BEFORE TAXI	
Generators .... On	FO
Prob heat .... ON	FO
Anti-ice ....___	FO
Packs .... AUTO	FO
Isolation valve ....AUTO	FO
Engine start switches ....CONT	FO
Recall .... Checked	C
Autobrake .... RTO	FO
Engine start levers .... IDLE detent	C
Flight controls .... Checked	C
Ground equipment .... Clear	B

BEFORE TAKE-OFF	
Recall .... Checked	C
Flaps ....___, Green light	C
Stabilizer trim ....___Units	C

AFTER TAKE-OFF	
Engine bleeds .... ON	PM
Packs .... AUTO	PM
Landing gear .... UP & OFF	PM
Flaps .... UP, No lights	PM

DESCENT	
Pressurization ....LAND ALT___	PM
Recall .... Checked	PF
Autobrake .... ___	PF
Landing data … Vref__, Minimums___	B
Approach briefing .... Completed	PF

APPROACH	
Altimeters ....QNH___ Set	B

LANDING	
Engine start switches.................CONT	PF
Speedbrake .... ARMED	PF
Landing gear .... Down, Green light	PF
Flaps .... ___Green light	PF

SHUTDOWN	
Fuel pumps .... OFF	FO
Prob heat ....___	FO
ANTI-ICE....................................... OFF	FO
Hydraulic panel .... Set	FO
Anti-collision light .... OFF	FO
Engine start switches ....___	FO
Auto brake .... OFF	FO
Flaps .... ___	FO
Parking brake ....___	C
Engine start levers .... CUTOFF	C
Weather radar .... Off	B
Transponder .... STBY	FO

SECURE	
IRSs .... OFF	FO
Emergency exit lights .... OFF	FO
Window heat ....OFF	FO
Packs .... OFF	FO
APU/External power .... OFF	FO
Wait for 2 min after APU SHUTDOWN	
Battery .... OFF	FO

Takeoff briefing	Approach briefing
Aircraft Data, Flight №, NOTAM, MEL, Weather, ALTN, Engine Start, Taxi, RW, SID, NADP, Airborne FREQ, NAV RAD, WXR/TERR disp, RTO, EOSID, EO ACCEL, Emergencies.	**Airport, NOTAM, MEL, Weather,STAR, Approach Procedure, GA procedure, Stabilized APP requirements, Lights system, WXR/TERR disp, LW, Flaps, APPspd, N1, A/BRK position, Taxi, ALTN.**

Cruise briefing
Actions in case of engine failure; Actions in case of emergency descent; MORA on the route over 10,000 feet; Weather at alternate aerodromes along the route; Escape route into RTE 2 (if applicable).

RC Description	ESF	Meas. FC	Norm. FC	RCC	Max. X-Wind (kts / m/sec)	
					T/O	LDG
Dry	---	≥ 0,51	≥ 0.50	**6**	34/17.5	34/17.5
Frost; Wet (includes damp and 3mm depth or less of water) **3mm depth or less**: Slush; Dry Snow; Wet Snow	**Good**	0,40-0,50	0,42-0,49	**5**	25/12.9	34/17.5
-15°C and colder: Compacted Snow	**Good to Medium**	0,36-0,39	0,40-0,41	**4**	22/11.3	34/17.5
Slippery when wet (wet runway); Dry or Wet Snow (any depth) over Compacted Snow **Greater then 3mm depth:** Dry Snow; Wet Snow **Warmer than -15°C:** Compacted snow	**Medium**	0,30-0,35	0,37-0,39	**3**	20/10.3	25/12.9
Greater then 3mm depth: Water; Slush	**Medium to Poor**	0,26-0,29	0,35-0,36	**2**	15/7.7	17/8.7
Ice	**Poor**	0,17-0,25	0,30-0,34	**1**	13/6.7	15/7.7
Wet Ice; Water on top of Compacted Snow; Dry Snow or Wet Snow over Ice	**Unreliable**	<0,17	<0,30	**0**	**---**	**---**

Reduce crosswind guidelines by 5 knots on wet or contaminated runways whenever asymmetric thrust is used.

A/C 15 kt Maximum Tail Wind Component:
RA-73313, RA-73316, RA-73317, RA-73318

2.9.3. Expanded Normal Checklist

PREFLIGHT		
Gear pins & Covers....................................	**Removed, Onboard**	**Captain**
Captain performing exterior preflight, ensures pins are removed and onboard the aircraft.		
Oxygen	**Tested, 100%**	**ALL**
All crew members verify that oxygen pressure drop test was performed and switch is set to 100%.		
Navigation transfer and display switches...........................	**NORMAL, AUTO**	**F/O**
Both verify VHF NAV & IRS transfer switches; FMC source select switch are in NORMAL position. SOURCE selector in AUTO position. CONTROL PANEL select switch is in NORMAL.		
Window Heat	**ON**	**F/O**
Both verify OVERHEAT lights are extinguished. Verifies that the ON lights are illuminated or OFF lights are extinguished (except at high ambient temperatures.)		
Pressurization Mode Selector	**AUTO**	**F/O**
Both verify that the ALTN and MANUAL lights are extinguished.		
Flight instruments	**HDG__ Altimeter__**	**Both**
Both verify that headings agree. QNH is set on main and standby altimeters, altitude readings correspond with aerodrome elevation.		
Speed brake	**DOWN**	**Captain**
Both verify speed brake lever DOWN detent. Verifies SPEED BRAKE ARMED, SPEED BRAKE DO NOT ARM and SPEEDBRAKES EXTENDED lights are extinguished.		
Parking brake	**Set**	**Captain**
Both verify parking brake set, PARKING BRAKE light ON and pressure normal.		
Engine Start Levers	**CUTOFF**	**Captain**
Both verify Engine Start Levers in CUTOFF position and ENG VALVE and SPAR VALVE are closed.		
BEFORE START		
Flight deck door	**Closed & Locked**	**F/O**
Both verify that Flight Deck Door closed and locked.		
Fuel	**__LBS or KG, Pumps ON**	**F/O**
Both verify Upper Display Unit fuel and CFP fuel quantities agree, distribution is within limits and fuel pumps are ON in accordance with fuel quantity and distribution.		
Passenger signs	**ON**	**F/O**
Both verify that Passenger Signs switch is in ON		
Hydraulic Panel...	**SET**	**F/O**
If pushback is not needed, or if pushback is needed and the nose gear steering lockout pin is installed: verify electric pump LOW PRESSURE lights are extinguished and system A and B pressures are 2800 psi. minimum.		
Doors	**Closed**	**F/O**
Both verify that the exterior doors are Closed.		
Windows..	**Locked**	**Both**
Both verify the lock lever is in the forward, locked position.		
MCP	**V2___, HDG___, ALT___**	**Captain**
Both verify that V2, initial heading and first cleared altitude are set.		

Takeoff speeds (CDU)	**V1__ VR__ V2__**	**Both**
Both pilots verify reference bugs on PFD and CDU TAKE OFF REF page speeds are set for calculated V1, VR and V2		
CDU preflight	**Completed**	**Captain**
Both verify that CDU preflight procedure has been completed.		
Rudder & Aileron Trim	**Free & 0**	**Captain**
Both verify that aileron and rudder trims set to zero.		
Transponder...	____	**F/O**
Both verify that the Squawk code is entered and at airports equipped to track airplanes on the ground proper transponder setting, but not a TCAS mode is selected.		
Taxi and takeoff briefing	**Completed**	**Captain**
Both verify that Pilot Flying has performed taxi and takeoff briefing.		
Anti-collision light	**ON**	**F/O**
Both verify that Anti-collision light switch is ON.		
BEFORE TAXI		
Generators	**On**	**F/O**
Both verify that the Engines Generators are on buses and GEN OFF BUS lights are extinguished.		
Prob Heat...	**ON**	**F/O**
Both verify that the PROB HEAT switches are ON.		
Anti-ice	____	**F/O**
Both verify the Engine Anti-Ice and Wing Anti-Ice switches are OFF/ON as required.		
Packs	**AUTO**	**F/O**
Both verify that PACKs control switches are set to AUTO.		
Isolation Valve	**AUTO**	**F/O**
Both verify that the Isolation Valve switch is AUTO		
Engine Start Switches	**CONT**	**F/O**
Both verify that Engine Start Switches are in CONT position.		
Recall	**Checked**	**Both**
Both check the appropriate system annunciator(s) and MASTER CAUTION lights illuminate.		
Autobrake	**RTO**	**F/O**
Both verify that the Autobrake Selector is set to RTO, and the AUTO BRAKE DISARM light is extinguished.		
Engine Start Levers	**IDLE detent**	**Captain**
Both verify that Engine Start Levers are in IDLE DETENT position.		
Flight controls	**Checked**	**Captain**
Both verify that flight controls check has been performed.		
Ground equipment	**Clear**	**Both**
Both verify that ground equipment is clear. Visual signal received from ground staff.		
BEFORE TAKEOFF		
Recall	**Checked**	**Captain**
Both check the appropriate system annunciator(s) and MASTER CAUTION lights illuminate.		
Flaps	**____, Green light**	**Captain**

Both verify that Flaps are in takeoff position. LE FLAP EXT light illuminated.		
Stabilizer trim	**__Units**	**Captain**
Both compare with the FMC CDU and verify stabilizer trim is set for takeoff.		
Before entering the departure runway, set the transponder mode selector to TA/RA		
AFTER TAKEOFF		
Engine bleeds	**ON**	**PM**
PF and PM ensure that ENG BLEED switches are in ON position		
Packs	**AUTO**	**PM**
PF and PM ensure that both PACK control switches in AUTO position.		
Landing gear	**UP & OFF**	**PM**
PF and PM verify that Landing gear are UP & OFF.		
Flaps	**UP, No lights**	**PM**
PF and PM verify that Flaps are UP.		
When climbing above transition altitude, set and crosscheck the altimeters to standard		
DESCENT		
Pressurization	**LAND ALT___**	**PM**
PF and PM verify that pressurization is set to landing altitude.		
Recall	**Checked**	**PF**
PF and PM check the appropriate system annunciator(s) and MASTER CAUTION lights illuminate.		
Autobrake	**____**	**PM**
PF and PM verify that Autobrake is set to desired position.		
Landing data	**Vref__, Minimums__**	**Both**
PF then PM verify and call VREF and minimums for expected approach.		
Approach briefing	**Completed**	**PF**
PF and PM verify that approach briefing has been completed.		
APPROACH		
Altimeters	**QNH___, set**	**PM**
When descending below the transition level, set and crosscheck the altimeters. PF then PM verify and call actual QNH on altimeters.		
LANDING		
Engine Start Switches	**CONT**	**PF**
PF and PM verify that Engine Start Switches are in CONT position.		
Speedbrake	**ARMED**	**PF**
PF and PM verify that Speedbrake Lever is in armed position and SPEED BRAKE ARMED light illuminated.		
Landing gear	**Down, Green light**	**PF**
PF and PM verify Landing Gear Lever in DOWN position and all green lights are ON		
Flaps	**__ Green light**	**PF**
PF and PM read flap setting from Flap Position Indicator and verifies Flap Handle in proper detent. LE FLAP EXT light illuminated		

SHUTDOWN		
Fuel pumps	**OFF**	**F/O**
Both ensure that all fuel pumps are OFF. If fuel is loaded in the center tank, position the left center tank fuel pump switch ON to prevent a fuel imbalance before takeoff.		
Prob. Heat	____	**F/O**
Both verify that the PROB HEAT switches are set to OFF or AUTO position.		
Anti-ice	**OFF**	**F/O**
Both ensure that ANTI ICE switches are OFF.		
Hydraulic panel	**Set**	**F/O**
Both ensure that ELEC hydraulic pumps are OFF and ENG driven pumps are ON.		
Anti-collision light	**OFF**	**F/O**
Both ensure that Anti-collision light switch is OFF.		
Engine Start Switches	____	**F/O**
Both verify that Engine Start Switches are in OFF or AUTO position		
Autobrake..	**OFF**	**F/O**
Both check the Autobrake Selector to OFF.		
Flaps	____	**F/O**
Both check flap setting is appropriate for the current conditions from Flap Position Indicator and ensures that Flap Handle in the same position.		
Parking brake	____	**Captain**
Both verify Parking Brake is SET and PARK BRAKE light is ON, or RELEASED if chocks are installed.		
Engine Start Levers...................................	**CUTOFF**	**Captain**
Both verify Engine Start Levers in CUT OFF position and ENG VALVE and SPAR VALVE are closed.		
Weather radar	**Off**	**Both**
Both verify that Weather Radar display is OFF on ND and that TEST mode is selected on Weather Radar Control Panel.		
Transponder	**STBY**	**F/O**
Both ensure that squawk code set as needed and transponder mode selector in STBY.		
SECURE		
IRSs...	**OFF**	**F/O**
Both ensure that IRS selectors are in OFF position.		
Emergency exit lights	**OFF**	**F/O**
Both verify that Emergency Lights Switch is OFF.		
Window Heat	**OFF**	**F/O**
Both verify that the ON lights are extinguished or OFF lights are illuminates		
Packs	**OFF**	**F/O**
Both verify that PACK Switches are OFF.		
If no qualified personal stays onboard:		
APU/External power	**OFF**	**F/O**
Both ensure that APU is OFF and External Power switch is OFF.		
Wait for 2 min. after APU SHUTDOWN		
Battery	**OFF**	**F/O**
Both ensure that Battery switch is OFF.		

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

2.10. BRIEFINGS

2.10.1. Preflight Briefing

Preflight briefing normally begins in special briefing rooms and it should be finished prior to an engine start. Topics are to be discussed in accordance with OM Part A.

2.10.2. Briefing with the Cabin Crew

On first flight with a new cabin crew or whenever a cabin crew change occurs, the Captain must brief the Purser.

The following topics are to be discussed, not limiting to:

- Logbook discrepancies that may affect flight attendant responsibilities or passenger comfort (e.g., coffee maker inoperative, broken seat backs, manual pressurization, etc.). Weather affecting the flight (e.g., turbulence, thunderstorms, weather near minimums, etc.). Provide the time after airborne when the weather may be encountered.
- Delays, unusual operations, non-routine operations (e.g., maintenance delays, ATC delays, re-routes, etc.).
- Possible supplementary procedures, which could be unusual to passengers as they could see them, for example, the deicing procedure.
- Shorter than normal (less than 10 minutes) taxi time or flight time which may affect preflight announcements or cabin service.
- Any other items that may affect the flight operation or in-flight service such as catering, fuel stops, armed guards, etc.
- Flight deck entrance procedures and communication between the flight deck and cabin.
- Actions in case of hijacking or unruled person.
- The importance of informing the flight crew about any non-standard situation in the cabin.
- A review of the sterile flight deck policy, responsibility for PA announcements when the Fasten Seat Belt sign is turned ON during cruise, emergency evacuation commands, or any other items appropriate to the flight.
- Anything that the flight crew or the cabin crews need to discuss related to the flight (for example: special cargo, flight crew meals etc.).

During the briefing, the PIC must ensure that the required information has been received and understood.

The Captain should inform the cabin crew if a circumstance has appeared in flight that could affect their service or that changes the already briefed plans.

2.10.3. Briefing for a Jumpseat Rider

When accepting a jumpseat rider (other than Boeing-737 NG rated pilot), the Captain is responsible to ensure the following items are briefed and understood by a rider:

- Use of seat-belts (including shoulder harness for take-off and landing and seat back all the way up on first observer seat).

- Using of interphone operations.
- How to stow the jumpseat in the event of an emergency evacuation.
- How to operate sliding windows and cockpit door emergency release handle.
- How to operate the oxygen mask.
- Sterile cockpit regulations.

For more detail, see JUMPSEAT PASSENGER SAFETY BRIEFING OM part A.

2.10.4. Takeoff Briefing

The takeoff briefing should be accomplished as soon as practical so it does not interfere with the final takeoff preparations.

The takeoff briefing is a description of the departure flight path with emphasis on anticipated track and altitude restrictions. It assumes normal operating procedures are used. Therefore, it is not necessary to brief normal or standard takeoff procedures. Additional briefing items may be required when any elements of the takeoff and/or departure are different from those routinely used.

Topics are to be discussed in accordance with OM Part A are presented in brief at the backside of the Normal Checklist and at the end of this chapter. It is strongly recommended to follow the topics order for standardization. Avoid interrupting the briefing. If it is not possible, clearly announce: **“STOP BRIEFING”,** remember the part on which it was interrupted, and then continue from this part, or, if the interruption has taken a long time, from the beginning of the part or even start the briefing from a very beginning.

Taxi briefing should include special procedures that will be used during taxi (e.g., delayed flap extension, anti-ice usage, etc.). Pay special attention to hot spots, if such are defined on taxi diagram.

On first flight with a new flight crew, review rejected take-off procedures, verbally and by touching the controls.

Engine-Failure intentions must always be briefed. Both pilots must clearly understand where the airplane should be flown and landed in case of engine failure on takeoff.

As part of the takeoff briefing for the first flight of the day and following a change of either flight crew member, cabin altitude warning indications and memory item procedures must be briefed on airplanes in which the CABIN ALTITUDE and TAKEOFF CONFIG lights are not installed, or are installed but not activated. The briefing must contain the following information:

Whenever the intermittent warning horn sounds in flight at an airplane flight altitude above 10,000 feet MSL:

- Immediately, don oxygen masks and set regulators to 100%.
- Establish crew communications.
- Do the CABIN ALTITUDE WARNING or Rapid Depressurization non-normal checklist.

Both pilots must verify on the overhead Cabin Altitude Panel that the cabin altitude is stabilized at or below 10,000 feet before removing oxygen masks.

Aircraft Data, Flight №, NOTAM, MEL, Weather, ALTN, Engine Start, Taxi, RW, SID, NADP, Airborne FREQ, NAV RAD, WXR/TERR disp, RTO, EOSID, EO ACCEL, Emergencies.
Aircraft Data – Check CDU IDENT page (SFP, ENG rating, FMC software version, NAV DATA validity date). **Flight №** - Number of current flight (cross check with CDU RTE page). **NOTAM** – Departure airport (Alt Dep) NOTAMs if any impact on current flight. **MEL** – Active AC MEL and special operations if any **Weather** – Current airport weather and associated procedures: Anti/De-Ice, use of Eng/Wing Anti-Ice, Eng. Static Runup, Windshear precautions, LVO **ALTN** - Choose alternate for takeoff if required by weather or other conditions **Engine Start** – SP procedures if any. **Taxi** – Expected Taxi route, hot spots on route, any restrictions. **RW** – Excepted RW for takeoff, RW condition, possible intersection takeoff **SID** – Expected or assigned SID, effective or issue date, requirements (ex. GNSS, DME/DME…), RNP/ANP, first alt, MSA, Transition Alt., route verification, chart remarks if any. **NADP** – according airport requirements if no special requirements according crew decision: Thrust reduction and acceleration alt. **Airborne FREQ** – Excepted or assigned airborne frequency. **NAV RAD** – required NAV radio frequencies (ILS, GLS, VOR, NDB…) and COURSE if requires. **WXR/TERR disp** – expected use of weather and terrain display according current airport weather and terrain situation. **RTO** – Rejected Takeoff maneuver, brake cooling schedule, sequence of action after RTO. **EOSID** – Route and crew action in case of engine failure after V1, Engine out acceleration altitude, holding areas that might be used for evaluating emergency situation. **Emergencies** – actions in case of immediate landing after takeoff, special airport procedures if any.

2.10.5. Cruise Briefing

When aircraft reached cruise FL, PF shall conduct CRUISE BRIEFING following cruise flight level.

The following topics are to be discussed, not limiting to:

- Actions in case of engine failure;
- Actions in case of emergency descent;
- MORA on the route over 10,000 feet;
- Weather at alternate aerodromes along the route
- Escape route into RTE 2 (if applicable)

2.10.6. Approach Briefing

Before the start of an instrument approach, the PF should brief the PM of his intentions in conducting the approach. Both pilots should review the approach procedure. All pertinent approach information, including minimums and missed approach procedures, should be reviewed and alternate courses of action considered. Topics are to be discussed in accordance with OM Part A are presented in brief at the backside of the Normal Checklist

Topics are to be discussed in accordance with OM Part A. These also may include:

- type of approach and the validity of the charts to be used;
- approach procedure including courses and heading;
- vertical profile including all minimum altitudes, crossing altitudes and approach minimums;
- speed restrictions;
- determination of the Missed Approach Point (MAP) and the missed approach procedure;
- other related crew actions such as tuning of radios, setting of course information, or other special requirements;
- any appropriate information related to a non-normal procedure, including non-normal configuration landing distance required versus landing distance available;
- management of AFDS.

Airport, NOTAM, MEL, Weather STAR, Approach Procedure, GA procedure Stabilized APP requirements, Lights system, WXR/TERR disp, LW, Flaps, APPspd, N1 A/BRK position, Taxi, ALTN.
Airport – Destination or alt. airport. **NOTAM** – Arrival airport and alt. airports NOTAMs if any impact on current flight **MEL** – Active AC MEL and special operations if any **Weather** – Destination and alternate airports weather and associated procedures: use of Eng/Wing Anti-Ice, LVO **STAR** – Expected arrival procedure (and expected RW if applicable), effective or issue date, AD elevation, requirements (ex. GNSS, DME/DME…), RNP/ANP, MSA, TL, route verification, chart remarks if any. **Approach Procedure** – Expected RW, type of approach, NAV radio FREQ (if applicable), Final App Course, route verification, FAF(P) alt and distance form RW, GS angle, chart remarks (if any), MDA(H)/DA(H) to be used (temperature correction if needed), alternate type of approach. **GA procedure -** GA actions, GA route according approach chart (route verification) or non-standard go-around route, GA altitude. **Stabilized APP requirements** – Minimum alt to be stabilized, expected deviations from requirements (if any). **Lights system** – If required: Approach light system, RW lights, TDZ lights, PAPI

WXR/TERR disp - expected use of weather and terrain display according current airport weather and terrain situation

LW – Expected Landing Weight,

Flaps – Landing flaps configuration to be use during landing.

App spd – Vref and Vapp speeds calculated for landing

N1 - Required N1 for current weight, flaps and environmental conditions.

A/BRK position - Autobrake position for Calculated factored landing distance for current RW conditions, REV usage (expected adverse weather).

Taxi - Expected Taxi route, hot spots on route any restrictions.

ALTN – Planned alternate, fuel and time for holding. Minimum and emergency expected fuel level.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

2.11. PROCEDURES GENERAL

2.11.1. Engine Start Procedure

Normal starter duty cycle:

Multiple consecutive start attempts are permitted. Each start attempt is limited to 2 minutes of starter usage.

A minimum of 10 seconds is needed between start attempts.

Extended engine motorings:

Starter usage is limited to 15 minutes for the first two extended engine motorings. A minimum of 2 minutes is needed between each attempt.

For the third and subsequent extended engine motorings, starter usage is limited to 5 minutes. A minimum of 10 minutes is needed between each attempt.

Normal engine start considerations:

- do not move an engine start lever to IDLE detent early or a hot start can occur;
- keep a hand on the engine start lever while monitoring RPM, EGT and fuel flow until stable;
- if fuel is shutoff accidentally (by closing the engine start lever) do not reopen the engine start lever in an attempt to restart the engine;
- failure of the ENGINE START switch to stay in GRD until the starter cutout RPM can cause a hot start. Do not re–engage the ENGINE START switch until engine RPM is below 20% N2.

If a fluid leak (other than a continuous stream) from any of the engine drains is discovered during the Exterior Inspection, the engine can be started. If during engine start, the ground crew reports a fluid leak from an engine drain, the engine start may be continued.

If the fluid leak continues after the engine is stable at idle, do one of the following:

- shut down the engine for maintenance action, or
- run the engine at idle thrust for up to 5 minutes. If the fluid leak stops during this time, no maintenance action is needed, or
- shut down and restart the engine. Run the engine at idle thrust for up to 5 minutes. If the fluid leak stops during this time, no maintenance action is needed.

For the first flight of the day, at airport elevations at or above 2,000 feet MSL, if the temperature is below 5°C, consider placing the Ignition select switch to BOTH before starting the engines. This may increase the likelihood of a successful engine start on the first attempt.

Do the ABORTED ENGINE START checklist for one or more of the following abort start conditions:

- the N1 or N2 does not increase or increases very slowly after the EGT increases;
- there is no oil pressure indication by the time that the engine is stable at idle;
- the EGT does not increase by 15 seconds after the engine start lever is moved to IDLE detent;
- the EGT quickly nears or exceeds the start limit.

Starting Windmilling Engines

A tailwind can cause the low pressure (N1) rotor to rotate in a reverse direction. During an engine start with the low pressure rotor windmilling in a reverse direction, the rotor will decelerate, stop, and then start rotating in the normal direction. Under these conditions the start will be slower than normal, with a higher than normal EGT as the tailwind velocity increases.

When the low pressure rotor is rotating in the reverse direction:

- With 25 knots or less of tailwind – Use normal starting procedures.
- With over 25 knots of tailwind – Turn the airplane into the wind.

2.11.2. Noise Abatement Procedure

ICAO Doc.8168 specifies two noise abatement procedures for takeoff.

NADP 1 (Noise Abatement Departure Procedure 1)

This procedure involves a power reduction at or above the prescribed minimum altitude and delaying flap/slat retraction until the prescribed maximum altitude is attained.

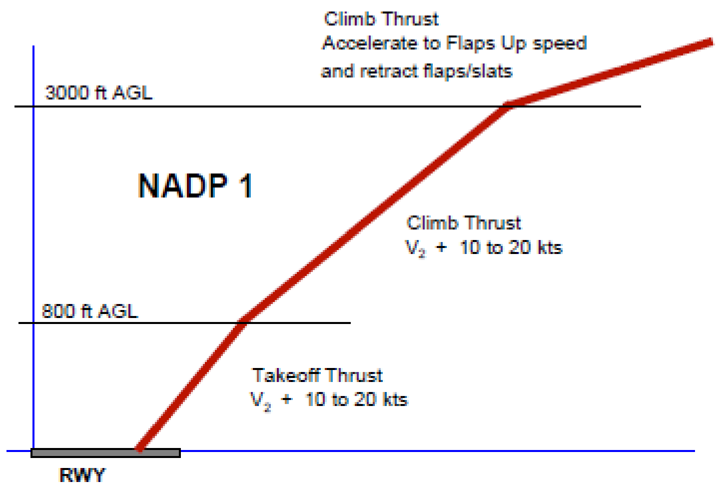
At the prescribed maximum altitude, accelerate and retract flaps/slats on schedule while maintaining a positive rate of climb and complete the transition to normal en-route climb speed.

The noise abatement procedure is not to be initiated at less than 800 feet AAL. The initial climbing speed to the noise abatement initiation point shall not be less than V2 + 10 knots.

On reaching an altitude at or above 800 feet AAL, adjust and maintain engine thrust in accordance with the noise abatement thrust schedule provided in the aircraft operating manual. Maintain a climb speed of V2 + 10 to 20 knots with flaps and slats in the take-off configuration.

At no more than an altitude equivalent to 3000 feet AAL, while maintaining a positive rate of climb, accelerate and retract flaps/slats on schedule.

At 3000 feet AAL, accelerate to normal en-route climb speed.



NADP 2 (Noise Abatement Departure Procedure 2)

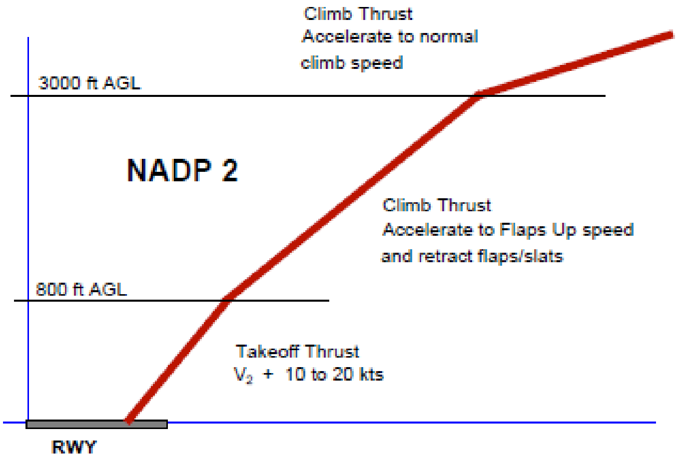
This procedure involves initiation of flap/slat retraction on reaching the minimum prescribed altitude. The flaps/slats are to be retracted on schedule while maintaining a positive rate of climb. The thrust reduction is to be performed with the initiation of the first flap/slat retraction or when the zero flap/slat configuration is attained. At the prescribed altitude, complete the transition to normal en-route climb procedures.

The noise abatement procedure is not to be initiated at less than 800 feet AAL. The initial climbing speed to the noise abatement initiation point is V2 + 10 to 20 knots.

On reaching an altitude equivalent to at least 800 feet AAL, decrease aircraft body angle whilst maintaining a positive rate of climb, accelerate towards Flaps Up speed and reduce thrust with the initiation of the first flaps/slats retraction or reduce thrust after flaps/slats retraction.

Maintain a positive rate of climb and accelerate to and maintain a climb speed equal to Flaps Up speed + 10 to 20 knots till 3000 feet AAL.

At 3000 feet AAL, accelerate to normal en-route climb speed.



Noise abatement shall not be determining factor in runway nomination under the following circumstances:

A. If the runway surface conditions are adversely affected (e.g. by snow, slush, ice, water, or other substances);

B. When the crosswind component, including gusts, exceeds 28 km/h (15 kt), or the tailwind component, including gusts, exceeds 9 km/h (5 kt).

C. For landing in conditions:

- when the ceiling is lower than 150 m (500 ft) above aerodrome elevation or the visibility is less than (1900 m); or,
- when the approach requires vertical minima greater than 100 m (300 ft) above aerodrome elevation and:
  - the ceiling is lower than 240 m (800 ft) above aerodrome elevation; or
  - the visibility is less than 3000 m;

D. For take-off when the visibility is less than 1900 m;

- when wind shear has been reported or forecast or when thunderstorms are expected to affect the approach or departure.

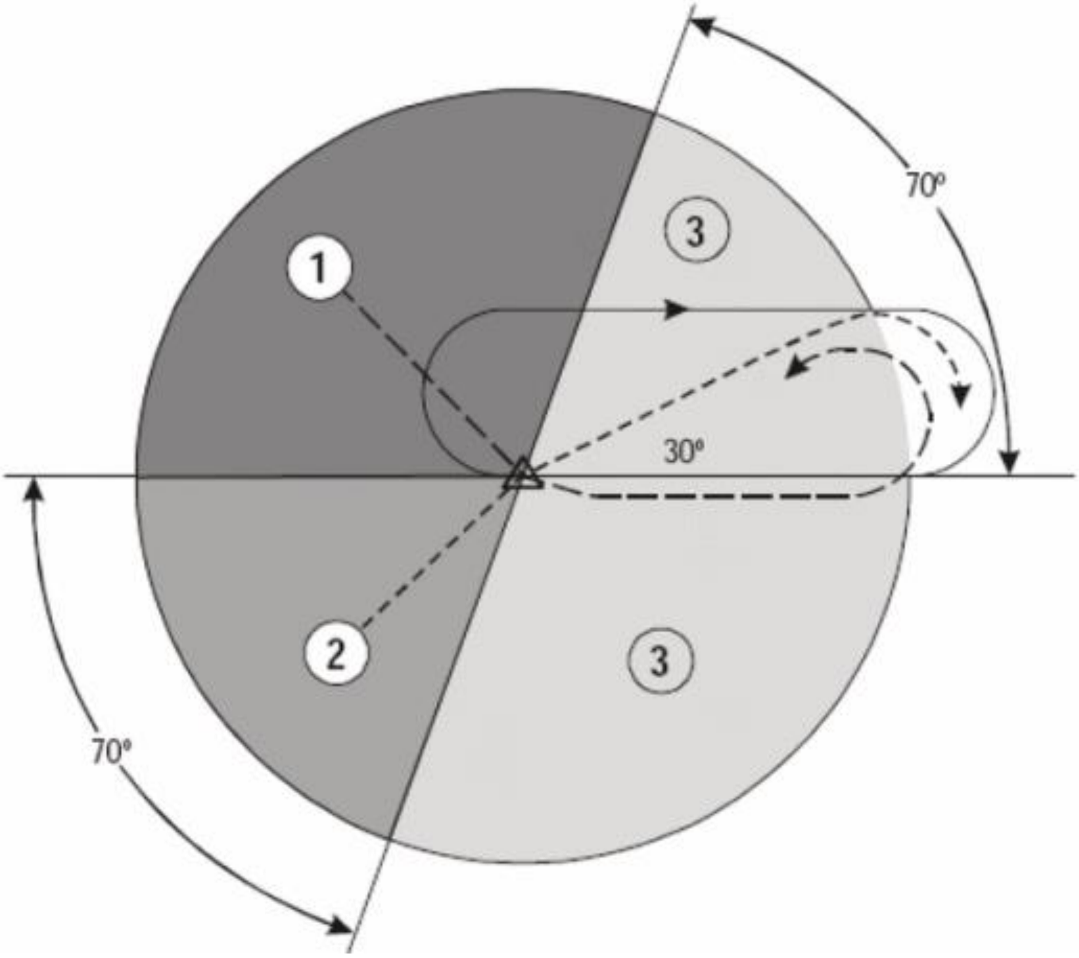
2.11.3. Holding

Start reducing to holding airspeed 3 minutes before arrival time at the holding fix so that the airplane crosses the fix, initially, at or below the maximum holding airspeed.

If the FMC holding speed is greater than the ICAO maximum holding speed, holding may be conducted at flaps 1, using flaps 1 maneuvering speed. Flaps 1 uses approximately 10% more fuel than flaps up. Holding speeds in the FMC provide an optimum holding speed based upon fuel burn and speed capability; but are never lower than flaps up maneuvering speed.

Maintain clean configuration if holding in icing conditions or in turbulence.

To enter the holding pattern use following schematic:



Sector 1 entry

Sector 1 procedure (parallel entry):

- at the fix, the aircraft is turned left onto an outbound heading for the appropriate period of time; then
- the aircraft is turned left onto the holding side to intercept the inbound track or to return to the fix; and then
- on second arrival over the holding fix, the aircraft is turned right to follow the holding pattern.

Sector 2 entry

Sector 2 procedure (offset entry):

- at the fix, the aircraft is turned onto a heading to make good a track making an angle of 30° from the reciprocal of the inbound track on the holding side; then
- the aircraft will fly outbound:
  - for the appropriate period of time (see “Time/distance outbound”), where timing is specified; or
  - until the appropriate limiting DME distance is reached, where distance is specified. If a limiting radial is also specified, then the outbound distance is determined either by limiting DME distance or the limiting radial, whichever comes first;
- the aircraft is turned right to intercept the inbound holding track; and
- on second arrival over the holding fix, the aircraft is turned right to follow the holding pattern.

Sector 3 entry

Sector 3 procedure (direct entry): Having reached the fix, the aircraft is turned right to follow the holding pattern

Time/distance outbound

The still air time for flying the outbound entry heading should not exceed:

- one minute if at or below 4250 m (14000 ft); or
- one and one-half minutes if above 4250 m (14000 ft).

Where DME is available, the length of the outbound leg may be specified in terms of distance instead of time.

ICAO Holding Airspeeds (Maximum)

Altitude	Speed Normal Conditions
Through 14,000 feet	230 knots
Above 14,000 to 20,000 feet MSL	240 knots
Above 20.000 to 34,000 feet MSL	265 knots
Above 34,000 feet MSL	0.82 Mach

Above FL250, use VREF 40+ 100 knots to provide adequate buffet margin.

2.11.4. Crosswind Landing Techniques

Three methods of performing crosswind landings are presented. They are the de-crab technique (with removal of crab in flare), touchdown in a crab, and the sideslip technique. Whenever a crab is maintained during a crosswind approach, offset the flight deck on the upwind side of centerline so that the main gear touches down in the center of the runway.

De-Crab During Flare

The objective of this technique is to maintain wings level throughout the approach, flare, and touchdown. On final approach, a crab angle is established with wings level to maintain the desired track. Just prior to touchdown while flaring the airplane, downwind rudder is applied to eliminate the crab and align the airplane with the runway centerline.

As rudder is applied, the upwind wing sweeps forward developing roll. Hold wings level with simultaneous application of aileron control into the wind. The touchdown is made with cross controls and both gears touching down simultaneously. Throughout the touchdown phase upwind aileron application is utilized to keep the wings level.

Touchdown in Crab

On dry RW, upon touchdown the airplane tracks toward the upwind edge of the RW while de-crabbing to align with the RW. Immediate upwind aileron is needed to ensure the wings remain level while rudder is needed to track the RW centerline. The greater the amount of crab at touchdown, the larger the lateral deviation from the point of touchdown. For this reason, touchdown in a crab only condition is not recommended when landing on a dry RW in strong crosswinds.

On very slippery runways, landing the airplane using crab only reduces drift toward the downwind side at touchdown, permits rapid operation of spoilers and autobrakes because all main gears touchdown simultaneously, and may reduce pilot workload since the airplane does not have to be de-crabbed before touchdown. However, proper rudder and upwind aileron must be applied after touchdown to ensure directional control is maintained.

Sideslip (Wing Low)

The sideslip crosswind technique aligns the airplane with the extended runway centerline so that main gear touchdown occurs on the runway centerline.

The initial phase of the approach to landing is flown using the crab method to correct for drift. Prior to the flare the airplane centerline is aligned on or parallel to the runway centerline. Downwind rudder is used to align the longitudinal axis to the desired track as aileron is used to lower the wing into the wind to prevent drift. A steady sideslip is established with opposite rudder and low wing into the wind to hold the desired course.

Touchdown is accomplished with the upwind wheels touching just before the downwind wheels. Overcontrolling the roll axis must be avoided because overbanking could cause the engine nacelle or outboard wing flap to contact the runway.

NOTE SIDESLIP ONLY (ZERO CRAB) LANDINGS ARE NOT RECOMMENDED WITH CROSSWIND COMPONENTS IN EXCESS OF 17 KNOTS AT FLAPS 15, 20 KNOTS AT FLAPS 30 AND 23 KNOTS AT FLAPS 40. THIS RECOMMENDATION ENSURES ADEQUATE GROUND CLEARANCE AND IS BASED ON MAINTAINING ADEQUATE CONTROL MARGIN.

Properly coordinated, this maneuver results in nearly fixed rudder and aileron control positions during the final phase of the approach, touchdown, and beginning of the landing roll. However, since turbulence is often associated with crosswinds, it is often difficult to maintain the cross control coordination through the final phase of the approach to touchdown.

If the crew elects to fly the sideslip to touchdown, it may be necessary to add a crab during strong crosswinds. Main gear touchdown is made with the upwind wing low and crab angle applied. As the upwind gear touches first, a slight increase in downwind rudder is applied to align the airplane with the runway centerline. At touchdown, increased application of upwind aileron should be applied to maintain wings level.

2.11.5. Flare and Touchdown

Unless an unexpected or sudden event occurs, such as windshear or collision avoidance situation, it is not appropriate to use sudden, violent or abrupt control inputs during landing. Begin with a stabilized approach on speed, in trim and on glide path.

NOTE When a manual landing is planned from an approach with the autopilot connected, the transition to manual flight should be planned early enough to allow the pilot time to establish airplane control before beginning the flare. The PF should consider disengaging the autopilot and disconnecting the autothrottle (if desired) 1 to 2 nm before the threshold, or approximately 300 to 600 feet above field elevation.

When the threshold passes out of sight under the airplane nose shift the visual sighting point to the far end of the runway. Shifting the visual sighting point assists in controlling the pitch attitude during the flare. Maintaining a constant airspeed and descent rate assists in determining the flare point. Initiate the flare when the main gear is approximately 20 feet above the runway by increasing pitch attitude approximately 2° - 3°. This slows the rate of descent.

After the flare is initiated, smoothly retard the thrust levers to idle, and make small pitch attitude adjustments to maintain the desired descent rate to the runway. A smooth thrust reduction to idle also assists in controlling the natural nose-down pitch change associated with thrust reduction. Hold sufficient back pressure on the control column to keep the pitch attitude constant. A touchdown attitude as depicted in the figure below is normal with an airspeed of approximately Vref. Ideally, main gear touchdown should occur simultaneously with thrust levers reaching idle.

Avoid rapid control column movements during the flare. If the flare is too abrupt and thrust is excessive near touchdown, the airplane tends to float in ground effect. Do not allow the airplane to float or attempt to hold it off. Fly the airplane onto the runway at the desired touchdown point and at the desired airspeed.

NOTE Do not trim during the flare. Trimming in the flare increases the possibility of a tail strike.

Prolonged flare increases airplane pitch attitude 2° to 3°. When prolonged flare is coupled with a misjudged height above the runway, a tail strike is possible. Do not prolong the flare in an attempt to achieve a perfectly smooth touchdown. A smooth touchdown is not the criterion for a safe landing.

Typically, the pitch attitude increases slightly during the actual landing, but avoid over-rotating. Do not increase the pitch attitude, trim, or hold the nose wheel off the runway after landing. This could lead to a tail strike.

Airspeed Control

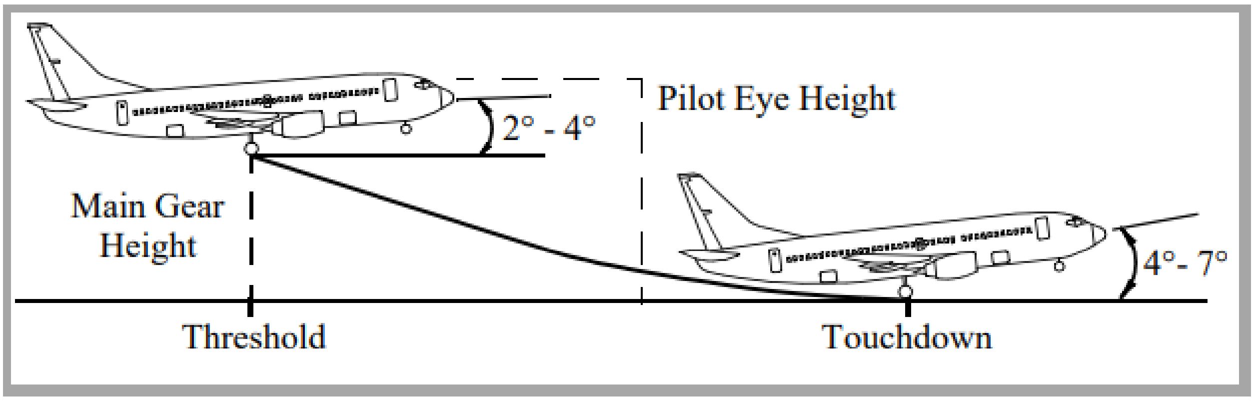
During an autoland, the autothrottle retards the thrust so as to reach idle at touchdown. The 5 knot additive is bled off during the flare. If the autothrottle is disconnected, or is planned to be disconnected prior to landing, maintain reference speed plus any wind additive until approaching the flare. Minimum command speed setting is VREF + 5 knots. With proper flare technique and thrust management the 5 knot additive and some of the steady wind additive may be bled off prior to touchdown. Plan to maintain gust correction until touchdown. Touchdown should occur at no less than VREF - 5 knots.

All approach speed additives should be accounted for when determining the landing distance by applying the Approach Speed Adjustment from the Normal or Non-Normal Landing Distance table.

Landing Flare Profile

The following diagrams use these conditions:

- 3° approach glide path;
- flare distance is approximately 1,000 to 2,000 feet beyond the threshold;
- typical landing flare times range from 4 to 8 seconds and are a function of approach speed;
- airplane body attitudes are based upon typical landing weights, flaps 30, V_{REF30} + 5 knots (approach) and V_{REF30} + 0 (touchdown), and should be reduced by 1° for each 5 knots above this speed.



NOTE Approaches at low gross weights can result in pitch attitudes that are lower than normal. This is especially noticeable when V_{REF} is limited for engine out controllability. As a result, the airplane could have a tendency to float unless it is flown onto the runway at the proper touchdown speed.

Bounced Landing Recovery

NOTE A bounced landing is defined as a landing where both main gears contact the ground and then both main gears leave the ground prior to landing.

If higher than idle thrust is maintained through initial touchdown, the automatic speedbrake deployment may be disabled even when the speedbrakes are armed. This can result in a bounced landing. During the resultant bounce, if the thrust levers are then retarded to idle, automatic speedbrake deployment can occur. This results in a loss of lift and nose up pitching moment which can result in a tail strike or hard landing on a subsequent touchdown.

If the airplane bounces during a landing attempt, hold or re-establish a normal landing attitude and add thrust as necessary to control the rate of descent. Thrust need not be added for a shallow bounce or skip. If a high, hard bounce occurs, initiate a go-around. Manually advance thrust levers to go-around thrust, and verify speedbrakes are retracted. Do not retract the flaps or landing gear until a positive rate of climb is established because a second touchdown may occur during the go-around.

NOTE When landing, the TO/GA switches are inhibited near touchdown and on the ground.

Rejected Landing / Balked Landing

Rejected Landing

A rejected landing is a discontinued landing attempt and go-around initiated at low altitude (below DA(H) or MDA(H)) but prior to touchdown. If a rejected landing becomes necessary, follow the Go-Around and Missed Approach procedure.

Balked Landing

A balked landing is a discontinued landing attempt and go-around initiated at or after touchdown, but prior to initiation of reverse thrust. The balked landing technique maintains landing flap configuration to expedite climb away from the runway environment. Considerations for a balked landing include clearance of an obstacle in the runway environment, insufficient runway for continued landing, or transitioning to a go-around from a low-energy state.

When performing a balked landing, disengage the autopilot, disconnect the autothrottle, smoothly advance thrust levers to go-around thrust, and verify speedbrakes are retracted. Maintain landing flaps configuration and smoothly rotate toward 15° pitch attitude at no less than VREF. Column forces during rotation can vary. When safely airborne with a positive rate of climb, continue the Go-Around and Missed Approach procedure.

WARNING: After reverse thrust is initiated, a full stop landing must be made. if an engine stays in reverse, safe flight is not possible

NOTE The takeoff configuration warning horn sounds when on the ground due to landing flap configuration.

Selecting the go-around flaps during a balked landing can cause a decrease in the airplane lift and can result in increased takeoff distance. However, if there is sufficient runway remaining to safely complete the balked landing with the go-around flap setting, the go-around flap setting can be selected.

Hard Landing

Boeing airplanes are designed to withstand touchdown rates well above typical touchdown rates seen during daily operations. Even a perceived hard landing is usually well below these design criteria. Boeing policy is that a pilot report is the only factor that consistently identifies a hard landing. If the pilot believes that a hard landing may have occurred, it should be reported. A maintenance inspection will determine if further maintenance action is needed.

NOTE When reporting hard landings, the report should specify if the landing was hard on the nose gear only, hard on the main gear only or hard on both main and nose gear. Specify if the landing was a hard bounced landing.

NOTE A bounced landing is defined as a landing where both main gears contact the ground and then both main gears leave the ground prior to landing.

2.11.6. Intercepting Glide Slope from Above

The following technique may be used for ILS, GLS, or approaches using IAN; however, it is not recommended for approaches using VNAV.

Normally the ILS profile is depicted with the airplane intercepting the glide slope from below in a level flight attitude. However, there are occasions when flight crews are cleared for an ILS approach when they are above the G/S. In this case, there should be an attempt to capture the G/S prior to the FAF. The map display can be used to maintain awareness of distance to go to the final approach fix. The use of autopilot is also recommended.

NOTE Before intercepting the G/S from above, the flight crew must ensure that the localizer is captured before descending below the cleared altitude or the FAF altitude.

NOTE In some instances, when intercepting the G/S from well above a 3° G/S, a false G/S capture can result in an unexpected rapid pitch-up command which can lead to a rapid loss of airspeed.

The following technique will help the crew intercept the G/S safely and establish stabilized approach criteria by 1,000 feet AAL:

- use autopilot strongly recommended;
- select APP on the MCP and verify that the G/S is armed;
- establish final landing configuration and set the MCP altitude no lower than 1,000 feet AAL;
- select the V/S mode and set -1000 to -1500 fpm to achieve G/S capture and be stabilized for the approach by 1,000 feet AAL. Use of the green altitude range arc may assist in establishing the correct rate of descent;
- monitor the rate of descent and airspeed to avoid exceeding flap placard speeds and flap load relief activation. At G/S capture observe the flight mode annunciations for correct modes and monitor G/S deviation;
- after G/S capture, continue with normal procedures.

NOTE. If the G/S is not captured or the approach not stabilized by 1,000 feet AAL, initiate a go-around. Because of G/S capture criteria, the G/S should be captured and stabilized approach criteria should be established by 1,000 feet AAL, even in VMC conditions.

2.11.7. Landing Procedure General

General

If flying on Radar Vectors, perform a “DIRECT TO” or “INTERCEPT COURSE TO” the FAF, OM, or appropriate fix, to simplify the navigation display. This provides:

- a display of distance remaining to the FAF, OM, or appropriate fix;
- a depiction of cross track error from the final approach course;
- LNAV capability during the missed approach procedure.

PF should keep hands on flight controls and thrust levers below 1,000 ft. even with autopilot and autothrottle engaged.

Before starting final approach, it is necessary to receive ATC (if applicable) clearance for any type of approach.

Approach Clearance

When cleared for an approach and on a published segment of that approach, the pilot is authorized to descend to the minimum altitude for that segment. When cleared for an approach and not on a published segment of the approach, maintain assigned altitude until crossing the initial approach fix or established on a published segment of that approach. If established in a holding pattern at the final approach fix, the pilot is authorized to descend to the procedure turn altitude when cleared for the approach.

Stabilized Approach Recommendations

All Instrument approaches should be stabilized by 1,000 feet AAL.

Visual and circling approaches should be stabilized by 500 feet AAL.

During circle-to-land approach aircraft must be stabilized on final approach course not below 300 ft AAL

In order to minimize risk of unstabilized approach, it is recommended to start landing flaps extension before reaching 1,500 feet (450 m) AAL.

An approach is considered stabilized when all of the following criteria are met:

- the airplane is on the correct flight path;
- only small changes in heading and pitch are required to maintain the correct flight path;
- the airplane should be at approach speed. Deviations of +15 knots to – 5 knots are acceptable if the airspeed is trending toward approach speed;
- the airplane is in the correct landing configuration;
- sink rate is no greater than 1,000 fpm; if an approach requires a sink rate greater than 1,000 fpm, a special briefing should be conducted;
- thrust setting is appropriate for the airplane configuration usually above IDLE;
- all briefings and checklists have been conducted.

Specific types of approaches are stabilized if they also fulfill the following:

- ILS and GLS approaches should be flown within one dot of the glide slope and localizer, or within the expanded localizer scale till 200 feet;
- approaches using IAN should be flown within one dot of the glide path and FAC;
- at the airports, where speed restrictions are applied on final approach for ATC separation purposes, approach speed should be stabilized by 500 feet AAL. This criterion is not applicable in adverse weather conditions and during LVO;
- during a circling approach, wings should be level on final when the airplane reaches 300 feet AAL.

Below 200 feet AAL:

- on the visual descent path it is acceptable to deviate more than one dot of glide slope during ILS or GLS approach if the flight crew has no doubt that aircraft will land at the touchdown zone of the RW and the rest of the approach criteria are met;
- thrust setting is appropriate for the airplane configuration and descend path, and above IDLE.

Unique approach procedures or abnormal conditions requiring a deviation from the above elements of a stabilized approach require a special briefing.

NOTE An approach that becomes unstabilized below prescribed altitude requires an immediate go-around.

These conditions should be maintained throughout the rest of the approach for it to be considered a stabilized approach. If the above criteria cannot be established and maintained until approaching the flare, initiate a go-around.

At 100 feet HAT for all visual approaches, the airplane should be positioned so the flight deck is within, and tracking to remain within, the lateral confines of the runway edges extended.

As the airplane crosses the runway threshold it should be:

- stabilized on approach airspeed to within + 10 knots until arresting descent rate at flare;
- on a stabilized flight path using normal maneuvering;

- positioned to make a normal landing in the touchdown zone (the first 3,000 feet or first third of the runway, whichever is less).
- Initiate a go-around if the above criteria cannot be maintained.

APPROACH REQUIREMENTS RELATING TO RNP

With appropriate operational approval, approaches requiring RNP alerting may be conducted in accordance with the following provisions:

- AFM indicates that the airplane has been demonstrated for selected RNP;
- at least one GPS or one DME is operational;
- any additional GPS or DME requirements specified by Operations Specification or by the selected terminal area procedure must be satisfied;
- no UNABLE REQD NAV PERF - RNP alert is displayed during the approach;
- when operating with the following RNP values, or smaller:

Approach Type	RNP
NDB, NDB/DME	0.6 NM
VOR, VOR/DME	0.5 NM
RNAV	0.5 NM
RNAV (GPS)/(GNSS)	0.3 NM

2.11.8. ILS or GLS Approach

The approach flight pattern assumes all preparations for the approach such as review of approach procedure, setting of minima, inbound course and radios are complete. It focuses on crew actions and avionics systems information. It also includes unique considerations during low weather minima operations. The flight pattern may be modified to suit local traffic and air traffic requirements.

The recommended operating practices and techniques in this section apply to both ILS and GLS approaches. Except for station tuning, pilot actions are identical. A sub-section titled GLS Approach containing information specific to the GLS is located at the end of this section.

ICAO defines the envelope of ILS quality where:

- G/S signal ensures a normal capture. This envelope is within 10 NM, $\pm 8^\circ$ of the centerline of the ILS glide path and up to 1.75 θ and down to 0.3 θ (θ = nominal glide path angle).
- LOC signal ensures a normal capture. This envelope is within 10 NM ±35° and within 18 NM ±10°.

FAIL PASSIVE

Fail passive refers to an AFDS which in the event of a failure, causes no significant deviation of airplane flight path or attitude. A DA(H) is used as approach minimums.

2.11.9. Non – ILS Instrument Approaches

Any time ILS equipment is installed and operative it is recommended to use ILS as the main type of approach. If the crew decided not to use ILS approach for any reason, follow the recommendations from the "Raw data monitoring requirements" section below.

Non-ILS approaches are defined as:

- RNAV approach - an instrument approach procedure that relies on airplane area navigation equipment for navigational guidance. The FMS on Boeing airplanes is FAA-certified RNAV equipment that provides lateral and vertical guidance referenced from an FMS position. The FMS uses multiple sensors (as installed) for position updating to include GPS, DME-DME, VOR-DME, LOC-GPS, and IRS.
- RNAV visual approach - a visual approach that relies on airplane navigation equipment to align the aircraft with a visual final. The approach is selected in the FMC and flown in the same way as an RNAV approach until reaching the visual segment.
- GPS approach - an approach designed for use by airplanes using stand-alone GPS receivers as the primary means of navigation guidance. However, Boeing airplanes using FMS as the primary means of navigational guidance, have been approved by the FAA to fly GPS approaches provided an RNP of 0.3 or smaller is used.

  NOTE: A manual FMC entry of 0.3 RNP is required if not automatically provided.
- VOR approach.
- NDB approach.
- LOC, LOC-BC, LDA, SDF, IGS, TACAN, or similar approaches.

Non-ILS approaches are normally flown using VNAV or V/S pitch modes or IAN.

TYPES OF APPROACHES

For airplanes not equipped with IAN, VNAV is the preferred method for accomplishing non-ILS approaches that have an appropriate vertical path defined on the FMC LEGS page. The section on Use of VNAV provides several methods for obtaining an appropriate path, to include published glide paths, and where necessary, a pilot constructed path. V/S may be used as an alternate method for accomplishing non-ILS approaches.

Airplanes with IAN are capable of using the MCP APP switch to fly non-ILS approaches that have an appropriate lateral and vertical path defined on the FMC LEGS page. Approaches using IAN provide the functions, indications, and alerting features similar to an ILS approach while following FMC glide path. Although non-ILS approaches using LNAV and VNAV can still be executed, IAN is normally used in place of LNAV and VNAV because of improved approach displays, alerts and standardized procedures.

Use of the Autopilot During Approaches

Automatic flight is the preferred method of flying non-ILS approaches. Automatic flight minimizes flight crew workload and facilitates monitoring the procedure and flight path. During non-ILS approaches, autopilot use allows better course and vertical path tracking accuracy, reduces the probability of inadvertent deviations below path, provides autopilot alerts and mode fail indications and enables lower RNP limits. Autopilot use is recommended until suitable visual reference is established on final approach.

Manually flying non-ILS approaches in IMC conditions increases workload and does not take advantage of the significant increases in efficiency and protection provided by the automatic systems. However, to maintain flight crew proficiency, pilots may elect to use the flight director without the autopilot when in VMC conditions.

NOTE Currently, the VNAV PTH mode contains no path deviation alerting. For this reason, the autopilot should remain engaged until suitable visual reference has been established.

RAW DATA MONITORING REQUIREMENTS

During localizer-based approaches (LOC, LOC-BC, LDA, SDF, and IGS) applicable localizer raw data must be monitored throughout the approach.

During non-localizer based approaches where the FMC is used for course or path tracking (VOR, TACAN, NDB, RNAV, GPS, etc.), monitoring raw data is recommended, if available. When using the FMCS (LNAV) without GPS updating to conduct a terminal area procedure or an instrument approach, raw data must be monitored and checked to ensure correct navigation. For an instrument approach, this check should be done no later than the FAF.

During VOR, DME or ADF approaches when only single FMC, single IRS or single GPS sources are available, radio data should be selected on both NDs and raw data must be monitored and checked for correct navigation. For an instrument approach, this check should be done no later than the FAF.

Checking raw data for correct navigation may be accomplished by either of the following:

- selecting POS and comparing the displayed raw data with the navaid symbols on the map.

  NOTE: The VOR radials and raw DME data should overlay the VOR/DME stations shown on the map and the GPS position symbol should nearly coincide with the tip of the airplane symbol (FMC position). Due to magnetic variation differences between the FMC database and the VOR station the VOR radial may not perfectly align with the FMC calculated course.

- displaying the VOR and/or ADF pointers on the map display and using them to verify your position relative to the map.

RNAV APPROACHES

RNAV approaches may be flown provided the RNP being used is equal to or less than the RNP specified for the approach and is consistent with the AFM demonstrated RNP capability.

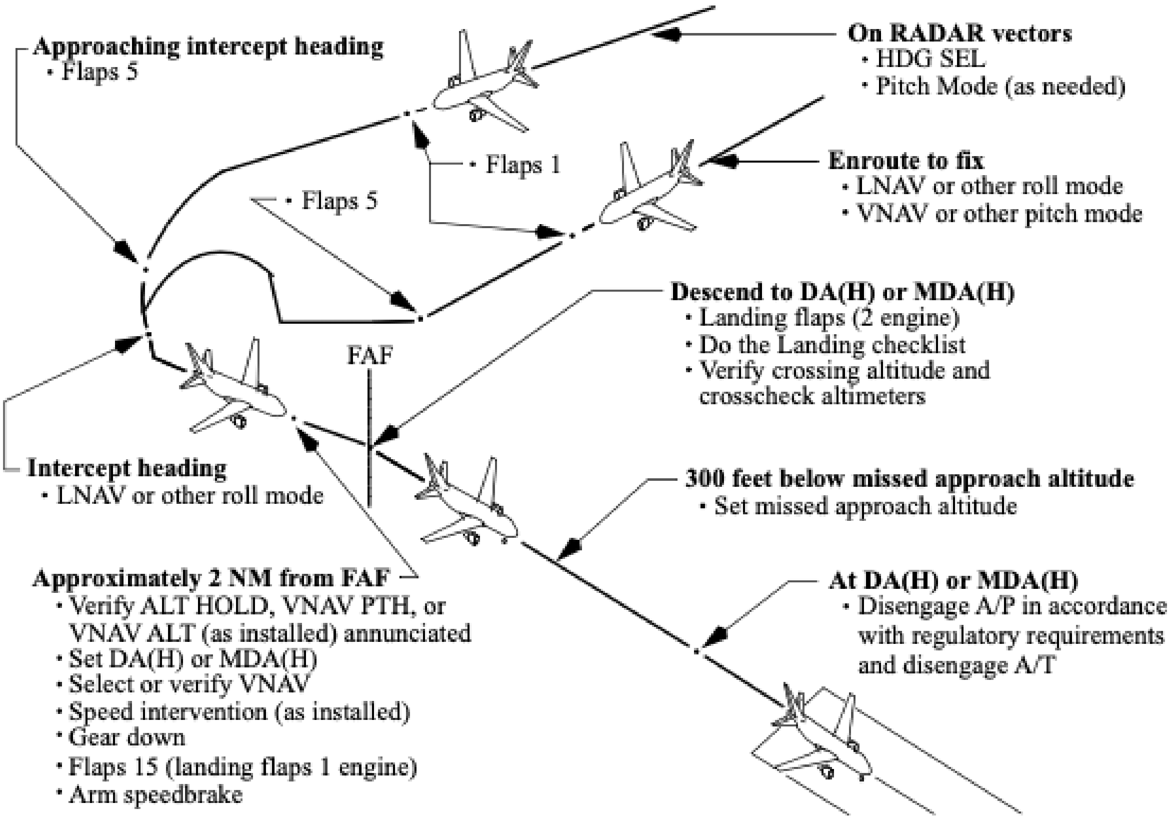
APPROACH REQUIREMENTS RELATING TO RNP

With appropriate operational approval, approaches requiring RNP alerting may be conducted in accordance with the following provisions:

- AFM indicates that the airplane has been demonstrated for selected RNP;
- at least one GPS or one DME is operational;
- any additional GPS or DME requirements specified by Operations Specification or by the selected terminal area procedure must be satisfied;
- no UNABLE REQD NAV PERF - RNP alert is displayed during the approach;
- when operating with the following RNP values, or smaller:

Approach Type	RNP
NDB, NDB/DME	0.6 NM
VOR, VOR/DME	0.5 NM
RNAV	0.5 NM
RNAV (GPS)/(GNSS)	0.3 NM

2.11.10. Instrument Approach Using VNAV



VNAV should be used only for approaches that have one of the following features:

- a published GP angle on the LEGS page for the final approach segment;
- an RW__ waypoint at the approach end of the runway;
- a missed approach waypoint before the approach end of the runway (for example, MX__).

This procedure is not authorized using QFE.

APPROACH PREPARATIONS FOR USING VNAV

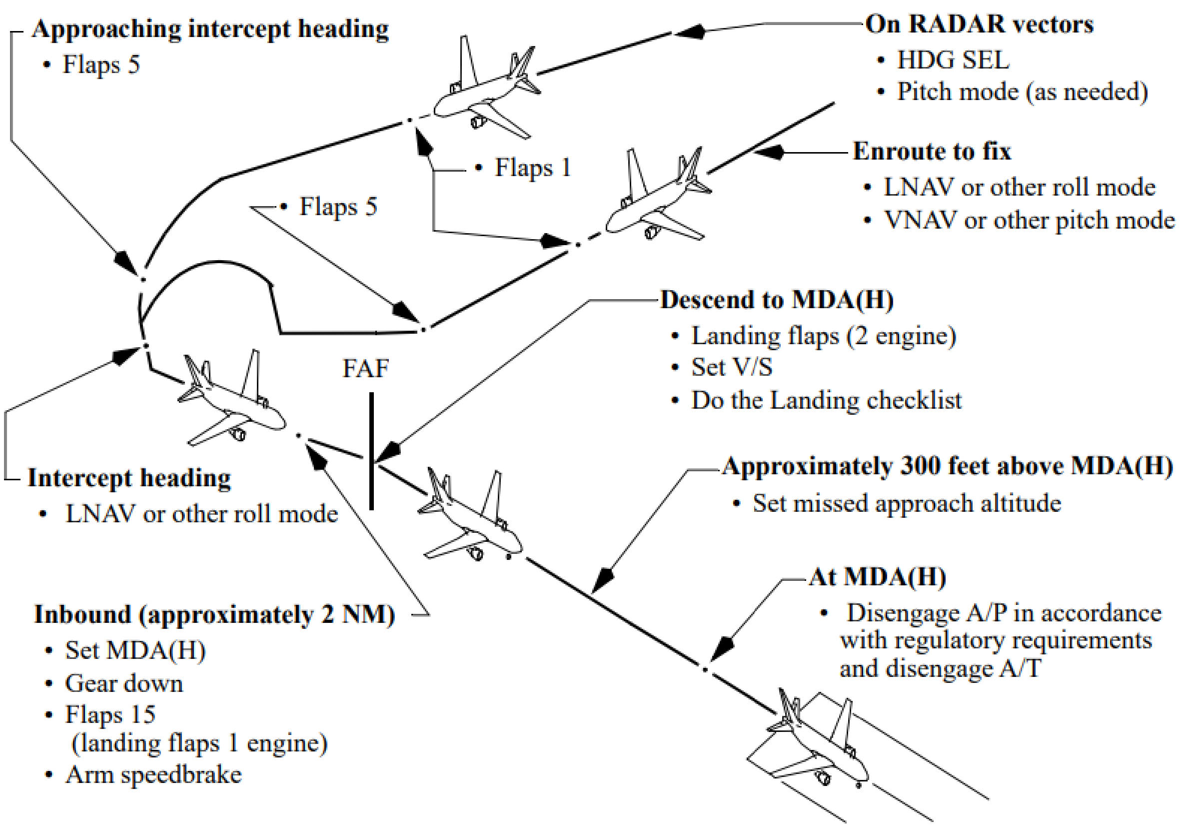
Select the approach procedure from the arrivals page of the FMC. Tune and identify appropriate navaids. Do not manually build the approach or add waypoints to the procedure. If additional waypoint references are desired, use the FIX page. To enable proper LNAV waypoint sequencing, select a straight-in intercept course to the FAF when being radar vectored to final approach.

Verify/enter the appropriate RNP and set the DA using the baro minimums selector. If required to use MDA(H) for the approach minimum altitude, set the barometric minimums selector at MDA(H) + 50 feet to ensure that if a missed approach is initiated, descent below the MDA(H) does not occur during the missed approach.

NOTE The approach RNP value is determined from one of three sources: manual entry by the flight crew, FMC default, or the navigation database. A manual entry overrides all others. If the navigation database contains an RNP value for the final approach leg, the RNP will appear when the leg becomes active, or up to 30 NM prior if the previous leg does not have an associated RNP value. The FMC default approach RNP will appear (no manual entry or navigation database value) when within 2 NM of the approach waypoint, including approach transitions, or when below 2,000 feet above the destination airport.

NOTE If desired altitude is not at an even 100 ft increment, set the MCP altitude to the nearest 100 ft increment above the altitude constraint or MDA(H).

2.11.11. Instrument Approach Using V/S



APPROACH PREPARATIONS FOR USING V/S

Select the approach procedure from the arrivals page of the FMC. Tune and identify appropriate navaids. If additional waypoint references are desired, use the FIX page. To enable proper LNAV waypoint sequencing, select a straight-in intercept course to the FAF when being radar vectored to final approach.

Verify/enter the appropriate RNP and set the MDA(H) + 50 ft using the baro minimums selector. If required to use MDA(H) for the approach minimum altitude, the barometric minimums selector may be set at MDA(H) + 50 ft to ensure that if a missed approach is initiated, descent below the MDA(H) does not occur during the missed approach.

If constraints or MDA(H)+50` do not end in zero (00) for example, 1820, set the MCP ALTITUDE window to the closest 100 foot increment above the constraint or the closest 100 foot increments above the MDA(H)+50`

Several techniques may be used to achieve a constant angle path that arrives at MDA(H) at or near the VDP:

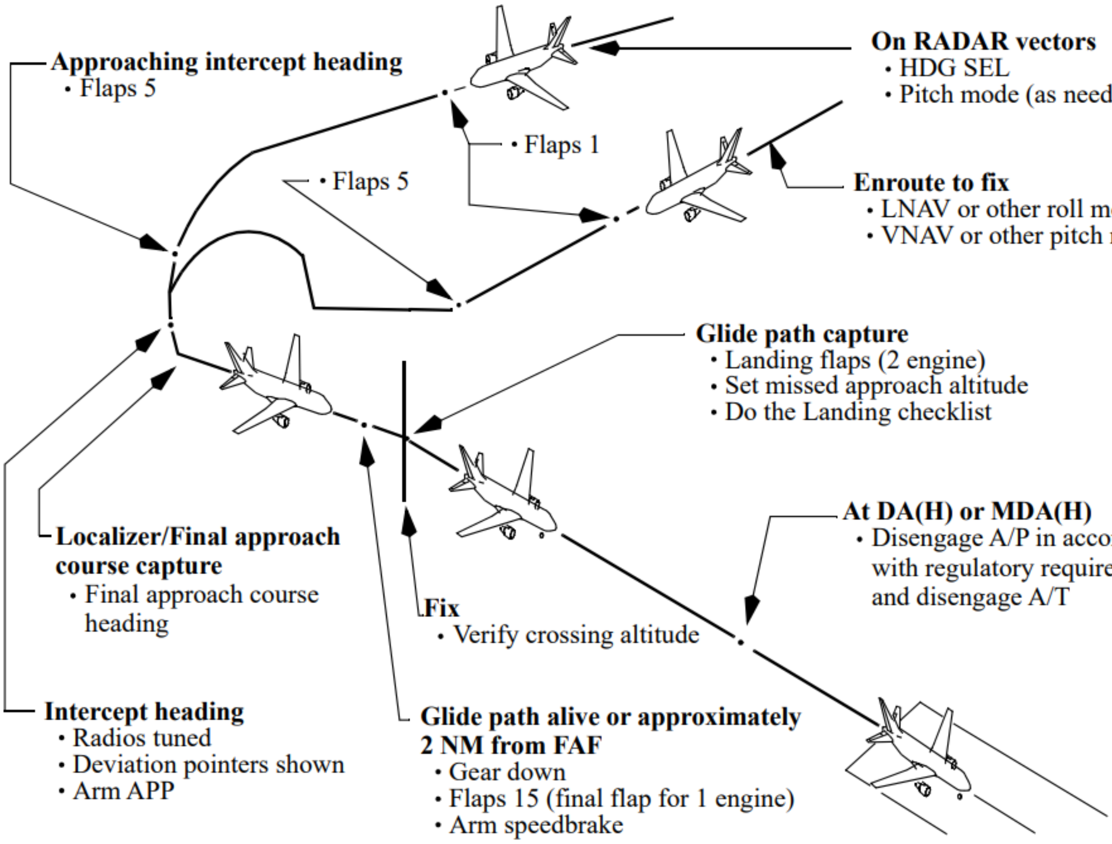
- the most accurate technique is to monitor the VNAV path deviation indication on the map display and adjust descent rate or FPA to maintain the airplane on the appropriate path. This technique requires the path to be defined appropriately on the legs page and that the header GPx.xx is displayed for the missed approach point or there is a RWxx or MXxx, or named waypoint on the legs page with an altitude constraint which corresponds to approximately 50 feet threshold crossing height. When this method is used, crews must ensure compliance with each minimum altitude constraint on the final approach segment (step-down fixes);
- select a descent rate or FPA that places the altitude range arc at or near the step-down fix or visual descent point (VDP). This technique requires the step-down fix or MDA(H) to be set in the MCP and may be difficult to use in turbulent conditions. See the Visual Descent Point section for more details on determining the VDP;
- using 300 feet per NM for a 3° path, determine the desired HAA which corresponds to the distance in NM from the runway end. The PM can then call out recommended altitudes as the distance to the runway changes (Example: 900 feet - 3 NM, 600 feet - 2 NM, etc.). The descent rate should be adjusted in small increments for significant deviations from the nominal path.

Be prepared to land or go-around from the MDA(H) at the VDP. Note that a normal landing cannot be completed from the published missed approach point on many instrument approaches.

Approximately 300 feet above the MDA(H), select the missed approach altitude. Leaving the MDA(H), be prepared to disengage the autopilot in accordance with regulatory requirements. Disconnect the autothrottle when disengaging the autopilot. Turn both F/Ds OFF, then place both F/Ds ON. This eliminates unwanted commands for both pilots and allows F/D guidance in the event of a go-around. Complete the landing.

On the V/S approach, the missed approach altitude is set when 300 feet above the MDA(H) to use the guidance of the altitude range arc during the approach and to prevent altitude capture and destabilizing the approach. Since there is no below path alerting, keeping the MDA(H) set as long as possible is recommended to help prevent inadvertent descent below MDA(H).

2.11.12. Instrument Approach Using IAN



GENERAL

For an instrument approach using IAN, select the approach procedure on the ARRIVALS page. Select the G/S prompt OFF if flying an ILS approach where the G/S transmitter is inoperative or when the G/S data is unreliable. Do not manually build the approach to add waypoints to the selected FMC procedure.

The approach profile illustrated depicts crew actions used during an approach using Integrated Approach Navigation (IAN). Since the techniques for approaches using IAN are similar to ILS approach techniques, only items considered unique to IAN are discussed in the remainder of this section. The approach profile illustrated assumes all preparations for the approach such as review of the approach procedure and setting of minima and radios, as required, are complete.

Airplanes with IAN are capable of using the MCP APP switch to execute instrument approaches based on flight path guidance from the FMC. Approaches using IAN provide the functions, indications and alerting features similar to an ILS approach. Although non-ILS approaches using LNAV and VNAV can still be performed, IAN is normally used in place of LNAV and VNAV because of standardized procedures.

NOTE: The glide path generated by the FMC is based on barometric data which is impacted by temperature deviation from ISA temperature. During approaches with temperatures above ISA, the generated IAN glide path can be slightly higher than the visual glide slope indicator path, for example from a PAPI or VASI. In these conditions, if the crew follows the visual glide slope the airplane will be below the IAN glide path which will cause a GLIDESLOPE alert to occur.

APPROACH TYPES THAT CAN USE IAN

- RNAV;
- GPS;
- VOR approach;
- NDB approach;
- LOC, LOC-BC, LDA, SDF, TACAN, or similar approaches.

IAN REQUIREMENTS AND RESTRICTIONS

- airplanes must be equipped with FMC U10.5 or later and IAN FMA displays;
- QFE operation is not authorized;
- RNP appropriate for the approach must be used;
- approaches using IAN are not authorized for RNAV (RNP) AR (airplanes with U12 and earlier);
- raw data monitoring is required during localizer based approaches;
- waypoints in the navigation database from the FAF onward may not be modified except as described below:
  - for airplanes with FMC update U10.7 or earlier, cold temperature altitude corrections are not permitted. This is because the AFM does not allow any altitude modification for the FAF or any waypoints between the FAF and the runway;
  - for airplanes with FMC update U10.8A, cold temperature altitude corrections are permitted for the FAF, or any waypoints between the FAF and the runway. However, the corrections must be entered prior to reaching the IAF, and only if the waypoints have an “at or above” altitude restriction;
  - for airplanes with FMC update U11.0 and later, cold temperature altitude corrections are permitted for the FAF, or any waypoints between the FAF and the runway.

IAN Recommendation

- the use of IAN is recommended for straight-in approaches only;
- during FMC based non-ILS approaches, raw data monitoring is recommended when available.

FLIGHT MODE ANNUNCIATIONS AND OTHER IAN FEATURES

FMAs vary depending on the source of the navigation guidance used for the approach, navigation radio or FMC.

FOR LOCALIZER BASED APPROACHES

Approach	FMA
ILS with G/S out, LOC, LDA, SDF	VOR/LOC and G/P
B/C LOC	B/CRS and G/P

IF THE FMC IS USED FOR LATERAL (COURSE) GUIDANCE

Approach	FMA
GPS, RNAV	FAC and G/P
VOR, NDB, TACAN	FAC and G/P

APPROACH PREPARATIONS FOR USING IAN

IAN may be used with the flight director, single autopilot, or flown with raw data. The procedure turn, initial approach, and final approach are similar to the ILS.

For FMC based approaches, a proper series of legs/waypoints describing the approach route including an appropriate vertical path or glide path (GP) angle must appear on the LEGS page. A GP angle displayed on the LEGS page means the vertical path complies with final approach step-down altitudes (minimum altitude constraints). A typical GP angle suitable for an approach using IAN is one that approximates 3° and crosses the runway threshold at approximately 50 feet.

The appropriate procedure must be selected in the FMC. If final approach course guidance is derived from the localizer, the radios must be tuned to the appropriate frequency. If final approach course guidance is derived from the FMC, the ILS or GLS should not be tuned. When flying a localizer approach with the glide slope out or unreliable (LOC GS out), the G/S prompt is selected OFF to ensure that the FMC generated glide path is flown.

NOTE: For all approaches, including B/C LOC approaches, the inbound front course must be set in the MCP.

FINAL APPROACH USING IAN

When on an intercept heading and cleared for the approach, select the APP mode.

Set missed approach altitude after glide path capture.

APP mode should not be selected until:

- the guidance to be used for the final approach is tuned and identified (as needed);
- the airplane is on an inbound intercept heading;
- both lateral and vertical deviation pointers appear on the attitude display in the proper position;
- clearance for the approach has been received.

NOTE: The IAN GP is a barometric vertical path and does not guarantee compliance with FMC altitude constraints prior to the FAF. However, the IAN GP will be at or above altitude constraints between the FAF and the missed approach waypoint that are included in the FMC database procedure.

Unlike an ILS approach, where configuration for landing is initiated when the glide slope comes alive, during an IAN approach, configuration for landing is initiated approximately 2 NM from the FAF. Waiting until the glide path is alive may give the crew too little time to accomplish all recommended actions before intercepting the glide path. This is because the IAN glide path full scale deflection is variable depending upon the vertical RNP value in use. At low vertical RNP values, the vertical deviation pointer does not move until just before intercepting the glide path. Initiating the landing configuration at approximately 2 NM from the FAF gives the crew time to slow and select landing flaps at the FAF.

Deviation scales are proportional to RNP. Similar to the ILS localizer and glide slope deviation scales, the IAN deviation scales become more sensitive as the airplane approaches the runway.

GPWS glide slope alerting is provided on IAN airplanes during ILS and approaches using IAN. GPWS glide slope alerts provide indications when the airplane deviates below the glide slope (ILS) or glide path (IAN) before reaching a full scale deflection.

During the approach, any time a full scale vertical or lateral deflection occurs or an UNABLE REQ'D NAV PERF - RNP alert occurs and suitable visual reference has not been established, a missed approach must be executed.

If the final approach course is offset from the runway centerline, maneuvering to align with the runway centerline is required. When suitable visual reference is established, continue following the glide path (GP) angle while maneuvering to align with the runway.

2.11.13. Circling Approach

GENERAL

Of all the different types of instrument approaches, those that end with circle-to-land maneuvers can pose the biggest challenges - and potential risks.

Visual maneuvering (circling) is the term used to describe the phase of flight after an instrument approach has been completed. It brings the aircraft into position for landing on a runway which is not suitably located for straight–in approach, i.e. one where the criteria for alignment or descent gradient cannot be met.

The circling approach should be flown with:

- landing gear down;
- flaps 15, and at flaps 15 maneuver speed;
- speedbrake armed.

Use the weather minima associated with the anticipated circling speed.

The circling approach may be flown following any instrument approach procedure. During the instrument approach, use VNAV or V/S modes to descend to the circling MDA(H).

NOTE: Do not use of the APP mode for descent to the circling MDA(H).

Maintain MCP altitude or MDA(H) using ALT HOLD or VNAV ALT (as installed) mode. Use HDG SEL for the maneuvering portion of the circling approach.

NOTE: If the MDA(H) does not end in “00”, set the MCP altitude to the nearest 100 feet above the MDA(H) and circle at MCP altitude.

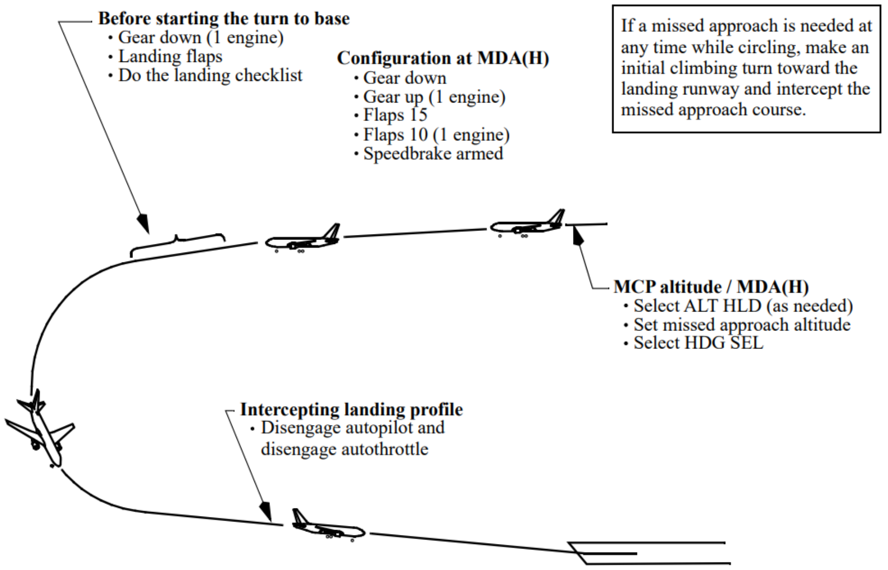
NOTE: If VNAV ALT (as installed) is allowed to remain as the pitch mode during the circling maneuver with a higher altitude, (e.g., missed approach altitude), set in the MCP, the VNAV ALT pitch mode will revert to CWS P.

When in ALT HOLD at MCP altitude or the MDA(H) and before commencing the circling maneuver, set the missed approach altitude.

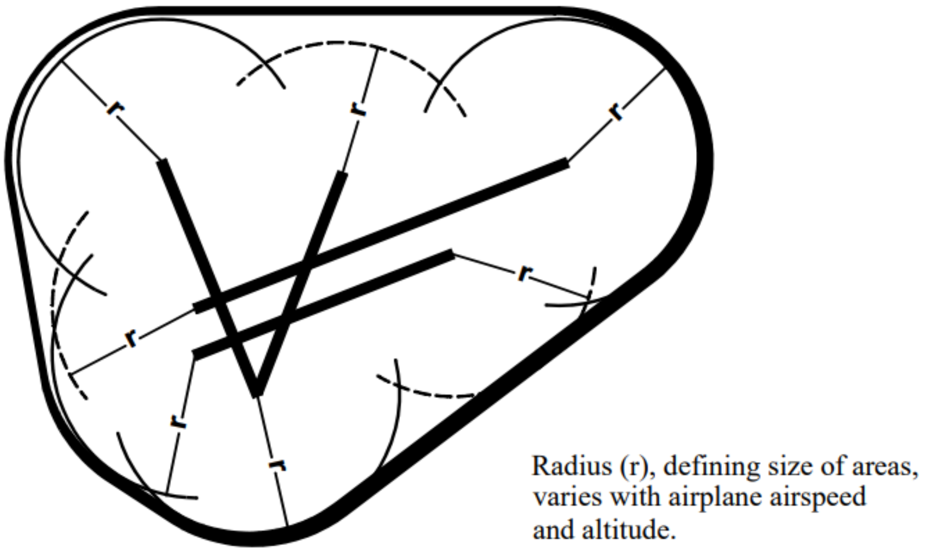
Before starting the turn to base leg, select landing flaps and begin decelerating to the approach speed plus wind additive. To avoid overshooting the final approach course, adjust the turn to final to initially aim at the inside edge of the runway threshold. Timely speed reduction also reduces turning radius to the runway. Do the Landing checklist. Do not descend below MDA(H) until intercepting the visual descent profile to the landing runway.

When intercepting the landing profile, disengage the autopilot, disconnect the autothrottle and continue the approach manually. At this point in the approach, the pilot's attention should be focused on intercepting the visual descent profile rather than attempting to set the MCP or FMC to allow continued use of the autopilot. After intercepting the visual descent profile, cycle both F/D to OFF, then to ON. This eliminates unwanted commands for both pilots and allows F/D guidance in the event of a go-around. Complete the landing.

If a missed approach is needed at any time while circling, make an initial climbing turn toward the landing runway and intercept the missed approach course.



PROTECTION AREA



	Category (Maximum IAS)	Circling Area Radius from Threshold
FAA	**D** (165 Kts)	**2.3** NM
ICAO	**D** (205 Kts)	**5.28** NM

NOTE: Adjust airplane heading and timing so that the airplane ground track does not exceed the obstruction clearance distance from the runway at any time during the circling approach.

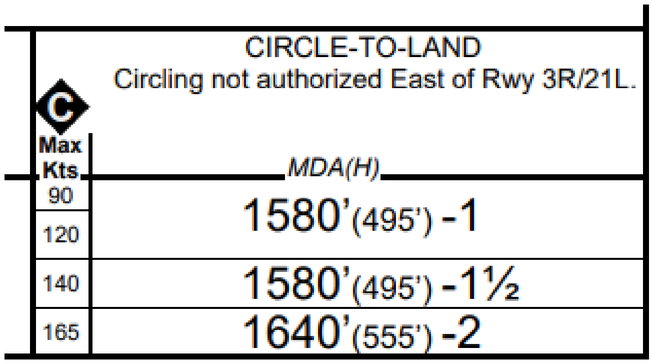
FAA EXPANDED CIRCLING MANEUVERING AIRSPACE RADIUS

The FAA has modified the criteria for circling approach areas via TERPS. Circling approach areas for approach procedures developed beginning in 2013 use the radius distances (in NM) as depicted in the following table. These distances, dependent on aircraft category, are also based on the circling altitude which accounts for the true airspeed increase with altitude.

Circling MDA in feet MSL	Circling Area Radius from Threshold (NM) Cat D Max 165 KIAS
1,000 or less	3.6
1,001 to 3,000	3.7
3,001 to 5,000	3.8
5,001 to 7,000	4.0
7,001 to 9,000	4.2
9,000 and above	4.4

EFFECT ON CHARTS

Charts where the new criteria have been applied can be identified by the “Inverse C” icon in the CIRCLE-TO-LAND minima box as shown in the following table.



NOTE: For detailed information concerning circling maneuver area refer to **OM PART A**.

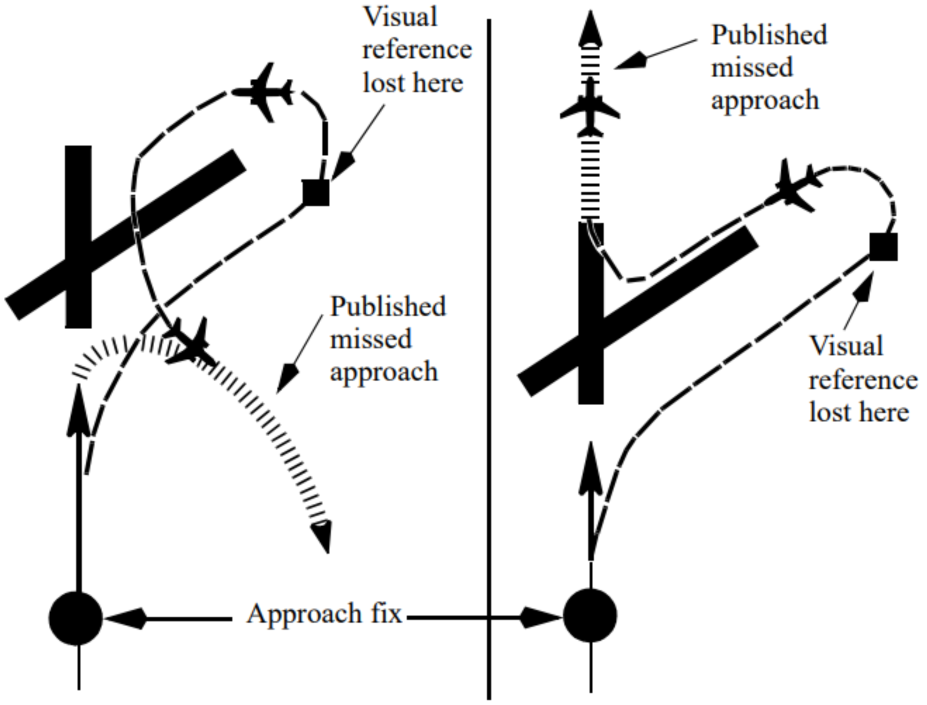
Descent below MDA(H) should not be made until:

- visual reference has been established and can be maintained;
- the pilot has the landing threshold in sight; and
- the required obstacle clearance can be maintained and the aircraft is in a position to carry out a landing.

MISSED APPROACH PROCEDURE

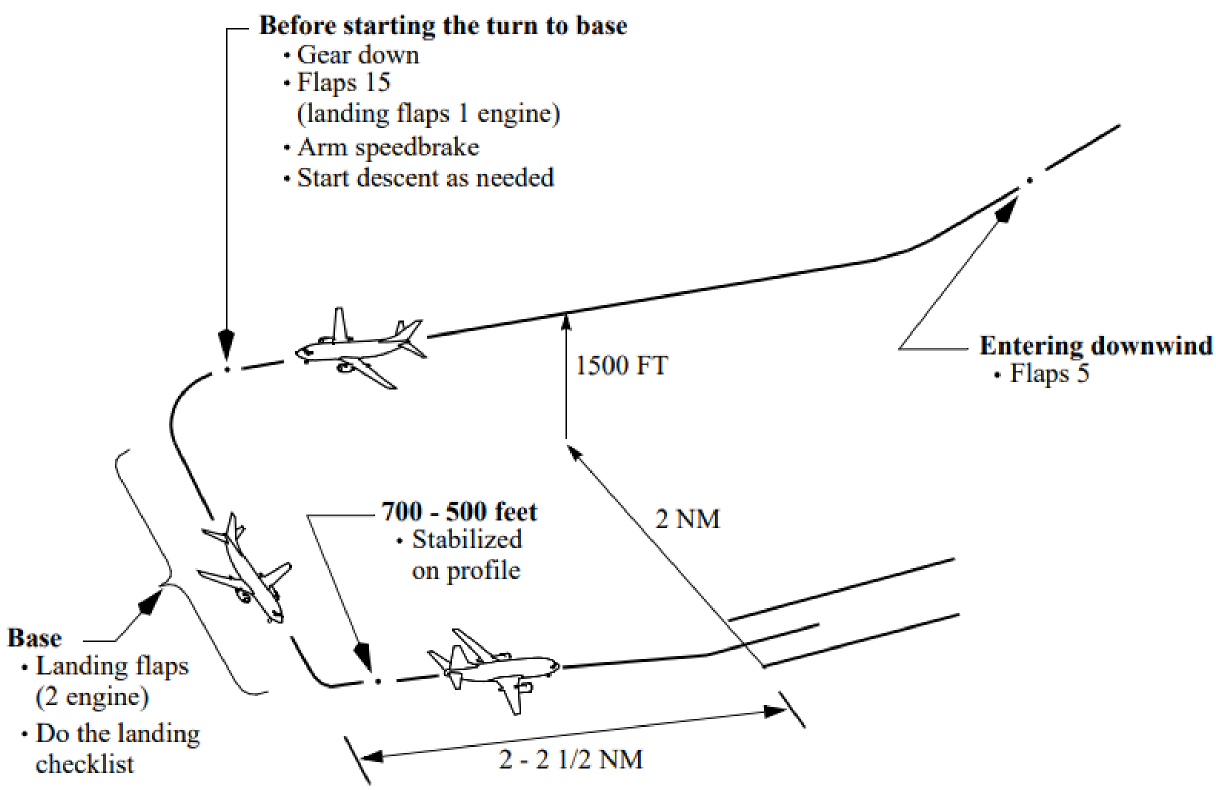
If a missed approach is required at any time while circling, make a climbing turn in the shortest direction toward the landing runway. This may result in a turn greater than 180° to intercept the missed approach course. Continue the turn until established on an intercept heading to the missed approach course corresponding to the instrument approach procedure just flown. Maintain the missed approach flap setting until close-in maneuvering is completed.

Different patterns may be required to become established on the prescribed missed approach course. This depends on airplane position at the time the missed approach is started. The following figure illustrates the maneuvering that may be required. This ensures the airplane remains within the circling and missed approach protection areas.



In the event that a missed approach must be accomplished from below the MDA(H), consideration should be given to selecting a flight path which assures safe obstacle clearance until reaching an appropriate altitude on the specified missed approach path.

2.11.14. Visual Approach



A visual approach is defined as an approach by an IFR flight where either part, or all, of an instrument approach procedure is not completed and the approach is executed in visual reference to terrain.

Use all available NAV aids to verify and support the visual approach.

Use PAPI (T-VASI) to verify visually the descent glide path.

Before executing a visual approach assure that:

− the visual reference can be maintained during the entire approach;
− ATC clearance is obtained;
− a short progressive briefing of the intended flight path in addition to the IFR approach briefing has been performed;
− actual weather at the airport must be:
  ▪ Clouds in excess of 2 octas are at or above 2000ft (600m);
  ▪ flight visibility 5 km or greater;
  ▪ surface in sight.

Ensure you have a clear understanding of the required missed approach procedure when accepting a clearance for a visual approach.

The recommended landing approach path is approximately 2.5° to 3°. Once the final approach is established, the airplane configuration remains fixed and only small adjustments to the glide path, approach speed, and trim are necessary. This results in the same approach profile under all conditions.

Maximum use of autopilot and autothrottle is recommended.

Typically fly at an altitude of 1,500 feet above the runway elevation and enter downwind with flaps 5 at flaps 5 maneuver speed. Maintain a track parallel to the landing runway approximately 2 NM abeam.

Before starting the turn to base leg, extend the landing gear, select flaps 15, arm the speedbrake, and slow to flaps 15 maneuver speed or approach speed plus wind additive if landing at flaps 15. If the approach pattern must be extended, delay lowering gear and selecting flaps 15 until approaching the normal visual approach profile. Turning base leg, adjust thrust as required while descending at approximately 600-700 fpm.

Extend landing flaps before turning final. Allow the speed to decrease to the proper final approach speed and trim the airplane. Do the Landing checklist. When established in the landing configuration, maneuvering to final approach may be accomplished at final approach speed (VREF plus wind additive).

Final Approach

Roll out of the turn to final on the extended runway centerline and maintain the appropriate approach speed. An altitude of approximately 300 feet AAL for each NM from the runway provides a normal approach profile. Attempt to keep thrust changes small to avoid large trim changes. With the airplane in trim and at approach airspeed, pitch attitude should be approximately the normal approach body attitude. At speeds above approach speed, pitch attitude is less. At speeds below approach speed, pitch is higher. Slower speed reduces aft body clearance at touchdown. Stabilize the airplane on the selected approach airspeed with an approximate rate of descent between 700 and 900 feet per minute on the desired glide path, in trim. Stabilize on the profile by 500 feet above touchdown.

NOTE: Descent rates greater than 1000 fpm should be avoided.

2.11.15. Go–Around and Missed Approach Procedure

GENERAL

If a missed approach is required following a dual autopilot approach with FLARE arm annunciated, leave the autopilots engaged. Push either TO/GA switch, call for flaps 15, ensure go-around thrust for the nominal climb rate is set and monitor autopilot performance. After a positive rate of climb is indicated on the baro altimeter retract the landing gear and check GA altitude.

At typical landing weights, actual thrust required for a normal go-around is usually considerably less than maximum go-around thrust. This provides a thrust margin for windshear or other situations requiring maximum thrust. If full thrust is desired after thrust for the nominal climb rate has been established, push TO/GA a second time.

If a missed approach is required following a single autopilot or manual instrument approach, or a visual approach, push either TO/GA switch, call for flaps 15, ensure/set go-around thrust, and rotate smoothly toward 15° pitch attitude. Then follow flight director commands. After a positive rate of climb is indicated on the baro altimeter retract the landing gear and check GA altitude.

NOTE: When performing a normal approach using flaps 15 for landing, if a go-around is required use flaps 15 for go-around. When performing a one-engine inoperative approach using flaps 15 for landing, if a go-around is required use flaps 1 for go-around. When using flaps 1 for go-around, limit bank angle to 15° when airspeed is less than VREF 15 + 15 knots or minimum maneuver speed, whichever is less.

During an automatic go-around initiated at 50 feet, approximately 30 feet of altitude is lost. If touchdown occurs after a go-around is initiated, the go-around continues. Observe that the autothrottle applies go-around thrust or manually apply go-around thrust as the airplane rotates to the go-around attitude.

NOTE: An automatic go-around cannot be initiated after touchdown.

The TO/GA pitch mode initially commands a go-around attitude and then transitions to speed as the rate of climb increases. Command speed automatically moves to a target airspeed for the existing flap position. The TO/GA roll mode maintains existing ground track. Above 400 feet AAL, verify that LNAV is engaged for airplanes equipped with the TO/GA to LNAV feature, or select a roll mode as appropriate.

NOTE: Route discontinuities after the missed approach point will prevent the TO/GA to LNAV function from engaging.

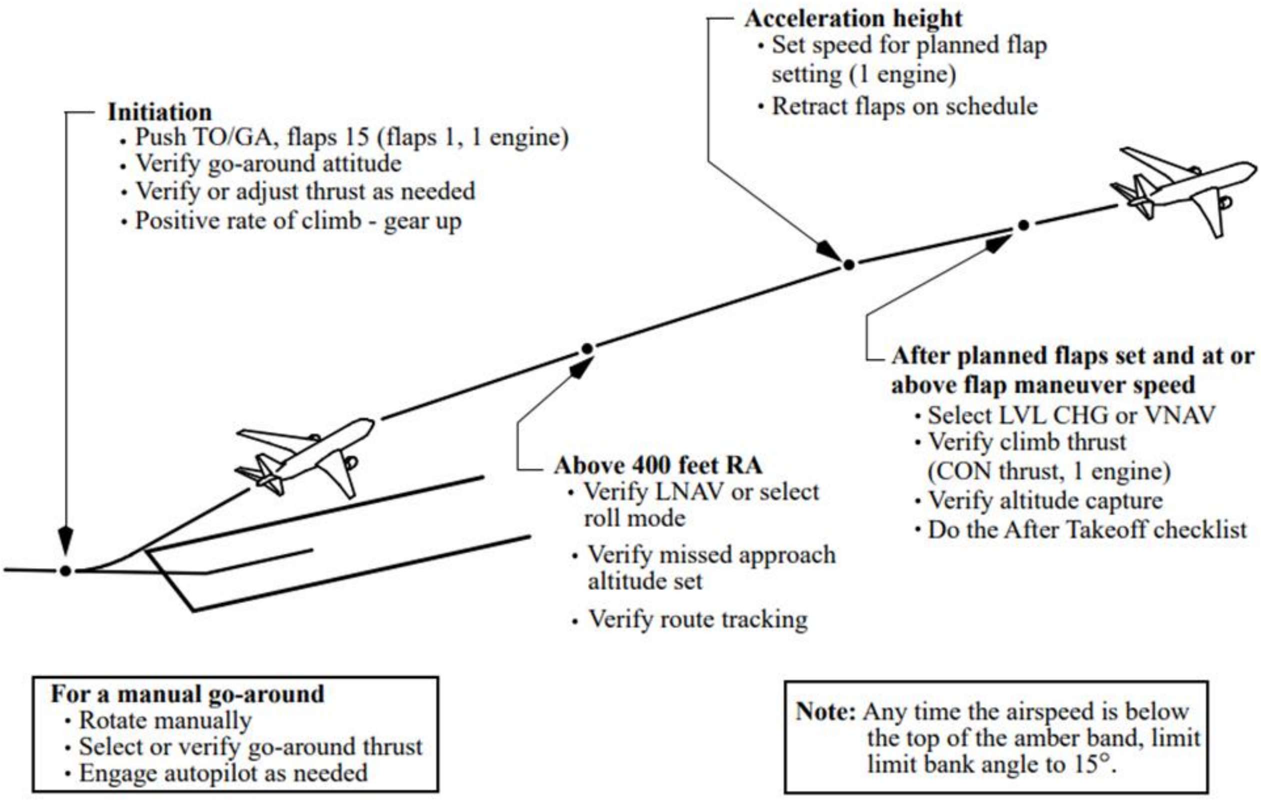
The minimum altitude for flap retraction during a normal takeoff is not normally applicable to a missed approach procedure. However, obstacles in the missed approach flight path must be taken into consideration.

NOTE: Other pitch and roll modes cannot be engaged until above 400 feet AAL.

NOTE: When accomplishing a missed approach from a dual-autopilot approach, initial selection of a pitch mode, or when altitude capture occurs above 400 feet AAL the AFDS reverts to single AP operation.

If initial maneuvering is required during the missed approach, do the missed approach procedure through gear up before initiating the turn. Delay further flap retraction until initial maneuvering is complete and a safe altitude and appropriate speed are attained.

If a go-around is initiated prior to the Missed Approach Point (MAP), do not initiate any turns until reaching the MAP.



2.11.16. Landing Roll Procedure

Avoid touching down with thrust above idle since this may establish an airplane nose up pitch tendency and increase landing roll.

After main gear touchdown, initiate the landing roll procedure. Fly the nose wheels smoothly onto the runway without delay. If the speedbrakes do not extend automatically move the speedbrake lever to the UP position without delay. Control column movement forward of neutral should not be required. Do not attempt to hold the nose wheels off the runway. Holding the nose up after touchdown for aerodynamic braking is not an effective braking technique and results in high nose gear sink rates upon brake application and reduced braking effectiveness.

To avoid possible airplane structural damage, do not make large nose down control column movements before the nose wheels are lowered to the runway.

To avoid the risk of a tail strike, do not allow the pitch attitude to increase after touchdown. However, applying excessive nose down elevator during landing can result in substantial forward fuselage damage. Do not use full down elevator. Use an appropriate autobrake setting or manually apply wheel brakes smoothly with steadily increasing pedal pressure as required for runway condition and runway length available. Maintain deceleration rate with constant or increasing brake pressure as required until stopped or desired taxi speed is reached.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

2.12. STANDARD OPERATION PROCEDURE

2.12.1. General

Предполетная подготовка проводится в порядке, определенном главой 8, части А РПП Авиапредприятия.

Before departure **captain** should:

- brief with the maintenance aircraft status;
- check ATL and Deferred Items List;
- receive purser's report about cabin preparation;
- check Passenger Address and communication with cabin;
- sign ATL and maintenance documents after refueling;
- check that actual fuel quantity is sufficient for flight;
- brief with maintenance staff de-icing procedure, if necessary;
- receive weather Information;
- prepare necessary charts and maps for flight.

Before departure **first officer** should:

- check the current revision of navigation manuals;
- check aircraft library and sealed document folder;
- prepare Runway Analysis Chart, if necessary;
- check that actual fuel quantity is sufficient for flight;
- receive weather Information;
- prepare necessary charts and maps for flight.

Cockpit preparation sequence:

- Electrical Power Up Procedure (if needed).
- Preliminary preflight procedure.
- Exterior inspection.
- CDU preflight procedures.
- Preflight procedures.

2.12.2. Electrical Power Up Procedure

The following procedure is accomplished to permit safe application of electrical power. The First Officer normally does this procedure. The Captain may do this procedure as needed. Refer to FCOM SP - Electrical - Electrical Power Up.

2.12.3. Preliminary Preflight Procedure – Captain or First Officer

The Preliminary Preflight Procedure assumes that the Electrical Power Up Supplementary Procedure is complete.

Normally preliminary preflight procedure performed by the First Officer.

A full IRS alignment is recommended before each flight. If time does not allow a full alignment, do the Fast Realignment supplementary procedure.

IRS mode selectors... OFF, then NAV

Verify that the ON DC lights illuminate then extinguish.

Verify that the ALIGN lights are illuminated.

The UNABLE REQD NAV PERF-RNP message may show until IRS alignment is complete.

VOICE RECORDER switch .... On

VOICE RECORDER Test switch .... Push

Hold switch for 5 seconds.

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Observe that the Test light illuminates. A tone may be heard through a headset jack.

RA73269 - 73313, RA73316 - 73321

Observe that the STATUS light flashes once. A tone may be heard through a headset jack.

Verify that the following are sufficient for flight:

- oxygen pressure (required pressure provided by FCOM, Performance Dispatch chapter, Enroute, Flight Crew Oxygen Requirements);
- hydraulic quantity (minimum – 76%, after «DAILY CHECK» – 92%);
- engine oil quantity (must be 60% full or 12.00 US quarts or more.)

Do the remaining actions after a crew change or maintenance action.

NOTE: The following oxygen pressure drop test only needs to be performed at one crewmember or observer station.

RA73269 - 73313, RA73316 - 73321

Oxygen pressure drop ....Test

Note the crew oxygen pressure.

Oxygen mask – Stowed and doors closed.

TEST/RESET switch ………………………………………………….. Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

EMERGENCY/Test selector ………………………………………… Push and hold

Continue to hold the TEST/RESET switch down and push the EMERGENCY/Test selector for 5 seconds. Verify that the yellow cross shows continuously in the flow indicator.

Verify that the crew oxygen pressure does not decrease more than 100 psig.

If the oxygen cylinder valve is not in the full open position, pressure can:

- decrease rapidly, or
- decrease more than 100 psig, or
- increase slowly back to normal.

Release the TEST/RESET switch and the EMERGENCY/Test selector. Verify that the yellow cross does not show in the flow indicator.

Normal/100% switch ……………………………………………………… 100%

Crew oxygen pressure ………………………………………………..……. Check

Verify that the pressure is sufficient for dispatch.

RA73269, RA73311, RA73312, RA73319, RA73315, RA73321

Oxygen pressure drop .... Test

Note the crew oxygen pressure.

Oxygen mask .... Stowed and doors closed

TEST/RESET switch .... Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

Regulator selector .... Rotate to EMER

Continue to hold the TEST/RESET switch down with the regulator selector in the EMER position for 5 seconds. Verify that the yellow cross shows continuously in the flow indicator.

Verify that the crew oxygen pressure does not decrease more than 100 psig.

If the oxygen cylinder valve is not in the full open position, pressure can:

- decrease rapidly, or
- decrease more than 100 psig, or
- increase slowly back to normal.

Release the TEST/RESET switch and rotate the regulator selector to 100%. Verify that the yellow cross does not show in the flow indicator.

Crew oxygen pressure .... Check.

Verify that the pressure is sufficient for dispatch using (1.5.14)

GEAR PINS ....ONBOARD

Pins should be located in the bin over the coat stowage.

MAINTENANCE and COMPANY DOCUMENTS.................................... Check

FLIGHT DECK ACCESS SYSTEM Switch .... As needed

Refer to FCOM SP for system test.

EMERGENCY EQUIPMENT .... Check

Fire extinguisher – Checked and stowed

Crash axe – Stowed

Escape ropes – Stowed

Other needed equipment – Checked and stowed

ELT switch .... Guard closed

Verify that the ELT light is extinguished.

PSEU Light .... VERIFY EXTINGUISHED

GPS Light .... VERIFY EXTINGUISHED

RA-73319

ILS light.. VERIFY EXTINGUISHED

RA-73319

GLS light... VERIFY EXTINGUISHED

SERVICE INTERPHONE switch.. OFF

ENGINE panel... Set

Verify that the REVERSER lights are extinguished.

Verify that the ENGINE CONTROL lights are extinguished.

EEC switches – ON

Oxygen panel .... Set

NOTE: PASSENGER OXYGEN switch activation causes deployment of the passenger oxygen masks.

PASSENGER OXYGEN switch - Guard closed

Verify that the PASS OXY ON light is extinguished.

Landing gear indicator lights .... Verify illuminated

FLIGHT RECORDER switch.. Guard closed

Verify that the OFF light is illuminated.

MACH AIRSPEED WARNING TEST switches.................. Push, one at a time

Verify that the clacker sounds.

STALL WARNING TEST switches ....Push, one at a time

Verify that each control column vibrates when the respective switch is pushed.

NOTE: The stall warning test requires that AC transfer busses are powered for up to 4 minutes.

With hydraulic power off, the leading edge flaps can droop enough to cause an asymmetry signal, resulting in a failure of the stall warning system test. Should this occur, obtain a clearance to pressurize the hydraulic system, place the “B” system electric pump ON and retract the flaps. When flaps are retracted repeat the test. At the completion of the test, turn the “B” system electric pump “OFF”.

Circuit breakers (P6 panel)... Check

Manual gear extension access door... Closed

Circuit breakers (control stand, P18 panel) .... Check

Parking brake... As needed

Set the parking brake if the brake wear indicators are to be checked during the exterior inspection

2.12.4. Exterior Inspection - Captain

Before each flight the Captain, and maintenance crew must verify that the airplane is satisfactory for flight.

Items at each location may be checked in any sequence.

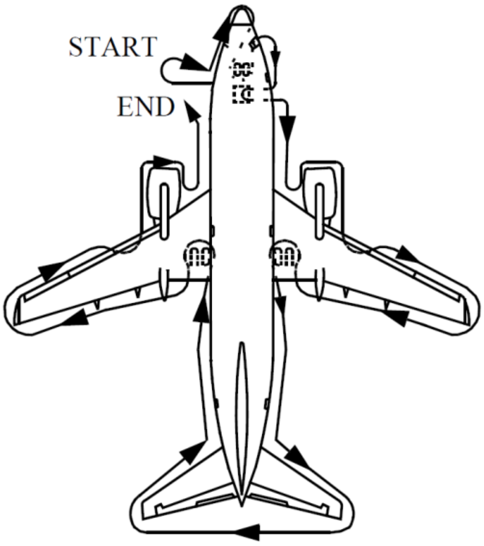
Use the detailed inspection route below to check that:

- The surfaces and structures are clear, not damaged, not missing parts and there are no fluid leaks*.
- The tires are not too worn, not damaged, and there is no tread separation.
- The gear struts are not fully compressed.
- The engine inlets and tailpipes are clear, the access panels are secured, the fan cowls are latched, the exterior, including the bottom of the nacelles, is not damaged, and the reversers are stowed.
- The doors and access panels that are not in use are latched.
- The probes, vents, and static ports are clear and not damaged.
- The skin area adjacent to the pitot probes and static ports is not wrinkled.
- The antennas are not damaged.
- The light lenses are clean and not damaged.

For cold weather operations see the Supplementary Procedures.

NOTE: * Fluid leaks from the engine drains are allowed provided the leaks are less than a continuous stream. Refer to the Engine Start Procedure for additional guidance.

Inspection Route



Inspection Route

Left Forward Fuselage

Probes, sensors, ports, vents, and drains (as applicable) .... Check
Doors and access panels (not in use) .... Latched
Oxygen pressure relief green disc .... In place
Forward outflow valve .... Check
Nose Radome .... Check
Conductor straps .... Secure
Forward E and E door .... Secure

Nose Wheel Well

Tires and wheels .... Check

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Exterior light .... Check
Gear strut and doors .... Check
Nose wheel steering assembly .... Check
Nose gear steering lockout pin .... As needed
Gear pin .... As needed
Nose wheel spin brake (snubbers) .... In place

Right Forward Fuselage

Probes, sensors, ports, vents, and drains (as applicable) .... Check
Oxygen pressure relief green disc .... In place
Doors and access panels (not in use) .... Latched

Right Wing Root, Pack, and Lower Fuselage

Ram air deflector door .... Extended
Pack and pneumatic access doors .... Secure
Probes, sensors, ports, vents, and drains (as applicable) .... Check
Exterior lights .... Check
Leading edge flaps .... Check

Number 2 Engine

Exterior surfaces (including the bottom of the nacelles)..... Check for damage
Access panels and fan cowl latches.. Latched
Probes, sensors, ports, vents, and drains (as applicable) .... Check
Fan blades, probes, and spinner .... Check
Thrust reverser .... Stowed
Exhaust area and tailcone .... Check

Right Wing and Leading Edge

Access panels .... Latched
Leading edge flaps and slats .... Check
Fuel measuring sticks .... Flush and secure
Wing Surfaces .... Check
Fuel tank vent .... Check

Right Wing Tip and Trailing Edge

Position and strobe lights .... Check
Static discharge wicks .... Check
Aileron and trailing edge flaps .... Check

Right Main Gear

Tires, brakes and wheels .... Check

Verify that the wheel chocks are in place as needed.

If the parking brake is set, the brake wear indicator pins must extend out of the guides.

Gear strut, actuators, and doors .... Check
Hydraulic lines .... Secure
Gear pins .... As needed

Right Main Wheel Well

APU FIRE CONTROL handle .... Up
NGS operability indicator light... Check

Verify that the light is green.

Wheel well .... Check

Right Aft Fuselage

Doors and access panels (not in use) .... Latched
Negative pressure relief door .... Closed
Outflow valve... Check
Probes, sensors, ports, vents, and drains (as applicable) .... Check
APU air inlet .... Check

Tail

Vertical stabilizer and rudder .... Check
Elevator feel probes .... Check
Tail skid .... Check

Verify that the tail skid is not damaged.

Horizontal stabilizer and elevator .... Check
Static discharge wicks .... Check

Strobe light .... Check
APU cooling air inlet and exhaust outlet .... Check

Left Aft Fuselage

Doors and access panels (not in use) .... Latched
Probes, sensors, ports, vents, and drains (as applicable) .... Check

Left Main Gear

Tires, brakes and wheels .... Check

Verify that the wheel chocks are in place as needed.

If the parking brake is set, the brake wear indicator pins must extend out of the guides.

Gear strut, actuators and doors .... Check
Hydraulic lines .... Secure
Gear pins .... As needed

Left Main Wheel Well

Wheel well .... Check
Engine fire bottle pressure .... Check

Left Wing Tip and Trailing Edge

Aileron and trailing edge flaps .... Check
Static discharge wicks Check
Position and strobe lights .... Check

Left Wing and Leading Edge

Fuel tank vent .... Check
Wing Surfaces .... Check
Fuel measuring sticks .... Flush and secure
Leading edge flaps and slats .... Check
Access panels .... Latched

Number 1 Engine

Exhaust area and tailcone .... Check
Thrust reverser .... Stowed
Fan blades, probes, and spinner .... Check
Probes, sensors, ports, vents, and drains (as applicable) .... Check
Access panels and fan cowl latches .... Latched
Exterior surfaces (including the bottom of the nacelles) ..... Check for damage

Left Wing Root, Pack, and Lower Fuselage

Leading edge flaps .... Check
Probes, sensors, ports, vents, and drains (as applicable) .... Check
Exterior lights .... Check
Pack and pneumatic access doors .... Secure
Ram air deflector door .... Extended

2.12.5. CDU Preflight Procedure – Captain and First Officer

Start the CDU Preflight Procedure any time after the Preliminary Preflight Procedure completed.

The Initial Data and Navigation Data entries must be completed before the flight instrument check during the Preflight Procedure.

The Performance Data entries must be completed before the Before Start Checklist.

The captain or first officer may make CDU entries. The other pilot must verify the entries.

Enter data in all the boxed items on the following CDU pages.

Enter data in the dashed items or modify small font items that are listed in this procedure. Enter or modify other items at pilot's discretion.

Failure to enter enroute winds can result in flight plan time and fuel burn errors.

Initial Data .... Set

IDENT page:

Verify that the MODEL is correct.

Verify that the ENG RATING is correct.

Verify that the navigation data base ACTIVE date range is current.

POS INIT page:

Verify that the time is correct.

Enter the present position on the SET IRS POS line. Use the most accurate latitude and longitude.

Navigation Data .... Set

RTE page:

RTE 1:

Enter the ORIGIN

Enter the route.

Enter the FLIGHT NUMBER (NWS____).

Activate and execute the route.

DEP/ARR page:

DEP

Select the runway and departure routing.

Execute the runway and departure routing.

ARR

Select the runway and arrival routing if needed.

Execute the runway and arrival routing.

Verify that the route is correct on the RTE page. Check the LEGS pages as needed to ensure compliance with the flight plan.

Verify the correct RNP for the departure.

RTE page:

RTE 2:

Set alternate RW for departure, SID, EOSID, route in case of diversion to alternate or return to departure airport (if required and available)

Performance Data .... Set

PERF INIT page:

RA-73269 - RA-73313, RA-73316 - 73321

CAUTION: Do not enter the ZFW into the GW boxes. The FMC will calculate performance data with significant errors.

Enter the ZFW.

Verify that the FUEL on the CDU, the dispatch papers, and the fuel quantity indicators agree.

RA-73269, RA-73314 - 73318, RA-73321

If refueling is not complete, enter the PLAN trip fuel as needed.

Verify that the fuel is sufficient for flight.

Thrust mode display:

RA-73269, RA-73319, RA-73321

Verify that TO shows.

RA-73311 - 73318

Verify that dashes are shown.

N1 LIMIT page:

RA-73314 - 73321

Confirm the OAT value is correct and reasonable for the ambient conditions.

RA-73311 - RA-73318

Enter or verify OAT. Confirm the OAT value is correct and reasonable for the ambient conditions.

TAKEOFF REF page:

Make data entries on page 2/2 before page 1/2.

Verify or enter a thrust reduction, acceleration and engine out acceleration height according to airport requirements or company policy.

Verify that the CDU preflight is complete.

Complete route verification.

2.12.6. Preflight Procedure – First Officer

Start Preflight procedure after CDU Preflight Procedure and route verification are completed.

The First Officer normally does this procedure.

Call: “**PREFLIGHT PROCEDURE**” C

Flight control panel.. Check

FLIGHT CONTROL switches – Guards closed

Verify that the flight control LOW PRESSURE lights are illuminated.

Flight SPOILER switches – Guards closed

YAW DAMPER switch – ON

Verify that the YAW DAMPER light is extinguished.

Verify that the standby hydraulic LOW QUANTITY light is extinguished.

Verify that the standby hydraulic LOW PRESSURE light is extinguished.

Verify that the STBY RUD ON light is extinguished.

ALTERNATE FLAPS master switch – Guard closed

ALTERNATE FLAPS position switch – OFF

Verify that the FEEL DIFF PRESS light is extinguished.

Verify that the SPEED TRIM FAIL light is extinguished.
Verify that the MACH TRIM FAIL light is extinguished.
Verify that the AUTO SLAT FAIL light is extinguished.

NAVIGATION panel .... Set
VHF NAV transfer switch – NORMAL
IRS transfer switch – NORMAL
FMC source select switch – NORMAL

DISPLAYS panel .... Set
SOURCE selector – AUTO
CONTROL PANEL select switch – NORMAL

Fuel panel .... Set
Verify that the ENG VALVE CLOSED lights are illuminated dim.
Verify that the SPAR VALVE CLOSED lights are illuminated dim.
Verify that the FILTER BYPASS lights are extinguished.
CROSSFEED selector – Closed
Verify that the VALVE OPEN light is extinguished.
FUEL PUMP switches – OFF
Verify that the center tank fuel pump LOW PRESSURE lights are extinguished.
Verify that the main tank fuel pump LOW PRESSURE lights are illuminated.

ELECTRICAL panel .... Set
BATTERY switch – Guard closed
CAB/UTIL power switch – ON
IFE/PASS SEAT power switch – ON
STANDBY POWER switch – Guard closed
Verify that the STANDBY PWR OFF light is extinguished.
Verify that the BAT DISCHARGE light is extinguished.
Verify that the TR UNIT light is extinguished.
Verify that the ELEC light is extinguished.
Generator drive DISCONNECT switches – Guards closed
Verify that the DRIVE lights are illuminated.
BUS TRANSFER switch – Guard closed
Verify that the TRANSFER BUS OFF lights are extinguished.
Verify that the SOURCE OFF lights are extinguished.
Verify that the GEN OFF BUS lights are illuminated.

Overheat and fire protection panel .... Check

Do this check if the flight crew did not do the Electrical Power Up supplementary procedure. This check is needed once per flight day. Verify that the engine No. 1, APU, and engine No. 2 fire switches are in.

Alert ground personnel before the following test is accomplished:

OVERHEAT DETECTOR switches – NORMAL
TEST switch – Hold to FAULT/INOP
Verify that the MASTER CAUTION lights are illuminated.
Verify that the OVHT/DET annunciator is illuminated.
Verify that the FAULT light is illuminated.

If the FAULT light fails to illuminate, the fault monitoring system is inoperative. Verify that the APU DET INOP light is illuminated.

Do not run the APU if the APU DET INOP light does not illuminate.

NOTE: The fire warning light flashes and the horn sounds on the APU ground control panel when this test is done with the APU running. This can be mistaken by the ground crew as an APU fire.

TEST switch – Hold to OVHT/FIRE

Verify that the fire warning bell sounds.

Verify that the master FIRE WARN lights are illuminated.

Verify that the MASTER CAUTION lights are illuminated.

Verify that the OVHT/DET annunciator is illuminated.

Master FIRE WARN light – Push

Verify that the master FIRE WARN lights are extinguished.

Verify that the fire warning bell cancels.

Verify that the engine No. 1, APU and engine No. 2 fire switches stay illuminated.

RA-73319

Verify that the engine No. 1 and engine No. 2 start lever lights stay illuminated.

Verify that the ENG 1 OVERHEAT and ENG 2 OVERHEAT lights stay illuminated.

Verify that the WHEEL WELL fire warning light stays illuminated.

EXTINGUISHER TEST switch – Check

TEST switch – Position to 1 and hold.

Verify that the three green extinguisher test lights are illuminated.

TEST switch – Release

Verify that the three green extinguisher test lights are extinguished.

Repeat for test position 2.

APU switch (as needed) .... START

NOTE: If extended APU operation is needed on the ground and the airplane busses are powered by AC electrical power, position an AC powered fuel pump ON. This extends the service life of the APU fuel control unit.

If fuel is loaded in the center tank, position the left center tank fuel pump switch ON to prevent a fuel imbalance before takeoff.

CAUTION: Position the center tank fuel pump switches ON only if the fuel quantity in the center tank exceeds 453 lkg.

CAUTION: Do not operate the center tank fuel pumps with the flight deck unattended.

When the APU GEN OFF BUS light is illuminated:

APU GENERATOR bus switches – ON

Verify that the SOURCE OFF lights are extinguished.

Verify that the TRANSFER BUS OFF lights are extinguished.

NOTE: Run the APU for two full minutes before using it as a bleed air source.

RA-73319

Lavatory SMOKE light .... Verify extinguished

EQUIPMENT COOLING switches .... NORM

Verify that the OFF light is extinguished.

EMERGENCY EXIT LIGHTS switch .... Guard closed

Verify that the NOT ARMED light is extinguished.

PASSENGER SIGNS.. Set

NO SMOKING switch – AUTO or ON

FASTEN BELTS switch – AUTO or ON

Windshield WIPER selectors.. PARK

Verify that the windshield wipers are stowed.

WINDOW HEAT switches... ON

Verify that the OVERHEAT lights are extinguished.

Verify that the ON lights are illuminated (except at high ambient temperatures.)

PROBE HEAT switches... AUTO

Verify that all lights are illuminated (or as needed by Cold Weather Operation SP).

WING ANTI–ICE switch .... OFF

Verify that the VALVE OPEN lights are extinguished.

ENGINE ANTI–ICE switches .... OFF

Verify that the COWL ANTI–ICE lights are extinguished.

Verify that the COWL VALVE OPEN lights are extinguished.

Hydraulic panel.. Set

ENGINE HYDRAULIC PUMPS switches .... ON

Verify that the LOW PRESSURE lights are illuminated.

ELECTRIC HYDRAULIC PUMPS switches .... OFF

Verify that the OVERHEAT lights are extinguished.

Verify that the LOW PRESSURE lights are illuminated.

RA-73319

High altitude landing switch... As needed

Verify that the INOP light is extinguished

Air conditioning panel .... Set

AIR TEMPERATURE source selector – As needed

TRIM AIR switch – ON

Verify that the ZONE TEMP lights are extinguished.

Temperature selectors – As needed

Verify that the RAM DOOR FULL OPEN lights are illuminated.

RECIRC FAN switches – AUTO

Air conditioning PACK switches – AUTO or HIGH

ISOLATION VALVE switch – OPEN

Engine BLEED air switches – ON

APU BLEED air switch – ON

Verify that the DUAL BLEED light is illuminated.

Verify that the PACK lights are extinguished.

Verify that the WING–BODY OVERHEAT lights are extinguished.

Verify that the BLEED TRIP OFF lights are extinguished.

Cabin pressurization panel .... Set

Verify that the AUTO FAIL light is extinguished.

Verify that the OFF SCHED DESCENT light is extinguished.

FLIGHT ALTITUDE indicator – Cruise altitude

LANDING ALTITUDE indicator – Destination field elevation

Pressurization mode selector – AUTO

Verify that the ALTN light is extinguished.

Verify that the MANUAL light is extinguished.

Lighting panel .... Set

RA-73269

All LANDING, TAXI and RUNWAY TURNOFF light switches - OFF

Ignition select switch .... IGN L or R

Alternate the ignition select switch position on subsequent starts.

The right igniter plug receives power from the AC standby bus.

ENGINE START switches .... OFF

Lighting panel .... Set

LOGO light switch .... As needed

POSITION light switch .... As needed

ANTI–COLLISION light switch .... OFF

WING illumination switch .... As needed

WHEEL WELL light switch .... As needed

Mode control panel .... Set

COURSE(S) .... Set

FLIGHT DIRECTOR switch .... ON

Move the switch for the pilot flying to ON first.

EFIS control panel .... Set

MINIMUMS reference selector .... RADIO or BARO

FLIGHT PATH VECTOR switch .... As needed

METERS switch .... As needed

BAROMETRIC reference selector .... IN or HPA

BAROMETRIC selector .... Set local altimeter setting

VOR/ADF switches .... As needed

Mode selector .... MAP

Range selector .... As needed

TRAFFIC switch .... As needed

WEATHER RADAR .... Off

Verify that the weather radar indications are not shown on the MAP.

Map switches .... As needed

NOTE: The oxygen test and set is not needed if the oxygen pressure drop test was done at this crewmember station during the Preliminary Preflight Procedure - Captain or First officer.

RA-73313, RA-73316 - 73319

Oxygen .... Test and set

Oxygen mask .... Stowed and doors closed

TEST/RESET switch .... Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

TEST/RESET switch .... Release

Normal/100% switch 100%

EMERGENCY/TEST selector .... Normal (non-emergency)

RA-73269 - 73312, RA-73314, RA-73315, RA-73321

Oxygen .... Test and set

Oxygen mask .... Stowed and doors closed

TEST/RESET switch .... Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

Release the TEST/RESET switch and rotate the regulator selector to 100%

Clock .... Set

Time/date selector .... UTC time

Display select panel .... Set

MAIN PANEL DISPLAY UNITS selector .... NORM

LOWER DISPLAY UNIT selector .... NORM

TAKEOFF CONFIG light .... Verify extinguished

CABIN ALTITUDE light .... Verify extinguished

Disengage light TEST switch .... Hold to 1

Verify that the A/P light is illuminated steady amber.

Verify that the A/T light is illuminated steady amber.

Verify that the FMC light is illuminated steady amber.

Disengage light TEST switch .... Hold to 2

Verify that the A/P light is illuminated steady red.

Verify that the A/T light is illuminated steady red.

Verify that the FMC light is illuminated steady amber.

Do the Initial Data and Navigation Data steps from the CDU Preflight Procedure and verify that the IRS alignment is complete before checking the flight instruments.

Flight instruments .... Check

Verify that the flight instrument indications are correct.

Verify that only these flags are shown:

- TCAS OFF
- NO VSPD until takeoff V–speeds are selected

Verify that the flight mode annunciations are correct:

- autothrottle mode is blank
- roll mode is blank
- pitch mode is blank
- AFDS status is FD.

Select the map mode.

GROUND PROXIMITY panel .... Check

FLAP INHIBIT switch – Guard closed

GEAR INHIBIT switch – Guard closed

TERRAIN INHIBIT switch – Guard closed

Verify that the GROUND PROXIMITY INOP light is extinguished.

GROUND PROXIMITY SYS TEST .... If required refer to FCOM SP

Landing gear panel .... Set

LANDING GEAR lever – DN

Verify that the green landing gear indicator lights are illuminated.

Verify that the red landing gear indicator lights are extinguished.

AUTO BRAKE select switch... RTO

Verify that the AUTO BRAKE DISARM light is extinguished

ANTISKID INOP light.. Verify extinguished

Engine display control panel .... Set

Verify that the primary and secondary engine indications show existing conditions.

Verify that no exceedance is shown.

Verify that the hydraulic quantity indications do not show RF.

CARGO FIRE panel .... Check

This check is needed once per flight day.

DETECTOR SELECT switches – NORM

TEST switch – Push

Verify that the fire warning bell sounds.

Verify that the master FIRE WARN lights are illuminated

Master FIRE WARN light – Push

Verify that the master FIRE WARN lights are extinguished.

Verify that the fire warning bell cancels.

Verify that the green EXTINGUISHER test lights stay illuminated.

Verify that the FWD and AFT cargo fire warning lights stay illuminated.

Verify that the DETECTOR FAULT light stays extinguished.

Verify that the DISCH light stays illuminated.

RA-73314, RA-73315

HUD system.. As needed

Radio tuning panel.. Set

WARNING: Do not key the HF radio while the airplane is being fueled. Injury to personnel or fire can occur.

Verify that the OFF light is extinguished.

VHF NAVIGATION radios .... Set for departure

Audio control panel .... Set

ADF radios.. Set

WEATHER RADAR control panel .... Set

On CAPT discretion do the Weather Radar Test before the first flight of the day or after the crew or aircraft change. If required refer to FCOM SP for system test.

Transponder panel.................................. If test is required refer to FCOM SP

STABILIZER TRIM override switch .... Guard closed

WARNING: Do not put objects between the seat and the aisle stand. Injury can occur when the seat is adjusted.

Seat.. Adjust

Rudder pedals .... Adjust

Adjust the rudder pedals to allow full rudder pedal and brake pedal movement.

CAUTION: Turn the rudder pedal adjust crank no faster than approximately one turn per second to avoid damage. Do not apply force to the pedals during adjustment.

Seat belt and shoulder harness .... Adjust

Call: “**PREFLIGHT PROCEDURE COMPLITED**”

Do the PREFLIGHT checklist on the captain’s command.

After PREFLIGHT checklist completed Call: “**PREFLIGHT CHECKLIST COMPLETE**”

2.12.7. Preflight Procedure – Captain

The Captain normally does this procedure. The First Officer may do this procedure if needed.

Lights .... Test

Master LIGHTS TEST and DIM switch.. TEST

The fire warning lights are not checked during this test. Use individual test switches or push to test features to check lights which do not illuminate during the light test. Use scan flow to verify that all other lights are flashing or illuminated. Verify that all system annunciator panel lights are illuminated.

Master LIGHTS TEST and DIM switch... As needed

EFIS control panel .... Set

MINIMUMS reference selector .... RADIO or BARO

FLIGHT PATH VECTOR switch .... As needed

METERS switch .... As needed

BAROMETRIC reference selector .... IN or HPA

BAROMETRIC selector .... Set local altimeter setting

VOR/ADF switches .... As needed

Mode selector .... MAP

Range selector .... As needed

TRAFFIC switch .... As needed

WEATHER RADAR .... Off

Verify that the weather radar indications are not shown on the MAP.

Map switches .... As needed

Mode control panel .... Set

COURSE(S) .... Set

FLIGHT DIRECTOR switch .... ON

Move the switch for the pilot flying to ON first.

BANK LIMIT selector .... As needed

Autopilot DISENGAGE bar.. Up

NOTE: The oxygen test and set is not needed if the oxygen pressure drop test was done at this crewmember station during the Preliminary Preflight Procedure - Captain or First Officer.

RA-73311 – 73313, RA-73316 - 73319,

Oxygen .... Test and set

Oxygen mask .... Stowed and doors closed

TEST/RESET switch .... Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

TEST/RESET switch .... Release

Normal/100% switch 100%

EMERGENCY/TEST selector .... Normal (non-emergency)

RA-73269, RA-73314, RA-73315, RA-73321

Oxygen .... Test and set

Oxygen mask .... Stowed and doors closed

TEST/RESET switch .... Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

Release the TEST/RESET switch and rotate the regulator selector to 100%

Clock .... Set

TIME/DATE pushbutton - UTC time

NOSE WHEEL STEERING switch .... Guard closed

Display select panel .... Set

MAIN PANEL DISPLAY UNITS selector........ NORM

LOWER DISPLAY UNIT selector........ NORM

TAKEOFF CONFIG light .... Verify extinguished

CABIN ALTITUDE light .... Verify extinguished

Disengage light TEST switch .... Hold to 1

Verify that the A/P light is illuminated steady amber.

Verify that the A/T light is illuminated steady amber.

Verify that the FMC light is illuminated steady amber.

Disengage light TEST switch .... Hold to 2

Verify that the A/P light is illuminated steady red.

Verify that the A/T light is illuminated steady red.

Verify that the FMC light is illuminated steady amber.

STAB OUT OF TRIM light .... Verify extinguished

Do the Initial Data and Navigation Data steps from the CDU Preflight Procedure and verify that the IRS alignment is complete before checking the flight instruments.

Flight instruments .... Check

Verify that the flight instrument indications are correct.

Verify that only these flags are shown:

- TCAS OFF
- NO VSPD until takeoff V–speeds are selected

Verify that the flight mode annunciations are correct:

- autothrottle mode is blank
- roll mode is blank
- pitch mode is blank
- AFDS status is FD

Select the map mode.

Integrated standby flight display .... Set

Verify that the approach mode display is blank.

Set the altimeter.

Verify that the flight instrument indications are correct.

Verify that no flags are shown.

Standby RMI.. Set

Select either VOR or ADF.

Verify only expected flags are shown.

SPEEDBRAKE lever .... DOWN detent

Verify that the SPEED BRAKE ARMED light is extinguished.

Verify that the SPEED BRAKE DO NOT ARM light is extinguished.

Verify that the SPEEDBRAKES EXTENDED light is extinguished.

Reverse thrust levers .... Down

Forward thrust levers .... Closed

Flap lever .... Set

Set the flap lever to agree with the flap position.

RA-73313, RA-73316 - 73318

Verify that the FLAP LOAD RELIEF light is extinguished.

Parking brake .... Set

Verify that the parking brake warning light is illuminated.

Check Hydraulic Brake Pressure Indicator readings is above 1000 psi.

NOTE: Do not assume that the parking brake can prevent airplane movement. Accumulator pressure can be insufficient.

Engine start levers .... CUTOFF

STABILIZER cutout switches .... NORMAL

RA-73314, RA-73315

HUD system .... As needed

Radio tuning panel .... Set

WARNING: Do not key the HF radio while the airplane is being fueled. Injury to personnel or fire can occur.

Verify that the OFF light is extinguished.

VHF NAVIGATION radios .... Set for departure

Audio control panel... Set

WARNING: Do not put objects between the seat and the aisle stand. Injury can occur when the seat is adjusted.

Seat .... Adjust

Rudder pedals .... Adjust

CAUTION: Turn the rudder pedal adjust crank no faster than approximately one turn per second to avoid damage. Do not apply force to the pedals during adjustment.

Seat belt and shoulder harness .... Adjust

Call: **“PREFLIGHT CHECKLIST”**.

2.12.8. Takeoff Briefing

Pilot flying Call: **“TAKEOFF BRIEFING”.**

After completion of takeoff briefing call: **“TAKEOFF BRIEFING COMPLETE”.**

2.12.9. After Obtaining Loadsheet

Captain	First Officer
Loadsheet data .... Check Check that following data are correct: flight route, flight number, A/C registration number, number of crew members, actual date and time of issue.	
ZFW .... Call Call actual ZFW from loadsheet.	Enter actual ZFW in to CDU **ZFW** .... Enter
CAUTION: DO NOT ENTER THE ZFW INTO THE GW BOXES. THE FMC WILL CALCULATE PERFORMANCE DATA WITH SIGNIFICANT ERRORS.	
Verify the FUEL on the CDU, Loadsheet (account for taxi fuel), OFP and UPPER DU agree. Verify that the fuel is sufficient for flight.	
Gross Weight .... Check Compare the called gross weight with gross weight from the loadsheet. Check that actual weight is equal or less than maximum takeoff weight.	Call actual gross weight from CDU. **Gross Weight** .... Call
Verify that the actual payload reported by the flight attendant is in agreement with the loadsheet.	
CRZ CG .... Call Call minimum of TOMAC and ZFMAC	**CRZ CG** .... Enter

<table>
<tr><th>Captain</th><th>First Officer</th></tr>
<tr><td colspan="2">Performance data (OPT or RWY analysis) .... Compute
Compute performance data using OPT EFB or RWY analysis.
Performance data .... Crosscheck
Crosscheck calculated performance data, CPT and F/O check data using their own sources.</td></tr>
<tr><td>Takeoff Thrust settings .... Call
Call an assumed temperature, or a fixed derate takeoff, or both as needed.</td><td>Select an assumed temperature, or a fixed derate takeoff, or both as needed.
Takeoff Thrust settings .... Select</td></tr>
<tr><td colspan="2">Verify N1 on CDU and OPT are agrees (a minor deviation can be accepted normally less than 1%)</td></tr>
<tr><td>CLIMB thrust .... Verify
Call a full or derated climb thrust as needed.
Verify that entered data are correct.</td><td>CLIMB thrust .... Verify/Select
Use automatically calculated CLB1 or CLB2 derated climb thrust. Select CLB climb thrust manually only for safety or performance reason.</td></tr>
<tr><td>Takeoff flap setting .... Call
Verify that entered data is correct.</td><td>On TAKEOFF REF page 1 verify or enter takeoff flap setting.
FLAPS .... Check / Enter</td></tr>
<tr><td>Takeoff CG .... Call
Call takeoff CG from the loadsheet.
STABILIZER TRIM .... Check
Verify that calculated trim and loadsheet trim data agree.</td><td>On TAKEOFF REF page 1/2 enter takeoff CG setting.
Takeoff CG .... Enter</td></tr>
<tr><td>Takeoff speeds .... Call
Call calculated takeoff speeds.</td><td>On TAKEOFF REF page 1/2 enter takeoff speeds.
V_1, V_R, V_2 .... Enter</td></tr>
<tr><td>MIN ACCEL HT .... Call
Call minimum acceleration height.</td><td>EO ACCEL HT .... Check/Enter</td></tr>
<tr><td colspan="2">NOTE: EO ACCEL HT CANNOT BE LESS THEN MIN ACCEL HT COMPUTED BY OPT.</td></tr>
</table>

2.12.10. Before Start Procedure

Start the Before Start Procedure after paper are on board and signed.

Establish communications with ground handling personnel..............................C

Obtain clearance to pressurize the hydraulic systems, verify if nose gear steering lockout pin is installed.

Call: "**BEFORE START PROCEDURE**" C

Flight deck door .... Closed and locked F/O

Verify that the LOCK FAIL light is extinguished.

Do the CDU Preflight Procedure – Performance Data steps before this procedure.

CDU display .... Set C, F/O

Normally the PF sets TAKEOFF REF page and PM sets the LEG page.

N1 bugs .... Check C, F/O

Verify that the N1 Reference bugs are correct.

MCP .... Set C

AUTOTHROTTLE ARM switch.. ARM

IAS/MACH selector .... Set V_2

Arm LNAV as needed.

Arm VNAV as needed.

Initial heading .... Set

Initial altitude .... Set

Taxi and Takeoff briefings .... Completed PF

Exterior doors .... Verify closed F/O

Flight deck windows .... Closed and locked C, F/O

Fuel panel .... Set F/O

If the center tank fuel quantity exceeds 453 kilograms:

LEFT and RIGHT CENTER FUEL PUMP switches .... ON

Verify that the LOW PRESSURE lights illuminate momentarily and then extinguish.

If the LOW PRESSURE light stays illuminated turn off the CENTER FUEL PUMPS switch.

AFT and FORWARD FUEL PUMPS switches... ON

Verify that the LOW PRESSURE lights are extinguished.

HYDRAULIC panel .... Set F/O

If pushback is needed and the nose gear steering lockout pin is not installed:

WARNING: Do not pressurize hydraulic system A. Unwanted tow bar movement can occur.

System A HYDRAULIC PUMP switches .... OFF

Verify that the system A pump LOW PRESSURE lights are illuminated.

System B electric HYDRAULIC PUMP switch.. ON

Verify that the system B electric pump LOW PRESSURE light is extinguished.

Check Hydraulic Brake Pressure Indicator readings is 2,800 psi minimum.

Verify that the system B pressure is 2,800 psi minimum.

If pushback is not needed, or if pushback is needed and the nose gear steering lockout pin is installed:

Electric HYDRAULIC PUMP switches .... ON

Verify that the electric pump LOW PRESSURE lights are extinguished.

Check Hydraulic Brake Pressure Indicator readings is 2,800 psi minimum.

Verify that the system A and B pressure is 2,800 psi minimum.

ANTI COLLISION light switch .... ON F/O

Trim .... Set C

Check each trim for freedom of movement.

Stabilizer trim .... ___ UNITS

Set the trim for takeoff.

Verify that the trim is in the green band.

Aileron trim .... 0 units

Rudder trim .... 0 units

Transponder .... As needed F/O

Call: **“BEFORE START PROCEDURE COMPLETE”** .... F/O

Call: **“BEFORE START CHECKLIST”** C

Do the “BEFORE START CHECKLIST” F/O

Call: **“BEFORE START CHECKLIST COMPLETE”**

Contact ATC .... F/O

Obtain clearance to start the engines and for pushback/towing (if needed)

2.12.11. Pushback or Towing Procedure

The Engine Start Procedure may be done during pushback or towing.

Establish communications with ground handling personnel...................... C

CAUTION: Do not hold or turn the nose wheel steering wheel during pushback or towing. This can damage the nose gear or the tow bar.

CAUTION: Do not use airplane brakes to stop the airplane during pushback or towing. This can damage the nose gear or the tow bar.

ANTI COLLISION light switch... As needed F/O

Transponder .... As needed F/O

Parking brake .... Set or release C

Set or release as directed by ground handling personnel.

When pushback or towing is complete:

Verify that the tow bar is disconnected C

Verify that the nose gear steering lockout pin is removed...................... C

System A HYDRAULIC PUMPS switches .... As needed ON, F/O

Verify that the system A pump LOW PRESSURE lights are extinguished

Verify that the system A pressure is 2800 psi minimum.

2.12.12. Engine Start Procedure

Captain	First Officer
Announce start sequence for the ground personnel: Call: **“START ENGINE № __ / __”** Normally №2 then №1.	Select the secondary engine display.
Call: **“START ENGINE №2”**	Air conditioning PACK switches ……. OFF Elapsed Time……..……………..……...Start CHRONO………………………………Start ENGINE START switch……...............GRD Call **"START VALVE OPEN”** Verify illumination Start Valve Open Light, verify increase in N2 RPM.
	Call “**OIL PRESSURE**” when Oil pressure is rising.
	Call «**N1 ROTATION**” when N1 rotation is increasing
When N2 is at 25% or when N2 acceleration is less than 1% in approximately 5 seconds (max motoring) and a minimum of 20% N2 Engine start lever..............................IDLE Call “**IDLE**” **CAUTION:** Do not apply rotational force when moving the engine start lever.	If there is no “IDLE” callout from Captain, call: “**25 (MAX MOTORING)**”
Verify EGT indication within 15 seconds after engine start lever rise to idle	Verify fuel flow and EGT indication. Call "**FUEL FLOW, EGT**"
Monitor fuel flow and EGT indications.	
	At 56% N2, verify that the ENGINE START switch moves to AUTO/OFF. If not, move the ENGINE START switch to AUTO/OFF.
	Verify that the START VALVE OPEN alert extinguishes when the ENGINE START switch moves to AUTO/OFF.
	CHRONO …………………………….. Stop Call: **“STARTER CUTOUT”**
	After the engine is stable at idle. Call: **“STABILIZED”** **NOTE:** The engine is stable at idle when the EGT start limit redline is no longer shown.

2.12.13. Before Taxi Procedure

Start “BEFORE TAXI PROCEDURE” when both engines are stabilized at IDLE.

Delay flap extension and flight control check if De/Anti-icing procedure is expected.

Call: **“FLAPS __ , BEFORE TAXI PROCEDURE”** C

Flap lever .... Set takeoff flaps F/O

GENERATOR 1 and 2 switches .... ON F/O

PROBE HEAT switches.. ON F/O

WING ANTI–ICE switch .... As needed F/O

ENGINE ANTI–ICE switches .... As needed F/O

PACK switches .... AUTO F/O

ISOLATION VALVE switch.. AUTO F/O

APU BLEED air switch .... OFF F/O

APU switch.. OFF F/O

ENGINE START switches... CONT F/O

Verify that the ground equipment is clear... BOTH

Flight controls .... Check C

RA-73313, RA-73316 - 73318

Make slow and deliberate inputs, one direction at a time.

Move the control wheel and the control column to full travel in both directions and verify:

- freedom of movement;
- that the controls return to center;

Hold the nose wheel steering wheel during the rudder check to prevent nose wheel movement.

Move the rudder pedals to full travel in both directions and verify:

- freedom of movement;
- that the rudder pedals return to center;

Blank the lower display unit .... F/O

RECALL .... Check C, F/O

Verify that all system annunciator panel lights illuminate and then extinguish.

Update changes to the taxi briefing, as needed. .... C or PF

Call: “**BEFORE TAXI PROCEDURE COMPLETE**” .... F/O

Call: “**BEFORE TAXI CHECKLIST**” C

Do the “BEFORE TAXI CHECKLIST” .... F/O

Call “**BEFORE TAXI CHECKLIST COMPLETE”** .... F/O

2.12.14. Taxi

<table>
<tr><th>Captain</th><th>First Officer</th></tr>
<tr><td colspan="2">Taxi clearance from ATC .... F/O
Obtain an ATC clearance to taxi.
Taxi briefing (if needed) .... Update C</td></tr>
<tr><td colspan="2">Hand signal from ground staff.. Receive
Verify that the pin is shown by ground staff.</td></tr>
<tr><td colspan="2">Clear of obstacles.. Check</td></tr>
<tr><td>Call: “LEFT SIDE CLEAR”</td><td>Call: “RIGHT SIDE CLEAR”
Call Captain for any unsafe obstruction</td></tr>
<tr><td>Call: “TAXI lights ON”
NOTE: Allowed to switch on TAXI and RUNWAY TURNOFF lights by captain</td><td>TAXI and RUNWAY TURNOFF light switches ON</td></tr>
<tr><td colspan="2">If ground/obstruction clearance is in doubt, stop the airplane and call for ATC.</td></tr>
<tr><td>PARKING BRAKE release</td><td></td></tr>
<tr><td>BREAKS.. Check
Apply the brakes smoothly make sure the plane is slowing down.</td><td></td></tr>
</table>

Taxi speeds described in OM part A.

If during taxing "HOLD POSITION" instruction received, after stopping, switch OFF TAXI and RWY TURN OFF lights to give the other aircraft or ground vehicles a clear signal that the aircraft is holding. Switch ON TAXI and RWY TURN OFF lights before resuming taxing.

Runway / Takeoff Data Change

If runway (intersection) or takeoff data (runway condition) has been changed during taxi, all necessary updates should be done before entering the runway. These actions may be done while taxiing, however the significant changes (RW change, departure route change…) should be done when the airplane is fully stopped.

When the new ATC clearance received the F/O should calculate performance data in accordance with the new clearance.

DEPARTURE page .... Set F/O

Select the runway and departure routing.

Execute the runway and departure routing.

Verify that the route is correct on the RTE page. Check the LEGS pages as needed to ensure compliance with the flight plan.

Verify the correct RNP for the departure.

Performance data .... Compute F/O

Compute performance data using OPT EFB or RWY analysis.

Thrust settings .... Select F/O

Select an assumed temperature, or a fixed derate takeoff, or both as needed.

Check/Select a full or derated climb thrust as needed on CDU.

EO ACCEL HT .... Check F/O

On TAKEOFF REF page 2/2 verify that engine out acceleration height is equal or more than MIN ACCEL HT.

ACCEL HT .... Check F/O

THR REDUCTION .... Check F/O

FLAPS .... Check / Enter F/O

On TAKEOFF REF page 1/2 verify or enter takeoff flap setting.

TRIM .... Call F/O

Verify that planned takeoff stab trim setting is in green band .... C, F/O

V_1, V_R, V_2 .... Enter

On TAKEOFF REF page 1/2 enter takeoff speeds.

Verify that data are correct .... C, F/O

When the new CDU data and MCP entries are set, check the airplane configuration agrees with the new data.

The PF should update the takeoff briefing.

2.12.15. Before Takeoff Procedure

Start Before Takeoff Procedure approaching to the takeoff RW and after “CABIN READY FOR TAKEOFF” report is received.

Engine warm up recommendations:

- verify an increase in engine oil temperature before takeoff;
- run the engines for at least 2 minutes;
- use a thrust setting normally used for taxi operations.

Captain	First Officer
Select SID charts	Select SID charts
Call: **”BEFORE TAKEOFF PROCEDURE”**	
	SEAT BELTS switch .... OFF then ON Notify the cabin crew to prepare for takeoff.
Check PFD altimeter setting, current altitude and announce: **“QNH ___”** **“Altitude ___”**	Call:**” ALTIMETERS”** Check PFD altimeter setting, current altitude and announce: **“QNH ___”** **“Altitude ___”**
Set the weather radar display as needed. Set the terrain display as needed. Do not use the terrain display for navigation. If the weather is a factor, both pilots select WXR on ND, if the weather is not a factor CPT selects WXR and FO selects TERR on ND	
	Call**: “BEFORE TAKEOFF PROCEDURE COMPLETE”**
Call: **“BEFORE TAKEOFF CHECKLIST”**.	Do the “BEFORE TAKEOFF CHECKLIST”.
	Call: “**BEFORE TAKEOFF CHECKLIST COMPLETE**”

2.12.16. Takeoff Procedure

Captain	First Officer
Before entering the departure runway, verify that the runway and runway entry point are correct. Entering the departure runway start Takeoff Procedure. Call **”RW ___ IDENTIFIED”**	
	STROBE light switch .... ON Use other lights as needed. **Transponder mode selector** .... TA/RA
Verify that there is no conflicting traffic on short final or on the departure runway.	
Align the airplane with the runway.	
	Takeoff clearance ……………….. Obtain
If First Officer will be Pilot Flying Captain transfers controls to First Officer.	
PF	**PM**
CHRONO Start …………………………………….C	
Call: **“TAKEOFF, LIGHTS ON”**.	
	Landing light switches all .... ON
Advance the thrust levers to approximately 40% N1. Allow the engines to stabilize. Call **“CHECK ENGINE STABILISED”**	
	When engines stabilized Call: **“STABILIZED”**.
TO/GA switch .... Push	
Verify that the correct takeoff thrust is set.	
	Monitor the engine instruments during the takeoff. Call out FMA changes and any abnormal indications. Adjust takeoff thrust before 60 knots as needed. During strong headwinds, if the thrust levers do not advance to the planned takeoff thrust, manually advance the thrust levers before 60 knots. Call: **“TAKEOFF THRUST SET”**. **NOTE:** Thrust should be adjusted, if required, to - 0% + 1% target N1

PF	PM
After takeoff thrust is set, the Captain's hand must be on the thrust levers until V_1.	
Verify 80 knots and call: **"CHECK"**.	Verify 80 knots. Call: **"80 KNOTS"**. Verify FMA change and call: **"THROTTLE HOLD".**
If NO A/T FMA change to **THR HLD**, **Captain** shall disengage A/T and call: **«MANUAL THRUST».**	
During takeoff roll the "**80 KNOTS**" call has three functions: 1. Incapacitation check. 2. It defines the high and low speed rejected takeoff. 3. Airspeed crosscheck to ensure pitot system is working properly.	
Verify V_1 speed.	Verify the automatic V_1 callout or call: **"V1"**.
Captain removes his hand from the thrust levers.	
At V_R, rotate toward 15° pitch attitude. After liftoff, follow F/D commands.	At V_R call: **"ROTATE"**. Monitor airspeed and vertical speed. Special attention should be focused on PFD pitch indication.
Establish a positive rate of climb.	Verify a positive rate of climb on the baro altimeter and call: **"POSITIVE RATE"**.
Verify a positive rate of climb on the baro altimeter and call: **"GEAR UP"**.	
	Call: **"GEAR UP"**. **Landing gear lever** .... UP Verify Landing gear lights off and call: **"GEAR UP"**
	At 400 feet radio altitude, call: **"400"**.
Engage the autopilot (if needed).	When autopilot engaged call **"COMMAND"**
Select, verify or call the roll mode as needed	Verify or select the roll mode if derected
Follow prescribed Noise Abatement Procedure as announce on pre-flight briefing:	
Verify automatic thrust reduction. If A/P and VNAV not engaged call **"N1"**	At thrust reduction altitude, call: **"THRUST REDUCTION"**. Push N1 if directed
Verify climb thrust is set	

PF	PM
Verify acceleration.	At acceleration altitude, call: **“ACCELERATION”**.
If A/P engaged without VNAV: - set the flaps up maneuvering speed and announce: “**FLAPS UP SPEED SET**”, - If A/P not engaged, without VNAV call: **“SET FLAPS UP SPEED”,** Call: **“FLAPS ___”** according to the flap retraction schedule. Verify flaps at directed position and call: “**CHECK**”	Check AFDS operation for flaps up maneuvering speed. Set the flaps up maneuvering speed (if directed). Call **“SPEED CHECKED, FLAPS ___”** **Flap lever**......................... Set as directed Monitor flaps and slats retraction After flaps retraction is set to desired position: Call **“FLAPS ___ SET”**
Call for pitch mode if needed.	After flap retraction is complete. Call **“FLAPS UP, NO LIGHTS”.** Select pitch mode if directed. **Retractable Lights .... OFF** **Engine bleeds** .... Operating **Air conditioning packs** .... Operating **AUTO BRAKE switch**....................... OFF **Engine Start Switches** .... As needed **Landing gear lever**.......................... OFF
Call: **“AFTER TAKEOFF CHECKLIST”**.	Do the “AFTER TAKEOFF CHECKLIST”.
	Call: **“AFTER TAKEOFF CHECKLIST COMPLETE”**.

2.12.17. Climb and Cruise Procedure

CDU display .... Set PF, PM

Normally the PF selects the VNAV or PROG page.

Normally the PM selects the LEGS page.

PF	PM
During climb and cruise, verify the RNP as needed.	
	Passing transition altitude call: **"TRANSITION ALTITUDE"**.
Call: **"SET STANDARD"**. Reset altimeters to standard.	Reset altimeter to standard, crosscheck the altimeters and call: **"STANDARD SET, CROSSCHECKED"**.
NOTE: Captain resets to standard the integrated standby flight display / standby altimeter.	
	At or above 10,000 feet MSL call: **"FL100** or **10,000 FEET"**.
Call: **"LIGHTS OFF"**.	ALL LANDING, TAXI and RUNWAY TURNOFF light switches .... OFF Other lights…………………… As needed
	Notify the cabin crew of cockpit status. SEAT BELTS……………..OFF, then ON
Shoulder harness might be disconnected at this point. Air conditioning, pressurization system, anti-ice switches position and engine parameters should be checked crossing FL100, 200 and 300. Verify that parameters are normal. Call out any abnormal indications.	
Verify altitude and call **"CHECKED"**	1000 ft before MCP selected altitude, call **"ONE THOUSAND TO LEVEL OFF"**
At Top of Climb	
Crosscheck altimeters indication. For RVSM operations all next equipment should operate properly: • two primary altimeters; • one autopilot; • one altitude–reporting transponder; • one altitude–alerting device. Verify level off and proper FMA annunciation.	
Aircraft trimming .... Check Use Primary Rudder Trim Technique for aircraft trimming	**TCAS Airspace Switch** .... BELOW, if installed

PF	PM
SEAT BELTS switch (as needed)....................................... C	
	Check systems: • Electrical, Hydraulic • Pressurization & Air Conditioning • Flight instruments and indicators • Engine parameters • FUEL (every 30 min) fuel T º, FF • Check 121,5 on VHF No 2.
Cruise briefing .... Complete.	
	RTE 2: Copy active route or put escape route or any other route for deviation in case of emergency.
	Flight documents...................... Fill out Fill out flight documents according the OM Chapter A–8 requirements.

Cruise briefing

The following topics are to be discussed, not limiting to:

- Actions in case of engine failure.
- Actions in case of emergency descent.
- MORA on the route over 10,000 feet.
- Weather at alternate aerodromes along the route.
- Escape route into RTE 2 (if applicable).

2.12.18. Descent Procedure

Descent preparation and approach briefing takes approximately 10 minutes so they should be initiated approximately 80–100 miles before top of descent.

Complete the Descent Procedure by 10,000 feet MSL.

CDU display .... Set PF, PM

Normally the PF selects the DESCENT page.

Normally the PM selects the LEGS page.

Losing airspeed can be difficult and may require a level flight segment. For planning purposes, it requires approximately 25 seconds and 2 NM to decelerate from 280 to 250 knots in level flight without speedbrakes. It requires an additional 35 seconds and 3 NM to decelerate to flaps up maneuver speed at average gross weights. Using speedbrakes to aid in deceleration reduces these times and distances by approximately 50%.

Altimeter Bug Setting

<table>
<tr><th>APPROACH TYPE</th><th>MINIMUM FOR</th><th>DESCENT LIMIT STANDARD</th><th>DA / DH / DA(H)/MDA(H) + 50</th></tr>
<tr><td colspan="4">PRECISION</td></tr>
<tr><td rowspan="2">ILS</td><td>ILS / ILS DME</td><td>DA(H)</td><td>Use DA(H)</td></tr>
<tr><td>CAT II / CAT IIIA</td><td>DH RA</td><td>Use DH RA</td></tr>
<tr><td colspan="4">APV (APPROACH PROCEDURE WITH VERTICAL GUIDANCE)</td></tr>
<tr><td>RNAV (GNSS)</td><td rowspan="4">LNAV/VNAV</td><td rowspan="4">DA(H)</td><td rowspan="4">Use DA(H)</td></tr>
<tr><td>RNAV (GPS)</td></tr>
<tr><td>RNP</td></tr>
<tr><td>RNAV</td></tr>
<tr><td colspan="4">Note RNAV (RNP) / RNAV AR AND LPV APPROACHES ARE NOT CERTIFIED.</td></tr>
<tr><td colspan="4">NON-PRECISION</td></tr>
<tr><td>LOC</td><td>LOC / LOC DME</td><td rowspan="7">DA(H)/MDA(H)</td><td rowspan="7">Use DA(H)/MDA(H) + 50</td></tr>
<tr><td>VOR</td><td>VOR / VOR DME</td></tr>
<tr><td>NDB</td><td>NDB / NDB DME / Lctr</td></tr>
<tr><td>RNAV (GPS)</td><td rowspan="4">LNAV</td></tr>
<tr><td>RNAV (GNSS)</td></tr>
<tr><td>RNP</td></tr>
<tr><td>RNAV</td></tr>
</table>

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td colspan="2">ATIS INFORMATION .... Obtain
Obtain weather at destination and alternate.
Analyze sufficient fuel to complete the flight.
EFB .... Prepare
Check landing performance.
Before commencing descent both pilots should have STAR charts on their EFB</td></tr>
<tr><td>Call: “YOU HAVE CONTROL”</td><td>Call: “I HAVE CONTROL”</td></tr>
<tr><td>Before the top of descent, modify the active route as needed for the arrival and approach.
Enter route to alternate into RTE 2 or as required.</td><td></td></tr>
<tr><td>RECALL ....Push</td><td></td></tr>
<tr><td colspan="2">Review the system annunciator lights.</td></tr>
<tr><td colspan="2">LANDING ALTITUDE Verify</td></tr>
<tr><td>Load FMC for approach:
– «DEP/APR» page: APP RWY, STAR, TRANSITION. Verify or enter the correct RNP for the arrival (if required).
– «LEGS» page –Verify LEGS page versus the approach chart. Use ND in PLN mode as necessary. Speed modifications are allowed as long as the maximum published speed is not exceeded.
– «APP REF» page – enter calculated Landing Gross Weight (omit if distance to DEST APT less than 200 nm), select desired Flap Setting for landing, check recommended VREF.
– «DES» page: Verify or adjust descent speed, speed restriction, descent forecast.
– «DES/FORECAST» page – enter the forecast descent wind, ISA deviation, QNH, Trans Level, anti-ice operation.</td><td></td></tr>
<tr><td colspan="2">Set or verify the NAVIGATION RADIOS and COURSES for the approach
Adjusts volume for markers check.</td></tr>
<tr><td>AUTOBRAKE selector........Set
Set to the needed brake setting</td><td></td></tr>
<tr><td colspan="2">RADIO/BARO minimums as needed for the approach........ Set
Insert DA(H) on EFIS control panel.
Use RADIO for CAT II/IIIa Approach.
QNH .... Preselect</td></tr>
</table>

PF	PM
STBY ILS (if needed) ……. SET APP mode	
Call**: “I HAVE CONTROL”**	Call**: “YOU HAVE CONTROL”**
	FMC loading …………………………Verify
Call**: “YOU HAVE CONTROL”**	Call**: “I HAVE CONTROL”**
Do the approach briefing.	

2.12.19. Approach Briefing

Before the start of an instrument approach, the PF should brief the PM of intentions in conducting the approach. Both pilots should review the approach procedure. All pertinent approach information, including minimums and missed approach procedures, should be reviewed and alternate courses of action considered.

Call: **“APPROACH BRIEFING”**.

After completion of approach briefing call: **“APPROACH BRIEFING COMPLETE”**

PF	PM
Call: “**I HAVE CONTROL**”	Call: “**YOU HAVE CONTROL**”
Call: **“DESCENT CHECKLIST”**.	Do the “DESCENT CHECKLIST”. Call: “DESCENT CHECKLIST COMPLETE”.
	Descent Clearance .... Obtain
SEAT BELTS selector ON C	
During the descent, verify the RNP as needed	

2.12.20. Approach Procedure

The Approach Procedure is normally started at transition level.

Complete the Approach Procedure before:

- the initial approach fix; or
- the start of radar vectors to the final approach course; or
- the start of a visual approach.

RA-73269, RA-73314, RA-73315, RA-73319, RA-73321

RA-73319

For a GLS approach, select the appropriate GLS channel.

For an ILS, LOC, BCRS, SDF or LDA approach, select the appropriate localizer frequency.

RA-73269, RA-73314, RA-73315, RA-73319, RA-73321

For all other approaches, select a VOR frequency in both VHF control panels.

PF	PM
During arrival and approach, verify the RNP as needed.	
	At or above 10,000 feet MSL call: **“FL100 or 10,000 FEET”**.
Call: **“LIGHTS ON”**.	
	LANDING, (retractable – OFF), TAXI and RUNWAY TURNOFF light switches ... **ON** **Other lights**……………………..As needed
	Notify the cabin crew of cockpit status. SEAT BELTS……………..OFF, then ON
	At transition level call: **“TRANSITION LEVEL”**.
Call: **“SET QNH___”**. Reset altimeters to QNH.	Reset altimeter to QNH, crosscheck the altimeters and call: **“QNH ___ SET, CROSSCHECKED”**.
Call: **“APPROACH CHECKLIST”**.	Do the “APPROACH CHECKLIST” Call: **“APPROACH CHECKLIST COMPLETE”**.

2.12.21. Flap Configurations for Approach and Landing

Flap Setting for Landing

For normal landings, use 30 or flaps 40. Flaps 30 provides better noise abatement and reduced flap wear/loads. When performance criteria are met, use flaps 40 to minimize landing speed, and landing distance. It is recommended to use flaps 40 for landing if:

- the runway is less than 2500m;
- the braking action is MEDIUM or less;
- when runway visibility (RVR) is less than 550.

Flap Extension

During flap extension, selection of the flaps to the next flap position should be made when approaching, and before decelerating below, the maneuver speed for the existing flap position.

NOTE: If the maneuver speed for the next flap setting is below VREF + wind additive, select the next flap position when approaching the VREF + wind additive speed. Maintain the VREF + wind additive speed rather than slowing to the maneuver speed for the selected flap setting.

Flap Extension Schedule.

Current Flap Position	At Speedtape “Display”	Select Flaps	Command Speed for Selected Flaps
UP	“UP”	1	“1”
1	“1”	5	“5”
5	“5”	15	“15”
15	“15”	30 or 40	(Vref 30 or Vref 40) plus wind additives

2.12.22. ILS or GLS Approach

APP mode should not be selected until:

- the ILS or GLS is tuned and identified;
- the airplane is on an inbound intercept heading;
- both localizer and glide slope pointers appear on the attitude display in the proper position;
- clearance for the approach has been received.

Check for correct crossing altitude, when crossing the final approach point (FAP) or OM.

False glide slope signals can be detected by crosschecking the final approach fix crossing altitude and VNAV path information before glide slope capture. Crosscheck distance to the runway with altitude. The altitude should be approximately 300 feet HAT per NM of distance to the runway for a 3° glide slope.

PF	PM
Initially: • If on radar vectors: • HDG SEL; • Pitch mode (as needed). • If enroute to a fix: • LNAV or other roll mode; • VNAV or other pitch mode.	
Call: **"FLAPS ___"** according to the flap extension schedule. When maneuvering to intercept the localizer, decelerate and extend flaps to 5. Attempt to be at flaps 5 and flaps 5 maneuver speed before localizer capture. Use other Flaps position if needed.	Call: **"SPEED CHECK FLAPS ___"** Set the flap lever as directed. Monitor flaps and slats extension.
When on localizer intercept heading: • Verify that the ILS or GLS is tuned and identified; • Verify that the LOC and G/S pointers are shown.	
	Call: **"ILS (GLS) IDENTIFIED"**
Arm APP mode. VOR/LOC mode might be armed first. Call: **"APPROACH ARMED (VOR/LOC ARMED)"** If a dual channel approach is desired, engage the second autopilot after APP mode is armed.	Use APP mode on ND display if needed. When first positive localizer pointer motion is observed. Call: **"LOCALIZER ALIVE"**.
NOTE: When using LNAV to intercept the final approach course, LNAV might parallel the localizer without capturing it.	
Verify that the localizer is captured. Verify the final approach course heading.	

PF	PM
Runway or go-around heading .... Set Call: **“RW (GO-AROUND) HDG SET”**	
WARNING: Interference with the glideslope signal can result in erroneous AFDS pitch guidance indicated by FMA mode degradation, the autopilot caution message, and removal of the F/D pitch bar. if this occurs, do a go-around unless suitable visual references can be established and maintained.	
Call: • **“GEAR DOWN”** • **“FLAPS 15”**	At glideslope alive (at first positive inward motion of G/S pointer), Call: **“GLIDESLOPE ALIVE”**. **Landing gear lever** .... DN Call: **“SPEED CHECK FLAPS 15”** **Flap lever** .... 15 Monitor flaps and slats extension. Verify that the green landing gear indicator lights are illuminated and call: **“GEAR DOWN”**. When flaps are 15 call: “**FLAPS 15 SET”**
Speedbrake lever .... ARM Verify that the SPEED BRAKE ARMED light is illuminated.	Notify the cabin crew to prepare for landing. **SEAT BELTS switch** .... OFF, then ON **Engine start switches**.................. CONT
At glideslope capture, call: **“FLAPS ___”** as needed for landing. Final flaps extention is recommended 1500 ft AAL or above. Landing C/L should be completed before 1000 ft. AAL	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed. Monitor flaps and slats extension.
Missed approach altitude .... Set	
Call: **“LANDING CHECKLIST”**.	Do the “LANDING CHECKLIST”
	At the final approach point call: **“PASSING FAP, ___ FEET”**.
	At 1,000 ft. AFE call: **“1000”**. If approach meets stabilized approach recommendations call: **“STABILIZED”**. Retractable lights……………………….ON
Perform go-around and missed approach procedure.	Any time if approach does not meet stabilized approach recommendations call: **“UNSTABILIZED, GO-AROUND”**.
	100 ft. above DA(H) call: **“APPROACHING MINIMUMS”**.

PF	PM
At DA(H) call decision: **“CONTINUE”** or **“GO-AROUND, FLAPS 15, CHECK THRUST”**.	At DA(H) call: **“MINIMUMS”**.
If suitable visual reference is established at DA(H), or the missed approach point, disengage the autopilot. Maintain the glidepath for landing.	

2.12.23. Instrument Approach Using VNAV

VNAV should be used only for approaches that have one of the following features:

- a published GP angle on the LEGS page for the final approach segment;
- an RW__ waypoint at the approach end of the runway;
- a missed approach waypoint before the approach end of the runway (for example, MX__).

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td colspan="2">It is not recommended to tune ILS Frequency during non-ILS APP as it can cause irrelevant below glideslope deviation EGPWS alert in some cases such as low temperature conditions, when vertical pass of non-ILS procedure does not consistent with the ILS GS or when visual aids vertical pass indication angle is different from ILS GS angle.</td></tr>
<tr><td>Initially:
• If on radar vectors:
• HDG SEL;
• Pitch mode (as needed).
• If enroute to a fix:
• LNAV or other roll mode;
• VNAV or other pitch mode.</td><td></td></tr>
<tr><td>Call: “FLAPS ___” according to the flap extension schedule.

When maneuvering to intercept the localizer, decelerate and extend flaps to 5. Attempt to be at flaps 5 and flaps 5 maneuver speed before localizer capture.
Use other Flaps position if needed</td><td>Call: “SPEED CHECK FLAPS ___”
Set the flap lever as directed.
Monitor flaps and slats extension</td></tr>
<tr><td colspan="2">When on the final approach course intercept heading for LOC, LOC-BC, SDF, or LDA approaches:
• verify that the localizer is tuned and identified
RA-73314, RA-73315, RA-73319
• verify that the anticipation cue or LOC pointer is shown
RA-73269, RA-73321
• verify that the LOC pointer is shown.</td></tr>
</table>

PF	PM
Select LNAV or arm the VOR/LOC mode.	Verify that the LNAV or VOR/LOC mode is armed.
WARNING: When using LNAV to intercept the final approach course, LNAV might parallel the localizer without capturing it. The airplane can then descend on the glidepath with the localizer not captured.	
Verify that LNAV is engaged or that VOR/LOC is captured. **RW (GO-AROUND) heading**.............. Set Call: **“RW (GO-AROUND) HDG SET”**	
Approximately 3 NM before the final approach fix: • set DA(H) or MDA(H) on the MCP • select or verify VNAV • verify VNAV PTH on FMA • select or verify speed intervention, as needed	Approximately 3 NM before the final approach fix, call: **“APPROACHING GLIDE PATH”**.
Approximately 2 NM before the final approach fix Call: • **“GEAR DOWN”** • **“FLAPS 15”**	**Landing gear lever** .... DN Call: **“SPEED CHECK FLAPS 15”** **Flap lever** .... 15 Monitor flaps and slats extension Verify that the green landing gear indicator lights are illuminated and call: **“GEAR DOWN”**. When flaps are 15 call: “**FLAPS 15 SET”**
Speedbrake lever .... ARM Verify that the SPEED BRAKE ARMED light is illuminated.	Notify the cabin crew to prepare for landing. **SEAT BELTS switch** .... OFF, then ON **Engine start switches**................... CONT
Beginning the final approach descent call: **“FLAPS ___”** as needed for landing. Final flaps extention is recommended 1500 ft AAL or above. Landing C/L should be completed before 1000 ft. AAL	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed. Monitor flaps and slats extension
Call: **“LANDING CHECKLIST”**.	Do the “LANDING CHECKLIST”.
	At the final approach fix call: **“PASSING FAF, ___ FEET”**.

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td colspan="2">At the final approach fix,
• verify the crossing altitude.
• crosscheck the altimeters. Verify they agree within 100 feet.
NOTE: PM should give this callout over FAF. If no radio navaid or distance from relevant navaid is established for FAF verification, use distance to the RW, the altitude should be approximately 300 feet per NM plus RW elevation for normal 3-degree approach profile.</td></tr>
<tr><td>When at least 300 feet below the missed approach altitude or FAF.

Missed approach altitude.................. Set</td><td></td></tr>
<tr><td></td><td>At 1,000 ft. AAL call: “1000”
If approach meets stabilized approach recommendations call: “STABILIZED”.
Retractable lights……………………….ON</td></tr>
<tr><td>Perform go-around and missed approach procedure.</td><td>Any time if approach does not meet stabilized approach recommendations call: “UNSTABILIZED, GO-AROUND”.</td></tr>
<tr><td></td><td>100 ft. above DA(H)(MDA(H)+50’) call: “APPROACHING MINIMUMS”.</td></tr>
<tr><td>At DA(H)(MDA(H)+50’) call decision: “CONTINUE” or “GO-AROUND, FLAPS 15, CHECK THRUST”.</td><td rowspan="2">At DA(H)(MDA(H)+50’) call: “MINIMUMS”.</td></tr>
<tr><td>If suitable visual reference is established at DA(H)(MDA(H)+50’), or the missed approach point, disengage the autopilot.
Maintain the glidepath for landing.</td></tr>
</table>

2.12.24. Instrument Approach Using V/S

PF	PM
It is not recommended to tune ILS Frequency during non-ILS APP as it can cause irrelevant below glideslope deviation EGPWS alert in some cases such as low temperature conditions, when vertical pass of non-ILS procedure does not consistent with the ILS GS or when visual aids vertical pass indication angle is different from ILS GS angle.	
Initially: • If on radar vectors: • HDG SEL; • Pitch mode (as needed). • If enroute to a fix: • LNAV or other roll mode; • VNAV or other pitch mode.	
Call: **“FLAPS ___”** according to the flap extension schedule. When maneuvering to intercept the final approach course, decelerate and extend flaps to 5. Attempt to be at flaps 5 and flaps 5 maneuver speed before final approach course. Use other Flaps positions if needed.	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed. Monitor flaps and slats extension
When on the final approach course intercept heading for LOC, LOC-BC, SDF, or LDA approaches: • verify that the localizer is tuned and identified **RA-73314, RA-73315, RA-73319** • verify that the anticipation cue or LOC pointer is shown **RA-73269, RA-73321** • verify that the LOC pointer is shown.	
Ensure appropriate navaids (VOR, LOC, or NDB) are tuned and identified before commencing the approach.	
Use LNAV or other roll mode to intercept the final approach course as needed.	
	Call: **“LOCALIZER ALIVE”**, as needed.
Select LNAV or arm the VOR/LOC mode.	Verify that the LNAV or VOR/LOC mode is armed.
WARNING: When using LNAV to intercept the final approach course, LNAV might parallel the localizer without capturing it. The airplane can then descend on the glidepath with the localizer not captured.	
Verify that LNAV is engaged or that VOR/LOC is captured.	
Approximately 3 NM before the final approach fix, set the first intermediate altitude constraint or MDA(H)+50`. Arm or verify V/S	Approximately 3 NM before the final approach fix, call: **“APPROACHING GLIDE PATH”**.

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td>Approximately 2 NM before the final approach fix call:
• “GEAR DOWN”
• “FLAPS 15”</td><td>Landing gear lever .... DN
Call: “SPEED CHECK FLAPS 15”
Flap lever 15
Monitor flaps and slats extension
Verify that the green landing gear indicator lights are illuminated and call: “GEAR DOWN”.
When flaps are 15 call: “FLAPS 15</td></tr>
<tr><td>Speedbrake lever .... ARMED
Verify that the SPEED BRAKE ARMED light is illuminated.</td><td>Notify the cabin crew to prepare for landing.
SEAT BELTS switch .... OFF, then ON
Engine start switches .... CONT
SET”</td></tr>
<tr><td>Beginning the final approach descent call: “FLAPS ___” as needed for landing.
Final flaps extention is recommended 1500 ft AAL or above.
Landing C/L should be completed before 1000 ft. AAL</td><td>Call: “SPEED CHECK FLAPS ___”
Set the flap lever as directed.
Monitor flaps and slats extension</td></tr>
<tr><td colspan="2">At descent point if no radio navaid or distance from relevant navaid is established for FAF verification, use distance to the RW, the altitude should be approximately 300 feet per 1 NM plus RW elevation for normal 3-degree approach profile.</td></tr>
<tr><td>Desired V/S... Set
Verify V/S mode annunciates.
Set desired V/S to descend to MDA(H)+50`.</td><td>Verify V/S mode annunciates.</td></tr>
<tr><td>Call: “LANDING CHECKLIST”.</td><td>Do the “LANDING CHECKLIST”.</td></tr>
<tr><td colspan="2">At the final approach fix (OM),
• verify the crossing altitude.
• crosscheck the altimeters. Verify they agree within 100 feet.</td></tr>
<tr><td>Approximately 300 feet above MDA(H)+50`
Missed approach altitude.................. Set</td><td></td></tr>
<tr><td></td><td>At 1,000 ft. AAL call: “1000”.
If approach meets stabilized approach recommendations call: “STABILIZED”.
Retractable lights……………………….ON</td></tr>
<tr><td>Perform go-around and missed approach procedure.</td><td>Any time if approach does not meet stabilized approach recommendations call: “UNSTABILIZED, GO-AROUND”.</td></tr>
</table>

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td></td><td>100 ft. above MDA(H) call:
“APPROACHING MINIMUMS”.</td></tr>
<tr><td>At MDA(H)+50` call decision:
“CONTINUE” or “GO-AROUND, FLAPS 15, CHECK THRUST”.</td><td>At MDA(H) call: “MINIMUMS”.</td></tr>
<tr><td colspan="2">At MDA(H)+50` / missed approach point:
If suitable visual reference is not established, execute missed approach.</td></tr>
<tr><td>If suitable visual reference is established disengage A/P and A/T</td><td></td></tr>
<tr><td>Call: “RECYCLE FD’s” or as needed.
Call “SET RUNWAY (GO-AROUND) HEADING”.</td><td>Turn OFF both flight directors and then place F/D ON (if called by PF).
RW (GO-AROUND) heading .... Set
Call: “RW (GO-AROUND) HDG SET”</td></tr>
<tr><td>Maintain the glidepath to landing.</td><td></td></tr>
</table>

2.12.25. Circling Approach

The circling approach should be flown with landing gear down, flaps 15, and at flaps 15 maneuver speed. Use the weather minima associated with the anticipated circling speed.

The circling approach may be flown following any instrument approach procedure. During the instrument approach, use VNAV or V/S modes to descend to the circling MDA(H).

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td colspan="2">Autopilot use is recommended until intercepting the landing profile.
PREPARATION:
Select the instrument approach procedure from the arrivals page of the FMC.
To enable proper VNAV path verify/enter appropriate altitude constraints for desired waypoints, but not below OCA as for instrument approach.
Tune and identify navaids for instrument approach. If additional waypoint references are desired, use the FIX page.
Verify/enter the appropriate RNP.
Set up MCP Courses and EFIS control panel for instrument approach.
Set the MDA(H) for circling using the baro minimums selector.
Do not use APPROACH mode.</td></tr>
</table>

PF	PM
STABILIZED APPROACH RECOMMENDATION: • Correct landing configuration should be established not later than 500 feet AFE. • Wings should be level on final when the airplane reaches 300 feet AFE.	
PF	**PM**
Accomplish an appropriate instrument approach. **Make final descent with gears down, flaps 15 and armed speedbrake (flaps 30 under TERPS)**	
	100 ft. above MDA(H) call: **“APPROACHING MINIMUMS”**.
Level off at MDA(H). Verify ALT HOLD or VNAV ALT on FMA. **Missed approach alt** on MCP – Set If suitable visual reference is established at circling MDA(H), call: **“CONTINUE CIRCLE”**. Select HDG SEL for the maneuvering portion of the approach. If suitable visual reference is not established or anytime when it’s lost, call: **“GO-AROUND, FLAPS 15, CHECK THRUST”**. Turn towards the landing runway. This may involve a turn of more than 180 degrees	At MDA(H) call: **“MINIMUMS”**. Verify ALT HOLD or VNAV ALT on FMA. **Missed approach alt** on MCP – Check
Before starting the turn to base call:	
“FLAPS ___” as needed for landing. Make a turn for DOWNWIND leg.	Call: **“SPEED CHECK FLAPS ___”** Monitor flaps and slats extension.
Heading for runway. Select or verify V/S is armed Set initial desired V/S Call: **“LANDING CHECKLIST”**.	Do the the “LANDING CHECKLIST”.
When intercepting the visual landing profile: • disengage the autopilot • disconnect the autothrottle • continue the approach manually	
Call: **“RECYCLE FD’s”**. Call **“SET RUNWAY (GO-AROUND) HEADING”**.	Turn OFF both flight directors and then place F/D ON. **RW heading (GO-AROUND)**.............. Set

PF	PM
Call: “**CONTINUE**”. Maintain the glide path to landing.	At 500 ft. AAL call: **“500 FEET”**. If approach meets stabilized approach recommendations, except final track and bank call: **“STABILIZED”**. Retractable lights……………………….**ON**
Perform go-around and missed approach procedure.	Any time if approach does not meet stabilized approach recommendations call: **“UNSTABILIZED, GO-AROUND”**
	At 300’ AAL verify all stabilized approach criteria are met.

2.12.26. Visual Approach

GENERAL

Before executing a visual approach assure that:

- the visual reference can be maintained during the entire approach;
- ATC clearance is obtained;
- a short progressive briefing of the intended flight path in addition to the IFR approach briefing has been performed;
- actual weather at the airport must be:
  - Clouds in excess of 2 octas are at or above 2000ft (600m),
  - flight visibility 5 km or greater;
  - surface in sight.

PF	PM
Autopilot use is recommended until intercepting the landing profile.	
Entering downwind leg call: **“FLAPS ___”** according to the flap extension schedule.	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed
Descend to 1,500 ft. AAL.	
Level off at 1,500 ft. AAL. Verify ALT HOLD mode annunciates.	
Missed approach altitude .... Set	
Abeam the landing runway threshold call: **“START TIME”**	**Chronograph** .... Start
	In 20 seconds call: **“20 SECONDS”**.
Call: • **“GEAR DOWN”** • **“FLAPS 15”**	

PF	PM
Speedbrake lever .... ARM	**Landing gear lever** .... DN Call: **“SPEED CHECK FLAPS 15”** **Flap lever** .... 15 Monitor flaps and slats extension. Notify the cabin crew to prepare for landing. **SEAT BELTS switch** .... OFF, then ON **Engine start switches**.................. CONT Verify that the green landing gear indicator lights are illuminated and call: **“GEAR DOWN”**. When flaps are 15 call: “**FLAPS 15 SET”**
While turning base leg, if autopilot is engaged: **V/S** .... Set Set V/S to descend at approximately 600–700 fpm.	
While turning base leg, if autopilot is not engaged call “**SET HEADING ___”** Heading to threshold Start descend at approximately 600–700 fpm	Heading to threshold**...........................** Set Call “**HEADING ___ SET”**
Before turning final. Call: **“FLAPS ___”** as needed for landing. Set or call for appropriate speed on MCP.	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed Monitor flaps and slats extension.
Call: **“LANDING CHECKLIST”**.	Do the “LANDING CHECKLIST”.
Intercept landing profile. Disengage autopilot.	
Call: **“RECYCLE FD’s”**. Call **“SET RW (GO-AROUND) HEADING**” or call for special heading, if required.	Turn OFF both flight directors and then place F/D ON. **RW (GO-AROUND) heading** .... Set Call: **“RW (GO-AROUND) HDG SET**”
Maintain the glidepath to landing.	
	At 500 ft. AAL call: **“500 FEET”**. If approach meets stabilized approach recommendations call: **“STABILIZED”**. Retractable lights………………………**ON**
Perform go-around and missed approach procedure.	Any time if approach does not meet stabilized approach recommendations call **“UNSTABILIZED, GO-AROUND”**.

2.12.27. Go–Around and Missed Approach Procedure

<table>
<tr><th>PF</th><th>PM</th></tr>
<tr><td>At the same time:
TO/GA switch .... Push
Call: “GO-AROUND, FLAPS 15, CHECK THRUST”.</td><td></td></tr>
<tr><td></td><td>Flap lever 15
Monitor flaps and slats retraction</td></tr>
<tr><td colspan="2">Initiate rotation towards 15 ° of pitch to get a positive rate of climb, then follow the Flight Director.
When near the ground, avoid excessive rotation rate in order to prevent a tail strike.
Verify:
• the rotation to go-around attitude (or rotate manually if A/P is not in use);
• that the thrust increases (or increase thrust manually if A/T is not in use).</td></tr>
<tr><td></td><td>Verify that the thrust is sufficient for the go-around or adjust as needed and call: “THRUST SET”</td></tr>
<tr><td>Verify a positive rate of climb on the baro altimeter and call: “GEAR UP”.</td><td>Verify a positive rate of climb on the baro altimeter and call: “POSITIVE RATE”.

Call: “GEAR UP”.
Landing gear lever .... UP
Verify Landing gear lights off
and call: “GEAR UP”</td></tr>
<tr><td>Engage the autopilot (if needed)
Engage A/T (if needed)</td><td>At 400ft AAL, call “400 FEET”.
Check Missed Approach Alt
Check CMD on PFD, and call: “COMMAND”.</td></tr>
<tr><td>Call for a roll mode.</td><td>Set appropriate roll mode</td></tr>
<tr><td colspan="2">Verify that the missed approach route is tracked</td></tr>
<tr><td colspan="2">NOTE: Route discontinuities after the missed approach point will prevent the TO/GA to LNAV function from engaging.</td></tr>
<tr><td colspan="2">The minimum altitude for flap retraction during a normal takeoff is not normally applicable to a missed approach procedure. However, obstacles in the missed approach flight path must be taken into consideration. Initiate flap retraction not lower then1000 ft AAL.</td></tr>
<tr><td>Call: “FLAPS ___” according to the flap retraction schedule.</td><td>At acceleration altitude , call: “ACCELERATION”.
Call “SPEED CHECKED, FLAPS ___”
Monitor flaps and slats retraction
Call “FLAPS ___ SET”</td></tr>
</table>

PF	PM
After flap retraction is complete (or desired flaps position).	
If A/P not engaged, call **“VNAV”** or **“LVL CHG, set speed ____”.** If A/P engaged, push VNAV or LVL CHG switch and announce: **“VNAV”** or **“LVL CHG”.**	Call **“FLAPS UP, NO LIGHT (FLAPS __)”** Push the VNAV or LVL CHG switch. If LVL CHG engaged, set requested speed **Engine bleeds**.......................... Operating **Air conditioning packs**............ Operating **AUTO BRAKE switch** .... OFF **Engine Start Switches**........... As needed **Landing gear lever** .... OFF
Verify that climb thrust is set.	
Verify that the missed approach altitude is captured.	
Call: **“AFTER TAKEOFF CHECKLIST”**.	Do the “AFTER TAKEOFF CHECKLIST”.

NON TO/GA GO–AROUND PROCEDURE (REJECTED APPROACH)

If a go-around is needed when both localizer and glideslope are armed, the approach mode can be disarmed by selecting APP, VOR/LOC, LNAV, or VNAV.

If a go-around is needed at altitude above 1,500 ft AAL, when VOR/LOC and / or G/S are active modes, use the following procedure to avoid high rates of climb and high thrust settings.

PF	PM
After VOR/LOC and G/S are both captured, the APP mode can be disengaged by: • pushing a TO/GA switch (do not use for this procedure); • disengaging A/P and turning off both F/D switches • retuning a VHF NAV receiver.	
Call **“ GO-AROUND, FLAPS__, CHECK THRUST”** Call **“Change ILS”** Call **HDG SEL** or **LNAV** Call **ALT HLD** or **LVL CHG** Engage Autopilot (if needed)	Verify that the thrust is sufficient for the go-around. VHF NAVIGATION radios .... retune Transfer frequency from STANDBY to ACTIVE position on both VHF Navigation Control panel. Push HDG SEL or LNAV Push ALT HLD or LVL CHG Check CMD on PFD, and call: “**COMMAND”**.
Verify that the missed approach altitude is captured.	
Call: **“AFTER TAKEOFF CHECKLIST”**.	Do the “AFTER TAKEOFF CHECKLIST”.

2.12.28. Landing Roll Procedure

PF	PM
Control the airplane manually. (If an autoland was accomplished, disengage the autopilot after touch down)	
Verify that the thrust levers are closed. Verify that the SPEEDBRAKE lever is UP. If speedbrakes not UP, extend Speedbrake Lever manually. Without delay, fly the nose wheel smoothly onto the runway.	Verify that the SPEEDBRAKE lever is UP. Call: **“SPEEDBRAKES UP”**. If the SPEEDBRAKE lever is not UP, call: **“SPEEDBRAKES NOT UP”**.
Verify correct autobrake operation	
Manual braking .... Use as needed Call: **"MANUAL BRAKING"**	Verify autobrake operation. When autobrake disengages, call: **“AUTOBRAKE DISARM”**. Monitor RW length and ground speed.
WARNING: After the reverse thrust levers are raised, a full stop landing must be made. If an engine remains in reverse, safe flight is not possible.	
Without delay, raise the reverse thrust levers to the interlocks and hold light pressure until the interlocks release. Apply reverse thrust as needed.	Verify that the forward thrust levers are closed. When both REV indications are green, call: **“REVERSERS NORMAL”**. If there is no REV indication(s) or the indication(s) stays amber Call: **“NO REVERSER ENGINE №1”** or **“NO REVERSER ENGINE№2”**, or **“NO REVERSERS”**.
By 60 knots, start movement of the reverse thrust levers to be at the reverse idle detent before taxi speed.	Call: **“60 KNOTS”**.
After the engines are at reverse idle, move the reverse thrust levers to full down.	
If captain was PM, transfer controls to captain call: **“I HAVE CONTROL”**	Call: **“YOU HAVE CONTROL”**

2.12.29. After Landing Procedure

Start the After Landing Procedure when clear of the active runway and receive ATC instruction.

After Landing:

- ensure taxi instructions are clearly understood, especially when crossing closely spaced parallel runways;
- delay non–essential radio or cabin communications until clear of all runways.

Engine cooldown recommendations:

- Run the engines for at least 3 minutes;
- Use a thrust setting normally used for taxi operations;
- Routine cooldown times less than 3 minutes are not recommended.

Captain	First Officer
When clear of the active runway and ATC instructions are received. Call: “**AFTER LANDING PROCEDURE**”	
Stop clock (flight time). **SPEEDBRAKE lever DOWN** Set or verify **WEATHER RADAR** (EFIS Control Panel) .... OFF	**Chronograph**.................................... Start **SPEEDBRAKE lever DOWN**.......... Verify **PROBE HEAT switches**........ AUTO/OFF **ENGINE ANTI–ICE** .... As required If used during landing, leave it ON until parked if OAT is below +10°C. **LANDING Lights** .... OFF **TAXI Light** .... ON **APU**... START **NOTE** If long taxi is expected, consider to start APU prior to entering the stand. **ENGINE START switches** .... OFF Leave switches CONT until ENGINE ANTI-ICE is ON **POSITION lights switch** .... STEADY **WEATHER RADAR** (EFIS Control Panel and Weather Radar Panel) .... OFF **AUTOBRAKE selector** .... OFF **FLAP lever** .... UP or as needed. **Transponder** .... XPNDR or ALT ON Call: “**AFTER LANDING PROCEDURE COMPLETE**”
ABSOLUTELY NO PAPER WORK UNTIL PARKED. F/O assist the Captain by reading the Taxi Chart.	
	When APU GEN OFF BUS (blue) illuminated call **“APU AVAILABLE”**
Taxi to the parking stand or gate. Call **“APU ON BUS”**	**APU GENERATOR Switch** .... ON Verify that the SOURCE OFF Lights are extinguished
Before entering the parking gate (stand)	

Captain	First Officer
Call: **"LEFT SIDE CLEAR"**	Call: **"Stand ___"** or verify VGSD **"Boeing 737 identified"** Call: **"RIGHT SIDE CLEAR"**
Turning into parking stand call: **"TAXI LIGHTS OFF"**.	**TAXI and RUNWAY TURNOFF light switches** .... OFF

2.12.30. Shutdown Procedure

Start the Shutdown Procedure after taxi is complete.

Call "**SHUTDOWN PROCEDURE**" C

Parking brake .... Set C

Electrical power .... Set F/O

If APU power is needed:

Verify that the APU GENERATOR OFF BUS light is illuminated.

APU GENERATOR Switch .... ON

Verify that the SOURCE OFF Lights are extinguished.

If external power is needed:

Verify that the PRIMARY GRD POWER AVAILABLE light is illuminated.

GRD POWER switch .... ON

Verify that the SOURCE OFF lights are extinguished.

If ENGINE ANTI–ICE and/or WING ANTI–ICE used during taxi in

ENGINE ANTI-ICE switches ....OFF F/O

Verify that the COWL VALVE OPEN lights illuminate bright, then extinguish.

ENGINE START switches ....OFF F/O

WING ANTI-ICE switch ....OFF F/O

Verify that the L and R VALVE OPEN lights illuminate bright, then dim.

Engine start levers .... CUTOFF C

Stop clock (block time) .... STOP FO

If towing is needed:

Ground handling personnel .... Establish communications C

WARNING: If the nose gear steering lockout pin is not installed and hydraulic system A is pressurized, any change to electrical or hydraulic power with the tow bar connected can cause unwanted tow bar movement.

Verify that the nose gear steering lockout pin is installed, or, if the nose gear steering lockout pin is not used C

System A HYDRAULIC PUMP switches .... OFF

Verify that the system A pump LOW PRESSURE lights are illuminated.

CAUTION: Do not hold or turn the nose wheel steering wheel during pushback or towing. This can damage the nose gear or the tow bar.

CAUTION: Do not use airplane brakes to stop the airplane during pushback or towing. This can damage the nose gear or the tow bar.

Parking brake .... Set or release C or F/O

Set or release as directed by ground handling personnel.

When aircraft is parked on stand:

Call PA: **"PARKING POSITION"**.. C

FUEL PUMP switches .... OFF F/O

CAB/UTIL power switch .... As needed F/O

IFE/PASS SEAT power switch .... As needed F/O

HYDRAULIC panel .... Set F/O

ENGINE HYDRAULIC PUMPS switches.. ON

ELECTRIC HYDRAULIC PUMPS switches.. OFF

Air conditioning PACK switches .... AUTO F/O

ISOLATION VALVE switch .... OPEN F/O

Engine BLEED air switches .... ON F/O

APU BLEED air switch.. ON F/O

Operate the APU for two full minutes before using it as a bleed air source.

Exterior lights switches.. As needed F/O

ANTI COLLISION light switch .... OFF F/O

Anti-collision OFF when N1<5%

FLIGHT DIRECTOR switches .... OFF F/O

MCP SPEED.. 100 F/O

FUEL FLOW Switch ....RESET F/O

Note the amount of fuel used.

STAB TRIM .... 5 units C

Transponder mode selector .... 2000 then STBY F/O

EFB CLOSE FLIGHT .... Select C, F/O

After information is received from ground personal that wheel chocks are in place:

Parking brake .... As needed C

APU switch .... As needed F/O

When cabin report **"DOORS IN DISARMED POSITION"** received.

FASTEN SEAT BELTS switch.. OFF C

Call: **"SHUTDOWN PROCEDURE COMPLETE"** C

Call: **"SHUTDOWN CHECKLIST"** C

Do the "SHUTDOWN CHECKLIST"

Call: "**SHUTDOWN CHECKLIST COMPLETE**"

After paper work completed F/O

FLIGHT DECK ACCESS SYSTEM switch .... OFF C or F/O

2.12.31. Secure Procedure

Call: "**SECURE PROCEDURE**".. C

Secure procedure may be done by read and do of secure checklist. Perform this procedure when:

- all passengers have left the aircraft;
- last flight of the day completed or in transit airport if leaving the aircraft for a long period of time;
- crew change.

During Hot or Cold weather operation use Supplementary Procedures for secure procedure.

EFB Display .... Off C, F/O

IRS mode selectors .... OFF F/O

EMERGENCY LIGHTS switch .... OFF F/O

WINDOW HEAT switches... OFF F/O

Air conditioning PACK switches .... OFF F/O

Call: **"SECURE CHECKLIST"** C

Do the SECURE CHECKLIST F/O

Call: "**SECURE CHECKLIST COMPLETE**"..F/O

2.12.32. Postflight External Inspection

After each flight the Captain must ensure that the aircraft has no visible damage, and surface is clean. Use standard Exterior Inspection procedure described previously in this chapter.

Write down remarks in ATL after exterior inspection any event that the pilot feels a maintenance inspection could be needed.

NOTE: If in doubt, the best course of action is to report it.

2.12.33. Electrical Power Down

This procedure should be accomplished every time the airplane is left unattended by qualified personnel.

This procedure assumes the Secure procedure is complete.

APU switch and/or GRD POWER switch .... OFF C or F/O

If APU was operating delay approximately 2 minutes after the APU GEN OFF BUS light extinguishes before placing the BATTERY switch OFF.

BATTERY switch .... OFF C or F/O

2.13. LOW VISIBILITY OPERATIONS

2.13.1. General

Low visibility operations consist of operating an aircraft in low visibility conditions. The term low visibility operations includes low visibility taxi, low visibility takeoff and lower than standard CATI landing.

Low visibility procedures (LVP) means procedures applied at an aerodrome for the purpose of ensuring safe operations during lower than standard CATI approaches and low visibility takeoffs. Low visibility procedures exist to support Low visibility operations at aerodromes when either surface visibility is sufficiently low to prejudice safe ground movement without additional procedural controls or the prevailing cloudbase is sufficiently low to preclude pilots obtaining the required visual reference to continue to landing at the equivalent of an ILS CATI DH/DA. It should be noted that in the latter case, surface visibility may be relatively good but the TWR visual control room may be in cloud/fog.

CREW DUTIES DURING LOW VISIBILITY OPERATIONS

- Captain – PF;
- First Officer – PM.

In order to operate below CAT I, the crew (both CPT and F/O) must be CAT II and / or CAT IIIA qualified.

SEAT POSITION

The correct seat adjustment is essential in order to have maximum view and visual segment.

The pilots must realize the importance of eye position during low visibility approaches and landing. A too-low seat adjustment may greatly reduce the visual segment. When the eye reference position is lower than intended, the already short visual segment is further reduced by the cut-off angle of the glare shield or nose.

CAUTION: Prior to approach, care should be taken to adjust the seat position to achieve the correct eye position.

USE OF LIGHTS

Adjust cockpit lights as necessary.

NOTE: Secondary images might occur on the flight deck windows during low visibility night operations.

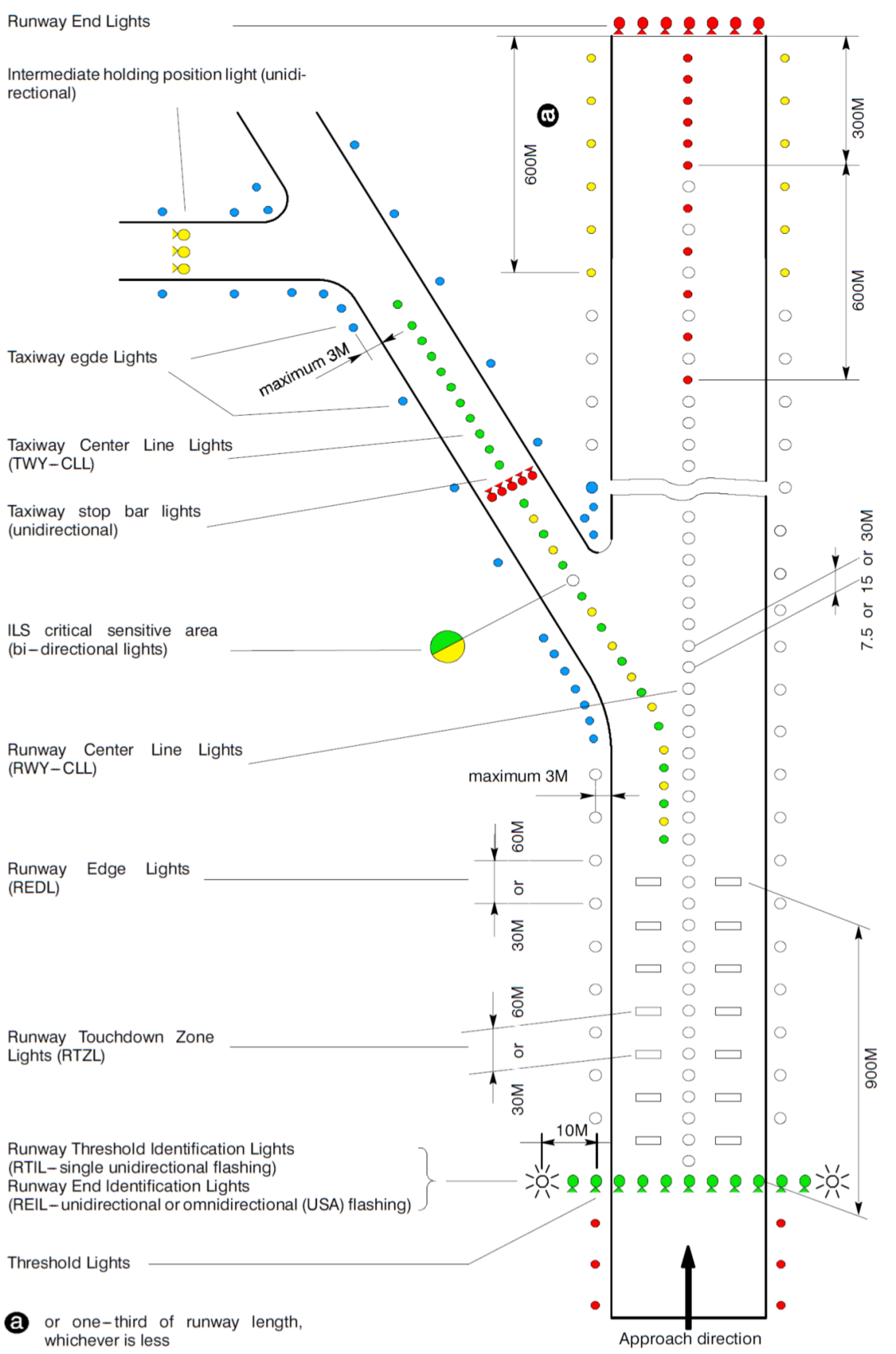
At night in low visibility conditions, landing lights can be detrimental to the acquisition of visual references.

NOTE: Reflected light from water droplets or snow may actually reduce visibility, landing lights would therefore not normally be used in CATII / III weather conditions.

2.13.2. Runway Lights System

Standard runway markings (day) and lights (night) include: RWY edge lights, RWY threshold lights, RWY end lights and TDZ lights and/or RWY center lights.

During night or any other operations when approach lights and RWY lights should operate, they should be switched ON and should be without faults.


Runway End Lights
Intermediate holding position light (unidi-rectional)
Taxiway egde Lights
maximum 3M
Taxiway Center Line Lights (TWY–CLL)
Taxiway stop bar lights (unidirectional)
ILS critical sensitive area (bi–directional lights)
Runway Center Line Lights (RWY–CLL)
maximum 3M
Runway Edge Lights (REDL)
Runway Touchdown Zone Lights (RTZL)
Runway Threshold Identification Lights (RTIL–single unidirectional flashing)
Runway End Identification Lights (REIL–unidirectional or omnidirectional (USA) flashing)
Threshold Lights
a or one–third of runway length, whichever is less
600M
300M
600M
7.5 or 15 or 30M
60M
or
30M
60M
or
30M
10M
900M
Approach direction

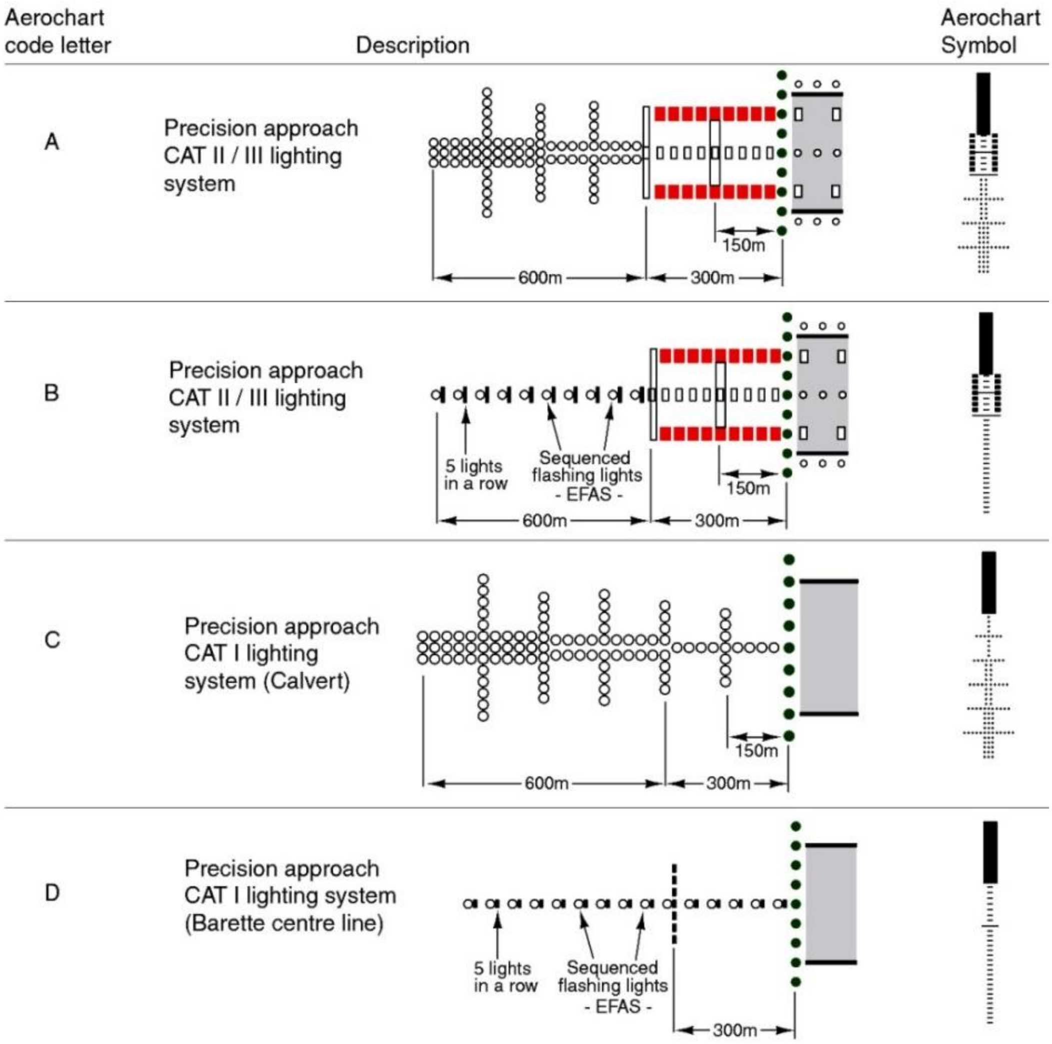
2.13.3. Approach Lighting Systems

A working knowledge of approach lighting systems and regulations as they apply to the required visual references is essential to safe and successful approaches.

A review of the approach and runway lighting systems available during the approach briefing is recommended as the pilot has only a few seconds to identify the lights required to continue the approach. For all low visibility approaches, a review of the airport diagram, expected runway exit, runway remaining lightingand expected taxi route during the approach briefing is recommended.

During night or any other operations when approach lights and RWY lights should operate, they should be switched ON and should be without faults.

Equipment category	Type of lights
FALS (full approach light system).	HIALS 720m or more, RWY CL.
IALS (intermediate approach light system).	HIALS 420m-719m, single power source, line lights.
BALS (basic approach light system).	HIALS 210m-419m, MIALS or ALS 210m or more.
NALS (no approach light system).	HIALS, MIALS or ALS less than 210m or without approach lights.



Failed or Downgraded Aerodrome Equipment Effect

Failed or downgraded equipment	Effect on landing minima		
	CAT IIIA	CAT II	CAT I
ILS stand-by transmitter	Not allowed	No effect	
Outer marker	No effect if replaced by published equivalent position		Not applicable
Middle marker	No effect		
TDZ RVR assessment system	May be temporarily replaced with Midpoint RVR if approved by the State of the Aerodrome. RVR may be reported by human observation.		No effect
Anemometer for runway in use	No effect if other ground source available		
Ceiling meter	No effect		
All approach lights	Not allowed for operations with DH >50ft	Not allowed	Minimums as for NALS
Approach lights except the last 210m	No effect	Not allowed	Minimums as for BALS
Approach lights except the last 420m	No effect	No effect	Minimums as for IALS
Standby power for approach lights	No effect		
Whole runway light system	Not allowed		Day: minimums as for NALS Night: not allowed
Edge lights	Day only Night: Not allowed		
Center line lights	Day: RVR 300m Night: Not allowed	Day: RVR 300m Night: RVR 550m	No effect
Center line lights spacing increased to 30 m	RVR 150 m	No effect	
TDZ lights	Day: RVR 200m Night: RVR 300m	Day: RVR 300m Night: RVR 550m	No effect
Standby power for runway lights	Not allowed		No effect
Taxiway light system	No effect - except delays due to reduced movement rate		

NOTE: Applicable conditions for the use of the table above:

- Multiple failures of runway lights are not acceptable.
- Deficiencies of approach and runway lights are treated separately.
- CATII/III operations: A combination of deficiencies in runway lights and RVR assessment equipment is not allowed.
- Failures other than ILS affect RVR only and not DH.

2.13.4. Low Visibility Flight Planning

Crew LVO Qualification .... Confirm

Airfield and runway LVO operational status…………………………………..Check

Takeoff Alternate Airport .... Select, if required

NOTAM (LVO related) .... Check

MEL status (LVO related) .... Check

LVO Departure Briefing .... Perform

Low Visibility Abnormals .... Review

Taxi / holding points .... Review

2.13.5. Low Visibility Taxi

Pilots need a working knowledge of airport surface lighting, markings, and signs for low visibility taxi operations. Understanding the functions and procedures to be used with stop bar lights, ILS critical area markings, holding points, and low visibility taxi routes is essential for conducting safe operations. Many airports have special procedures for low visibility operations.

Extreme caution should be exercised by Captain and First Officer. Taxy speed should be limited by 10 knots.

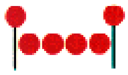
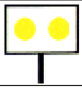
If there is any doubt about the position of the aircraft whilst taxiing before takeoff or after landing, STOP and inform ATC immediately.

Be careful not to cross the CAT II / CAT III holding point: green taxiway centerline lights may be illuminated beyond the CAT II / CAT III holding point if it does not coincide with a taxiway stop bar or if a preceding aircraft is just lining up.

Any checklist must be done when the aircraft is stationary only.

Entering the runway and lining up for takeoff, double check runway heading reference and make sure that the aircraft is on the correct runway and on a runway centerline. Use the ILS Localizer scale as a reference. This also be verified by ND especially when lining up and there is parallel runway.

2.13.6. Taxiway Arrangement Of SMGCS (Surface Movement Guidance And Control System) Features

	Feature	Description
	Stop Bar Lights.	Row of red, in-pavement lights that when illuminated designate a runway hold position. **Never cross an illuminated red stop bar.**
	Runway Guard Lights.	Elevated or in-pavement yellow flashing lights installed at runway holding positions.
	Taxiway Centerline Lights.	Green in-pavement lights to assist taxiing aircraft in darkness and in low visibility conditions.
	Clearance Bar Lights.	In-pavement yellow lights. When installed with geographic position markings they indicate designated aircraft or vehicle hold points.
	Geographic Position Marking (pink spot).	Indicates a specific location on the airport surface.
	Taxiway Centerline Marking.	Provides a visual cue to permit taxiing along a designated path. Marking may be enhanced on light-colored pavement by outlining with a black border.

2.13.7. Low Visibility Takeoff

Low visibility takeoff (LVTO) means a takeoff with an RVR lower than 400 m but not less than 125 m.

Low visibility takeoff operations, below landing minima, require a takeoff alternate. When selecting a takeoff alternate, consideration should be given to unexpected events such as an engine failure or other non-normal situation that could affect landing minima at the takeoff alternate.

Use the centerline lights and/or markings as the primary directional guidance: as speed increases, the streaming effect of these improves guidance (offsetting somewhat the restricted view from the high flight deck). Also, the noise of the nose wheels running over the centerline lights is a confirmation that the Take-Off Run is straight.

Localizer guidance would be helpful in maintaining the runway centerline. Manual tuning of the ILS Frequency / Course is required.

If the visual references are lost below 80 kts, the takeoff may be rejected.

In case of rejected takeoff, it is very important to report to the tower as soon as possible after aircraft stopped.

2.13.8. Low Visibility Approach

Category II/III operations are based on an approach to touchdown using the automatic landing system. Normal operations should not require pilot intervention. However, pilot intervention should be anticipated in the event inadequate airplane performance is suspected, or when an automatic landing cannot be safely accomplished in the touchdown zone. Guard the controls on approach through the landing roll and be prepared to take over manually, if required.

During an autoland with crosswind conditions, fail passive airplanes will touchdown in a crab. After touchdown, the rudder must be applied to maintain runway centerline. The autopilot must be disengaged immediately after touchdown. The control wheel should be turned into the wind as the autopilot is disengaged. The A/T disconnects automatically two seconds after touchdown.

2.13.9. Minimum Equipment Requirements for LVO Approach

CAT IIIa operations are based on the approach to touchdown using the automatic landing system.

- Dual Channel Autopilot engaged;
- Autothrottle at the start of final approach;
- Low Range Radio Altimeter and display for each Pilot;
- Decision Height (DH) Display for each Pilot;
- Two Digital Air Data Computer Systems;
- Windshield Wipers for each Pilot;
- ILS Receiver and display for each Pilot;
- Flight Mode Annunciator for each Pilot;
- ADIRU's (associated with the engaged autopilots) in NAV mode;
- Dual Hydraulic Systems (Sys A + Sys B);
- Two sources of electrical power. (APU generator);
- Both Engines Operating;
- Yaw Damper;
- Antiskid & Autobrake Systems;
- Automatic Ground Spoiler System.

2.13.10. Approach Preparation

Destination airport .... Confirm LVP in Force

Aircraft limitation .... Check

Crew LVO qualification .... Confirm

Aircraft status .... Check

Airfield equipment .... Check

Seat position .... Adjust

Flight deck lighting .... Adjust

Landing light usage .... Review

Task sharing and standard callouts .... Review

Procedures in case of AFDS malfunction .... Review

FAF altitude, expected vertical speed on glideslope .... Review

TASK SHARING

At 100ft above DH, the PM will begin to look inside and monitor the instruments reading all FMA's and the PF will be heads out looking for visual cues.

2.13.11. Final Approach

The pilots should monitor the quality of the approach, flare, and landing including speedbrake deployment and autobrake application.

NOTE: The APP mode should be selected, both autopilots engaged in CMD, and the airplane stabilized on the localizer and glide path before descending below 800 feet RA.

When established on the glide slope, set the missed approach altitude in the altitude window of the MCP. Extension of landing flaps at speeds in excess of flaps 15 speed may cause flap load relief activation and large thrust changes.

Check for correct crossing altitude, when crossing the final approach fix (FAF or OM).

There have been incidents where airplanes have captured false glide slope signals and maintained continuous on glide slope indications as a result of an ILS ground transmitter erroneously left in the test mode. False glide slope signals can be detected by crosschecking the final approach fix crossing altitude and VNAV path information before glide slope capture. A normal pitch attitude and descent rate (vertical speed) should also be indicated on final approach after glide slope capture. Further, if a glide slope anomaly is suspected, an abnormal altitude range-distance relationship may exist. This can be identified by crosschecking distance to the runway with altitude or crosschecking the airplane position with waypoints indicated on the navigation display. The altitude should be approximately 300 feet HAT per NM of distance to the runway for a 3° glide slope.

If a false glide slope capture is suspected, perform a missed approach if visual conditions cannot be maintained.

Below 1,500 feet radio altitude, the flare mode is armed. The FLARE annunciation indicates the second autopilot is fully engaged. As the lowest weather minimums are directly related to the system status, both pilots must observe the FLARE annunciation.

NOTE: During a dual autopilot approach and after FLARE ARM annunciation, any attempted manual override of the autopilots may result in an autopilot disengage.

Approaching Decision Height

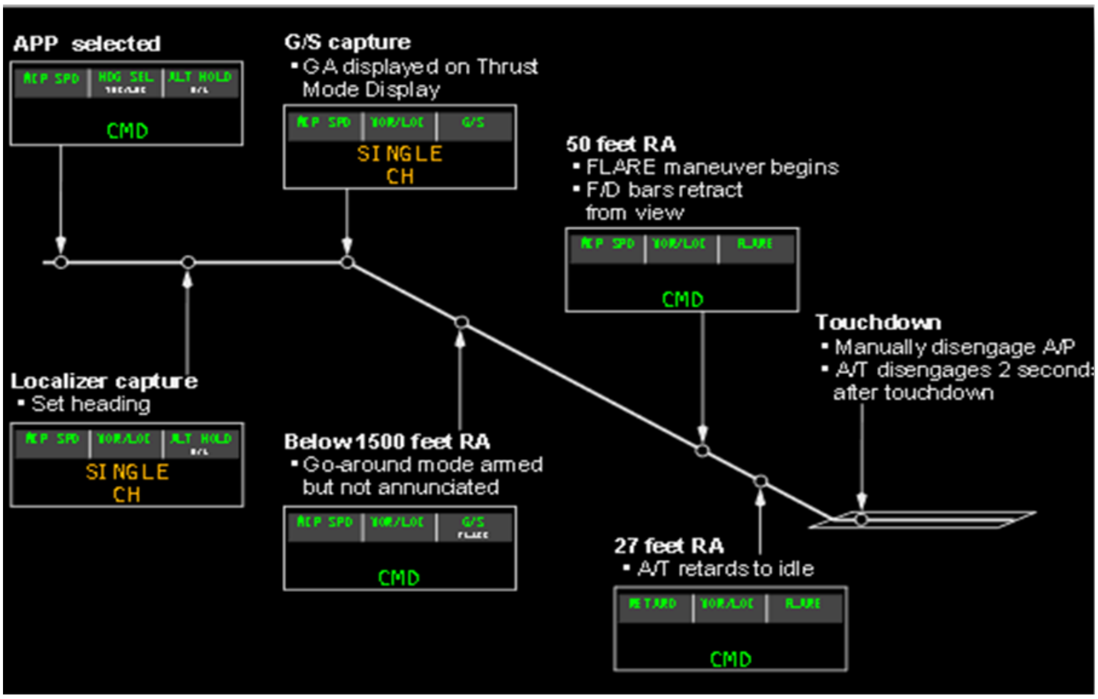
For most Category II and Category III fail passive approaches, the Decision Height is the controlling minima and the altitude value specified is advisory. A Decision Height is usually based on a specified radio altitude above the terrain on the final approach or touchdown zone.

FLARE

The flare mode brings the airplane to a smooth automatic landing touchdown. The flare mode is not intended for single autopilot or flight director only operation.

The A/P flare maneuver starts at approximately 50 feet RA and is completed at touchdown:

- FLARE engaged is annunciated and F/D command bars retract.
- the A/T begins retarding thrust at approximately 27 feet RA so as to reach idle at touchdown. A/T FMA annunciates RETARD.
- the A/T automatically disengages approximately 2 seconds after touchdown.
- the A/P must be manually disengaged after touchdown. Landing roll-out is executed manually after disengaging the A/P.



PF	PM
Initially: • If on radar vectors: • HDG SEL; • Pitch mode (as needed). • If enroute to a fix: • LNAV or other roll mode; • VNAV or other pitch mode.	
Call: **“FLAPS ___”** according to the flap extension schedule. When maneuvering to intercept the localizer, decelerate and extend flaps to 5. Attempt to be at flaps 5 and flaps 5 maneuver speed before localizer capture. Use other Flaps position if needed	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed. Monitor flaps and slats extension.
When on localizer intercept heading: • Verify that the ILS is tuned and identified; • Verify that the LOC and G/S pointers are shown and in proper position.	
	Call: **“ILS IDENTIFIED”**.
Arm APP mode. VOR/LOC mode might be armed first. Call: **“APPROACH ARMED (VOR/LOC ARMED)”** Engage the second autopilot.	Use APP mode on ND display if needed. When first positive localizer pointer motion is observed. Call: **“LOCALIZER ALIVE”**.
	When first positive localizer pointer motion is observed. Call: **“LOCALIZER ALIVE”**.

PF	PM
NOTE When using LNAV to intercept the final approach course, LNAV might parallel the localizer without capturing it.	
Verify that the localizer is captured. Verify final approach course heading.	
Runway or go-around heading......... Set Call: **“RW (GO-AROUND) HDG SET”**	
	Call: **“GLIDESLOPE ALIVE”**.
WARNING: Interference with the glideslope signal can result in erroneous AFDS pitch guidance indicated by FMA mode degradation, the AUTOPILOT caution message, and removal of the F/D pitch bar. If this occurs, do a go-around unless suitable visual references can be established and maintained.	
Call: • **“GEAR DOWN”** • **“FLAPS 15”**	At glideslope alive (at first positive inward motion of G/S pointer), Call: **“GLIDESLOPE ALIVE”**. **Landing gear lever** .... DN Call: **“SPEED CHECK FLAPS 15”** **Flap lever** .... 15 Monitor flaps and slats extension. Verify that the green landing gear indicator lights are illuminated and call: **“GEAR DOWN”**. When flaps are 15 call: “**FLAPS 15 SET”**
Speedbrake lever .... ARMED Verify that the SPEED BRAKE ARMED light is illuminated.	Notify the cabin crew to prepare for landing. **SEAT BELTS switch** .... OFF, then ON **Engine start switches**................... CONT
At glideslope capture call: **“FLAPS ___”** as needed for landing. Final flaps extention is recommended 1500 ft AAL or above. Landing C/L should be completed before 1000 ft. AAL	Call: **“SPEED CHECK FLAPS ___”** Set the flap lever as directed. Monitor flaps and slats extension.
Missed approach altitude .... Set.	
Call: **“LANDING CHECKLIST”**.	Do the “LANDING CHECKLIST”.
	At the final approach point call: **“PASSING FAP, ___ FEET”**.
Verify altitude and call **“CHECK”**	At 1500’ AAL call: **“SYSTEM TEST”**
	At 1,000 ft. AAL call: **“1000 FEET”**. If approach meets stabilized approach recommendations call: **“STABILIZED”**.

PF	PM
Perform go-around and missed approach procedure.	Any time if approach does not meet stabilized approach recommendations call: **“UNSTABILIZED, GO-AROUND”**.
Call **“CHECK, AUTOLAND”**.	Call **“500 feet, FLARE armed”.**
Verify the AFDS (autoland) status at 500 feet AAL. • FLARE armed • NO steady A/P WARNING light	
At 400-350 feet RA check stabilizer nose-up-trim.	
	100 ft. above DA(H) call: **“APPROACHING MINIMUMS”**.
By reaching DA(H) call decision: **“CONTINUE”** or **“GO-AROUND, FLAPS 15, CHECK THRUST”**.	At DA(H) call: **“MINIMUMS”**.
	At 50-40 feet RA check FLARE is activated on FMA and call **“FLARE”** or **“NO FLARE”** if not.
In case of “NO FLARE”, perform the Go-Around Procedure.	
	At 27 feet RA check RETARD on FMA, call **“RETARD”** or **“NO RETARD”** if not.
Verify throttles are moving to idle or set the idle thrust manually.	
Disengage the autopilot after touchdown. The Autothrottle automatically disengages 2 seconds after touchdown.	

2.13.12. Autoland on runways with CAT I facilities

Autoland in CAT I or better weather conditions is possible on runways with CAT I facilities for training purpose, data gathering (maintenance request) or on Captain decision, if the following precautions are taken into account:

- all limitations from chapter 1.5.3 are satisfied;
- the final approach course path is aligned with the runway centerline;
- the RW is equipped with the CAT I ILS facilities (SkyBag – Charts – RADIO NAVIGATION – Type of aid / Type of Supported ops);
- no NOTAM about the degradation of ILS facilities exists;
- crew and aircraft are authorized for CATII or III operations;
- crew must be aware that LOC or G/S beam fluctuations, independently of the aircraft system, may occur and must be ready to immediately disconnect the autopilot, and to perform manual landing or go-around;
- visual references should be obtained at an altitude appropriate for the CATI approach, otherwise go-around should be initiate.

СТРАНИЦА НАМЕРЕНО ОСТАВЛЕНА ПУСТОЙ

Глава 3. Supplementary Procedures

3.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

3.1. GENERAL

For supplementary procedures refer to FCOM Supplementary Procedures Chapter.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 4. Abnormal and Emergency

4.0. СОДЕРЖАНИЕ

4.1. РЕКОМЕНДАЦИИ ЭКИПАЖУ ПО ДЕЙСТВИЯМ ПРИ АВИАЦИОННОМ СОБЫТИИ

Детализированные рекомендации экипажу по действиям при авиационном событии содержится в РПП Части А.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.2. EVENTS REQUIRING MAINTENANCE INSPECTION

During ground or flight operations, events may occur which require a maintenance inspection after the flight. Use the following guidance to determine which events require a maintenance inspection:

- hard landing - specify if the landing was hard on the nose gear only, hard on the main gear only or hard on both main and nose gear. Specify if the landing was a hard-bounced landing;

NOTE: A bounced landing is defined as a landing where both main gears contact the ground and then both main gears leave the ground prior to landing.

NOTE: A nose first landing is considered to be a hard nose gear landing.

- overweight landing - if the overweight landing was not a hard landing the flight crew should record that the landing was not a hard landing;
- high drag/high side load event - one or more of the following conditions occurred:
  - airplane ran off the prepared surface;
  - airplane landed short of prepared surface;
  - two or more tires were blown during landing;
  - one or more landing gear hit an obstacle or were hit by an obstacle;
  - airplane landed with a large crab or high bank angle resulting in abnormally large side-to-side forces as the airplane aligns with the runway after touchdown.
- severe turbulence;
- overspeed - flap/slat, MMO/VMO, landing gear, landing gear tires;
- high-energy stop (refer to the AMM for guidance);
- lightning strike;
- extreme dust;
- tail strike;
- any event that the pilot feels a maintenance inspection could be needed. An example of such an event is an overly aggressive pitch up during a TCAS event or a Terrain Avoidance maneuver that could cause structural damage;
- high levels of reverse thrust were used at low speed;
- operator-specific procedures or policies may include additional events that require a maintenance inspection.

NOTE: If in doubt, the best course of action is to report it.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.3. ABNORMAL AND EMERGENCY PROCEDURE GENERAL

4.3.1. Preface

The purpose of this chapter not to substitute the relevant sections of AFM and QRH, but to arrange the responsibilities and task sharing between the crew members, which are not described in the Non-Normal checklist.

In the event of a conflict between the given Chapter and QRH Non-Normal checklist, the last take precedence.

Aircrews are expected to accomplish NNCs listed in the QRH.

These checklists ensure maximum safety until appropriate actions are completed and a safe landing is accomplished. Techniques discussed in this chapter minimize workload, improve crew coordination, enhance safety, and provide a basis for standardization.

4.3.2. Non-Normal Situation Guidelines

When a non-normal situation occurs, the following guidelines apply:

- **NON-NORMAL RECOGNITION:** The crewmember recognizing the malfunction calls it out clearly and precisely.
- **MAINTAIN AIRPLANE CONTROL:** It is mandatory that the Pilot Flying (PF) fly the airplane while the Pilot Monitoring (PM) maintains situational awareness and supports the Pilot Flying to ensure obstacle clearance and airplane control. Maximum use of the autoflight system is recommended to reduce crew workload.
- **ANALYZE THE SITUATION:** NNCs should be accomplished only after the desired flight path and appropriate configuration are correctly established. Review all warning lights, caution lights and other alerts to positively identify the malfunctioning system(s). In the event of multiple malfunctions, the flight crew needs to analyze the situation, use good judgment and determine which NNC takes priority.

  NOTE: Pilots should don oxygen masks and establish crew communications anytime oxygen deprivation or air contamination is suspected, even though an associated warning has not occurred.

- **TAKE THE PROPER ACTION:** Although some in-flight non-normal situations require immediate corrective action, difficulties can be compounded by the rate the PF issues commands and the speed of execution by the PM. Commands must be clear and concise, allowing time for acknowledgment of each command prior to issuing further commands. The PF must exercise positive control by allowing time for acknowledgment and execution. The other crewmembers must be certain their reports to the PF are clear and concise, neither exaggerating nor understating the nature of the non-normal situation. This eliminates confusion and ensures efficient, effective, and expeditious handling of the non-normal situation. In the event of multiple malfunctions, consider doing memory items first followed by reference steps. Upon completion of an NNC, review all warning lights, caution lights and other alerts to determine the need to do other NNCs.

- **EVALUATE THE NEED TO LAND:** If the NNC directs the crew to plan to land at the nearest suitable airport, or if the situation is so identified in the QRH section CI.2, (Checklist Instructions, Non-Normal Checklists), diversion to the nearest airport where a safe landing can be accomplished is required. If the NNC or the Checklist Instructions do not direct landing at the nearest suitable airport, the pilot must determine if continued flight to destination may compromise safety.

4.3.3. Crew Coordination in Non-Normal Situation

PF	PM
Fly the aircraft. Use the autopilot and autothrottle, if applicable.	Recognize the malfunction. Silence the aural warning, if needed. Call out malfunction clearly and precisely.
Analyze the situation.	
Call for respective memory items, if needed.	Do requested memory items, if needed.
	Advice ATC (MAYDAY / PAN PAN), if needed
Call for respective Non-Normal Checklist, if needed.	Do the requested Non-Normal Checklist, if needed.
Normally the PF becomes responsible for the ATC communication when PM performs NNC. Use standard call-outs **“I HAVE COMMUNICATIONS …. YOU HAVE COMMUNICATIONS”.**	
Evaluate the need to land.	
Advise Purser (NITS), if needed. Advice passengers, if needed.	Advise ATC of intentions, if needed.

4.3.4. Procedures Initiation

No action shall be taken (except audio warning cancel) until:

- the appropriate flight path is stabilized;
- at least 400 ft AAL, in case of failure during takeoff, approach or go-around.

4.3.5. Task Sharing

Each crewmember is responsible for moving the controls and switches in their area of responsibility.

The phase of flight areas of responsibility for both normal and non-normal procedures are shown in the Area of Responsibility illustration. Typical panel locations are shown.

NOTE: During: rejected takeoff, engine fire (on ground), emergency evacuation, smoke, fumes and emergency descent, a CPT-F/O task sharing applies.

In case of abnormal situation during flight, which may cause or have already caused deviation from the flight plan. The flight crew should inform company of the situation and captain's intentions. For analyzing and transferring information to all concerned departments and to help flight crew in decision making.

In case of technical problems, the flight crew should contact MAINTROL, to receive maintenance recommendations.

4.3.6. Checklists with Memory Steps

After flight path control has been established, do the memory steps of the appropriate NNC. The emphasis at this point should be on containment of the problem. Reference steps are initiated after the airplane flight path and configuration are properly established.

Complete all applicable NNCs prior to beginning final approach. Exercise common sense and caution when accomplishing multiple NNCs with conflicting directions. The intended course of action should be consistent with the damage assessment and handling evaluation.

4.3.7. Situations Beyond the Scope of Non-Normal Checklists

It is rare to encounter in-flight events which are beyond the scope of the Boeing recommended NNCs. These events can arise as a result of unusual occurrences such as a midair collision, bomb explosion or other major malfunction. In these situations the flight crew may be required to accomplish multiple NNCs, selected elements of several different NNCs applied as necessary to fit the situation, or be faced with little or no specific guidance except their own judgment and experience. Because these situations are rare, it is not practical or possible to create definitive flight crew NNCs to cover all events.

4.3.8. Landing at the Nearest Suitable Airport

"Plan to land at the nearest suitable airport" is a phrase used in the QRH.

In a non-normal situation, the pilot-in-command, having the authority and responsibility for operation and safety of the flight, must make the decision to continue the flight as planned or divert. In an emergency situation, this authority may include necessary deviations from any regulation to meet the emergency. In all cases, the pilot-in-command is expected to take a safe course of action.

The QRH assists flight crews in the decision making process by indicating those situations where "landing at the nearest suitable airport" is required. These situations are described in the Checklist Instructions or the individual NNC.

The regulations regarding an engine failure are specific. Most regulatory agencies specify that the pilot-in-command of a twin engine airplane that has an engine failure or engine shutdown should land at the nearest suitable airport at which a safe landing can be made

A suitable airport is defined by the operating authority for the operator based on guidance material but, in general, must have adequate facilities and meet certain minimum weather and field conditions. If required to divert to the nearest suitable airport, the guidance material typically specifies that the pilot should select the nearest suitable airport "in point of time" or "in terms of time." In selecting the nearest suitable airport, the pilot-in-command should consider the suitability of nearby airports in terms of facilities and weather and their proximity to the airplane position. The pilot-in-command may determine, based on the nature of the situation and an examination of the relevant factors, that the safest course of action is to divert to a more distant airport than the nearest airport. For example, there is not necessarily a requirement to spiral down to the airport nearest the airplane's present position if, in the judgment of the pilot-in-command, it would require equal or less time to continue to another nearby airport

For persistent smoke or a fire which cannot positively be confirmed to be completely extinguished, the safest course of action typically requires the earliest possible descent, landing and evacuation. This may dictate landing at the nearest airport appropriate for the airplane type, rather than at the nearest suitable airport normally used for the route segment where the incident occurs.

4.3.9. Approach and Landing

When a non-normal situation occurs, a rushed approach can often complicate the situation. Unless circumstances require an immediate landing, complete all corrective actions before beginning the final approach.

For some non-normal situations, the possibility of higher airspeed on approach, longer landing distance, a different flare profile or a different landing technique should be considered.

Plan an extended straight-in approach with time allocated for the completion of any lengthy NNC steps such as the use of alternate flap or landing gear extension systems. Arm autobrakes and speedbrakes unless precluded by the NNC.

NOTE: The use of autobrakes is recommended because maximum autobraking may be more effective than maximum manual braking due to timely application upon touchdown and symmetrical braking. However, the Advisory Information in the PI chapter of the QRH includes Non-Normal Configuration Landing Distance data specific to the use of maximum manual braking. When used properly, maximum manual braking provides the shortest stopping distance.

Fly a normal glide path and attempt to land in the normal touchdown zone. After landing, use available deceleration measures to bring the airplane to a complete stop on the runway. The captain must determine if an immediate evacuation should be accomplished or if the airplane can be safely taxied off the runway.

4.3.10. Command Speed – Non-Normal Conditions

Occasionally, a non-normal checklist instructs the flight crew to use a V_{REF} speed that also includes a speed additive such as V_{REF15} + 15 knots. When V_{REF} has been adjusted by a NNC, this becomes the V_{REF} used for landing. This V_{REF} does not include wind additives. For example, if a non-normal checklist specifies “Use flaps 15 and V_{REF15} + 10 knots for landing”, the flight crew would select flaps 15 as the landing flaps and look up the V_{REF15} speed in the FMC or QRH and add 10 knots to that speed.

When using the autothrottle, position command speed to V_{REF} + 5 knots. Sufficient wind and gust protection is available with the autothrottle connected that no further wind additives are needed.

If the autothrottle is disconnected, or is planned to be disconnected prior to landing, appropriate wind additives must be added to the V_{REF} to arrive at command speed, the speed used to fly the approach. For example, if the checklist states “use V_{REF40} + 30 knots”, command speed should be positioned to V_{REF} (V_{REF40} + 30 knots) plus wind additive (5 knots minimum, 15 knots maximum).

4.4. NON-NORMAL CHECKLISTS INSTRUCTIONS

4.4.1. Introduction

For NON-NORMAL CHECKLISTS INSTRUCTIONS refer to QRH Checklist Instructions Chapter.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.5. NON-NORMAL MANEUVERS

4.5.1. Approach to Stall or Stall Recovery

Do all recoveries from approach to stall as if an actual stall has occurred.

NOTE: Do not use flight director commands during the recovery.

Pilot Flying	Pilot Monitoring
Call **“STALL”** Initiate the recovery: • Hold the control column firmly. • Disengage autopilot and autothrottle. • Smoothly apply nose down elevator to reduce the angle of attack until buffet or stick shaker stops. Nose down stabilizer trim may be needed. *	• Monitor altitude and airspeed • Verify all needed actions have been done and call out any omissions • Call out any trend toward terrain contact
Continue the recovery: • Roll in the shortest direction to wings level if needed. ** • Advance thrust levers as needed. • Retract the speedbrakes. • Do not change gear or flap configuration, except • During liftoff, if flaps are up, call for flaps 1.	• Monitor altitude and airspeed. • Verify all needed actions have been done and call out any omissions. • Call out any trend toward terrain contact. Set the FLAP lever as directed.
Complete the recovery: • Check airspeed and adjust thrust as needed. • Establish pitch attitude. • Return to the desired flight path. Re-engage the autopilot and autothrottle if desired.	• Monitor altitude and airspeed. • Verify all needed actions have been done and call out any omissions. • Call out any trend toward terrain contact.
WARNING: *If the control column does not provide the needed response, stabilizer trim may be needed. excessive use of pitch trim can aggravate the condition, or can result in loss of control or in high structural loads.	
WARNING: ** Excessive use of pitch trim or rudder can aggravate the condition, or can result in loss of control or in high structural loads.	

4.5.2. Ground Proximity Warning System (GPWS) Response

GPWS CAUTION

Accomplish the following maneuver for any of these aural alerts:

- SINK RATE;
- TERRAIN;
- DON'T SINK;
- TOO LOW FLAPS;
- TOO LOW GEAR;
- TOO LOW TERRAIN;
- GLIDESLOPE;
- BANK ANGLE;
- AIRSPEED LOW (airplanes with AIRSPEED LOW aural);
- CAUTION TERRAIN;

RA-73314, RA-73315, RA-73316, RA-73317, RA-73318, RA-73319

- CAUTION OBSTACLE;

Pilot Flying	Pilot Monitoring
Correct the flight path, airplane configuration, or airspeed.	Advice ATC if unable to follow prescribed procedure.

The below glideslope deviation alert can be cancelled or inhibited for:

- localizer or backcourse approach;
- circling approach from an ILS;
- when conditions require a deliberate approach below glideslope;
- unreliable glideslope signal.

NOTE: If a terrain caution occurs when flying under daylight VMC, and positive visual verification is made that no obstacle or terrain hazard exists, the alert may be regarded as cautionary and the approach may be continued.

NOTE: Some aural alerts repeat.

GPWS WARNING

Accomplish the following maneuver for any of these conditions:

- Activation of the "PULL UP" or "TERRAIN, TERRAIN PULL UP" aural warning;
- Activation of the "PULL UP" or "OBSTACLE, OBSTACLE PULL UP" aural warning;
- Other situations resulting in unacceptable flight toward terrain.

Pilot Flying	Pilot Monitoring
Call: **"PULL UP, MAXIMUM THRUST"**. • Disengage autopilot. • Disengage autothrottle. • Aggressively apply maximum* thrust. • Simultaneously roll wings level and rotate to an initial pitch attitude of 20°. • Retract speedbrakes.	• Assure maximum* thrust. • Verify all needed actions have been completed and call out any omissions.

• If terrain remains a threat, continue rotation up to the pitch limit indicator (if available) or stick shaker or initial buffet.	
• Do not change gear or flap configuration until terrain separation is assured. • Monitor radio altimeter for sustained or increasing terrain separation. • When clear of terrain, slowly decrease pitch attitude and accelerate.	• Advice ATC • Monitor vertical speed and altitude (radio altitude for terrain clearance and barometric altitude for a minimum safe altitude.) • Call out any trend toward terrain contact.
When clear of terrain: - MSA has been reached or, - warning stopped when on Final approach.	
• Flight Path ....Adjust • Configuration & Speed ..As required	• ATC instructions ....Request

NOTE: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be needed to obtain a positive terrain separation. Use smooth, steady control to avoid a pitch attitude overshoot and stall.

NOTE: Do not use flight director commands.

NOTE: *Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

NOTE: Except for mountain area, if the RW is in-sight and positive visual verification is made that no obstacle or terrain hazard exists when flying under daylight VMC conditions before a terrain or obstacle warning, the alert may be regarded as cautionary and the approach may be continued.

ПРЕДУПРЕЖДЕНИЕ: при заходе на посадку и срабатывании звуковой сигнализации:

“PULL UP” or “TERRAIN, TERRAIN PULL UP”

“PULL UP” or “OBSTACLE, OBSTACLE PULL UP”

На горном аэродроме – **ВЫПОЛНИТЬ GPWS ESCAPE MANEUVER!**

В остальных случаях, только **днём в визуальных метеоусловиях при устойчивой видимости ВПП** заход может быть продолжен при условии, что экипаж визуально имел полную уверенность в отсутствии опасности от препятствий до срабатывания сигнализации.

В случае наличия проблем с сигналом GPS для исключения ложного срабатывания сигнализации GPWS КВС может рассмотреть возможность перевода переключателя «TERR INHIBIT» в положение «TERR INHIBIT» перед выполнением захода на посадку в следующих случаях:

– FMC message “GPS_L/R INVALID” и/или “UNABLE REQD NAV PERF – RNP” и/или “VERIFY POS: FMC-GPS”;
– “TERR POS” на ND;
– активация сигнального табло GPS;
– после ухода на второй круг в случае, если у экипажа имеется обоснованная уверенность в том, что срабатывание сигнализации было ложным (визуальные условия и/или захват LOC & G/S при заходе по ILS).

4.5.3. Traffic Avoidance

Immediately accomplish the following by recall whenever a TCAS traffic advisory (TA) or resolution advisory (RA) occurs.

WARNING: Comply with the RA if there is a conflict between the RA and air traffic control.

WARNING: Once an RA has been issued, safe separation could be compromised if current vertical speed is changed, except as needed to comply with the RA. This is because TCAS II-to-TCAS II coordination can be in progress with the intruder aircraft, and any change in vertical speed that does not comply with the RA can negate the effectiveness of the other aircraft's compliance with the RA.

NOTE: If stick shaker or initial buffet occurs during the maneuver, immediately accomplish the APPROACH TO STALL RECOVERY procedure.

NOTE: If high speed buffet occurs during the maneuver, relax pitch force as needed to reduce buffet, but continue the maneuver.

NOTE: Do not use flight director pitch commands until clear of conflict

FOR TA

Pilot Flying	Pilot Monitoring
TA's are indicated by the aural "TRAFFIC, TRAFFIC" which sounds once and is then reset until the next TA occurs. The TRAFFIC annunciation appears on the navigation display.	
Call: **"TCAS "**	
Look for traffic using traffic display as a guide (adjust ND scale). Attempt to establish visual contact. Call out any conflicting traffic.	
NOTE: Maneuvers based solely on a TA can result in reduced separation and are not recommended.	

FOR RA, EXCEPT A CLIMB IN LANDING CONFIGURATION

WARNING: Do not follow a DESCEND (fly down) RA issued below 1000 feet AAL.

Pilot Flying	Pilot Monitoring
Call: **"TCAS "** if not announced during TA.	
Look for traffic using traffic display as a guide (adjust ND scale). Attempt to establish visual contact. Call out any conflicting traffic.	
If maneuvering is needed, disengage the autopilot and disengage the autothrottle. Smoothly adjust pitch and thrust to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	Call ATC: **"TCAS RA"**
Attempt to establish visual contact. Call out any conflicting traffic.	

FOR A CLIMB RA IN LANDING CONFIGURATION

Pilot Flying	Pilot Monitoring
Call: **"TCAS GO AROUND "**	
Disengage the autopilot and disengage the autothrottle. Advance thrust levers forward to ensure maximum thrust is attained and call “FLAPS 15”. Smoothly adjust pitch to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action	Call ATC: **"TCAS RA"** Verify maximum thrust set. Position flap lever to 15 detent.
Verify a positive rate of climb on the baro altimeter and call **“GEAR UP”**	Verify a positive rate of climb on the altimeter and call **“POSITIVE RATE”** Set the landing gear lever to UP
Attempt to establish visual contact. Call out any conflicting traffic.	

СТАНДАРТНАЯ ФРАЗЕОЛОГИЯ РАДИООБМЕНА ПРИ СРАБАТЫВАНИИ TCAS RA

CIRCUMSTANCES	PHRASEOLOGIES
If maneuvering is required:	TCAS RA
When TCAS commands "CLEAR OF CONFLICT"	CLEAR OF CONFLICT, RESUMING CLIMB / DESCEND / FL___ (assigned clearance);
After an ATC clearance or instruction contradictory to the TCAS RA is received, the flight crew will follow the RA and inform ATC directly	UNABLE, TCAS RA;

4.5.4. Upset Recovery

Historically, an upset was defined as unintentionally exceeding one or more of the following conditions:

- Pitch attitude greater than 25 degrees nose up.
- Pitch attitude greater than 10 degrees nose down.
- Bank angle greater than 45 degrees.
- Less than above parameters but flying at an airspeed inappropriate for the conditions.

An upset condition is now considered any time an airplane is diverting from the intended airplane state. An airplane upset can involve pitch or roll angle deviations as well as inappropriate airspeeds for the conditions.

The following actions represent a logical progression for recovering the airplane. The sequence of actions is for guidance only and represents a series of options to be considered and used dependent on the situation. Not all actions can be needed once recovery is under way. If needed, use minimal pitch trim during initial recovery. Careful use of rudder to aid roll control should be considered only if roll control is ineffective and the airplane is not stalled.

These actions assume that the airplane is not stalled. A stalled condition can exist at any attitude and can be recognized by one or more of the following:

- Stick shaker
- Buffet that can be heavy at times.
- Lack of pitch authority.
- Lack of roll control.
- Inability to stop a descent.

If the airplane is stalled, first recover from the stall by applying and maintaining nose down elevator until stall recovery is complete and stick shaker stops.

NOSE HIGH RECOVERY

Pilot Flying	Pilot Monitoring
Recognize and confirm the developing situation	
Call **"UPSET RECOVERY"** Disengage autopilot. Disengage autothrottle. Recover: • Apply nose-down elevator. Apply as much elevator as needed to obtain a nose down pitch rate. • Apply appropriate nose down stabilizer trim*. • Reduce thrust. • Roll (adjust bank angle) to obtain a nose down pitch rate*. Complete the recovery: • When approaching the horizon, roll to wings level. • Check airspeed and adjust thrust. • Establish pitch attitude.	Call out attitude, airspeed and altitude throughout the recovery. Verify all needed actions have been done and call out any continued deviation.

WARNING: * Excessive use of pitch trim or rudder can aggravate an upset, result in loss of control, or result in high structural loads.

NOSE LOW RECOVERY

Pilot Flying	Pilot Monitoring
Recognize and confirm the developing situation	
Call **“UPSET RECOVERY”** Disengage autopilot. Disengage autothrottle. Recover: • Recover from stall, if needed. • Roll in the shortest direction to wings level. If bank angle is more than 90 degrees, unload and roll. Complete the recovery: • Apply nose up elevator. • Apply nose up trim, if needed*. • Adjust thrust and drag, if needed.	Call out attitude, airspeed and altitude throughout the recovery. Verify all needed actions have been done and call out any continued deviation.

WARNING: *Excessive use of pitch trim or rudder can aggravate an upset, result in loss of control or result in high structural loads.

4.5.5. Windshear

WINDSHEAR CAUTION

For predictive windshear caution alert: (**“MONITOR RADAR DISPLAY”** aural).

Pilot Flying	Pilot Monitoring
Maneuver as needed to avoid the windshear.	

WINDSHEAR WARNING

Predictive windshear warning during takeoff roll: (**“WINDSHEAR AHEAD, WINDSHEAR AHEAD”** aural)

- Before V1, reject takeoff.
- After V1, perform the Windshear Escape Maneuver.

Windshear encountered during takeoff roll:

- If windshear is encountered before V1, there may not be sufficient runway remaining to stop if an RTO is initiated at V1. At VR, rotate at a normal rate toward a 15 degree pitch attitude. Once airborne, perform the Windshear Escape Maneuver.
- If windshear is encountered near the normal rotation speed and airspeed suddenly decreases, there may not be sufficient runway left to accelerate back to normal takeoff speed. If there is insufficient runway left to stop, initiate a normal rotation at least 2,000 feet before the end of the runway, even if airspeed is low. Higher than normal attitudes may be needed to lift off in the remaining runway. Ensure maximum thrust is set.

Predictive windshear warning during approach: (**“GO–AROUND, WINDSHEAR AHEAD”** aural)

- perform the Windshear Escape Maneuver;
- at pilot’s discretion, perform a normal go–around.

Windshear encountered in flight:

− perform the Windshear Escape Maneuver.

NOTE: The following are indications the airplane is in windshear:

- windshear warning (two–tone siren followed by “WINDSHEAR, WINDSHEAR, WINDSHEAR”) or
- unacceptable flight path deviations.

NOTE: Unacceptable flight path deviations are recognized as uncontrolled changes from normal steady state flight conditions below 1000 feet AAL, in excess of any of the following:

- 15 knots indicated airspeed
- 500 fpm vertical speed
- 5° pitch attitude
- 1 dot displacement from the glideslope
- unusual thrust lever position for a significant period of time.

WINDSHEAR ESCAPE MANEUVER

Pilot Flying	Pilot Monitoring
Call: **“WINDSHEAR, MAXIMUM THRUST”**	
MANUAL FLIGHT	
• Disengage autopilot. • Push either TO/GA switch. • Aggressively apply maximum thrust* • Disengage autothrottle. • Simultaneously roll wings level and rotate toward an initial pitch attitude of 15°. • Retract speedbrakes. • Follow flight director TO/GA guidance (if available) **	• Verify maximum* thrust. • Verify all needed actions have been completed and call out any omissions.
AUTOMATIC FLIGHT	
• Push either TO/GA switch*** • Verify TO/GA mode annunciation. • Verify GA thrust. • Retract speedbrakes. • Monitor system performance****	• Verify GA* thrust. • Verify all needed actions have been completed and call out any omissions.
MANUAL OR AUTOMATIC FLIGHT	
• Do not change flap or gear configuration until windshear is no longer a factor. • Monitor vertical speed and altitude. • Do not attempt to regain lost airspeed until windshear is no longer a factor.	• Monitor vertical speed and altitude. • Call out any trend toward terrain contact, descending flight path, or significant airspeed changes.

NOTE: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be needed to obtain a positive terrain separation. Use smooth, steady controls to avoid a pitch attitude overshoot and stall.

NOTE: *Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

NOTE: **Do not exceed the Pitch Limit Indication

NOTE: *** If TO/GA is not available, disengage autopilot and autothrottle and fly manually.

WARNING: **** Severe windshear can exceed the performance of the AFDS. Be prepared to disengage the autopilot and autothrottle and fly manually.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.6. ENGINES

4.6.1. Rejected take off

The Captain has the sole responsibility for the decision to reject the takeoff. The decision must be made in time to start the rejected takeoff maneuver by V1. If a malfunction occurs before V1, for which the Captain does not intend to reject the takeoff, he will announce his intention by calling **"GO"**. If the decision is to reject the takeoff, the Captain must clearly announce **"REJECT"**. This call both confirms the decision to reject the takeoff and also states that the Captain assumes control of the airplane. Immediately start the rejected takeoff maneuver. If the First officer is making the takeoff, the First officer must maintain control of the airplane until the Captain makes a positive input to the controls.

Prior to 80 knots	Above 80 knots
• activation of the master caution system; • system failure(s); • unusual noise or vibration; • tire failure; • abnormally slow acceleration; • takeoff configuration warning; • fire or fire warning; • engine failure; • predictive windshear warning; • if a side window opens; • if the airplane is unsafe or unable to fly.	• fire or fire warning; • engine failure; • predictive windshear warning; • if the airplane is unsafe or unable to fly.

During the takeoff, the crew member observing the non-normal situation will immediately call it out as clearly as possible.

Captain	First officer
Without delay: **“REJECT (I HAVE CONTROL)”** .... Call Simultaneously: **Thrust levers** .... Close **Autothrottles** .... Disengage If **RTO** autobrakes is selected, monitor system performance and apply manual wheel brakes if the **AUTOBRAKE DISARM** light illuminates or deceleration is not adequate **SPEED BRAKE lever**...................... Raise **Reverse thrust**............................... Apply Apply up to the maximum amount consistent with conditions. **Maximum braking** .... Use Continue until certain the airplane can stop on the runway	Verify actions as follows: **Thrust levers**............................... Closed **Autothrottles**........................ Disengaged **Maximum brakes** .... Applied **SPEED BRAKE lever UP** .... Verify **SPEED BRAKE UP** .... Call If SPEED BRAKE lever is not UP call **“SPEEDBRAKES NOT UP”** **Reverse thrust** .... Applied When both REV indications are green. **“REVERSERS NORMAL”** .... Call If there is no REV indication(s) or the indication(s) stays amber, call **“NO**

	REVERSER ENGINE № 1”, or **“NO REVERSER ENGINE № 2”**, or **“NO REVERSERS”.** Call out omitted action items.
When stopping is assured: Start movement of the reverse thrust levers to reach the reverse idle detent before taxi speed.	**60 knots** .... Call
After aircraft has stopped:	
Make PA**: “ATTENTION CREW, AT STATIONS. ATTENTION CREW, AT STATIONS”**	Notify ATC: **"MAYDAY, MAYDAY, MAYDAY** or **PAN-PAN, PAN-PAN, PAN-PAN**, **NWS___ REJECTED TAKE-OFF, STANDBY"**
If Evacuation is Required:	
EVACUATION NNC .... Call	Do the EVACUATION NNC from QRH Back Cover.2.
If Evacuation is Not Required:	
Make PA**: “CANCEL ALERT, CANCEL ALERT”**	
Review Brake Cooling Schedule for brake cooling time and precautions (refer to the QRH Performance Inflight chapter). Do not attempt to clear the runway, until it is absolutely clear that an evacuation is not necessary and that it is safe to do so. Consider the following: • the possibility of wheel fuse plugs melting; • the need to clear the runway; • the requirement for remote parking; • wind direction in case of fire;	
• alerting fire equipment; • not setting the parking brake unless passenger evacuation is necessary; • advise the ground crew of the hot brake hazard; • advise passengers of the need to remain seated or evacuate.	
If able to vacate the runway	
	"NWS__ ready to vacate RW" Notify ATC
RECOMMENDATIONS AFTER RTO OM Part A	

4.6.2. Engine Failure After V1

GENERAL

Differences between normal and engine out profiles are few. One engine out controllability is excellent during takeoff roll and after liftoff. Minimum control speed in the air is below V_R and V_{REF}.

ENGINE FAILURE RECOGNITION

An engine failure at or after V1 initially affects yaw much like a crosswind effect. Vibration and noise from the affected engine may be apparent and the onset of the yaw may be rapid.

The airplane heading is the best indicator of the correct rudder pedal input. To counter the thrust asymmetry due to an engine failure, stop the yaw with rudder. Flying with lateral control wheel displacement or with excessive aileron trim causes spoilers to be raised.

ROTATION AND LIFTOFF - ONE ENGINE INOPERATIVE

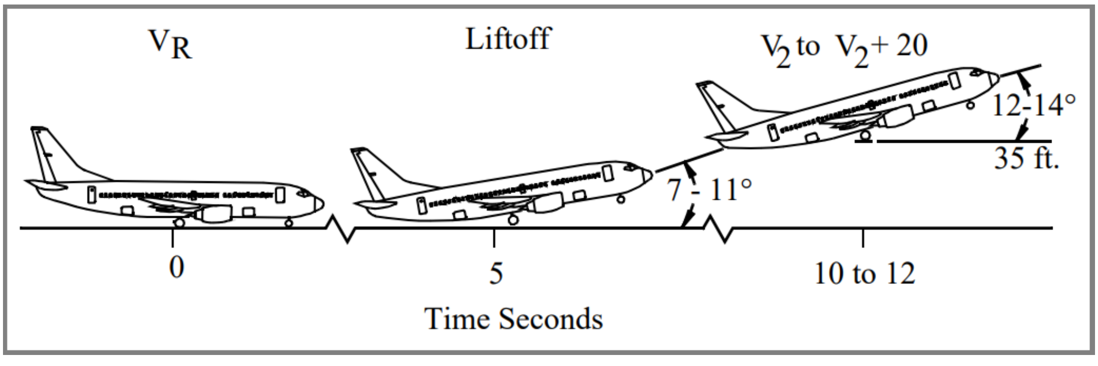
If an engine fails between V1 and liftoff, maintain directional control by smoothly applying rudder proportionate with thrust decay.

During a normal all engine takeoff, a smooth continuous rotation toward 15° of pitch is initiated at VR. With an engine inoperative, a smooth continuous rotation is also initiated at VR; however, the target pitch attitude is approximately 2° to 3° below the normal all engine pitch attitude resulting in a 12° to 14° target pitch attitude. The rate of rotation with an engine inoperative is also slightly slower (1/2° per second less) than that for a normal takeoff, resulting in a rotation rate of approx 1.5° to 2.5° per second. After liftoff adjust pitch attitude to maintain the desired speed.

If the engine failure occurs at or after liftoff apply rudder and aileron to control heading and keep the wings level. In flight, correct rudder input approximately centers the control wheel. To center the control wheel, rudder is required in the direction that the control wheel is displaced. This approximates a minimum drag configuration.

TYPICAL ROTATION - ONE ENGINE INOPERATIVE

Liftoff attitude depicted in the following tables should be achieved in approximately 5 seconds. Adjust pitch attitude, as needed, to maintain desired airspeed of V_2 to V_2 + 20 knots.

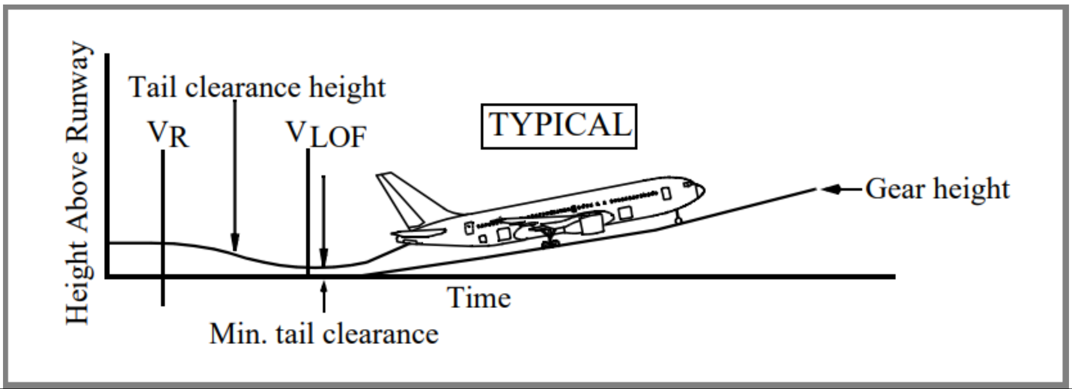


Retract the landing gear after a positive rate of climb is indicated on the altimeter.

Retract flaps in accordance with the technique described in this chapter.

TYPICAL TAKEOFF TAIL CLEARANCE - ONE ENGINE INOPERATIVE

The minimum tail clearance remains constant for all takeoff flap settings. The rotation speed schedules were developed to maintain a constant tail clearance.



Model	Flaps	Liftoff Attitude (degrees)	Minimum Tail Clearance inches (cm)	Tail Strike Pitch Attitude (degrees)
737-800	1 5 10 15 25	9.0 8.7 8.4 8.2 8.0	8 (20) 11 (28) 14 (36) 15 (38) 17 (43)	11.0

INITIAL CLIMB - ONE ENGINE INOPERATIVE

The initial climb attitude should be adjusted to maintain a minimum of V2 and a positive climb. After liftoff the flight director provides proper pitch guidance for the engine inoperative condition. Crosscheck indicated airspeed, vertical speed and other flight instruments. The flight director commands a minimum of V2, or the existing speed up to a maximum of V2 + 20 knots.

If the flight director is not used, attitude and indicated airspeed become the primary pitch references.

Retract the landing gear after a positive rate of climb is indicated on the baro altimeter. The initial climb attitude should be adjusted to maintain a minimum of V2. If an engine fails at an airspeed between V2 and V2 + 20 knots, climb at the airspeed at which the failure occurred. If engine failure occurs above V2 + 20 knots, increase pitch to reduce airspeed to V2 + 20 knots and maintain V2 + 20 knots until reaching acceleration height. After flap retraction and all obstructions are cleared, on the FMC ACT ECON CLB page, select ENG OUT followed by the prompt corresponding to the failed engine. This displays the MOD ENG OUT CLB page (ENG OUT CLB for FMC U10.3 and later) which provides advisory data for an engine out condition.

The flight director roll mode commands wings level or HDG SEL (as installed) after liftoff until LNAV engagement or another roll mode is selected. If ground track is not consistent with desired flight path, use HDG SEL/LNAV to achieve the desired track.

Indications of an engine fire, impending engine breakup or approaching or exceeding engine limits, should be dealt with as soon as possible. Accomplish the appropriate memory checklist items as soon as the airplane is under control, the gear has been retracted and a safe altitude (typically 400 feet AAL or above) has been attained. Accomplish the reference checklist items after the flaps have been retracted and conditions permit.

If an engine failure has occurred during initial climb, accomplish the appropriate checklist after the flaps have been retracted and conditions permit.

IMMEDIATE TURN AFTER TAKEOFF - ONE ENGINE INOPERATIVE

Obstacle clearance or departure procedures may require a special engine out departure procedure. If an immediate turn is required, initiate the turn at the appropriate altitude (normally at least 400 feet AAL). Maintain V2 to V2 + 20 knots with takeoff flaps while maneuvering.

NOTE: Limit bank angle to 15° until V2 + 15 knots. Bank angles up to 30° are permitted at V2 + 15 knots with takeoff flaps. With LNAV engaged, the AFDS may command bank angles greater than 15°.

After completing the turn, and at or above acceleration height, accelerate and retract flaps.

4.6.3. Crew coordination on engine out takeoff

PF	PM
Directional control .... Maintain Maintain directional control by smoothly applying rudder proportionate with thrust decay.	
At Vr speed	
Rotation .... .Initiate With an engine inoperative, a smooth continuous rotation is also initiated at VR; however, the target pitch attitude is approximately 2° to 3° below the normal all engine pitch attitude resulting in a 12°-14° target pitch attitude. The rate of rotation with an engine inoperative is also slightly slower (1/2° per second less) than that for a normal takeoff, resulting in a rotation rate of approx.1.5° to 2.5° per second. After liftoff adjust pitch attitude to maintain the desired speed of V2 to V2 +20 knots.	**Airspeed and vertical speed**..... . Monitor
Verify positive rate of climb on the baro altimeter and call **“Gear Up”**	Verify positive rate of climb on the baro altimeter and call **“Positive rate”** Call: **“GEAR UP”**. **Landing gear lever** .... UP Verify Landing gear lights off and call: **“GEAR UP”**
Initial climb attitude .... Adjust To maintain a minimum of V2 and a positive climb. After liftoff the flight director provides proper pitch guidance.	
Crosscheck indicated airspeed, vertical speed and other flight instruments. The flight director commands a minimum of V2, or the existing speed up to a maximum of V2 + 20 knots. If engine failure occurs above V2 + 20 knots, increase pitch to reduce airspeed to V2 + 20 knots and maintain V2 + 20 knots until reaching acceleration height.	

Roll control .... Establish Apply rudder and aileron to control heading and keep the wings level. The correct rudder input approximately centers the control wheel.	
The Flight Director roll mode commands heading after liftoff until LNAV engagement or another roll mode is selected. If ground track is not consistent with desired flight path, use HDG SEL/LNAV to achieve the desired track. Limit bank angle to 15 until V2+15 knots. Bank angles up to 30 are permitted at V2+15 knots with takeoff flaps. With LNAV engaged, the AFDS may command bank angles greater than 15°. When aircraft stabilized, the crew must execute one of the procedures according to the situation (see below) and the takeoff briefing: • ENGINE OUT SID (if any); • climb straight ahead (up to Minimum Radar Altitude, MSA or 25 n.m what is first, then vectors);	
At 400 feet AAL	
Call for proper ROLL MODE.	Call: “**400 feet”** **Proper ROLL MODE**.......................... Set Verify proper mode annunciation on FMA
Engage the autopilot (if needed). When at a safe altitude above 400 feet AAL with correct rudder pedal or trim input, if VNAV is armed for takeoff the autopilot may be engaged. If VNAV is not armed for takeoff autopilot engagement should be delayed until the flaps are up and LVL CHG is selected. This allows the AFDS to remain in the TO/GA mode during flap retraction.	When autopilot engaged call **“COMMAND”**
Call: “**(NNC title) MEMORY ITEMS”**	In some specific situations (immediate turns after takeoff or airport regulations) ATC contact may be needed as soon as practical (before NNC memory items)
Perform memory items according to areas of responsibility	
NOTE: During a Reduced Thrust (ATM) Takeoff it is not necessary to increase thrust beyond the reduced level on the operating engine in the event of an engine failure. However, if more thrust is needed during an ATM takeoff, thrust on the operating engine may be increased to full rated takeoff thrust by manually advancing the thrust lever. **NOTE:** During a fixed derate takeoff, a thrust increase following an engine failure could result in loss of directional control and should not be accomplished unless, in the opinion of the Captain, terrain contact is imminent.	

NOTE: If an engine failure occurs during takeoff when using both the reduced thrust (ATM) and fixed derate methods thrust may be advanced to the fixed derate limit only.	
	ATC... Advise Advise ATC about nature of failure and intentions using proper urgency or distress call (PAN-PAN or MAYDAY). ATC may be advised earlier if time permits and aircraft is stabilized
At Engine Out Acceleration Altitude In any case EO acceleration height must not be less than 400 AAL or Min EO ACC ALT (use EFB). If an immediate turn is required, initiate the turn at the appropriate altitude (normally at least 400 feet AAL). Maintain V2 to V2 + 20 knots with takeoff flaps while maneuvering. Radar vectoring by ATC is available above MRVA (Minimum Radar Vectoring Altitude). **NOTE:** Limit bank angle to 15° until V2 + 15 knots. Bank angles up to 30° are permitted at V2 + 15 knots with takeoff flaps. With LNAV engaged, the AFDS may command bank angles greater than 15°. After completing the turn, and at or above acceleration height, accelerate and retract flaps. Maintain V2 to V2 + 15 knots with takeoff flaps while maneuvering.	
At engine out acceleration height, if VNAV is engaged, a near-level climb segment is commanded for acceleration. If VNAV is not engaged, leave the pitch mode in TO/GA and select flaps up maneuvering speed on the MCP. Engine-out acceleration and climb capability for flap retraction are functions of airplane thrust to weight ratio. The flight director commands a near level or a slight climb (0-200 fpm) flap retraction segment. Accelerate and retract flaps on the takeoff flap retraction speed schedule.	
Acceleration (VNAV engage) .... Verify If flying manually: Call: **“Set FLAPS UP speed”** If using autopilot: Set speed for desired flap setting. Call: **“FLAPS UP speed set”** Check aircraft acceleration.	Call: **“Acceleration”** Set commanded speed on MCP and call: **“FLAPS UP speed set**”
Call **“FLAPS ___”** according to the flap retraction schedule.	Call: **"SPEED Checked, FLAPS ___ "** Set the flap lever as directed. Monitor flaps and slats extension.

Flaps are in desired position	
With VNAV engaged and flaps up, check that FMC commands a climb at flaps up maneuver speed. If VNAV is not engaged, select LVL CHG Call: **"N1 LIMIT MAX CONTINUOUS"**	Select N1 page on CDU Activate CON. Assist PF to set the correct thrust
Call: **"(NNC title) CHECKLIST"**	
Before commencing NNC both pilots should verify that correct C/L is going to be done.	
	Perform requested NNC
	When NNC complete Call: **"(NNC title) checklist complete"**
Call: **"AFTER TAKE-OFF checklist"**	Do the “AFTER TAKEOFF CHECKLIST”. Call: **“AFTER TAKEOFF CHECKLIST COMPLETE”**.
Make decision	

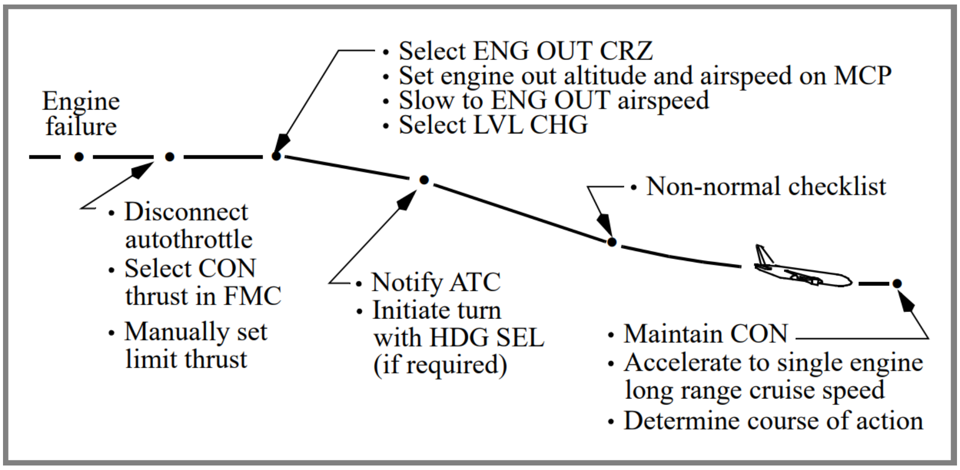
4.6.4. Engine Inoperative Cruise/Driftdown

Execution of a non-normal checklist or sudden engine failure may lead to the requirement to perform an engine inoperative driftdown and diversion to an alternate airport. Engine inoperative cruise information is available from the FMC. An analysis of diversion airport options is available using the FMC ALTN DEST page (as installed).

If an engine failure occurs while at cruise altitude, it may be necessary to descend. The autothrottle should be disconnected and the thrust manually set to CON. On the FMC CRZ page, select the ENG OUT prompt, followed by the prompt corresponding to the failed engine. This displays MOD ENG OUT CRZ (ENG OUT CRZ for FMC U10.3 and later) and the FMC calculates engine out target speed and maximum engine out altitude for that speed at the current gross weight. The fields are updated as fuel is burned.

Set the nearest FL below MAX altitude in the MCP altitude window and the engine out target airspeed in the MCP IAS window. Allow airspeed to slow to engine out speed then engage LVL CHG. If the engine out target airspeed and maximum continuous thrust (MCT) are maintained, the airplane levels off above the original MAX altitude. However, the updated MAX altitude is displayed on the ENG OUT CRZ page. After viewing engine out data, select the ERASE prompt to return to the active CRZ page. With the MOD ENG OUT CRZ (ENG OUT CRZ for FMC U10.3 and later) page selected, no other FMC data pages can be executed.

After level off at the target altitude, maintain MCT and allow the airplane to accelerate to the single engine long range cruise speed. Maintain this speed with manual thrust adjustments. Entering the new cruise altitude and airspeed on the ECON CRZ page updates the ETAs and Top of Descent predictions.



NOTE: If the airplane is at or below maximum ENG OUT altitude when an engine becomes inoperative, select the MOD ENG OUT CRZ (ENG OUT CRZ for FMC U10.3 and later). Maintain engine out cruise speed using manual thrust adjustments.

In the unlikely event severe engine damage occurs, additional drag associated with severe internal damage or a damaged engine cowl may prevent the airplane from maintaining the FMC calculated engine out cruise altitude. If severe internal engine damage or severe damage to the engine cowl is confirmed or suspected and driftdown is required to reach the intended point of landing, fly the FMC calculated engine out target speed and use MCT on the operating engine. Descend until the airplane is able to maintain altitude at the target speed. Engine Electronic Control logic may allow for additional thrust, if needed, from the operating engine below 15,000 feet pressure altitude and 0.45 Mach by advancing the thrust lever to the forward mechanical stop.

If required to cruise at maximum altitude, set MCT and establish a climb, decelerating slowly to EO CLB speed. At level off select EO LRC for best fuel economy.

ETOPS engine-out procedures may be different from standard non-normal procedures. Following an engine failure, the crew performs a modified driftdown procedure determined by the ETOPS route requirements. This procedure typically uses higher descent and cruise speeds, and a lower cruise altitude following engine failure. This allows the airplane to reach an alternate airport within the specific time limits authorized for the operator. These cruise speeds and altitudes are determined by the operator and approved by its regulatory agency and usually differ from the engine-out speeds provided by the FMC. These cruise speeds and altitudes can be founded in **FPPM-ENROUTE**-Engine inoperative section. The captain, however, has the discretion to modify this speed if actual conditions following the diversion decision dictate such a change.

	Speed (1 eng inop)	60 min	120 min
Boeing 737 NG	0.79M/330KIAS	427 nm	835 nm

PF	PM
MASTER CAUTION ENGINE 1(2) FAIL……………………………………… Announce When the malfunction verified by another pilot by response **"CHECK":** **MASTER CAUTION**……………………………………………………..…. Reset	
Call: **"N1 LIMIT MAX CONTINUOUS"**	Select N1 page on CDU Activate CON.
AUTOTHROTTLE .... Disconnect **CON thrust** .... Set	Assist PF to set the correct thrust
Call: **"SELECT ENGINE OUT CRUISE PAGE"**	**ENG OUT CRZ page**..................... Select Call: **"MAX altitude ___, speed___ "**
If the airplane is above maximum engine out altitude perform drift down procedure	
If the turn from airway or track is required: **HDG approx. 30º R/L** .... Set **HDG SEL** .... Push To be away from the airway or track, taking into consideration other aircraft position, obstacles, the weather, boundaries, national flight rules, direction to alternate, etc.	Verify settings on MCP and FMA change
ENG OUT altitude on MCP................ Set **ENG OUT speed on MCP** .... Set Decelerate to ENG OUT airspeed **LVL CHG** .... Push	Verify settings on MCP and FMA change **ATC (PAN-PAN or MAYDAY)**......... Notify
Call: **(NNC title) memory items (if required)**	
Perform memory items according to areas of responsibility	
When stable descent is established and if driftdown procedure no longer required adjust HDG, SPEED and THRUST as needed to reach route alternate airport.	

4.6.5. Single engine approach and landing

For detailed information refer to QRH **“One Engine Inoperative Landing”** NNC.

The single engine approach and landing profiles are similar to the two engine approach.

Pilots must follow all standard callouts and procedures. However, flaps 15 provide the best performance capability, and reduced available thrust requires careful drag management.

The autopilot is certified for single engine CAT I ILS approaches if desired.

During a single autopilot or flight director (manual) approach, the pilot must use rudder pedal pressure to control yaw, followed by rudder trim to maintain an in–trim condition during the entire approach. A centered control wheel indicates proper trim.

NOTE: Use of the autothrottle for an approach with an engine inoperative is not recommended.

Intercept the localizer with flaps 5 at flaps 5 speed. When the glide slope is alive, lower the landing gear, extend flaps to 15, set final approach speed, and decelerate.

ENGINE FAILURE ON FINAL APPROACH

If an engine failure should occur on final approach with the flaps in the landing position, the decision to continue the approach or execute a go–around should be made immediately. If the approach is continued and sufficient thrust is available, continue the approach with landing flaps. If the approach is continued and sufficient thrust is not available for landing flaps, retract the flaps to 15 and adjust thrust on the operating engine. Command speed should be increased to 20 knots over the previously set flaps 30 or 40 Vref. This sets a command speed that is equal to at least VREF for flaps 15 and is represented by the white colored bug on the airspeed tape. Wind additives should be added as needed, if time and conditions permit.

If a go–around is required, follow the Go–Around and Missed Approach procedures except use flaps 15 initially if trailing edge flaps are at 30 or 40. Subsequent flap retraction should be made at a safe altitude and in level flight or a shallow climb.

4.6.6. Engine Oil System Indications

Oil pressure is considered as the most significant of several oil system indicators.

Oil temperature, oil quantity and oil pressure indications enable the flight crew to recognize a deteriorating oil system. While engine operation is governed by both oil pressure and oil temperature limits, there is no minimum oil quantity limit. Therefore, there is no low oil quantity NNC in the QRH.

If abnormal oil quantity indications are observed, check oil pressure and oil temperature. If oil pressure and oil temperature indications are normal, operate the engine normally. Accomplish the appropriate NNC for any non-normal oil pressure or oil temperature indications.

4.6.7. Fire Engine Tailpipe

On the ground, there is always a concern about identifying a tailpipe fire. The flight crew needs to use information from outside observers and flight deck indications to quickly determine if the fire is a tailpipe fire or not. A tailpipe fire is likely to **occur only during engine start or shutdown**. After the fuel is cutoff, the flight crew determines whether a fire is a tailpipe fire or not. If it is a tailpipe fire, the flight crew exits the checklist and goes to the Tailpipe Fire checklist. This method stops fuel flowing to the fire before any discussion about the type of fire. Also, this ensures the engine is still available to blow out a tailpipe fire.

Engine tailpipe fires can occur due to the following:

- engine control malfunction;
- excess fuel in the combustor, turbine or exhaust nozzle;
- oil accumulation in the hot section flow path or exhaust system.

If a tailpipe fire is reported, the crew should accomplish the NNC without delay. Flight crews should consider the following when dealing with this situation:

- motoring the engine is the primary means of extinguishing the fire;

- to prevent an inappropriate evacuation, flight attendants should be notified without significant delay;
- communications with ramp personnel and the tower are important to determine the status of the tailpipe fire and to request fire extinguishing assistance;
- the engine fire checklist is inappropriate because the engine fire extinguishing agent is not effective against a fire inside the tailpipe.

4.6.8. Loss of Engine Thrust Control

All turbo fan engines are susceptible to this malfunction whether engine control is hydro-mechanical, hydro-mechanical with supervisory electronics (e.g. PMC) or Full Authority Digital Engine Control (FADEC). Engine response to a loss of control varies from engine to engine. Malfunctions have occurred in-flight and on the ground. The major challenge the flight crew faces when responding to this malfunction is recognizing the condition and determining which engine has malfunctioned. The Engine Limit or Surge or Stall NNC is written to include this malfunction. This condition can occur during any phase of flight.

Failure of engine or fuel control system components or loss of thrust lever position feedback has caused loss of engine thrust control. Control loss may not be immediately evident since many engines fail to some fixed RPM or thrust lever condition. This fixed RPM or thrust lever condition may be very near the commanded thrust level and therefore difficult to recognize until the flight crew attempts to change thrust with the thrust lever. Other engine responses include: shutdown, operation at low RPM, or thrust at the last valid thrust lever setting (in the case of a thrust lever feedback fault) depending on altitude or air/ground logic. In all cases, the affected engine does not respond to thrust lever movement or the response is abnormal.

Since recognition may be difficult, if a loss of engine control is suspected, the flight crew should continue the takeoff or remain airborne until the NNC can be accomplished. This helps with directional control and may preclude an inadvertent shutdown of the wrong engine. In some conditions, such as during low speed ground operations, immediate engine shutdown may be necessary to maintain directional control.

4.6.9. Loss of Thrust on Both Engines

Dual engine failure is a situation that demands prompt action regardless of altitude or airspeed. Accomplish memory items and establish the appropriate airspeed to immediately attempt a windmill restart. There is a higher probability that a windmill start will succeed if the restart attempt is made as soon as possible (or immediately after recognizing an engine failure) to take advantage of high engine RPM. Establishing airspeeds above the crossbleed start envelope and altitudes below 30,000 feet improves the probability of a restart. Loss of thrust at higher altitudes may require descent to a lower altitude to improve windmill starting capability.

The in-flight start envelope defines the region where windmill starts were demonstrated during certification. It should be noted that this envelope does not define the only areas where a windmill start may be successful. The LOSS OF THRUST ON BOTH ENGINES NNC is written to ensure that flight crews take advantage of the high RPM at engine failure regardless of altitude or airspeed. Initiate the memory portion of the LOSS OF THRUST ON BOTH ENGINES NNC before attempting an APU start for the reasons identified above. If the windmill restart is not successful, an APU start should be initiated as soon as practical to provide electrical power and starter assist during follow-on engine start attempts.

During in-flight restart attempts use the lower of the EGT redline and the EGT start limit redline, if displayed. A hung or stalled in-flight start is normally indicated by stagnant RPM and increasing EGT. During start, engines may accelerate to idle slowly but action should not be taken if RPM is increasing and EGT is not near or rapidly approaching the limit.

NOTE: When electrical power is restored, do not confuse the establishment of APU generator power with the establishment of engine generator power at idle RPM and advance the thrust lever prematurely.

4.6.10. Airframe Vibration Due to Engine Severe Damage or Separation

Certain engine failures, such as fan blade separation can cause high levels of airframe vibration. Although the airframe vibration may seem severe to the flight crew, it is extremely unlikely that the vibration will damage the airplane structure or critical systems. However, the vibration should be reduced as soon as possible by reducing airspeed and descending. In general, as airspeed decreases vibration levels decrease. As airspeed or altitude change the airplane can transition through various levels of vibration. It should be noted that the vibration may not completely stop.

If vibration remains unacceptable, descending to a lower altitude (terrain permitting) allows a lower airspeed and normally lower vibration levels. Vibration will likely become imperceptible as airspeed is further reduced during approach.

The impact of a vibrating environment on human performance is dependent on a number of factors, including the orientation of the vibration relative to the body. People working in a vibrating environment may find relief by leaning forward or backward, standing, or otherwise changing their body position.

Once airframe vibration has been reduced to acceptable levels, the crew must evaluate the situation and determine a new course of action based on weather, fuel remaining, and available airports.

4.6.11. Recommended Technique for an In-Flight Engine Shutdown

Any time an engine shutdown is needed in flight, good crew coordination is essential. Airplane incidents have turned into airplane accidents as a result of the flight crew shutting down the incorrect engine.

When the flight path is under complete control, the crew should proceed with a deliberate, systematic process that identifies the affected engine and ensures that the operating engine is not shut down. Do not rush through the shutdown checklist, even for a fire indication. The following technique is an example that could be used:

When an engine shutdown is needed, the PF disconnects the A/T. The PF then verbally coordinates confirmation of the affected engine with the PM and then slowly retards the thrust lever of the engine that will be shutdown.

Coordinate activation of the fuel control switch as follows:

- PM places a hand on and verbally identifies the start lever for the engine that will be shutdown
- PF verbally confirms that the PM has identified the correct start lever
- PM moves the start lever to cutoff.PM moves the fuel control switch to cutoff.

If the NNC requires activation of the engine fire switch, coordinate as follows:

- PM places a hand on and verbally identifies the engine fire switch for the engine that is shutdown
- PF verbally confirms that the PM has identified the correct engine fire switch
- PM pulls the engine fire switch.

4.6.12. Engine Fire. Company policy.

The following detailed instructions for the pilots were discussed and approved on Company instructors meeting on Nov'21:

- If information about Engine Fire confined to the engine core exhaust has been received from ground staff during engine start or shutdown, but no any indication in the cockpit exists, then follow "Engine Tailpipe Fire" NNC.
- If aircraft is on the ground and there is any indication of the engine fire in the cockpit or engine fire which is not confined to the engine core exhaust during engine start or shutdown reported by someone on the ground, then follow "Engine Fire on the Ground" NNC.
- If engine fire indication has been received at take off after V1, continue take off, start "Engine Fire or Engine Severe Damage or Separation" NNC memory items when aircraft under control and above 400' AAL, using two bottles by memory (if needed). Continue "Engine Fire or Engine Severe Damage or Separation" NNC after flap retraction completed. If Engine Fire has not been extinguished after using two bottles, be ready for immediate landing and no need to retract the flaps.
- If engine fire indication has been received on final approach and "Engine Fire or Engine Severe Damage or Separation" NNC memory items have not been started in the air, follow "Engine Fire on the Ground" NNC after landing and complete stop on the ground. If the crew has started "Engine Fire or Engine Severe Damage or Separation" NNC memory items in the air, then continue "Engine Fire or Engine Severe Damage or Separation" NNC after stop on the ground.
- In case when fire has not been extinguished after "Engine Fire or Engine Severe Damage or Separation" NNC memory items were completed in the air start "EVACUATION" NNC after full stop on the ground.

4.6.13. Engine fire on the ground

Engine Fire on the Ground

Condition: One or more of these occur on the ground:
- An engine fire is observed
- Engine fire warning

1 Thrust levers (both) . . . . Close — C

2 PARKING BRAKE. . . . . Set — C

3 Advise the cabin. — C or F/O

4 Engine start lever (affected engine) . . . . CUTOFF — C

5 Engine fire switch (affected engine) . . . . Pull — F/O

To manually unlock the engine fire switch, press the override and pull.

6 Engine fire switch (affected engine) . . . . Rotate to the stop and hold for 1 second — F/O

7 Choose one:

◆Evacuation is **needed**:

►►**Go to the Evacuation checklist on page Back Cover.4**

■ ■ ■ ■

◆Evacuation is **not** needed:

►►**Go to step 8**

8 Advise the cabin. — C

9 Advise ATC. — F/O

The two main goals of this checklist are:
- to prevent passengers from starting an evacuation on their own, and
- to respond to an engine fire that is not annunciated, but is observed and reported by ground crew, the cabin crew, the crew of another airplane, or Air Traffic Control.

The new checklist is intended to be used during ground operations when an engine fire occurs. It assumes the airplane can be anywhere on the ground when the fire occurs:
- gate,
- ramp,
- parking area,
- taxiway,
- runway.

For airplanes with an engine fire alerting system such as the Engine Indicating, the existing annunciated engine fire checklist has been modified to accommodate an engine fire on the ground. The new unannunciated Engine Fire on the Ground checklist located near the back cover of the QRH, just before the Evacuation checklist.

4.6.14. Engine Fire After Takeoff (in flight)

In case of an engine fire, when the flight path is under control, the gear has been retracted, and a safe altitude has been attained (minimum 400 feet AAL) accomplish the NNC memory items. Due to asymmetric thrust considerations, Boeing recommends that the PF retard the affected thrust lever after the PM confirms that the PF has identified the correct engine.

ENGINE FIRE
or
Engine Severe Damage or Separation

Condition: One or more of these occur:
- Engine fire warning
- Airframe vibrations with abnormal engine indications
- Engine separation.

1 Autothrottle (if engaged) . . . . .Disengage

2 Thrust lever
(affected engine) . . . . . . Confirm . . . . Close

3 Engine start lever
(affected engine) . . . . . . Confirm . . . . . CUTOFF

4 Engine fire switch
(affected engine) . . . . . . Confirm . . . . Pull

To manually unlock the engine fire switch, press the override and pull.

5 **If** the engine fire switch or ENG OVERHEAT light is illuminated:

Engine fire switch . . . . . Rotate to the stop and hold for 1 second

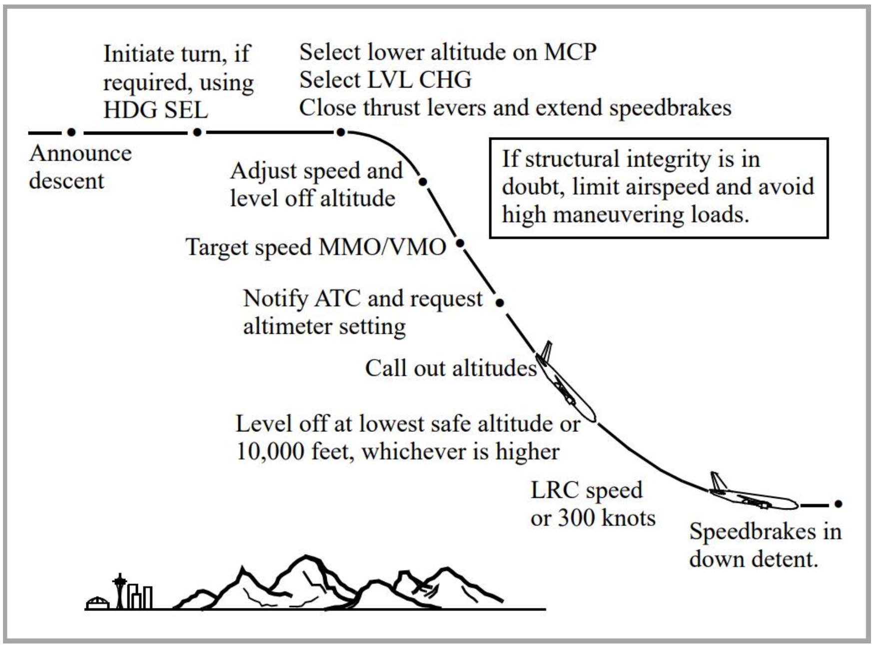
If after 30 seconds the engine fire switch or ENG OVERHEAT light stays illuminated:

Engine fire switch. . . . . . . .Rotate to the other stop and hold for 1 second

4.7. EMERGENCY DESCENT (CABIN ALTITUDE)

This section addresses basic techniques and procedures for a rapid descent. Some routes over mountainous terrain require careful operator planning to include carrying additional oxygen, special procedures, higher initial level off altitudes, and emergency routes in the event a depressurization is experienced. These requirements are normally addressed in an approved company route manual or other document that addresses route specific depressurization procedures. This maneuver is designed to bring the airplane down smoothly to a safe altitude, in the minimum time, with the least possible passenger discomfort.

NOTE: Use of the autopilot is recommended.



If the descent is performed because of a rapid loss of cabin pressure, crewmembers should place oxygen masks on and establish communication at the first indication of a loss of cabin pressurization. Verify cabin pressure is uncontrollable, and if so begin descent. If structural damage exists or is suspected, limit airspeed to current speed or less. Avoid high maneuvering loads.

Perform the maneuver deliberately and methodically. Do not be distracted from flying the airplane. If icing conditions are entered, use anti-ice and thrust as required.

NOTE: Rapid descents are normally made with the landing gear up.

The PM checks the lowest safe altitude, notifies ATC, and obtains an altimeter setting (QNH). Both pilots should verify that all memory items have been accomplished and call out any items not completed. The PM calls out 2,000 feet and 1,000 feet above the level off altitude.

Level off at the lowest safe altitude or 10,000 feet, whichever is higher. Lowest safe altitude is the Minimum Enroute Altitude (MEA), Minimum Off Route Altitude (MORA), or any other altitude based on terrain clearance, navigation aid reception, or other appropriate criteria.

If severe turbulent air is encountered or expected, reduce to the turbulent air penetration speed.

PF	PM
Call (first who notices alert or decompression): **"OXYGEN MASK ON"**	
Oxygen masks (without control transfer)……………………..……………. Done **Crew communication**…………………..……………………………………... Establish	
Call: **"CABIN ALTITUDE memory items"**	Perform memory items
IF CABIN ALTITUDE IS CONTROLLABLE	
Call: **"CABIN ALTITUDE WARNING OR RAPID DEPRESSURIZATION CHECKLIST"**	Call: **"Cabin Altitude Controllable"** Perform CABIN ALTITUDE WARNING or Rapid Depressurization NNC
IF CABIN ALTITUDE IS UNCONTROLLABLE	
	Call: **"Cabin Altitude Uncontrollable"**
Call**: "EMERGENCY DESCENT MEMORY ITEMS"**	
Alert cabin crew: **" EMERGENCY DESCENT, EMERGENCY DESCENT"**	
Without delay: **HDG SEL knob**................................ Turn **HDG SEL** .... Push **Lower altitude on MCP**..................... Set **LVL CHG** .... Push **THRUST levers** .... Close **SPEEDBRAKE** .... Flight detent **Speed to MMO/VMO** .... Adjust If structural integrity is in doubt, limit airspeed and avoid high maneuvering loads **HEADING**..................................... Adjust **ALTITUDE**.................................... Adjust Descend to lowest safe altitude or 10,000 feet, whichever is higher.	**PASS OXYGEN switch** .... ON **FASTEN BELTS switch** .... ON **LANDING LIGHTS** .... ON **ENGINE START switches (both)**.. CONT **ATC SQUAWK 7700** .... Set Advice ATC: **"MAYDAY, MAYDAY, MAYDAY, NWS ___, EMERGENCY DESCENT".** **Lowest safe altitude** .... Check **Altimeter setting** .... Obtain
• To be able to find the accurate MORA (Greed MORA), at least one pilot should have a CFP and Enroute chart correctly opened on the part of the route flown by the airplane.	

<table>
<tr><td colspan="2">

- The MCP speed has to be adjusted either to M.82/340 kts or to be limited to current speed (or less) if structural integrity exists or is suspected. Do not assume structural integrity unless it can be positively assured from the flight deck
- High maneuvering loads should be avoided
- During descent, the IAS/MACH speed window changes from MACH to IAS at approximately 330 KIAS, manually reset to VMO as needed
- During descent both pilots should watch for conflicting traffic both visually and by reference to TCAS

</td></tr>
<tr><td colspan="2">FMA and flight parameters ……………………..………………………………. Check</td></tr>
<tr><td>Call: "CABIN ALTITUDE WARNING or Rapid Depressurization Checklist"</td><td>Perform CABIN ALTITUDE WARNING or Rapid Depressurization NNC</td></tr>
<tr><td colspan="2">2000 ft above level off altitude</td></tr>
<tr><td>Speed LRC or 300 .... Set</td><td>Call: “2000 TO LEVEL OFF”</td></tr>
<tr><td colspan="2">1000 ft above level off altitude.</td></tr>
<tr><td>Speedbrake .... Retract</td><td>Call: “1000 TO LEVEL OFF”</td></tr>
<tr><td colspan="2">After level off</td></tr>
<tr><td colspan="2">Flight crew must use oxygen when cabin altitude is above 10,000 feet</td></tr>
<tr><td>Make PA: “EMERGENCY DESCENT IS COMPLETED. PURSER TO THE COCKPIT”</td><td></td></tr>
</table>

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.8. PASSENGER EVACUATION

NOTE: Main door emergency evacuation slide system must be armed prior to taxi, takeoff, and landing whenever passengers are carried.

Evacuation has to be done according to QRH “EVACUATION” non-normal check list which can be found on Back Cover.2.

FOR DETAILED INFORMATION REFER TO OM PART A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.9. PILOT INCAPACITATION

FOR DETAILED INFORMATION REFER TO OM PART A

СТРАНИЦА НАМЕРЕННО ОСТАЛЕНА ПУСТОЙ

4.10. OVERWEIGHT LANDING

Overweight landings may be safely accomplished by using normal landing procedures and techniques. There are no adverse handling characteristics associated with overweight landings. Landing distance is normally less than takeoff distance for flaps 30 or 40 landings at all gross weights. However, wet or slippery runway field length requirements should be verified from the landing distance charts in the PI chapter of the QRH. Brake energy limits will not be exceeded for flaps 30 or 40 normal landings at all gross weights.

NOTE: Use of flaps 30 rather than flaps 40 is recommended to provide increased margin to flap placard speed.

If stopping distance is a concern, reduce the landing weight as much as possible. At the captain's discretion, reduce weight by holding at low altitude with a high drag configuration (gear down) to achieve maximum fuel burn-off.

Analysis has determined that, when landing at high gross weights at speeds associated with non-normal procedures requiring flaps set at 15 or less, maximum effort stops may exceed the brake energy limits. The gross weights where this condition can occur are well above maximum landing weights. For these non-normal landings, maximize use of the available runway for stopping.

Observe flap placard speeds during flap extension and on final approach. In the holding and approach patterns, maneuvers should be flown at the normal maneuver speeds. During flap extension, airspeed can be reduced by as much as 20 knots below normal maneuver speeds before extending to the next flap position. These lower speeds result in larger margins to the flap placards, while still providing normal bank angle maneuver capability, but do not allow for a 15° overshoot margin in all cases.

Use the longest available runway, and consider wind and slope effects. Where possible avoid landing in tailwinds, on runways with negative slope, or on runways with less than normal braking conditions. Do not carry excess airspeed on final. This is especially important when landing during an engine inoperative or other non-normal condition. At weights above the maximum landing weight, the final approach maximum wind additive may be limited by the flap placards and load relief system.

Fly a normal profile. Ensure that a higher than normal rate of descent does not develop. Do not hold the airplane off waiting for a smooth landing. Fly the airplane onto the runway at the normal touchdown point. If a long landing is likely to occur, go-around. After touchdown, immediately apply maximum reverse thrust using all of the available runway for stopping to minimize brake temperatures. Do not attempt to make an early runway turnoff.

Autobrake stopping distance guidance is contained in the PI chapter of the QRH. If adequate stopping distance is available based upon approach speed, runway conditions, and runway length, the recommended autobrake setting should be used.

OVERWEIGHT AUTOLANDS POLICY

Overweight autolands are not recommended**.** Autopilots on Boeing airplanes are not certified for automatic landings above maximum landing weight. At higher than normal speeds and weights, the performance of these systems may not be satisfactory and has not been thoroughly tested. An automatic approach may be attempted, however the pilot should disengage the autopilot prior to flare height and accomplish a manual landing.

In an emergency, should the pilot determine that an overweight autoland is the safest course of action, the approach and landing should be closely monitored by the pilot and the following factors considered:

- touchdown may be beyond the normal touchdown zone; allow for additional landing distance;
- touchdown at higher than normal sink rates may result in exceeding structural limit;
- plan for a go-around or manual landing if autoland performance is unsatisfactory; automatic go-arounds can be initiated until just prior to touchdown, and can be continued even if the airplane touches down after initiation of the go-around.

4.11. LOW FUEL MANAGEMENT

Refer to FOM Part A Chapter 8.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.12. EMERGENCY LANDING

Emergency landing can be prepared and unprepared.

When emergency landing is unprepared there is no sufficient time to brief cabin crew and passengers, however flight crew must take any possible effort by sharing information with cabin crew about imminent landing and nature of emergency.

When emergency landing is necessary to perform and time permits, share as much as possible information with ATC, cabin crew and passengers to make a best coordination for survival, search and rescue after emergency landing completed.

Consider burning off fuel prior to emergency landing, if the situation permits. This provides a lower approach speed and reduces possibility of fire in case of airplane disintegration.

NOTE: Do not reduce fuel to a critical amount, as emergency landing with engine power available improves ability to properly control touchdown.

Due to high potential risk and crew duties after touch down and following Evacuation, the Captain always performs PF duties during emergency landing.

Captain	First officer
Call: **"I HAVE CONTROL"**. Fly the airplane to the nearest airfield or populated locality and call for corresponding non-normal checklist. Make PA: **“PURSER TO THE COCKPIT”**. Brief the purser about emergency landing NITS: • Nature of the problem • Intention • Time needed – to sort out the problem • Special instructions if required.	Send distress signals. Determine position, course, speed, altitude, situation, intention, time and position of intended touchdown and transmit mayday. Set transponder code 7700.
At 2000 feet AAL	
Make PA: **“FINISH PREPARATION, FINISH PREPARATION”**	
At 500 feet AAL	
Make PA: **“BRACE FOR IMPACT, BRACE FOR IMPACT”**	
After aircraft comes to a full stop	
If evacuation is needed it has to be done according to QRH “EVACUATION” non-normal check list which can be found on Back Cover.2.	
	Proceed to nearest available exit and evacuate as soon as possible.

Ensure all passengers are out of the airplane, proceed to nearest available exit and evacuate.	
NOTE: The captain will be the last person to leave an aircraft alive before its utter destruction!	

4.13. DITCHING

SEND DISTRESS SIGNALS

Transmit Mayday, current position, course, speed, altitude, situation, intention, time and position of intended touchdown, and type of airplane using existing air-to-ground frequency. Set transponder code 7700 and, if practical, determine the course to the nearest ship or landfall.

ADVISE CREW AND PASSENGERS

Alert the crew and the passengers to prepare for ditching. Assign life raft positions (as installed) and order all loose equipment in the airplane secured. Put on life vests, shoulder harnesses, and seat belts. Do not inflate life vests until after exiting the airplane.

FUEL BURN-OFF

Consider burning off fuel prior to ditching, if the situation permits. This provides greater buoyancy and a lower approach speed. However, do not reduce fuel to a critical amount, as ditching with engine thrust available improves ability to properly control touchdown.

PASSENGER CABIN PREPARATION

Confer with cabin personnel either by interphone or by having them report to the flight deck in person to ensure passenger cabin preparations for ditching are complete.

DITCHING FINAL

Transmit final position. Select flaps 40 or landing flaps appropriate for the existing conditions. Advise the cabin crew of imminent touchdown. On final approach announce ditching is imminent and advise crew and passengers to brace for impact. Maintain airspeed at VREF. Maintain 200 to 300 fpm rate of descent. To accomplish the flare and touchdown, rotate smoothly to touchdown attitude of 10° to 12°. Maintain airspeed and rate of descent with thrust.

INITIATE EVACUATION

After the airplane has come to rest, proceed to assigned ditching stations and deploy slides/rafts. Evacuate as soon as possible, ensuring all passengers are out of the airplane.

NOTE: Be careful not to rip or puncture the slides/rafts. Avoid drifting into or under parts of the airplane. Remain clear of fuel-saturated water

Captain	First officer
Call: **"I HAVE CONTROL, DITCHING NNC"**. Fly the airplane to the nearest ship or landfall and verify all actions done properly.	**DITCHING NNC** .... Accomplish Send distress signals. Determine position, course, speed, altitude, situation, intention, time and position of intended touchdown and transmit mayday. Set transponder code 7700.

Make PA: **“PURSER TO THE COCKPIT”**. Brief the purser about emergency landing NITS: • Nature of the problem • Intention • Time needed – to sort out the problem • Special instructions if required.	
Put on life vests, shoulder harnesses, and seat belts. Do not inflate life vests until after exiting the airplane.	
At 2000 feet AAL	
Make PA: **“FINISH PREPARATION, FINISH PREPARATION”**	
At 500 feet AAL	
Make PA: **“BRACE FOR IMPACT, BRACE FOR IMPACT”**	
Maintain airspeed at VREF. Flare the airplane to achieve the minimum rate of descent at touchdown. Maintain 200–300 fpm rate of descent until the start of the flare. At flare, rotate smoothly to a touchdown attitude of 10–12°. Maintain airspeed and rate of descent with thrust. At touchdown, reduce thrust to idle.	
After impact set both engine start levers to CUTOFF. This closes fuel shutoff valves to prevent discharge of fuel from ruptured fuel lines.	
Open flight deck windows. This ensures no cabin differential pressure prevents the opening of the doors or emergency exits. **CAUTION: Do not open aft entry or service doors as they may be partially submerged.**	
	Proceed to nearest available exit and evacuate as soon as possible.
Ensure all passengers are out of the airplane, proceed to nearest available exit and evacuate.	
NOTE: The captain will be the last person to leave an aircraft alive before its utter destruction!	

4.14. SICK PASSENGER

FOR DETAILED INFORMATION REFER TO OM PART A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

4.15. LITHIUM-BATTERIES FIRE IN THE COCKPIT

If lithium-battery fire occurs in the cockpit - do the **Smoke, Fire or Fumes NNC**.

While doing the checklist consider the following:

− if necessary, transfer control to the flight crew member seated on the opposite side of the fire;

− establish cabin crew communication and report the situation in the cockpit to get assistance from cabin crew;

− if there are flames: put on the smoke hood, fire gloves and use cockpit fire extinguisher;

- If the fire source can be removed from the cockpit: coordinate actions with cabin crew, disconnect electrical cable, unlock and release the clamps, remove the source;

− if the situation becomes uncontrollable – immediate landing must be considered!!!

Several electronic devices contain lithium batteries, for example:

− laptop computers;

− mobile phones;

− portable electronic tablets, etc.

Fire or smoke from lithium battery is due to thermal runaway in the battery cells.

It is important to know that fire extinguishers are efficient on flames but cannot stop thermal runaway. The treatment for thermal runaway of lithium battery is to cool the battery by pouring water or non-alcoholic liquid on the device.

A lithium battery fire should not be treated as a Class D fire.

Halon or Halon replacement and/or water fire extinguishers can be used to control the fire and prevent its spread to surrounding flammable materials. This should be followed by immediate dousing with water or other non-flammable liquid from any available source to douse the fire. Monitor for re-occurrence and continue to pour non-flammable liquids to cool cells until cooled.

Examples of non-flammable liquids: water, juice, coffee, tea, etc.

Do not use flammable liquids: alcohol, oils, or products containing alcohol and/or oils.

Dousing (Extinguishing) with non-flammable liquids prevents adjacent cells from overheating. A battery pack involved in a fire can reignite multiple times as the heat is transferred to other cells in the pack. The device should be continually cooled with non-flammable liquids.

USE CAUTION TO ENSURE NOT TO USE LIQUIDS ON, OR IN THE IMMEDIATE VICINITY OF ANY OTHER PIECE OF ELECTRICAL EQUIPMENT.

It is important:

− Not to pick up and attempt to move a burning device or a device that is emitting smoke.

− Not to cover the device or use ice to cool the device. This would insulate the device increasing the overheating of device and resulting thermal runway.

In case of a fire involving a portable electronic device in which a passenger or crew member sustains a burn, it should be treated as a chemical burn.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 5. Performance

5.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

5.1. GENERAL

DEFINITIONS

AFM – airplane Flight Manual. The manual furnished by the manufacturer for each airplane in accordance with the rules of the certifying authority.

ASDA – accelerate-stop distance available. The sum of the available runway and stopway distances.

Assumed temperature reduced thrust analysis – a method of calculating takeoff performance that assumes a temperature greater than the actual temperature. This method, which permits the use of lower thrust settings when takeoff weight is less than the performance limited weight, can reduce wear on engines.

Brake configuration – a functional combination of braking variables, such as antiskid inoperative and autobrakes inoperative. Many airplanes are certified with various braking types and configurations that have different stopping capabilities. For example, some airplanes can be operated with one or two brakes deactivated, which results in a reduced stopping capability.

Brake energy – kinetic braking energy level that results from the braking force during a landing or aborted takeoff. The energy level is a function of weight, true airspeed, altitude, runway slope, runway conditions, wind component, and brake configuration.

Brake energy limited weight – for takeoff, the brake release gross weight, as limited by the capability of the brakes to absorb the energy during an aborted takeoff initiated at V_1. For landing, the gross weight, as limited by the capability of the brakes to absorb the energy during landing.

Clearway – an obstacle-free area beyond the takeoff runway that can be used as a part of the takeoff distance available.

Climb limited weight –brake release gross weight, as limited by the ability to take off and climb, maintaining the regulatory-required gradient in first segment, second segment, and final segment climb.

Contaminated runway – a deposit (such as snow, slush, ice, or standing water) on the runway pavement that effects on both acceleration and stopping capability.

A runway is considered to be contaminated when more than 25% of the runway surface area (whether in isolated areas or not) within the required length and width being used is covered by the following:

- Surface water more than 3 mm (0.125 in) deep, or by slush, or loose snow, equivalent to more than 3 mm (0.125 in) of water;
- Snow which has been compressed into a solid mass which resists further compression and will hold together or break into lumps if picked up (compacted snow); or
- Ice, including wet ice.

Dry runway – a dry runway is one which is neither wet nor contaminated, and includes those paved runways which have been specially prepared with grooves or porous pavement and maintained to retain ‘effectively dry’ braking action even when moisture is present.

Dump runway – a runway is considered damp when the surface is not dry, but when the moisture on it does not give it a shiny appearance.

Environmental envelope – the upper and lower limits of outside air temperature and pressure altitude within which the airplane is certified to operate.

Extended second segment – the segment of takeoff climb, beginning at gear up, during which continued climb to level-off height can occur within the takeoff thrust time limit.

Final segment – the segment of takeoff climb from flap retraction height to 1,500 feet above reference zero.

First segment – the segment of takeoff climb from screen height to gear up.

Gradient A ratio, expressed as a percentage, of the change in geometric height divided by the horizontal distance traveled in flight.

Improved climb – a performance option that increases V_2 to achieve greater climb gradients. Improved climb is typically used when performance is limited by takeoff climb or obstacles.

Level-off height – height of third segment.

Obstacle limited weight – brake release gross weight, as limited by the most restrictive obstacle in the regulatory takeoff flight path.

Reference zero – the point on the runway surface at the end of the takeoff distance required.

Screen height – the minimum height, specified by the regulatory agency, that must be attained at the end of the takeoff distance required.

Slippery Runway – a runway with deposit affecting on stopping capability only.

Second segment – the segment of takeoff climb from gear up to the level-off height.

Stopway – an area beyond the takeoff runway capable of supporting the airplane in an aborted takeoff. The stopway can be used as part of the accelerate-stop distance.

Takeoff distance required – the greater of the following two distances: (1) the distance to take off and climb to the screen height, assuming an engine failure prior to V_1, or (2) a regulatory factor (typically 115%) times the distance to take off and climb to the screen height with all engines operating. Regulatory requirements state that these distances must be within the takeoff distance available

Takeoff thrust – the amount of thrust the engine can produce at the takeoff rating

Third segment – the segment of takeoff climb that starts when level-off height has been reached and flap retraction begins and ends when flaps are up and final climb speed has been reached.

Tire speed limited weight – the maximum weight, as limited by tire speed rating.

Tire speed rating – the maximum ground speed permissible for structural integrity of the tires.

TODA – takeoff distance available. The total length of the usable runway plus the clearway.

TORA – takeoff runway length available

V_1– takeoff decision speed. Following a failure of the critical engine, the speed from which (1) a decision to continue the takeoff results in a takeoff distance to screen height at V_2 that will not exceed the usable takeoff distance, and (2) a decision to bring the airplane to a full stop will not exceed the accelerate-stop distance available.

V_{1MCG} – takeoff decision speed for minimum control speed on the ground.

V_2 – takeoff safety speed. The target speed to be attained at the screen height, assuming an engine failure during the takeoff.

V_{MBE} – maximum brake energy speed. The highest speed from which the airplane can be brought to a stop without exceeding the maximum energy absorption capability of the brakes.

V_{MCA} – the minimum speed in the air at which directional control of the airplane can still be maintained with a critical engine inoperative.

V_{MCG} – the minimum speed on the ground at which directional control of the airplane can still be maintained with a critical engine inoperative.

V_R – takeoff rotation speed. The speed at which rotation from the three-point attitude to the takeoff attitude is initiated.

V_{REF} – reference speed, which can be used to define flap retraction and final climb speeds and the speed at the threshold for landing.

V_S – stall speed.

Wet runway – a runway is considered wet when the runway surface is covered with water, or equivalent, less than specified in subparagraph above or when there is sufficient moisture on the runway surface to cause it to appear reflective, but without significant areas of standing water.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

5.2. TAKEOFF PERFORMANCE

Takeoff Weight Limitations when operating on a Dry Runway

There are 6 limitations affecting on takeoff gross weight:

- field Length;
- climb;
- obstacle/Level-off Height;
- tire Speed;
- brake Energy;
- structural.

5.2.1. Field Length Limited Performance

The takeoff gross weight is Field Length limited when the field length required by the FAR is equal to field length available. The field length required under FAR Part 25 considers both the all-engines-operative and critical engine-inoperative cases. The field length to be considered in the all-engine-operative case is defined as the actual takeoff distance to a screen height of 35 feet multiplied by the factor 1.15. The critical-engine-inoperative case, the field length required is the greater of the accelerate-go and accelerate-stop distances. These two distances are related by V_1, the takeoff decision speed. By selecting V_1 so that the accelerate-go and the accelerate-stop distance are each equal to the distances available, the resulting takeoff weight is maximized for a particular runway situation. At most airports, the field length available is synonymous with the actual length of the runway; but it may also include clearway, stopway, or combination of both. Clearway is defined as an area extending beyond the end of the runway above which no part of the terrain protrudes. Up to one-half of the air distance may extend over a clearway. A stopway is an area extending beyond the end of the runway which may be used in decelerating the airplane during a refused takeoff (accelerate-stop) and, therefore, must be capable of supporting the airplane without causing structural damage.

The Runway Analysis software provides the takeoff performances based on balanced field length without clearway and stopway considerations.

Limit by Field Length is indicated in the Runway Analysis charts by the letter “**F**”.

5.2.2. Climb Limited Performance

Takeoff weight becomes climb limited when the available climb gradient equals the minimum gradient specified by the FAR for critical-engine-inoperative case. Minimum gradients are defined for the first, second, and final segments of the takeoff flight path; however, the second segment is generally the most limiting. At the climb limited takeoff weight the 2.4% gradient requirement is met at the gear-up height.

Limit by Climb is indicated in the Runway Analysis charts by the letter “**C**”.

5.2.3. Obstacle Clearance/Level-Off Height Limited Performance

The takeoff gross weight becomes obstacle limited when the net takeoff flight path just clears an obstacle by the required 35 feet. The takeoff flight path is the calculated height versus distance relationship beginning at the point where the airplane achieves a height of 35 feet. The critical engine is assumed inoperative.

The flight path extends to a point where a gross height of 1,500 feet has been attained, and the transition to the enroute configuration is completed unless a distant obstacle requires an extension of the net flight path. The gross or actual flight path is used to determine the actual height at the transition from one segment to another. Net flight path is the gross path reduced 0.8 percent gradient capability, and the regulations require that the net flight path minus 35 feet be used to demonstrate vertical obstacle clearance. The takeoff gross weight becomes level-off limited when, with critical engine assumed inoperative, the airplane can not achieve the level-off height within 5-minute Takeoff Thrust time limit. Minimum level-off height is 400 feet.

Limit by Obstacle clearance or Level-off height is indicated in the Runway Analysis charts by the asterisk ("*****").

5.2.4. Tire Speed Limited Performance

The tires are certified to a maximum speed. The rating is given in miles per hour. The airplane must not exceed the maximum rated tire speed. The airplane reaches its maximum ground speed at liftoff. If this speed exceeds the tire speed limitation, takeoff gross weight must be reduced. In this case takeoff gross weight becomes tire speed limited.

Limit by Tire Speed is indicated in the Runway Analysis charts by the letter "**T**".

5.2.5. Brake Energy Limited Performance

The brake energy limitation is associated with the maximum amount of energy that the brakes may absorb and still be effective. It is expressed as a maximum brake energy speed, V_{MBE}, for a give takeoff weight. If the amount of energy greater than the limitation, takeoff gross weight must be reduced. In this case takeoff gross weight becomes brake energy limited. Brake energy limit should be checked at hot/high airports or when operating with tailwinds and/or improved climb where higher V_1's are involved.

Limit by Brake Energy is indicated in the Runway Analysis charts by the letter "**B**".

The maximum allowable takeoff weight is the least of the field length, climb, obstacle/level-off, tire speed, brake energy and structural limit weight.

On a Dry Runway the reverse thrust credit is assumed unavailable for engine-out accelerate-stop calculations.

5.2.6. V1(MCG) Limited Takeoff

Regulations prohibit scheduling takeoff with a V_1 less than minimum V_1 for control on the ground, $V_{1(MCG)}$. The Runway Analysis software does not provide the weights and speeds when V_1 is less than $V_{1\ (MCG)}$. This typically occurs at full rated thrust and light weights. In this case the weight and speed entries in the Runway Analysis charts are blank. **TAKEOFF IS NOT ALLOWED** in this condition.

5.2.7. Takeoff Weight Limitations when Operating on Slippery and Contaminated Runways

Takeoff weight limitations on Slippery and Contaminated runways are the same of Dry Runway case, but in this case the screen height for the engine-out accelerate-go case is reduced from 35 feet to 15 feet. Reverse thrust credit assumed unavailable for engineout accelerate-stop calculations.

Non-Dry Runways are divided into two categories:

- Slippery runways having surface conditions affecting on the airplane's deceleration capability only.
- Contaminated runways having surface conditions affecting on both the airplane's deceleration and acceleration capabilities.

The slippery runways are the runways covered with the solid contaminants (ice or compacted snow) or covered with the thin layer of loose contaminant having little effect on airplane acceleration capability (for example, standing water with depth less than 3 mm or dry snow with depth less than 10 mm). So the wet runway may also be considered as the slippery runway.

For these runways the airplane's deceleration capabilities describes in terms of slippery runway (braking action and friction coefficient). To obtain takeoff performance on slippery runways the Slippery tables in the Takeoff Analysis or in the FCOM (chapter PI) should be used.

The contaminated runways are the runways covered by relatively thick layer of loose contaminant having large effect on airplane acceleration capability (for example, standing water with depth 3 mm and more or dry snow with depth 10 mm and more).

For these runways the "equivalent depth of standing water" parameter is used. To obtain takeoff performance on contaminated runways the Slush/Standing Water tables in the Takeoff Analysis or in the FCOM (chapter PI) should be used. In accordance to FAA AC 25-13, the slippery runways are considered as contaminated, but the wet runways are not considered as contaminated.

TAKEOFF ON RUNWAYS COVERED BY DRY/COMPACTED SNOW (ADVISORY INFORMATION)

If the depth of the dry snow is less than 10 mm, the Runway Condition Assessment Table presented below is used to obtain takeoff performance. This table is also used when taking off from runway covered by the compacted snow.

Runway Condition Assessment	Braking Action
3 mm or less of Dry Snow Good	Good
3-10 mm of Dry Snow (OAT above -3°C)	Medium to Poor
3-10 mm of Dry Snow (OAT at or below -3°C)	Medium
Compacted Snow (OAT above -3°C)	Medium to Poor
Compacted Snow (OAT above -13°C and at or below -3°C)	Medium
Compacted Snow (OAT at or below -13°C)	Good to Medium

If the depth of the dry snow is greater than 10 mm, Equivalent Standing Water Depth Table is used to obtain takeoff performance.

Dry Snow Depth	Equivalent Standing Water Depth
10-25 mm	3 mm
26-50 mm	6 mm
51-100 mm	13 mm

Russian regulatory for airport operations prohibits the takeoff when depth of the dry snow is greater than 50 mm.

MINIMUM LINE-UP DISTANCE CORRECTIONS

Regulations require that the runway length be adjusted to account for alignment of the airplane prior to takeoff. The table below provides TORA, TODA and ASDA adjustments for both 90 degree taxiway entry and 180 degree turnaround. These values may be used when obtaining takeoff weights from the Airplane Flight Manual or a takeoff module OPT.

When using line-up allowances with the Field Length Limit chart, the field length available must be reduced by the ASDA adjustment.

	90 DEGREE TAXIWAY ENTRY	180 DEGREE TURNROUND	
		45 M (200 FT) RW	60 M (200 FT) RW
	LINE-UP DISTANCE – M(FT)	LINE-UP DISTANCE – M(FT)	LINE-UP DISTANCE – M(FT)
TORA & ASDA	11 (35)	19 (64)	19 (64)
ASDA	26 (86)	35 (115)	35 (115)

5.2.8. Reduced and Derated Takeoff Thrust

INTRODUCTION

Normally, takeoffs are conducted with less than full rated takeoff thrust whenever performance capabilities permit. Lower takeoff thrust reduces EGT, improves engine reliability, and extends engine life.

Takeoff thrust reduction can be achieved using reduced takeoff thrust (Assumed Temperature Method or ATM), derated takeoff thrust (fixed derate), or a combination of these two methods. Regardless of the method, takeoff speeds based on the selected rating (full rated or fixed derate) and the selected assumed temperature should be used. These takeoff speeds may be obtained from the takeoff analysis (runway/airport analysis) or another approved source. Takeoff with less than full rated takeoff thrust using any of these methods complies with all regulatory takeoff performance requirements.

NOTE: Takeoff with full rated takeoff thrust is recommended if windshear conditions are suspected, unless the use of a fixed derate is required to meet a dispatch performance requirement

REDUCED TAKEOFF THRUST (ATM)

Reduced takeoff thrust (ATM) is a takeoff thrust level less than the full rated takeoff thrust. Reduced takeoff thrust is achieved by selecting an assumed temperature higher than the actual ambient temperature.

When using ATM, the takeoff thrust setting is not considered a takeoff operating limit since minimum control speeds (VMCG and VMCA) are based on the full rated takeoff thrust. At any time during takeoff, thrust levers may be advanced to the full rated takeoff thrust.

NOTE: Reduced takeoff thrust (ATM) may be used for takeoff on a wet runway if approved takeoff performance data for a wet runway is used. However, reduced takeoff thrust (ATM) is not permitted for takeoff on a runway contaminated with standing water, slush, snow, or ice.

NOTE: For reduced takeoff thrust (ATM) takeoffs, slightly more back pressure may be needed during rotation and initial climb.

DERATED TAKEOFF THRUST (FIXED DERATE)

Derated takeoff thrust (fixed derate) is a certified takeoff thrust rating lower than full rated takeoff thrust. In order to use derated takeoff thrust, takeoff performance data for the specific fixed derate level is required. Derated takeoff thrust is obtained by selection of a fixed takeoff derate in the FMC.

When using derated takeoff thrust, the takeoff thrust setting is considered a takeoff operating limit since minimum control speeds (VMCG and VMCA) and stabilizer trim setting are based on the derated takeoff thrust. Thrust levers should not be advanced unless conditions are encountered during the takeoff where additional thrust is needed on both engines, such as a windshear condition.

NOTE: Derated takeoff thrust (fixed derate) may be used for takeoff on a wet runway and on a runway contaminated with standing water, slush, snow, or ice

Derated takeoff thrust (fixed derate) may permit a higher takeoff weight when performance is limited by VMCG, such as on a runway contaminated with standing water, slush, snow, or ice. This is because derated takeoff thrust allows a lower VMCG.

Derated takeoff thrust (fixed derate) may permit a lower takeoff weight when takeoff weight is limited by the Minimum Takeoff Weight requirement.

COMBINATION ATM AND FIXED DERATE

NOTE: All limitations and restrictions for reduced takeoff thrust (ATM) and derated takeoff thrust (fixed derate) must be observed.

Reduced takeoff thrust (ATM) and derated takeoff thrust (fixed derate) may be combined by first selecting a fixed derate and then an assumed temperature higher than the actual ambient temperature. Thrust levers should not be advanced unless conditions are encountered during the takeoff where additional thrust is needed on both engines, such as a windshear condition

THRUST CONTROL

When conducting a reduced thrust (ATM) takeoff, if more thrust is needed (up to full rated thrust) when thrust is in THR HLD mode, thrust levers must be advanced manually. If conditions are encountered during the takeoff where additional thrust is needed, such as a windshear condition, the crew should not hesitate to manually advance thrust levers to full rated thrust.

When conducting a derated thrust (fixed derate) takeoff or a takeoff with a combination ATM and fixed derate, takeoff speeds consider VMCG and VMCA only at the fixed derate level of thrust. Thrust levers should not be advanced beyond the fixed derate limit unless conditions are encountered during the takeoff where additional thrust is needed on both engines, such as a windshear condition.

NOTE: If an engine failure occurs during takeoff, any thrust increase beyond the fixed derate limit could result in loss of directional control.

When combining a high level of derate with a high assumed temperature, or if a climb thrust rating higher than the automatically selected climb thrust rating is selected, it is possible that the climb thrust may be higher than the takeoff thrust. In such case, thrust levers will advance forward upon reaching thrust reduction altitude.

If more thrust is needed (up to full rated thrust) when THR HLD mode is displayed, the thrust levers must be manually advanced. When the airplane is below 800 feet RA, full GA N1 can be determined by pushing a TO/GA switch a second time. This will set the reference N1 bugs for full GA thrust.

When the airplane is above 800 feet RA, pushing a TO/GA switch advances the thrust levers to full GA thrust.

IMPROVED CLIMB PERFORMANCE TAKEOFF

When not field length limited, an increased climb limit weight is achieved by using the excess field length to accelerate to higher takeoff and climb speeds. This improves the climb gradient, thereby raising the climb and obstacle limited weights. V_1, V_R and V_2 are increased and must be obtained by airport analysis or OPT.

5.3. RUNWAY ANALYSIS

The takeoff analysis tables allow to determine:

- Maximum allowable takeoff weight and the takeoff speeds for given conditions.
- Assumed Temperature when Assumed Temperature method is used.

Takeoff analysis table sheet is presented below

Table sheet consists of header, main part and footer.

```
Elevation: 387.0 ft                                                    UWOO
737-800(W)/CFM56-7B26                                              ORENBURG
                                                               ORENBURG,RUS
                                                           Dated 12-Apr-19

      Derate = Full Rated Thrust                   Flaps = Flaps 05
         A/C = Auto                                  A/I = Off
 Runway Cond = Dry                                Brakes = All Operative
        Gear = Normal

*A* indicates OAT outside Environmental Envelope
```

Power %N1	OAT °C	Climb (kg)	Runways 08	Runways 26
-	60A	65400	64400F 140-140-144	64400F 140-140-144
-	58A	66400	65200F 141-141-145	65200F 141-141-145
-	56A	67500	66000F 141-142-146	65900F 141-142-146
-	54A	68600	66800F 142-143-147	66700F 142-143-147
97.0	52	69800	67600F 143-143-148	67600F 143-143-148
97.2	50	71100	68500F 143-144-148	68400F 143-144-148
97.5	48	72300	69300F 144-145-149	69300F 144-145-149
97.8	46	73700	70200F 144-146-150	70200F 144-146-150
98.0	44	74900	71100F 145-146-151	71100F 145-146-151
98.4	42	76200	72000F 146-147-152	72000F 146-147-152
98.7	40	77500	72900F 146-148-153	72900F 146-148-153
99.0	38	78900	73800F 147-149-154	73800F 147-149-154
99.3	36	80200	74600F 147-149-155	74600F 147-149-155
99.6	34	81600	75500F 148-150-156	75500F 148-150-156
99.9	32	82900	76300F 148-151-157	76300F 148-151-157
100.3	30	84200	77200F 149-151-157	77100F 149-151-157
99.7	25	84900	78000F 149-152-158	78000F 149-152-158
99.0	20	85000	78700F 150-153-159	78700F 150-153-159
98.2	15	85200	79400F 151-153-159	79300F 151-153-159
97.5	10	85300	80000F 151-154-160	80000F 151-154-160
96.7	5	85500	80700F 152-155-160	80700F 152-155-160
95.9	0	85600	81400F 153-155-161	81400F 153-155-161
94.3	-10	85700	82800F 154-157-162	82800F 154-157-162
92.7	-20	85700	84200F 156-158-163	84200F 156-158-163
91.0	-30	85800	85700F 158-160-164	85700F 158-160-164

```
Wind Correction (kg/kts), Ref 0.0 kts (based on -10.0 and 20.0)
Above Ref            0       +23                  +25
Below Ref            0      -588                 -587

QNH Correction (kg/mbar), Ref 1013.0 mbar (based on 1003.0 and 1023.0)
Above Ref          +38       +46                  +50
Below Ref          -81       -72                  -72

Min Flap Ret Ht-ft          1000                 1000

MAX BRAKE RELEASE WEIGHT MUST NOT EXCEED MAX CERT WEIGHT OF     86182 KG
Limit Codes: B=Brake Energy F=Field *=Obstacle/Level-off S=Structural T=Tire
             V=VMCG X=Takeoff not allowed for selected corrections
             W=Takeoff not allowed below ref. wind
08     *** NO EMERGENCY TURN *** 26 JUN 2017
26     *** NO EMERGENCY TURN *** 26 JUN 2017
```

HEADER OF THE TAKEOFF ANALYSIS TABLE

```
Elevation: 387.0 ft                                          UWOO
737-800(W)/CFM56-7B26                                    ORENBURG
                                                    ORENBURG, RUS
                                                  Dated 12-Apr-19

      Derate = Full Rated Thrust          Flaps = Flaps 05
         A/C = Auto                         A/I = Off
 Runway Cond = Dry                       Brakes = All Operative
        Gear = Normal
```

Header contains following items:

- Airport name and ICAO airport code.
- Airport elevation in feet.
- Date.
- Runway surface condition.
- Flaps setting.
- Air Conditioning, Anti Ice and Anti Skid systems status.
- Wind condition and Derate setting.
- Airplane and engine models and Structural Takeoff Weight value in kg.
- Bar of the limit codes.

MAIN PART OF THE TAKEOFF ANALYSIS TABLE

A indicates OAT outside Environmental Envelope

Power %N1	OAT °C	Climb (kg)	Runways 08	Runways 26
-	60A	65400	64400F 140-140-144	64400F 140-140-144
-	58A	66400	65200F 141-141-145	65200F 141-141-145
-	56A	67500	66000F 141-142-146	65900F 141-142-146
-	54A	68600	66800F 142-143-147	66700F 142-143-147
97.0	52	69800	67600F 143-143-148	67600F 143-143-148
97.2	50	71100	68500F 143-144-148	68400F 143-144-148
97.5	48	72300	69300F 144-145-149	69300F 144-145-149
97.8	46	73700	70200F 144-146-150	70200F 144-146-150
98.0	44	74900	71100F 145-146-151	71100F 145-146-151
98.4	42	76200	72000F 146-147-152	72000F 146-147-152
98.7	40	77500	72900F 146-148-153	72900F 146-148-153
99.0	38	78900	73800F 147-149-154	73800F 147-149-154
99.3	36	80200	74600F 147-149-155	74600F 147-149-155
99.6	34	81600	75500F 148-150-156	75500F 148-150-156
99.9	32	82900	76300F 148-151-157	76300F 148-151-157
100.3	30	84200	77200F 149-151-157	77100F 149-151-157
99.7	25	84900	78000F 149-152-158	78000F 149-152-158
99.0	20	85000	78700F 150-153-159	78700F 150-153-159
98.2	15	85200	79400F 151-153-159	79300F 151-153-159
97.5	10	85300	80000F 151-154-160	80000F 151-154-160
96.7	5	85500	80700F 152-155-160	80700F 152-155-160
95.9	0	85600	81400F 153-155-161	81400F 153-155-161
94.3	-10	85700	82800F 154-157-162	82800F 154-157-162
92.7	-20	85700	84200F 156-158-163	84200F 156-158-163
91.0	-30	85800	85700F 158-160-164	85700F 158-160-164

Main part contains following items:

- Outside air temperature in degrees Celsius. Letter “A” associated with temperature means this temperature value is out of operational envelope but can be used to performance calculations.

- Maximum performance limited takeoff weights in kg (to the nearest 100 kg) with limit codes and related takeoff speeds in knots (presented V = actual V – 100 kt) for each runway.

FOOTER OF THE TAKEOFF ANALYSIS TABLE

```
Wind Correction (kg/kts), Ref 0.0 kts (based on -10.0 and 20.0)
Above Ref          0       +23                        +25
Below Ref          0      -588                       -587

QNH Correction (kg/mbar), Ref 1013.0 mbar (based on 1003.0 and 1023.0)
Above Ref        +38       +46                        +50
Below Ref        -81       -72                        -72

Min Flap Ret Ht-ft        1000                       1000

MAX BRAKE RELEASE WEIGHT MUST NOT EXCEED MAX CERT WEIGHT OF     86182 KG
Limit Codes: B=Brake Energy F=Field *=Obstacle/Level-off S=Structural T=Tire
             V=VMCG X=Takeoff not allowed for selected corrections
             W=Takeoff not allowed below ref. wind
08      *** NO EMERGENCY TURN *** 26 JUN 2017
26      *** NO EMERGENCY TURN *** 26 JUN 2017
```

Footer contains following items:

- wind corrections in kg/knot;
- level-off (minimum flap retraction) height value in feet;
- runway length in feet;
- clearway/stopway length in feet;
- runway slope value in percents;
- line-up distances in feet;
- pressure altitude corrections.

AVAILABLE TAKEOFF CONFIGURATION

Takeoff Analysis Tables provide takeoff performance data for:

- Dry runway.
- Contaminated runway (3 mm slush and 3 mm standing water).
- Slippery runway (airplane braking coefficient MuB = 0.2, 0.10 and 0.05).
- Air Conditioning packs is ON and OFF.
- Engine and Wing Anti-Ice is OFF.
- Antiskid is ON.
- All brakes are operating.
- Flaps setting is 5 and 15.
- No derates.

Takeoff Analysis Tables also provide corrections for non-zero wind, non-standard QNH, air conditioning packs OFF and anti-ice ON.

To obtain takeoff performance for the other takeoff configurations use FCOM (chapters PD, PI).

NOTE: Slippery runway conditions are provided in terms of airplane braking coefficient MuB. To obtain actual MuB values use Slippery Runway Parameters Table presented below

PRESSURE ALTITUDE CORRECTIONS

Airfield elevation as printed with the data has been assumed as standard pressure altitude for analysis. Crew will correct the allowable takeoff weight for lower than standard station pressure by reducing by the values printed under the columns. Do not take credit when station pressure (QNH) is above standard.

WIND CORRECTIONS

For wind correction use all the steady wind, either for head- or tailwind. The values indicate weight corrections, kg per knot. Only headwind correction can be omitted if not needed (no headwind corrections should be made for reduced thrust takeoff). Do not apply wind corrections to the Climb Limit Weight, as this limit assures a climb gradient in still air in accordance with regulations regarding one engine inoperative climb in the takeoff segment.

MEL/CDL ITEMS CORRECTIONS

To obtain MEL/CDL items corrections refer to Flight Crew Operations Manual (chapter PI) or Quick Reference Handbook (chapter PI).

5.3.1. The Maximum Allowable Takeoff Weight and the Takeoff Speeds Determination Procedure

- Obtain QNH, OAT, wind and runway surface condition from ATC/ATIS.
- If the runway surface is slippery, obtain the airplane braking coefficient MuB from the Slippery Runway Parameters Table. If the runway is covered by dry or compacted snow use the Runway Condition Assessment Table and the Equivalent Standing Water Depth Table (page VII) then use the Slippery Runway Parameters Table.
- If the runway surface is contaminated with slush or standing water and depth of contamination is more than 3 mm, refer to FCOM (chapter PI) to obtain takeoff performance. Takeoffs in slush/standing water depths greater than 13 mm (0.5 inches) are not recommended because of possible airplane damage as a result of slush impingement on the airplane structure.
- Determine the wind component parallel to runway using wind component chart.
- Multiply the wind component parallel to runway by the value “ADD KG/KT HEADWIND” or “SUB KG/KT TAILWIND” from the footer of the Takeoff Analysis Table to obtain wind correction by headwind or tailwind.
- Determine QNH/1013.2 difference.
- Multiply (6) by the value “BELOW REF –KG/MB” from the footer of the Takeoff Analysis Table to obtain pressure altitude correction. If QNH is above 1013.2 the pressure altitude correction should not be applied.
- Find the brake release weight and takeoff speeds for the given temperature and runway in the main part of the Takeoff Analysis Table.
- If planning to activate anti-ice and (or) to deactivate air conditioning system in the takeoff phase obtain the corrections for air-conditioning packs OFF and (or) antiice ON using MTOW Corrections Table depending of limit factor.
- If the limit factor is “Field”, “Obstacle”, “Level-off”, “Tire speed”, “Break energy” or “VMCG” adjust the brake release weight by wind correction, pressure altitude correction and corrections for air-conditioning packs OFF and anti-ice ON. If the limit factor is “Climb”, adjust the brake release weight by the same corrections except the wind correction.

- Find the maximum certified takeoff weight in the header of the Takeoff Analysis Table.
- Select the minimum of Steps 10 and 11. This is your Maximum Allowable Takeoff Weight.
- The takeoff speeds shown in the Takeoff Analysis Tables are valid for zero wind, standard barometric pressure, air-conditioning packs ON and engine and wing anti-ice OFF configurations only. If you have corrected the brake release weight for the off-standard barometric pressure, non-zero wind, air-conditioning packs OFF or anti-ice ON configurations, the speeds shown in the Takeoff Analysis Tables do not apply to the corrected brake release weight. Consult Flight Crew Operations Manual (chapter PI) to get the takeoff speeds.
- If the actual takeoff weight is lower than the Maximum Allowable Takeoff Weight the proper takeoff speeds are not shown. Use FCOM, QRH or FMC to obtain proper takeoff speeds.

THE ASSUMED TEMPERATURE DETERMINATION PROCEDURE

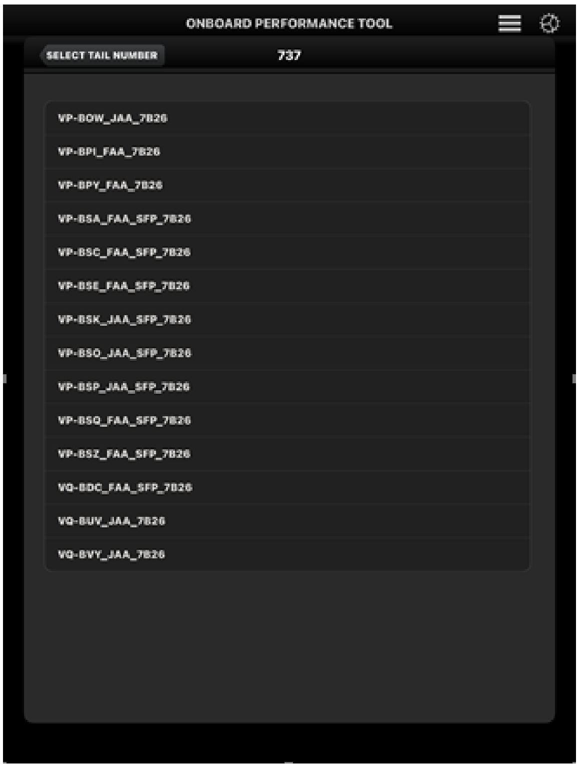
- Obtain QNH, OAT, wind and runway surface condition from ATC/ATIS.
- If the runway surface condition is “Contaminated” or “Slippery” the using of Assumed Temperature Method is NOT ALLOWED!
- Determine the wind component parallel to runway using wind component chart.
- Multiply the wind component parallel to runway by value “ADD KG/KT HEADWIND” or “SUB KG/KT TAILWIND” from the footer of the Takeoff Analysis Table to obtain wind correction by headwind or tailwind.
- Determine QNH/1013.2 difference.
- Multiply (5) by value “BELOW REF –KG/MB” from the footer of the Takeoff Analysis Table to obtain pressure altitude correction. If QNH is above 1013.2 the pressure altitude correction should not be applied.
- Adjust the actual takeoff weight by wind correction, pressure altitude correction and air-conditioning packs OFF/anti-ice ON corrections.
- Note that “ADD KG/KT HEADWIND” correction must be subtracted from, “SUB KG/KT TAILWIND” correction must be added to and pressure altitude correction must also be added to the actual takeoff weight value to determine the Assumed Temperature.
- To determine the Assumed Temperature, move up runway column in the main part of the Takeoff Analysis Table until the lowest weight which just equals or exceeds the corrected actual takeoff weight from step 9 is located. The temperature associated with this weight is your Assumed Temperature which must then be entered into FMC.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

5.4. USING THE ONBOARD PERFORMANCE TOOL FOR TAKEOFF – APPLE IOS

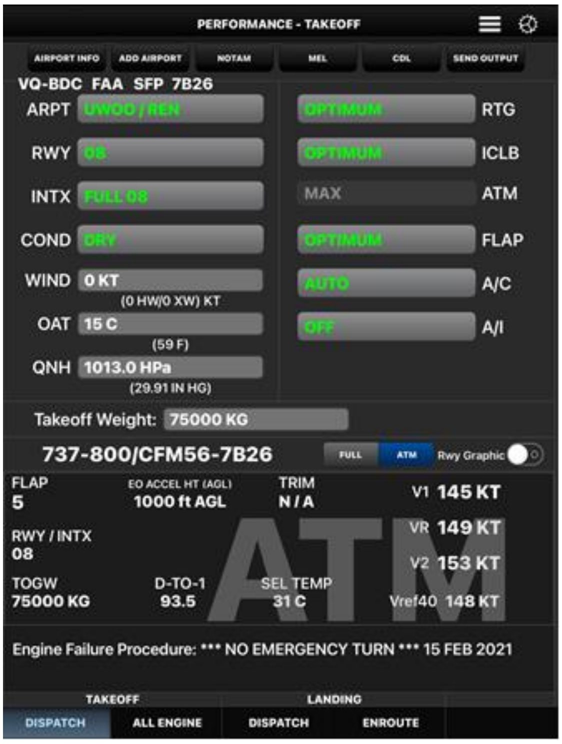
Depending on how your administrator has set up the application, OPT may or may not display an airplane selection menu when the application is launched. If your OPT installation has more than one airplane tail or type loaded, then you will be presented with a list of tails, or airplanes, to choose from. This list might look like that shown below.

Selecting the appropriate tail number or airplane description will load the appropriate
databases and start OPT. Once the loading process is complete, the main screen for
takeoff appears, entitled at the top of the page PERFORMANCE - TAKEOFF.

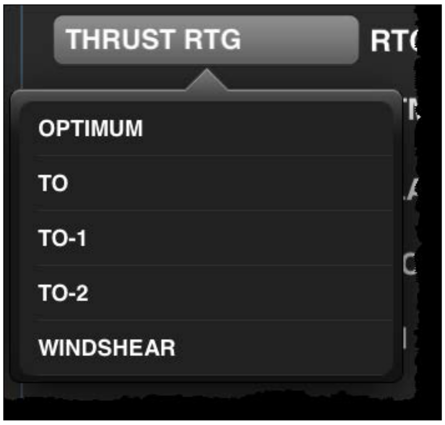


Selecting any of the functions shown on the Tab Bar will switch to the appropriate OPT screen and lighten the background color of that button slightly to denote the active screen.

The Tool Bar near the top of the screen is used to display various context sensitive actions that are dependent upon which screen is currently being displayed. Selecting any of the items, when active, will display a new screen to make further selections or input. Items on the tool bar which have a name that has been "grayed-out" are inactive and not available for selection.



Popovers are denoted by a gray button with rounded corners and a name to describe its function. When selected, the application will display a menu to make a selection from. For instance, the THRUST RTG popover button will display a menu similar to this when selected:



Simply selecting any of the items will enable that menu option, while touching somewhere off the menu will cause the menu to close with no change made to the current selection.

The main crew screen, for the most part, is set up in three basic functional areas. These areas are divided by cyan group boxes. In portrait mode, the upper left portion is devoted mainly to crew input of airport and atmospheric information, the upper right portion is devoted to crew input of airplane configuration information, while the lower portion is devoted to output and results. Some buttons and/or input boxes may not be available on your screen, depending on the airplane model and other options controlled by the administrator.

5.4.1. Runway/Atmospheric Inputs

As noted above, the crew input is mainly divided up into airport/atmospheric inputs and airplane configuration inputs. The airport/atmospheric inputs would typically vary little between different OPT installations and consists at a minimum of the airport search button, the runway search button, and edit boxes for wind, OAT, and altimeter setting (QNH). In addition, there is an optional button to make selections for intersection calculations.

NOTE: The tail number or configuration is also generally listed in the upper left corner of this area.

5.4.2. Selecting the Runway

Once the airport has been selected, selecting the RWY button will display a dropdown button selection that might look like that shown:

Selecting any of the available runways will load that runway for use, change the button name to the (green) runway ID, and check for existing departure and intersection information in the airport database.

5.4.3. Selecting Intersections

If your administrator has included intersection data in your airport database and, if there is intersection data available for the selected runway, the INTX button will become active.

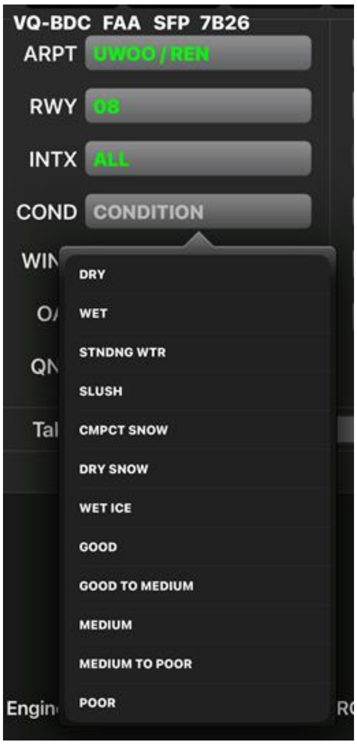
A typical selection is shown:

Selecting the ALL button will calculate takeoff information for the full length runway (08) and each of the available intersections. Selecting any single entry will limit the calculation to just that entry, such as intersection A.

If your administrator has excluded the ALL selection, ALL will not be shown as an available selection and only individual calculations are available.

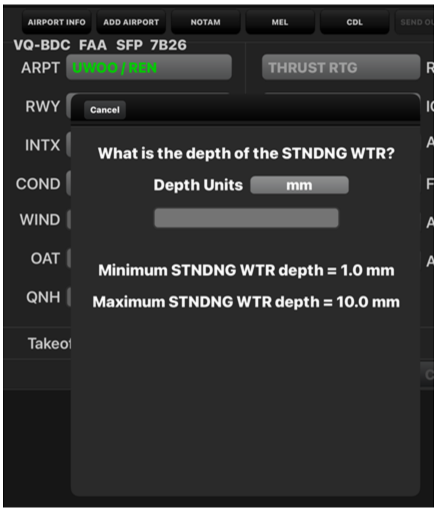
5.4.4. Selecting Runway Condition

Selecting the CONDITION button will display the list of runway conditions that your administrator has set up. A typical selection might look like this:



In this example, the possible choices are for dry, wet, standing water, and slippery runway conditions of good, medium, and poor.

If the user selects STANDING WATER, OPT will then display a window to ask how deep the standing water is.



5.4.5. Entering Wind

Wind inputs may be made in either wind component or direction and magnitude into the WIND edit box. Depending on how your installation of OPT is set up, you may also see the head/tail and crosswind components displayed under the wind input box.

If the wind is input in wind component, headwinds are considered to be positive and tailwinds negative. The correct format for wind in the direction/magnitude format is just that, direction/magnitude, e.g. 040/20 would be a 20 knot wind coming from a magnetic heading of 40 degrees. The wind component is then resolved by comparing this information with the runway heading. After the cursor has left the input box, units are appended to the user’s input to show how OPT has interpreted the input. If only inputting the headwind or tailwind magnitude, one may change the input units by adding, for instance, an “M” (for meters) to the input component, such as -5M. This would then be interpreted as a 5 m/s tailwind. An input of 5K would be interpreted as a 5 kt headwind. Additionally, a headwind or tailwind may be denoted in the entry by appending an “H” or a “T” to the wind magnitude, such as 5T for a 5 kt tailwind.

5.4.6. Entering OAT

OAT is entered into the OAT edit box. Again, depending on your installation, the alternate unit (C or F) may be shown under your input. If the user desires to input temperature in units not expected by OPT, however, simply place a ‘C’ or ‘F’ after the temperature (e.g. 59F) and OPT will make the correct conversion. Similar to the wind input, after the cursor leaves the edit box, the units that are assumed are displayed in the edit box and the alternate units are shown below the edit box.

5.4.7. Entering QNH

QNH, or pressure variation, may also be input in two different units, namely HPa (mb), or inches of Mercury (Hg). OPT will check the magnitude of this input and convert accordingly. Any inputs greater than 100 will be interpreted as HPa, while anything less will be interpreted as inches Hg. Depending on your installation, the converted values may displayed under your input.

5.4.8. Airplane Configuration Inputs

Once the airport, runway, and atmospheric inputs are made, the crew may specify the required airplane configuration inputs. When there are selectable derates available, the OPT user interface (UI) will include a rating dropdown button. If it is allowed by the administrator, this list may also include Optimum and Windshear selections. These selections will compute the best combination of fixed derate plus assumed temperature for maximum derate or special windshear guidance information respectively.

Other common airplane configuration and policy inputs are made through several other available dropdown buttons. Depending on your airplane type, these buttons might include:

- flaps;
- air conditioning;
- anti-ice;
- APU;
- V1 policy;
- improved climb policy;
- reverse thrust availability.

If allowed by the administrator, the flap selection pull down menu may include an Optimum selection. If this option is chosen, the software will find the optimum takeoff flap position between the highest and lowest setting available in the allowable list set up by the administrator. If optimum is the only setting allowed, the Onboard Performance Tool will choose between all of the certified flap positions for that airplane. In some cases, when a departure procedure is selected that is valid for only a specific flap(s), the flap selection list will only contain that flap position(s).

If selected by the administrator, the flap selection pull down menu may include a Preferred selection. If this option is chosen, the software will attempt to use a single, preferred flap position specified by the administrator. If the performance available is not sufficient, OPT will revert to using Optimum flaps for its calculation.

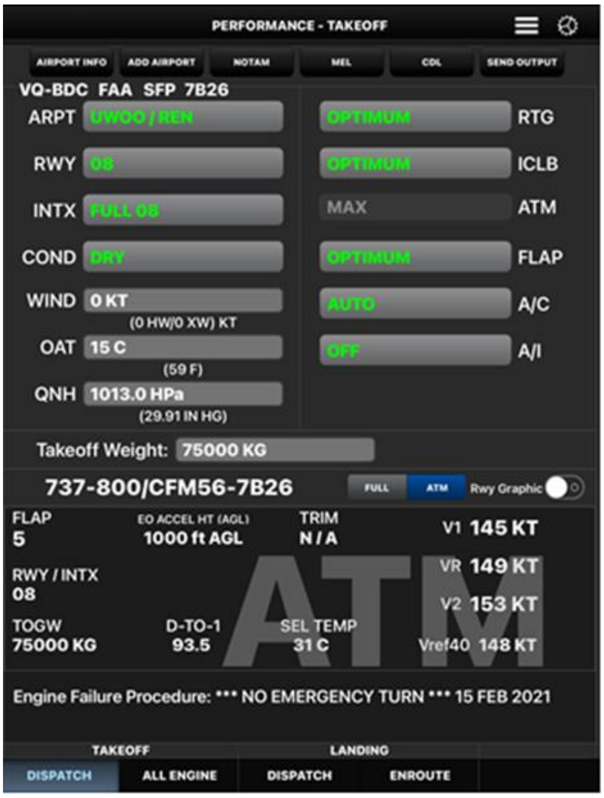
5.4.9. Calculating Takeoff Performance

Once all of the required inputs are made and the CALC button becomes active (blue), the user may perform a calculation by selecting this button. If the takeoff weight has not been input, then OPT will calculate the maximum takeoff weight for each of the runways/intersections selected.

In the IOS environment, this means that the user should “swipe” across the screen with their finger to scroll across and reveal more pages. The current runway/intersection being viewed is displayed on the left side of the output.

For this example, the results shown are for the full length runway 08, the optimum flap position (for weight 75000kg) is flaps 5, and derated 1 (D-TO 1) thrust was used.

If the planned takeoff weight has been entered in the Takeoff Weight edit box, then OPT will calculate all required parameters for both the maximum takeoff thrust and best assumed temperature cases.



When results are initially displayed, OPT will display the ATM results for the first runway/intersection listed. In the example above, the ATM results for runway 08 are shown.

By swiping across the output area, one may display the results for the next runway, in either the ATM mode or the full thrust mode, as currently selected.

5.4.10. Adjustments

The takeoff data in the Operations Manual and the Flight Planning and Performance Manual are based on a dry runway with all equipment operating at the start of takeoff.

Data is also provided for wet or contaminated runways and for cases where equipment is inoperative or deactivated.

Before starting into the details of the takeoff performance adjustments here is the difference between a slippery and a contaminated runway.

A slippery runway is a runway with a reduced tire to ground friction capability due to the presence of water, ice, compact snow or other substances. A slippery runway only affects the airplane's ability to decelerate. A slippery runway does NOT affect the airplane's ability to accelerate.

On the other hand, a contaminated runway affects the ability of the airplane to accelerate as well as decelerate. This would include runways contaminated by a measurable depth of slush, snow, or standing water. This runway contamination creates a drag force, which resists the airplane's ability to accelerate. Runway contamination also reduces the tire to ground friction and therefore affects the airplane's ability to decelerate and stop.

Both JAA and FAA Airplane Flight Manuals contain field length limited takeoff weights based on a dry, smooth runway, with maximum manual braking. They do not include credit for the use of reverse thrust.

Since the available runway cannot be increased to accommodate these different conditions weight reductions and V_1 adjustments to determine a reduced takeoff weight, which can safely be operated from the same runway are provided.

The weight adjustments result in a reduced weight and adjusted V_1 speed. This reduced weight and adjusted V_1 speed require the same distance as the baseline dry runway condition.

The FAA or JAA approved Airplane Flight Manuals contain dry runway performance.

The FAA and JAA approved Airplane Flight Manuals may contain field length limited takeoff weights which take into account a wet runway and a wet skid-resistant runway.

The JAA approved Airplane Flight Manual also includes advisory data which accounts for a runway covered with slush, snow and wet ice.

This slush snow and ice covered runway data is not part of the FAA approved Airplane Flight Manual, but is still provided to the operator in the form of weight reductions and V_1 adjustments in the Performance Inflight section of the Quick Reference Handbook as tabular data, as well as in an operational computer program.

Any of these data sources can be used to create required operational data.

WET

As discussed earlier, the basic dry runway field length limited weight performance is based on the longer of the all engine accelerate-go distance to 35 feet plus fifteen percent, or the engine inoperative accelerate-go distance to 35 feet or the accelerate-stop distance assuming a maximum effort stop on a dry runway with no credit for reverse thrust.

If the runway is WET, not dry, and the dry runway performance weight and V_1 speed is used, the following would be the result. The all engine distance to 35 feet would not change because a wet runway does not affect the airplane's acceleration. The engine inoperative accelerate-go distance to 35 feet would not change for the same reason. But, the accelerate-stop distance would increase because the stopping capability is degraded due to the reduction in tire to ground friction on a wet runway.

If the Airplane Flight Manual dry runway limited weight and V_1 speed are used on a wet runway, the airplane may not have the ability to stop on the runway.

In the Airplane Flight Manual calculations for wet runway, we are allowed to use different assumptions in computing the performance.

The wet runway engine inoperative accelerate-go distance is computed based on a reduced screen height of 15 feet.

As mentioned before, the accelerate-stop distance required is increased due to the reduction of tire to ground friction on a wet runway surface.

But, the Airplane flight Manual wet runway stopping performance does take credit for reverse thrust on one engine.

Taking credit for 15 foot screen height and one engine reverse thrust reduces the effect that the wet runway has on the field length required.

To further reduce the effect of wet runway takeoff performance, the calculation rebalances go and stop to obtain a new, lower, wet runway V_1 speed.

The combination of a wet runway, 15 feet screen height, credit for one engine reverse thrust, and the rebalancing to a wet runway V_1 speed still results in a takeoff distance longer than the Airplane flight Manual dry runway calculation.

Therefore, the allowable field length limited takeoff weight on a wet runway will be less than the normal field length limited weight on a dry runway.

Separate takeoff performance charts are provided in the Operations Manual and the flight Planning and Performance Manual to calculate the wet runway field length limited weight and the wet runway V_1 speeds.

The wet runway field length limit weight will typically be lower than the dry runway field length limit weight.

The wet runway balanced V_1 speed will typically be lower than dry runway balanced V_1 speed by 5 to 12 knots at the same weight, altitude, and temperature.

Because V_1 speed is reduced for the wet runway performance, the chance of V_1 speed being limited by V_{1MCG} is increased.

Even when the runway is wet, the airplane would be limited by minimum field length only on a VERY short runway.

The charts in the Operations Manual and the flight Planning and Performance Manual are used in the same way as the charts for dry runway takeoff calculations.

As mentioned earlier, unlike older models, the Airplane Flight Manual now contains field length limited takeoff performance based on a wet skid-resistant runway. A wet skid-resistant runway is commonly known as a wet grooved or a porous friction course runway.

The Airplane Flight Manual allows use of the wet skid-resistant performance on runways constructed and maintained to meet FAA advisory guidelines.

Separate takeoff performance charts are provided in the Operations Manual and the flight Planning and Performance Manual to calculate the wet skid-resistant field length limited weight and the wet skid-resistant V_1 speeds.

The wet skid-resistant runway field length limit weight is typically lower than the dry runway field length limit weight by less than 1,000 kilograms.

In fact, in many cases there will not be any weight reduction due to this wet skid-resistant performance this is particularly true at low altitudes and normal temperatures.

The wet skid-resistant balanced V_1 speed will typically be less than dry runway balanced V_1 speed by 4 to 8 knots.

Because V_1 speed is reduced with the wet skid-resistant runway performance, the chance of V_1 speed being limited by V_{1MCG} is increased.

Even when the runway is a wet skid-resistant runway, the airplane would be limited by minimum field length only on a VERY, VERY short runway.

The charts in the Operations Manual and the Flight Planning and Performance Manual are used in the same way as the charts for dry runway takeoff calculations.

As stated earlier, both the wet runway and wet skid-resistant takeoff performance are part of the Airplane Flight Manual.

Because both the wet runway and wet skid-resistant takeoff performance are based on a 15 feet screen height, the use of clearway in the analysis of wet runway and wet skid-resistant takeoff performance is not allowed.

The obstacle clearance margin in the Airplane Flight Manual results in the net flight path clearing the obstacle by a minimum of 15 feet.

Improved climb is allowed with wet or wet skid-resistant performance.

Both the assumed temperature and derate method of reducing takeoff thrust may be used with wet and wet skid-resistant performance.

SLIPPERY

As with the wet runway data, an airplane dispatched assuming a dry runway, but operating on a slippery runway, would not have the ability to stop within the normal dry runway distances.

The Quick Reference Handbook and the Flight Planning and Performance Manuals contain slippery runway data as a function of reported braking action.

This data is presented as weight adjustments and V_1 adjustments from the normal dry runway performance.

$V_{1(MCG)}$ limit weights are also presented.

The data is presented at three levels of reported braking action.

This reported braking action is based upon subjective evaluation of actual braking performance, as reported by flight crews.

Good is a performance level that is comparable to a wet runway and has also been used in JAA certifications for compact snow.

Poor is a performance level that is used in JAA certifications for wet, melting ice.

Boeing does not correlate the airplane's performance to the runway friction measured by the runway friction measuring vehicles.

The use of this slippery runway data is the same whether the data comes from the Operations Manual Performance Inflight section or the Flight Planning and Performance Manual.

There are two data adjustments plus $V_{1(MCG)}$ limit weight information provided. The data adjustments are to takeoff weight and V_1 speed.

These adjustments represent how much the takeoff weight must be reduced and the V_1 adjusted so that the distance required for accelerate-go and accelerate-stop does not change.

To minimize the weight adjustment, the screen height above the end of the runway is reduced from 35 to 15 feet and credit for one engine at reverse thrust is assumed during the stop calculation.

The $V_{1(MCG)}$ limit weight is the maximum weight at which the airplane can accelerate to $V_{1(MCG)}$ speed and still stop within the field length available for the reported braking action.

An example will be used to show how slippery runway data adjustments are applied.

The slippery runway adjustments are valid for all flap settings.

CONTAMINATED

As shown earlier, an airplane dispatched assuming dry runway, but operating on a slush covered runway, would not have the ability to stop from V_1 within the normal dry runway distances nor would it reach 35 feet within the available distance following a failure of the critical engine.

The Quick Reference Handbook and the Flight Planning and Performance Manual contain takeoff performance data for runways contaminated with slush or standing water.

Slush on the runway will reduce the airplane's capability to accelerate and it will also reduce the airplane's ability to stop. While accelerating, the aircraft tires displace the slush. The displacement of the slush creates a drag force which reduces the airplane's acceleration.

As the airplane accelerates to a higher speed, the spray from the displaced slush will start impinging on parts of the fuselage or wings creating an even higher slush drag force. The slush drag force goes up with the square of the ground speed. The slush drag force peaks just above the hydroplaning speed and then decreases with further speed increase.

Hydroplaning speed is the speed when the tire starts to lift out of the slush. The drag forces due to slush displacement and impingement begin to decrease at this speed.

If the airplane must do an emergency stop, the stopping force due to the wheel brakes will be greatly reduced due to the reduction in tire to ground friction.

This is particularly true at high speeds where the tires will hydroplane.

A runway is contaminated when over 25% of the available runway is covered with a measurable depth of slush or standing water.

The Quick Reference Handbook and the Flight Planning and Performance Manual contain data for slush depths of 2mm, 6mm and 13mm.

Takeoff is not recommended in depths greater than 13mm due to the possibility of damage as a result of spray impingement on the airplane structure.

The takeoff data provided for slush and standing water take into account the effect that the slush has on the airplane's ability to accelerate as well as the effect on the airplane's ability to stop.

As with the slippery and wet runway data, the effect of slush takeoff weight is minimized by using a reduced screen height of 15 feet for the engine inoperative accelerate-go distance and credit for one engine reverse thrust on the stop.

A new V_1 speed is also computed.

Even with the reduced screen height, credit for reverse thrust and with a new V_1 speed, the weight reductions due to slush are extremely large.

The slush and standing water data is presented in the Quick Reference Handbook and the Flight Planning and Performance Manual in the Performance Inflight section. It is in the same format as the slippery runway data.

There are two data adjustments plus $V_{1(MCG)}$ limit weight information provided.

The data adjustments are for takeoff weight and the V_1 speed.

These adjustments represent how much the takeoff weight must be reduced and V_1 adjusted, so that the distance required for accelerate-go and accelerate-stop will not change.

As mentioned previously, to minimize the weight reduction the screen height above the end of the runway is reduced from 35 to 15 feet and credit for one engine at reverse thrust is assumed during the stop calculation.

The $V_{1(MCG)}$ limit weight is the maximum weight at which the airplane can accelerate to $V_{1(MCG)}$ speed and still stop within the field length available for the assumed reported braking action.

The slush weight reductions and V_1 adjustments take into account the reduction in tire to ground friction and the effect of hydroplaning on the airplane's stopping performance.

There is no additional adjustment required for reported braking action.

5.5. LANDING PERFORMANCE

5.5.1. Landing Performance Requirements

Landing performance requirements include landing field length, approach climb, and landing climb considerations. The most limiting of these requirements defines the landing performance limited weight for dispatch.

5.5.2. Landing Field Length

Landing field length performance is developed from flight test demonstrated landing distances. These distances are measured from a height of 50 feet above the landing surface. The flight test landings are performed on a dry runway using an aggressive touchdown technique. Maximum manual wheel braking and speed brakes are used during the landing ground roll. Reverse thrust is not used or credited in the flight test demonstrated landing distance.

The certified landing distance on a dry runway is the flight test demonstrated distance, plus an additional margin of 67 percent. For a wet runway, the certified dry runway landing distance is increased by an additional margin of 15 percent.

The landing field limit weight is the maximum weight for which the landing distance available meets certified landing field length requirements. This weight accounts for landing flap setting, wind and pressure altitude, dry or wet runway surface condition, and airplane stopping configuration.

The landing field limit weight does not directly account for runway slope, non-standard temperature, or approach speed additives. These considerations are protected by the margins used to define certified landing distance.

5.5.3. Climb Performance

Climb performance requirements are based on the certified approach and landing flap combinations for each airplane model. Climb performance limits for landing ensure a minimum climb gradient capability in the certified approach and landing configurations in case a go-around becomes necessary at any point during the landing approach.

For a two engine airplane, the minimum climb gradient required in the approach configuration is 2.1 percent. The approach climb configuration is based on the selected approach flap setting, with one engine inoperative, landing gear up, and the operating engine at go-around thrust.

The approach climb capability is calculated at a speed not greater than 1.4 times the stall speed for the selected approach flap.

The minimum climb gradient required in the landing configuration is 3.2 %. The landing climb configuration is based on the selected landing flap setting, with all engines operating and landing gear down.

The speed used for landing climb calculations may not exceed the certified landing reference speed for the selected landing flap setting. The certified landing reference speed must be at least 23 percent greater than the associated stall speed.

Landing climb capability is based on thrust available eight seconds after the thrust levers are moved from minimum flight idle to the takeoff position.

The most limiting of the approach and landing climb requirements defines the climb limited landing weight. For a two engine airplane, approach climb is typically the more limiting requirement. Whereas for a 4 engine airplane, landing climb is typically the more limiting requirement.

The climb limited landing weight for the selected approach and landing flap settings accounts for airport temperature and pressure altitude, as well as engine bleed for air conditioning and anti-ice.

Ice accumulation on unheated surfaces is also accounted for when operating in icing conditions during any part of the flight with forecast landing temperature below 10 degrees Celsius.

For operators who are subject to Joint Aviation Authority regulations there is an additional climb performance requirement for low visibility instrument approaches. The JAA minimum gradient requirement in the go-around configuration is the greater of 2.5 percent, or the published gradient for the specific airport, based on one engine inoperative and landing gear up. This requirement only applies to low visibility instrument approaches with a decision height of less than 200 feet.

The performance requirements which we have been discussed here, and the maximum certified structural landing weight, define the maximum allowable scheduled landing weight for dispatch.

5.5.4. Landing Performance Data

Landing performance information can be found in two separate sections of the Operations Manual.

The Performance Dispatch section of Volume 1 provides landing performance information for dispatch in tabular format.

The Performance Inflight section of the Quick Reference Handbook provides advisory landing performance information for flight crew reference.

The Flight Planning and Performance Manual is an additional source of landing performance data, which includes both dispatch information and advisory information in graphical format. The Flight Planning and Performance Manual includes landing performance information for some conditions and airplane configurations which are not available in the Operations Manual.

The Performance Dispatch section of the Operations Manual provides the landing performance information. This information includes landing field limit weight and landing climb limit weight.

For JAA operators, go-around climb gradient capability is also included for low visibility operations.

Landing field limit weight information is provided in tabular format for one landing flap setting, with antiskid operative, and for airport pressure altitudes up to 3000 feet.

Two separate tables are used to determine the landing field limit weight.

The first Landing Field Limit Weight table adjusts landing field length for the effect of wind. The table is entered with landing field length available and wind component to determine the wind corrected landing field length.

The second Landing Field Limit Weight table is entered with wind corrected landing field length, airport pressure altitude and runway surface condition to determine the landing field limit weight based on automatic speed brakes and all wheel brakes operating. A weight adjustment is provided at the bottom of the table for manual speed brakes. For some airplane models, a separate weight adjustment is provided for brake deactivated performance.

Landing climb limit weight information is provided in tabular format for one landing flap setting, the associated approach flap setting, and for airport pressure altitudes up to 3000 feet.

When a common approach flap setting defines the most limiting condition for the climb limit weight, multiple landing flap settings may be included in a single table.

The landing climb limit weight table is entered with airport outside air temperature and airport pressure altitude to determine the landing climb limit weight for air conditioning packs ON.

Weight adjustments are provided at the bottom of the table for air conditioning packs OFF, anti-ice ON, and ice accumulation on unheated surfaces.

For JAA operators only, climb gradient capability in the go-around configuration is also included in the Performance Dispatch section of the Operations Manual.

This group of tables can be used to evaluate JAA minimum gradient requirements for low visibility operations.

The first table is used to determine reference climb gradient based on airport outside air temperature and pressure altitude up to 3000 feet.

The second table provides a gradient adjustment for weight, and the third table provides a gradient adjustment for speed based on the weight adjusted gradient from the first and second tables. Finally, adjustments for air conditioning packs OFF, anti-ice ON and icing conditions are applied as appropriate to determine go-around climb gradient capability.

The Flight Planning and Performance Manual includes additional landing performance information for dispatch which is not available in the Operations Manual. The Flight Planning and Performance Manual includes pressure altitudes up to the maximum certified for landing, as well as limit weight information for each certified landing flap setting. Landing field limit weights are also included for anti-skid inoperative performance for airplane models which allow anti-skid inoperative dispatch relief.

The Flight Planning and Performance Manual information is typically presented in graphical format. It may be necessary to refer to this information for airplane configurations and environmental conditions which are not available in the Operations Manual.

5.5.5. Quick Turnaround

The quick turnaround limit weight is the maximum landing weight for which there is no mandatory minimum ground time prior to the next departure. For landing weights above the quick turnaround limit weight, a mandatory minimum ground time must be observed. This mandatory ground time protects against wheel fuse plug melt and loss of tire pressure during the next takeoff.

As a certification requirement, quick turnaround limit weight is conservatively based on maximum manual wheel braking with no credit for reverse thrust. Note however that quick turnaround limits must be respected in operation, regardless of actual stopping technique.

Quick turnaround limit weight information can be found in the Performance Dispatch section of the Operations Manual for one landing flap setting and for airport pressure altitudes up to 3000 feet.

The Flight Planning and Performance Manual provides information for all certified landing flap settings and airport pressure altitudes.

Both documents use a similar tabular presentation format for quick turnaround limit weight.

The Quick Turnaround Limit Weight table is entered with airport outside air temperature and airport pressure altitude to determine quick turnaround limit weight. Notes at the bottom of the table provide weight adjustments for slope and wind. The required ground time, which varies by airplane model, is also noted at the bottom of the table for cases when the quick turnaround limit weight is exceeded. For some airplane models, an alternate procedure for evaluating quick turnaround limits may be available.

Operators may choose either the Quick Turnaround Limit Weight table information or an approved alternate procedure, if available, to comply with quick turnaround requirements.

Advisory brake cooling information is provided in the Performance Inflight section of the Quick Reference Handbook and in the Flight Planning and Performance Manual. This advisory information is intended to assist operators in avoiding brake problems that could result from repeated landings at short time intervals or a rejected takeoff.

The first QRH brake cooling schedule table is entered with brakes on speed, airplane gross weight, airport pressure altitude up to 4000 feet and outside air temperature to determine reference brake energy per brake based on a single stop with all brakes operating. Note that the brakes on speed used in the table may be based on either indicated airspeed adjusted for wind or ground speed.

The brakes on indicated airspeed during landing will normally be five to seven knots lower than the approach speed when recommended braking procedures are followed.

The reference brake energy per brake from the first Brake Cooling Schedule table is used in a second table to adjust for the type of event, and the wheel braking method used in the case of landing.

A separate brake energy adjustment table is used when considering the effect of reverse thrust.

The third Brake Cooling Schedule table is used to determine recommended cooling time, either in flight or on the ground, based on total energy absorbed per brake.

Additional procedural guidance is provided in the notes at the bottom of the table when in the caution zone, or in the fuse plug melt zone.

Note that maximum quick turnaround limits must be observed regardless of the guidance information obtained from the advisory Brake Cooling Schedule tables.

It is important to understand the differences between the published Quick Turnaround Limit Weight and Recommended Brake Cooling Schedule tables, so that they are used correctly in operation.

The quick turnaround limit weight is a regulatory requirement which may result in operational dispatch restrictions, whereas the recommended brake cooling schedule is advisory information which provides additional brake cooling guidance to the flight crew.

The quick turnaround limit weight restriction only guarantees that fuse plugs will not melt during the next takeoff. The recommended brake cooling schedule provides additional brake energy protection if it becomes necessary to reject the takeoff.

Finally, the Recommended Brake Cooling Schedule provides the only means of evaluating brake cooling requirements following repeated landings at short time intervals or a rejected takeoff.

5.5.6. Advisory Landing Distance Information

Advisory information includes autobrake landing distance, slippery runway landing distance and landing distance for non-normal configurations that affect the landing performance of the airplane. Advisory landing distance calculations assume that the airplane crosses the runway threshold at a height of 50 feet and include an air distance allowance of at least 305 meters to touchdown. The use of maximum available reverse thrust is assumed during the landing ground roll, when stopping performance is based on maximum manual wheel braking.

It is important to remember that the published advisory landing distances are generally actual distances with no additional margins of conservatism.

For JAA operators, the published advisory landing distance for slippery runways is the actual distance increased by 15 percent.

The autobrake system commands a constant airplane deceleration during the landing ground roll for improved passenger comfort. Autobrake landing distance is provided in the Quick Reference Handbook as advisory information to assist in the selection of the most suitable autobrake setting for a given available field length.

The Quick Reference Handbook tables provide actual landing distance on a dry runway from a 50 foot threshold height to a full stop, based on autobrake setting and approach speed. These distances are valid with or without reverse thrust and for all landing flap settings.

A second table provides autobrake landing distance adjustments for altitude, wind and temperature deviation from standard day. These adjustments are ignored when the first table is entered with ground speed.

When autoland is used in conjunction with autobrakes, touchdown should be assumed to occur 760 meters from the runway threshold. The increased air distance allowance can be accounted for by adjusting the advisory landing distance read from the QRH autobrake landing distance tables.

Landing distance on slippery and contaminated runways is affected by the reduced friction between the airplane's tires and the runway surface. This reduced friction results in degraded wheel braking effectiveness and longer landing distances when using manual braking, or when using autobrakes.

Slippery runway landing distance is included in the Quick Reference Handbook as advisory information to assist flight crews in evaluating the expected performance of the airplane when landing on slippery runways. The QRH information is provided for one landing flap setting, and for pilot reported braking actions of good, medium and poor. The first slippery runway landing distance table is entered with reported braking action and wheel braking configuration to determine reference landing distance required.

A second table provides landing distance adjustments for weight, airport pressure altitude, wind, approach speed, runway slope and available reverse thrust. These adjustments are added as necessary to the reference slippery runway landing distance read from the first table.

It is important to remember that the autobrake system may not be capable of producing target decelerations on slippery runways due to the reduced wheel braking effectiveness.

The non-normal configuration landing distance advisory information can be used to determine actual landing distance requirements for non-normal configurations and system failures that affect the landing performance of the airplane. Separate Quick Reference Handbook tables are provided for dry runway surface conditions, and for reported braking actions of good, medium and poor. The appropriate table for the reported runway surface condition is entered with non-normal landing configuration to determine reference landing distance required based on the recommended final approach speed and landing flap setting called out in the non-normal checklist. Landing distance adjustments for weight, altitude, wind, slope and approach speed additives are also provided. Remember that the published non-normal configuration landing distances are actual distances and assume a maximum effort stop.

Глава 6. Flight Planning

Refer to OM part A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 7. Weight and Balance

Refer to OM part A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 8. Loading

8.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

8.1. GENERAL

All operations with embarkation and disembarkation of passengers, baggage, cargo and mail are organized by Company Ground Operations Department or Company representatives in accordance with respective Weight and Balance Control and Loading Manual.

Proper embarkation and disembarkation of baggage, cargo and mail should be supervised by assigned cabin crew member.

Loading of an aeroplane is to be in accordance with the aeroplane's weight and balance limits.

For detailed loading instructions refer to WEIGHT AND BALANCE CONTROL AND LOADING MANUAL.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 9. MEL CDL

Refer to OM part A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 10. Aircraft Systems

For information about aircraft systems, refer to FCOM Systems Description.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 11. Emergency Equipment

For complete information about emergency equipment, refer to FCOM Chapter 1 - Airplane General, Emergency Equipment, Doors, Windows and OM PART A.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 12. Evacuation Procedure

12.0. СОДЕРЖАНИЕ

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

12.1. EVACUATION

A successful outcome for an emergency situation depends, first of all, upon each crew member's perfect knowledge and execution of the duties assigned to him.

The captain should check frequently that all crew members know exactly their assigned positions and their specific duties, as well as the duties of the other crew members, in case of an abnormal or an emergency condition.

Since it is not possible to cover all the situations that may occur, the captain will be responsible for adapting the following instructions to obtain the best coordination of the emergency operation.

Should it be physically impossible for the captain to carry out his duties, another crew member will substitute for him according to the chain of command.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

12.2. COMMUNICATIONS

12.2.1. Callouts

There is typical phraseology used by the crew members in case of emergency situations:

PHRASE (PA)	MEANING
PURSER TO THE COCKPIT.	Purser must immediately go to the cockpit.
FINISH PREPARATION, FINISH PREPARATION.	The cockpit crew gives this order approximately at 2000 FT AAL. Should be used if Purser was previously notified about state of emergency and the following emergency landing.
BRACE FOR IMPACT, BRACE FOR IMPACT.	The cockpit crew announces to brace for impact approximately 500 FT AAL. Should be used in case of emergency landing or possible hard landing.
ATTENTION CREW, AT STATIONS, ATTENTION CREW, AT STATIONS.	The cockpit crew makes a short and precise announcement to warn that an emergency evacuation may soon be required. Cabin crews must proceed to their emergency stations. This command may be announced both on the ground and in the air at any time without further explanations. Usually used if time is a factor and there is no time to brief Purser about state of emergency.
EVACUATE, EVACUATE.	The cockpit crew orders an immediate evacuation, and the cabin crew directs passengers to all available exists. DO NOT ANNOUNCE SIDE OF EVACUATION.
CANCEL ALERT.	When the Captain decides that an evacuation is not required, the cockpit crew makes an immediate announcement to this effect.
CONTROLLED DISEMBARKATION.	The cockpit crew announces this command if controlled disembarkation procedure was previously discussed with Purser, passengers and cabin crew are properly notified about further actions.

12.2.2. Captain to Purser Briefing

The flight crew briefing to the Purser should be clear, precise and short, and it should provide the following information:

- nature of the emergency (the situation);
- intentions (landing/ditching, diversion, etc.);
- time available to prepare the cabin (watch synchronization between flight and cabin crew);

- special Instructions (signal for brace position, signal to remain seated if no evacuation is needed, who will inform the passengers and when, re-sitting etc).

The purser should repeat all the briefing items to be sure that everything was understood in the right way.

If time permits, it’s strongly recommended to the Captain to be the first person who announces about planning emergency landing to the passengers.

If the aircraft has the malfunction significantly deteriorating it’s stability or controllability characteristics, it highly recommended

12.3. PLANNED GROUND OR WATER EVACUATION

If the crew expect evacuation after landing or ditching, the following flow should be applied.

CAPTAIN TO PURSER BRIEFING
Refer to FOM Part B, Chapter 12, 12.2.2 Captain to Purser Briefing.

⇩

PASSENGER ANNOUNCMENT
Preferably should be done by Captain. If time is limited, may be performed by Purser.

⇩

AT 2000 FT AAL
FINISH PREPARATION, FINISH PREPARATION.

⇩

500 FT AAL
BRACE FOR IMPACT, BRACE FOR IMPACT.

⇩

AFTER COMPLETION OF EMER EVAC C/L

⇩ ⇩

IF EVACUATION REQUIRED	IF EVACUATION IS NOT REQUIRED
EVACUATE, EVACUATE.	CANCEL ALERT.

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12.4. UNPLANNED GROUND OR WATER EVACUATION

If time is limited and there is no possibility to prepare the cabin for planned evacuation, the following flow should be applied.

WHEN DECISION ABOUT EMERG LANDING HAS BEEN MADE
ATTENTION CREW, AT STATIONS, ATTENTION CREW, AT STATIONS. Preferably should be done in the air. If not possible – should be accomplished on the ground when aircraft stopped.

⇩

500 FT AAL
BRACE FOR IMPACT, BRACE FOR IMPACT.

⇩

AFTER COMPLETION OF EMER EVAC C/L

⇩ ⇩

IF EVACUATION REQUIRED	IF EVACUATION IS NOT REQUIRED
EVACUATE, EVACUATE.	CANCEL ALERT.

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

12.5. CONTROLLED DISEMBARKATION

If evacuation is not required, but the passengers should be disembarked with escape slides, the following flow should be applied.

AIRCRAFT PREPARATION
The aircraft must be prepared for the procedure: – both engines are switched off; – APU is started (if available). DO NOT START THE APU IF ACFT IS DAMAGED; – SHUTDOWN CHECKLIST has been done; – ATC is notified (if possible).

⇩

CAPTAIN TO PURSER BRIEFING
The Captain should cooperate with the Purser and determine the details of controlled disembarkation procedure (which sides, which exits may be used).

⇩

PASSENGER ANNOUNCEMENT
Preferably should be done by Captain. If required, may be implemented by Purser.

⇩

PURSER REPORT
When cabin is ready for controlled disembarkation procedure, the Purser must notify the Captain.

⇩

INITIATION OF PROCEDURE
CONTROLLED DISEMBARKATION *The Captain should announce via PA.*

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

12.6. EMERGENCY EVACUATION PROCEDURE

12.6.1. Evacuation Checklist

The Evacuation checklist implementation is requested by Captain.

It may be initiated by FO if Captain can’t implement his/her responsibilities.

The flight crew will keep in mind that as long as the evacuation order is not triggered, the flight crew may defer or cancel the passenger evacuation. As soon as the evacuation order is triggered, this decision is irreversible.

12.6.2. Cockpit-Assigned Duties for Evacuation

- **If it is POSSIBLE to reach the passenger cabin:**

CAPTAIN	– **Is the last person to leave the cockpit**. Proceeds to the cabin, and helps with passenger evacuation, as necessary. – **Is the last person to leave the aircraft**. Checks that all persons have evacuated the aircraft. – Evacuates the aircraft, via the rear door, or any other available exit, if he/she cannot reach the rear door. – On ground, he/she takes command of operations until rescue units arrive.
FIRST OFFICER	– Proceeds to the cabin, and takes the emergency equipment. – Evacuates the aircraft, using any available exit. – Helps passengers on ground, and directs them away from the aircraft.

- **If it is NOT POSSIBLE to reach the passenger cabin:**

The cockpit crew should evacuate the aircraft via the cockpit clearview windows, by using the escape ropes.

On ground, each crewmember must help passengers, and direct them away from the aircraft.

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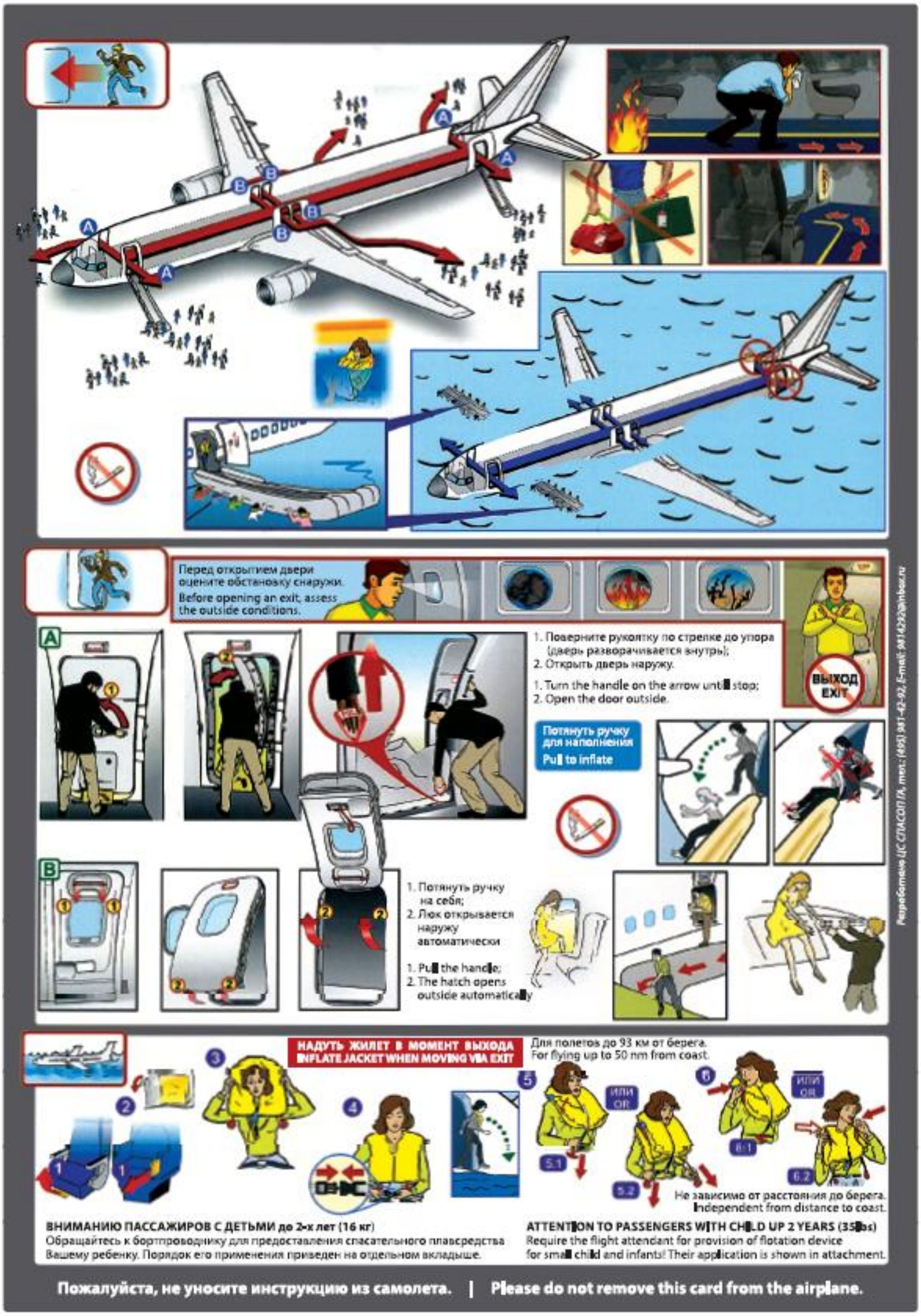
12.7. PASSENGER BRIEFING CARD



Перед открытием двери оцените обстановку снаружи.
Before opening an exit, assess the outside conditions.
A
1. Поверните рукоятку по стрелке до упора (дверь разворачивается внутрь);
2. Открыть дверь наружу.
1. Turn the handle on the arrow until stop;
2. Open the door outside.
ВЫХОД EXIT
Потянуть ручку для наполнения
Pull to inflate
B
1. Потянуть ручку на себя;
2. Люк открывается наружу автоматически
1. Pull the handle;
2. The hatch opens outside automatically
НАДУТЬ ЖИЛЕТ В МОМЕНТ ВЫХОДА
INFLATE JACKET WHEN MOVING VIA EXIT
Для полетов до 93 км от берега.
For flying up to 50 nm from coast.
ИЛИ OR
Не зависимо от расстояния до берега.
Independent from distance to coast.
ВНИМАНИЮ ПАССАЖИРОВ С ДЕТЬМИ до 2-х лет (16 кг)
Обращайтесь к бортпроводнику для предоставления спасательного плавсредства Вашему ребенку. Порядок его применения приведен на отдельном вкладыше.
ATTENTION TO PASSENGERS WITH CHILD UP 2 YEARS (35 lbs)
Require the flight attendant for provision of flotation device for small child and infants! Their application is shown in attachment.
Пожалуйста, не уносите инструкцию из самолета. | Please do not remove this card from the airplane.
Разработано ЦС СПАСОП ГА, тел.: (495) 981-42-92, E-mail: 9814292@inbox.ru

12.8. EVACUATION IN THE EVENT OF A WATER LANDING

After ditching cabin crew starts Evacuation just after aircraft stop without command from the cockpit.

Ditching is an emergency landing on water. Complete knowledge of procedures and equipment is necessary. All loose objects must be secured.

Passengers should be instructed to put on life vests and then follow inflation instructions after leaving the airplane.

When the aircraft has come to a complete stop, cabin crew must start emergency evacuation without any delay.

WARNING:

- **Do not inflate the life vest prior to exiting the airplane. Passenger inflation of the life vest before exiting the airplane may prevent the wearer from being able to leave the airplane.**
- **Before opening the door, verify that the actual water level is below the door sill. If the door is unusable, the slide/raft may be re-positioned at another door and manually deployed.**

DITCHING ATTITUDE

Ditching studies for the 737 airplanes indicate that, with an optimum center of gravity and normal gross weight, the airplane should come to rest slightly nose high in the water:

- the forward doors should be over 1,5 meters above the water
- the aft doors approximately 1,0 meters above the water

UNPLANNED DITCHING

In an unplanned (heavy gross weight) ditching, such as a rejected or aborted takeoff or soon after takeoff, the airplane may come to rest in a tail low attitude with:

- the forward doors approximately 1,6 meters above the water;
- the aft doors approximately 0,5 meters above the water.

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Глава 13. Ground – Air Visual Signals

Refer to OM part A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ

Глава 14. Co-operation between the Cockpit and Cabin crew

Refer to OM part A

СТРАНИЦА НАМЕРЕННО ОСТАВЛЕНА ПУСТОЙ